Network Theory

Detailed Solutions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Network Theory (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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SOLUTIONS

Basic Concepts of Networks

Topics

1. KCL and KVL
2. Source Transformation
3. Network Reduction

Unit Snapshot*

Stream: EC	**Questions:** 31	**MCQ:** 23	**MSQ:** 0	**NAT:** 8
1M: 24	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–6M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT | 2026 | 1M | Ans: 1.00 | ● ○ ○

Since

$$\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega} = \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega},$$

the bridge is balanced.

Use superposition. With only the 6 V source acting (the 1 mA source open), the two equal arms share the 6 V identically, so the top and bottom terminals are at the same potential and this source adds nothing to V_o. (The two midpoints themselves are not at the same potential: the battery keeps them 6 V apart and a current does flow through it.)

With only the 1 mA source acting (the 6 V source replaced by a short), the balanced bridge puts both midpoints at the same potential, so no current flows in that short and it can be removed:

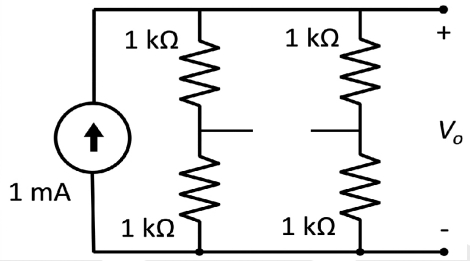


Fig. Solution Q1.26.1

Hence the 1 mA source current divides equally between the two identical branches:

Therefore, the output voltage is

$$V_o = (0.5\,\text{mA})(2\,\text{k}\Omega) = 1\,\text{V}.$$

1.00

Q1.25.1 MCQ | 2025 | 1M | Ans: A | ● ○ ○

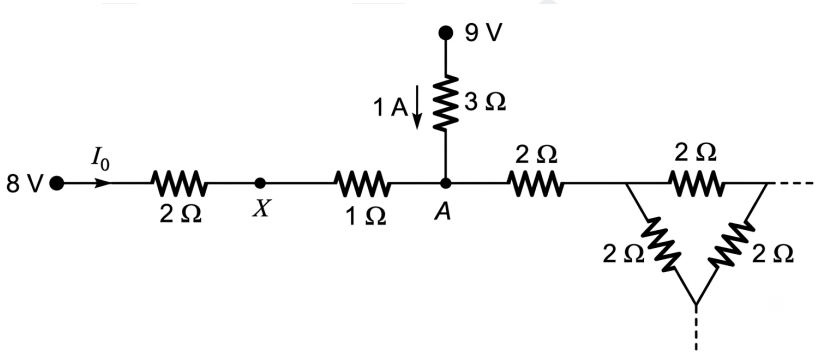


Fig. Solution Q1.25.1

$$V_A = 9 - (1)(3) = 6\,\text{V}$$

The current flowing from the 8 V node to node A is

$$I_o = \frac{8-6}{2+1} = \frac{2}{3}\,\text{A}$$

Therefore, the voltage at node X is

$$V_X = 8 - 2\left(\frac{2}{3}\right) = \frac{20}{3}\,\text{V}$$

Hence, the correct answer is

A. $\frac{20}{3}$

Q1.24.1 NAT | 2024 | 1M | Ans: 2 | ● ● ○

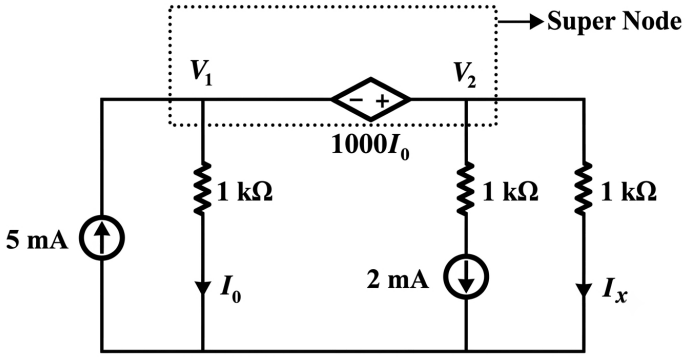


Fig. Solution Q1.24.1

Applying KCL to the supernode,

$$\frac{V_1}{1\,\text{k}\Omega} + \frac{V_2}{1\,\text{k}\Omega} + 2\,\text{mA} - 5\,\text{mA} = 0$$

$$V_1 + V_2 = 3 \qquad \cdots (1)$$

Since

$$I_0 = \frac{V_1}{1\,\text{k}\Omega},$$

the dependent source gives

$$V_2 - V_1 = 1000I_0 = 1000\left(\frac{V_1}{1000}\right) = V_1$$

$$V_2 = 2V_1 \qquad \cdots (2)$$

From (1) and (2),

$$V_1 = 1\text{ V}, \qquad V_2 = 2\text{ V}$$

Hence,

$$I_x = \frac{V_2}{1\,\text{k}\Omega} = \frac{2}{1000} = 2\text{ mA}$$

2

Key Points

- A **supermesh** is formed when a current source lies on the common branch between two adjacent meshes. Since the voltage across a current source is unknown, KVL cannot be applied to the individual meshes.
- To identify a supermesh, look for a current source shared by two neighboring meshes. Combine those meshes into a single larger loop excluding the current-source branch, apply KVL around the outer perimeter, and use the current-source relation as an additional constraint equation.
- A **supernode** is formed when a voltage source lies between two non-reference nodes. Since the current through a voltage source is unknown, KCL cannot be applied directly at the individual nodes.
- To identify a supernode, look for a voltage source connecting two non-reference nodes. Treat the two nodes and the

voltage source as a single extended node, apply KCL to the entire supernode, and use the voltage-source relation as an additional constraint equation.

Q1.23.1 NAT | 2023 | 1M | Ans: 1.36 | ● ● ○

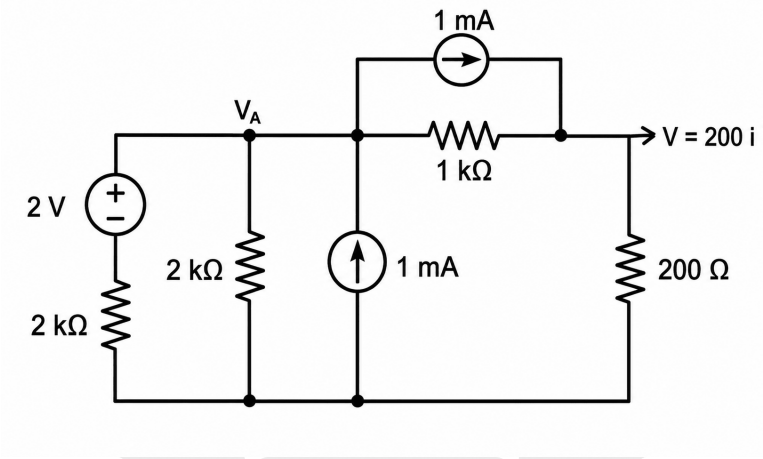


Fig. Solution Q1.23.1

Applying KCL at node V_A,

$$\frac{V_A - 2}{2\,\text{k}\Omega} + \frac{V_A}{2\,\text{k}\Omega} + i = 1\,\text{mA}$$

Multiplying by 2000,

$$V_A - 2 + V_A + 2000i = 2 \;\Rightarrow\; 2V_A + 2000i = 4$$

$$V_A + 1000i = 2 \qquad \cdots (1)$$

Applying KCL at the right node V,

$$\frac{V_A - 200i}{1\,\text{k}\Omega} + 1\,\text{mA} = i \;\Rightarrow\; V_A - 200i + 1 = 1000i$$

$$V_A + 1 = 1200i \qquad \cdots (2)$$

Substituting (1) into (2),

$$2 - 1000i + 1 = 1200i$$

$$i = \frac{3}{2200} = 1.364 \times 10^{-3}\ \text{A}$$

$$\boxed{1.36}$$

Q1.22.1 MCQ | 2022 | 1M | Ans: B | ● ○ ○

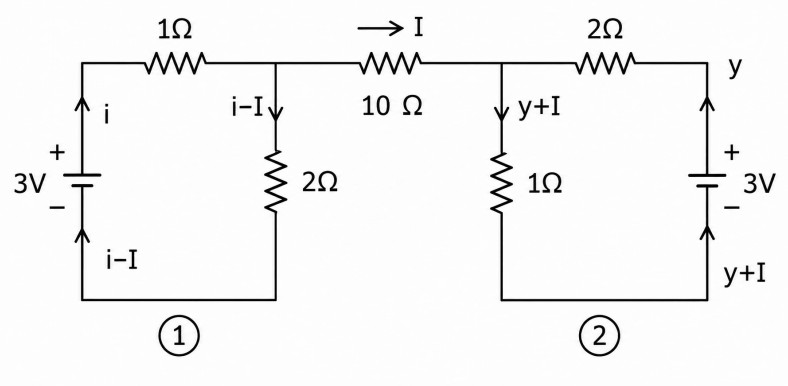


Fig. Solution Q1.22.1

Observe that the 10Ω resistor is the *only* connection between the left and right subnetworks.
Assume a current I flows through the 10Ω resistor from left to right.
For the left loop, applying KCL at the node connected to the 10Ω resistor,

$$i = i - I \Rightarrow I = 0$$

Similarly, for the right loop,

$$y = y + I \Rightarrow I = 0$$

Physically, current must flow in a closed path. Since there is no return path between the left and right subnetworks other than the 10Ω resistor itself, no current can flow through the 10Ω resistor.
Therefore, the correct answer is

B. 0 A

Q1.21.1 NAT 2021 1M Ans: 1 ● ○ ○

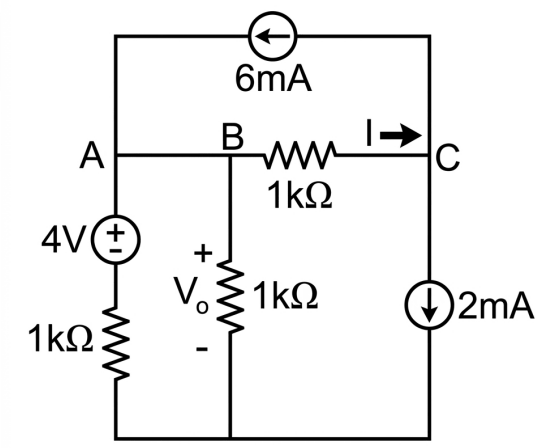


Fig. Solution Q1.21.1

Consider current I through the 1 kΩ resistor between nodes B and C.
Since nodes A and B are directly connected,

$$V_A = V_B = V_0$$

Applying KCL at node C,

$$I = 6 + 2 = 8 \text{ mA}$$

Applying KCL at node B,

$$\frac{V_0 - 4}{1} + \frac{V_0}{1} + I = 6$$

$$2V_0 + I = 10$$

Substituting $I = 8$,

$$2V_0 + 8 = 10 \Rightarrow V_0 = 1$$

$$\boxed{1}$$

Q1.21.2 NAT | 2021 | 1M | Ans: 0.5 | ●●○

Rearrange the circuit to understand it properly;

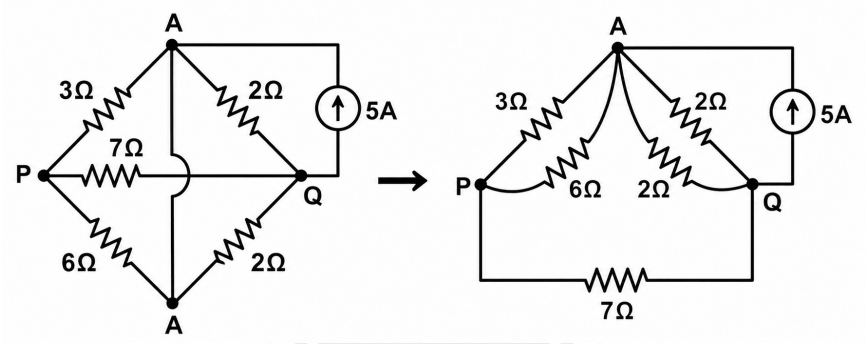


Fig. Solution Q1.21.2

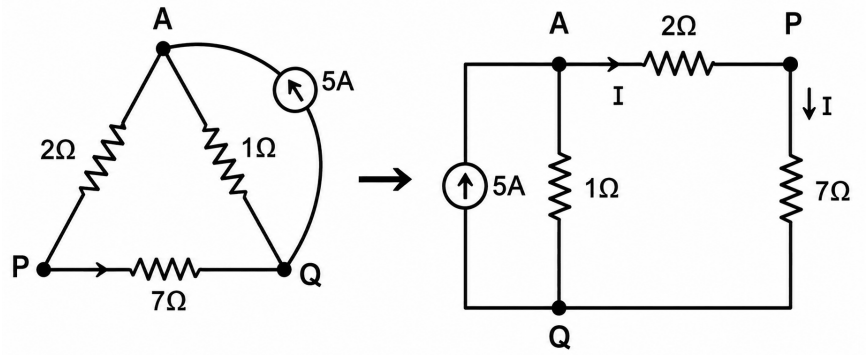


Fig. Solution Q1.21.2

Hence, using current division;

$$I = 5\left(\frac{1}{1+9}\right) = 0.5 \text{ A}.$$

Therefore,

$$\boxed{0.5}$$

Key Points

- **Voltage division rule:** If two resistors R_1 and R_2 are connected in series across a voltage source V, then the voltages across R_1 and R_2 are

$$V_{R_1} = V\frac{R_1}{R_1 + R_2}, \qquad V_{R_2} = V\frac{R_2}{R_1 + R_2}$$

- **Current division rule:** If two resistors R_1 and R_2 are connected in parallel across a current source I, then the currents through R_1 and R_2 are

$$I_{R_1} = I\frac{R_2}{R_1 + R_2}, \qquad I_{R_2} = I\frac{R_1}{R_1 + R_2}$$

Q1.17.1 NAT | 2017 | 1M | Ans: 2.14 ● ● ○

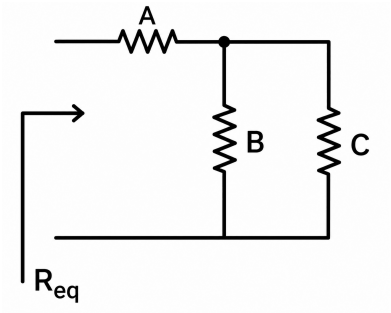


Fig. Solution Q1.17.1

The equivalent resistance is

$$R_{eq} = R_A + \left(\frac{R_B R_C}{R_B + R_C}\right).$$

The three available resistances are

$$10\Omega, \quad 5\Omega, \quad 2\Omega.$$

To maximize R_{eq}, the largest resistance should be placed in series:

$$R_A = 10\Omega, \qquad R_B = 5\Omega, \qquad R_C = 2\Omega.$$

Hence,

$$R_{eq(\max)} = 10 + \frac{5 \times 2}{5 + 2} = 10 + \frac{10}{7} = \frac{80}{7}\Omega.$$

To minimize R_{eq}, the smallest resistance should be placed in series:

$$R_A = 2\Omega, \qquad R_B = 10\Omega, \qquad R_C = 5\Omega.$$

Thus,

$$R_{eq(\min)} = 2 + \frac{10 \times 5}{10 + 5} = 2 + \frac{10}{3} = \frac{16}{3}\Omega.$$

Therefore,

$$\frac{R_{eq(\max)}}{R_{eq(\min)}} = \frac{\frac{80}{7}}{\frac{16}{3}} = \frac{80}{7} \cdot \frac{3}{16} = \frac{15}{7} \approx 2.14.$$

2.14

Q1.16.1 NAT | 2016 | 2M | Ans: 5.0 ● ● ○

Method Mesh Analysis

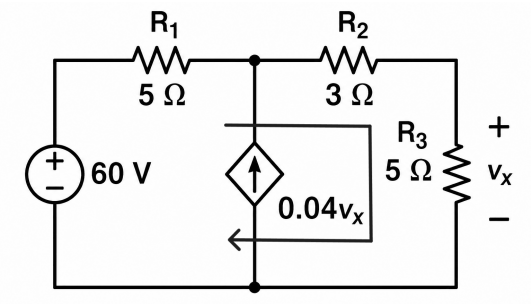


Fig. Solution Q1.16.1

The current through the 5Ω resistor is $\frac{v_x}{5} = 0.2v_x$

Since the dependent current source supplies $0.04v_x$ the current through R_1 is $0.2v_x - 0.04v_x = 0.16v_x$

Applying KVL around the outer loop,

$$-60 + 5(0.16v_x) + 3(0.2v_x) + v_x = 0$$

$$-60 + 0.8v_x + 0.6v_x + v_x = 0 \Rightarrow v_x = 25 \text{ V}$$

The current through R_2 is the same as the current through the 5Ω resistor:

$$I_{R_2} = \frac{v_x}{5} = \frac{25}{5} = 5 \text{ A}$$

Method Nodal Analysis

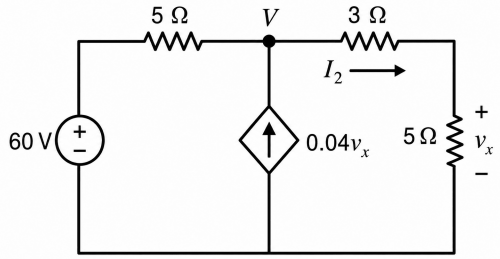


Fig. Solution Q1.16.1

Applying KCL at node V,

$$\frac{V - 60}{5} - 0.04v_x + \frac{V}{3+5} = 0.$$

Multiplying by 40,

$$8(V - 60) - 1.6v_x + 5V = 0$$

$$13V - 1.6v_x = 480 \quad \cdots (1)$$

Since the 3Ω and 5Ω resistors are in series,

$$I_{R_2} = \frac{V}{8} = \frac{v_x}{5}$$

Hence,

$$V = 1.6v_x \quad \cdots (2)$$

Substituting (2) into (1),

$$13(1.6v_x) - 1.6v_x = 480 \Rightarrow v_x = 25 \text{ V}$$

Therefore,

$$I_{R_2} = \frac{v_x}{5} = 5 \text{ A}.$$

5.0

Q1.15.1 NAT 2015 1M Ans: 8.00 ● ○ ○

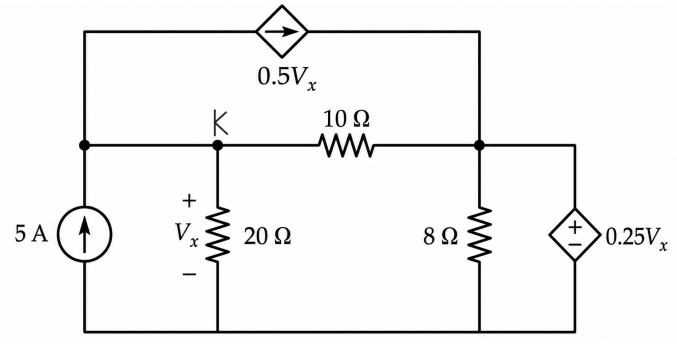


Fig. Solution Q1.15.1

Applying KCL at node K,

$$-5 + \frac{V_x}{20} + \frac{V_x - 0.25V_x}{10} + 0.5V_x = 0$$

$$-5 + \frac{V_x}{20} + \frac{0.75V_x}{10} + \frac{0.5V_x}{1} = 0$$

$$5 = \frac{12.5V_x}{20} \Rightarrow V_x = 8V$$

8.00

Q1.15.2 MCQ 2015 1M Ans: A ● ○ ○

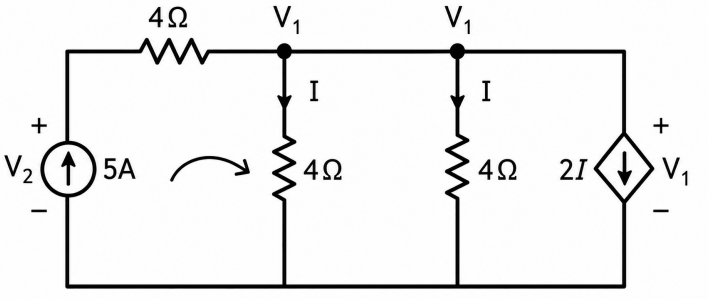


Fig. Solution Q1.15.2

Applying KCL at the node V_1,

$$5 = I + I + 2I \Rightarrow I = \frac{5}{4}\text{ A}$$

Voltage across the parallel branches is;

$$V_1 = 4I = 5\text{ V}$$

Applying KVL in the left loop,

$$V_2 = (5)(4) + V_1 = 25\text{ V}$$

Hence, the correct answer is

A. 5 V, 25 V

Q1.14.1 MCQ | 2014 | 1M | Ans: C | ● ○ ○

The dependent source has output current

$$I_o = A_I I_i$$

where I_i is the controlling variable.
Thus,
i) The **output quantity** is a current (I_o).
ii) The **controlling quantity** is also a current (I_i).
Hence, the dependent source is a

Current Controlled Current Source (CCCS).

Therefore, the correct answer is

C. Current Controlled Current Source

Key Points

- **Dependent Sources Summary**

Source Type	Output	Controlled By	Equation
CCCS	Current	Current	$I_o = \beta I_i$
VCCS	Current	Voltage	$I_o = g_m V_i$
CCVS	Voltage	Current	$V_o = r_m I_i$
VCVS	Voltage	Voltage	$V_o = A_v V_i$

Q1.14.2 MCQ | 2014 | 1M | Ans: A | ● ○ ○

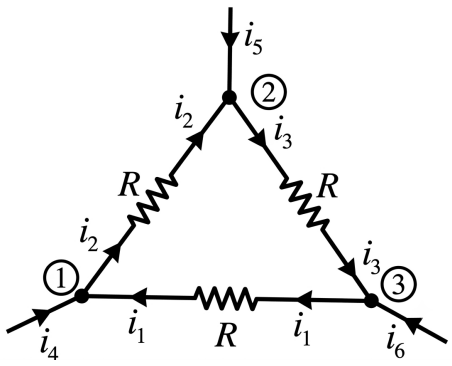


Fig. Solution Q1.14.2

Given,

$$i_1 = 2 \text{ A}, \qquad i_4 = -1 \text{ A}, \qquad i_5 = -4 \text{ A}.$$

Applying KCL at node 1,

$$-i_1 - i_4 + i_2 = 0$$

$$i_2 = 2 + (-1) = 1 \text{ A}.$$

Applying KCL at node 2,

$$-i_2 - i_5 + i_3 = 0$$

$$i_3 = 1 + (-4) = -3 \text{ A}.$$

Applying KCL at node 3,

$$i_1 - i_3 - i_6 = 0$$

$$i_6 = 2 - (-3) = 5 \text{ A}$$

Hence, the correct answer is

A. $i_6 = 5$ A

Q1.13.1 MCQ 2013 2M Ans: D ● ○ ○

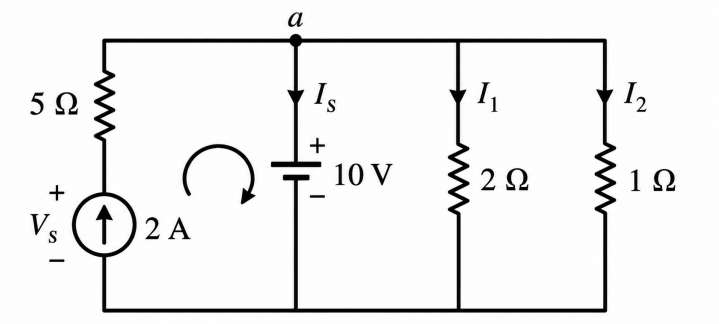


Fig. Solution Q1.13.1

$$I_1 = \frac{10}{2} = 5 \text{ A}, \qquad I_2 = \frac{10}{1} = 10 \text{ A}.$$

Applying KCL at node a,

$$I_s + I_1 + I_2 - 2 = 0$$

$$I_s + 5 + 10 - 2 = 0 \Rightarrow I_s = -13 \text{ A}$$

Applying KVL in left loop,

$$-V_s + 5(2) + 10 = 0 \Rightarrow V_s = 20 \text{ V}.$$

Hence, the correct answer is;

D. − 13, 20

Q1.13.2 MCQ 2013 2M Ans: C ● ○ ○

The 10 V source is connected in parallel with the 1Ω resistor.

Using Ohm's law,

$$I_2 = \frac{V}{R} = \frac{10}{1} = 10 \text{ A}.$$

Therefore, the correct answer is

C. 10

Q1.13.3 MCQ 2013 2M Ans: C ● ● ○

Given, $C_1 = 10\mu F,\ V_{10} = 10V, \qquad C_2 = 5\mu F,\ V_2 = 5V, \qquad C_3 = 2\mu F,\ V_{30} = 2V$

Method Using Maximum Charge Rating

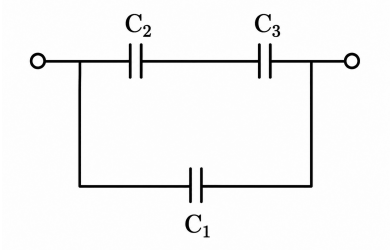


Fig. Solution Q1.13.3

The maximum charge that each capacitor can safely store is;

$$Q_{1(\max)} = C_1 V_{10} = 10 \times 10 = 100\mu C$$

$$Q_{2(\max)} = C_2 V_{20} = 5 \times 5 = 25\mu C$$

$$Q_{3(\max)} = C_3 V_{30} = 2 \times 2 = 4\mu C$$

Since C_2 and C_3 are connected in series, the charge on both capacitors must be the same.
Therefore, the maximum permissible series charge is limited by the smaller value:

$$Q_{23} = 4\mu C$$

The equivalent capacitance of the series combination is

$$C_{23} = \frac{C_2 C_3}{C_2 + C_3} = \frac{5 \times 2}{5 + 2} = \frac{10}{7}\mu F$$

Hence, the maximum voltage across the series branch is

$$V_{eq} = V_{23} = \frac{Q_{23}}{C_{23}} = \frac{4}{10/7} = 2.8V$$

Since C_1 is connected in parallel with the series branch, it also has the same voltage across it:

$$V_{C_1} = 2.8V$$

Therefore,

$$Q_1 = C_1 V_{C_1} = 10 \times 2.8 = 28\mu C$$

Total charge stored across the terminals is

$$Q_T = Q_1 + Q_{23} = 28 + 4 = 32\mu C$$

Thus,

$$V_{\max} = 2.8V, \qquad Q_T = 32\mu C$$

Method Using Voltage Division

Since C_2 and C_3 are in series, capacitor C_3 reaches its breakdown voltage first because it has the smallest voltage rating.

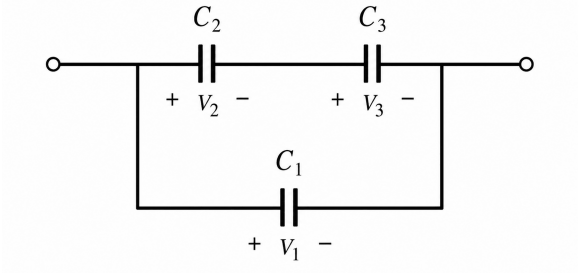


Fig. Solution Q1.13.3

Let the maximum safe voltage across the terminals be V_1.
Using voltage division in series capacitors,

$$V_3 = \frac{C_2}{C_2 + C_3} V_1$$

At the limiting condition, $V_3 = 2V$
Hence,

$$2 = \frac{5}{5 + 2} V_1 \Rightarrow V_1 = 2.8V$$

The equivalent capacitance across the terminals is

$$C_{eq} = C_1 + \frac{C_2 C_3}{C_2 + C_3}$$

$$C_{eq} = 10 + \frac{5 \times 2}{5 + 2} = 10 + \frac{10}{7} = \frac{80}{7}\mu F$$

Therefore, the total charge stored is

$$Q = C_{eq} V_1$$

$$Q = \frac{80}{7} \times 2.8 = 32\mu C$$

Thus,

$$V_{max} = 2.8V, \qquad Q = 32\mu C$$

Hence, the correct answer is

C. 2.8 V and 32 μC

Q1.09.1 MCQ | 2009 | 1M | Ans: C | ● ○ ○

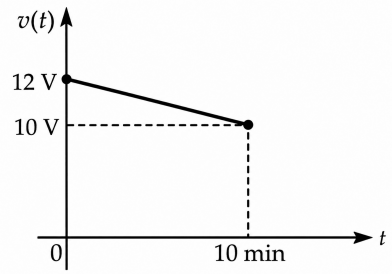


Fig. Solution Q1.09.1

The power delivered by the battery is

$$p(t) = v(t)\, i(t)$$

Given that the current is constant,

$$i(t) = 2 \text{ A}$$

and the voltage decreases linearly from 12 V to 10 V in

$$T = 10 \text{ min} = 600 \text{ s}.$$

The energy delivered by the battery is

$$E = \int_0^T p(t)\, dt = \int_0^{600} v(t) i(t)\, dt$$

$$E = 2 \int_0^{600} v(t)\, dt$$

The integral of voltage with respect to time equals the area under the v-t graph.
The area under the graph consists of:

$$A_1 = 10 \times 600 = 6000, \qquad A_2 = \frac{1}{2}(12 - 10) \times 600 = 600.$$

Therefore,

$$E = 2(A_1 + A_2) = 13200 \text{ J} = 13.2 \text{ kJ}.$$

Hence, the correct answer is

C. 13.2 kJ

Q1.09.2 MCQ 2009 1M Ans: A ● ○ ○

It is given that the 60 V source is *absorbing* power.

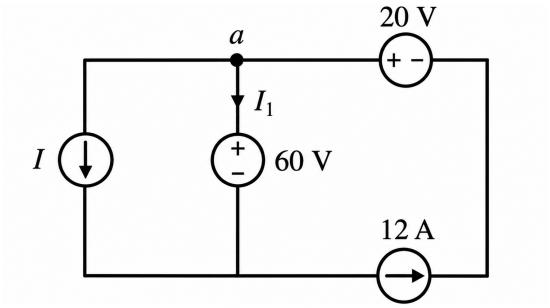


Fig. Solution Q1.09.2

For a voltage source to absorb power, current must enter its positive terminal.
Let the current through the 60 V source be I_1 as shown downward through the source.
Applying KCL at node a,

$$I + I_1 - 12 = 0 \Rightarrow I = 12 - I_1$$

Since the 60 V source is absorbing power,

$$I_1 > 0$$

Therefore,

$$I < 12 \text{ A}$$

Therefore, the correct answer is

A. 10 A

Q1.05.1 MCQ 2005 2M Ans: C ● ○ ○

An ideal voltmeter has infinite resistance.
Therefore, no current flows through the voltmeter branch

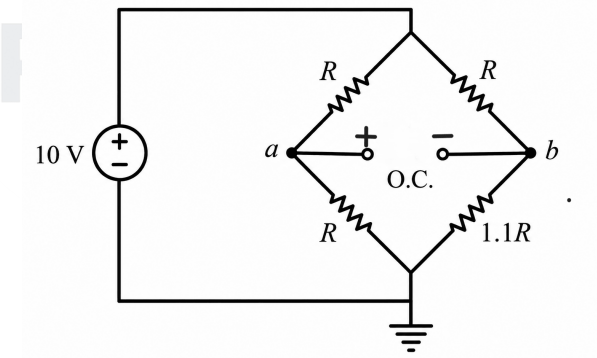


Fig. Solution Q1.05.1

The voltage at node a is obtained using the voltage divider rule:

$$V_a = \frac{R_2}{R_1 + R_2} \times 10$$

Since $R_1 = R_2 = R$;

$$V_a = \frac{R}{R + R} \times 10 = 5 \text{ V}$$

Similarly, the voltage at node b is

$$V_b = \frac{R_3}{R_3 + R_4} \times 10$$

$$V_b = \frac{1.1R}{1.1R + R} \times 10 = \frac{11}{21} \times 10 = 5.238 \text{ V}$$

Therefore,

$$V_{ab} = V_a - V_b$$

$$V_{ab} = 5 - 5.238 = -0.238 \text{ V}$$

The correct answer is

C. -0.238 V

Q1.02.1 MCQ | 2002 | 1M | Ans: A | ● ○ ○

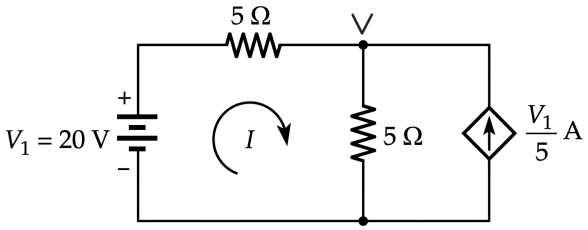


Fig. Solution Q1.02.1

The dependent current source value is

$$I_s = \frac{V_1}{5} = \frac{20}{5} = 4 \text{ A}$$

Applying KCL at node V,

$$\frac{V - 20}{5} + \frac{V}{5} - 4 = 0$$

$$\frac{2V - 20}{5} = 4 \Rightarrow V = 20 \text{ V}$$

Hence, the power associated with the dependent current source is

$$P = VI_s = 20 \times 4 = 80 \text{ W}$$

Since the dependent current source current flows upward while the upper terminal is at positive potential. Therefore, the source is delivering power to the circuit.

Hence, the correct answer is

A. delivers 80 W

Key Points

- Power sign convention for a voltage source:

Current Direction	Power
Entering $+ve$ terminal	Absorbed
Leaving $+ve$ terminal	Delivered
Entering $-ve$ terminal	Delivered
Leaving $-ve$ terminal	Absorbed

Q1.01.1 MCQ 2001 1M Ans: C ● ○ ○

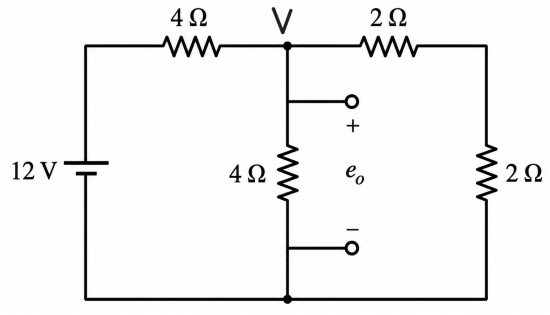


Fig. Solution Q1.01.1

Applying KCL at the node V;

$$\frac{e_0 - 12}{4} + \frac{e_0}{4} + \frac{e_0}{4} = 0$$

Multiplying throughout by 4,

$$e_0 - 12 + e_0 + e_0 = 0 \Rightarrow e_0 = 4 \text{ V}$$

Hence, the correct answer is

C. 4 V

Q1.00.1 MCQ 2000 1M Ans: D ● ○ ○

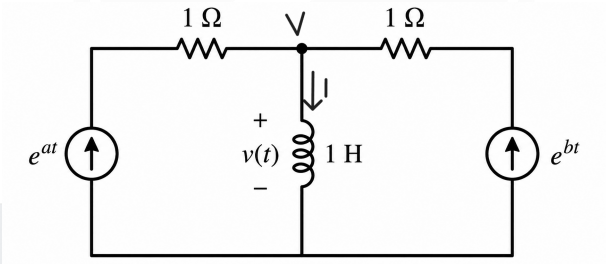


Fig. Solution Q1.00.1

Applying KCL at node V,

$$-I + e^{at} + e^{bt} = 0 \Rightarrow I = e^{at} + e^{bt} \quad (1)$$

The voltage across the inductor is

$$v(t) = L\frac{dI}{dt}$$

Substituting L=1H and I from (1),

$$v(t) = \frac{d}{dt}\left(e^{at} + e^{bt}\right)$$

$$v(t) = ae^{at} + be^{bt}$$

Hence, the correct answer is

D. $ae^{at} + be^{bt}$

Q1.00.2 MCQ 2000 1M Ans: A ● ○ ○

Applying KVL along the right side,

$$0 + 1 + E + 5 + 10 = 0$$

$$\Rightarrow E = -16 \text{ V}$$

Hence, the correct answer is

A. -16 V

Q1.98.1 MCQ 1998 1M Ans: C ● ○ ○

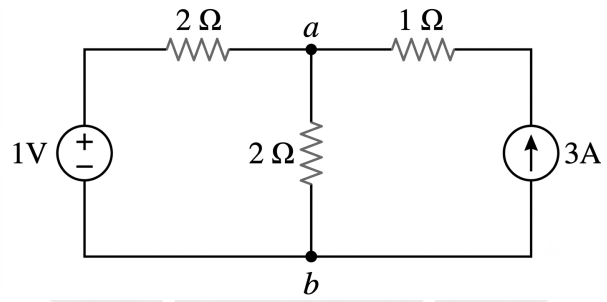


Fig. Solution Q1.98.1

Applying KCL at node a,

$$\frac{V_{ab} - 1}{2} + \frac{V_{ab}}{2} = 3$$

$$V_{ab} = \frac{7}{2} = 3.5 \text{ V}$$

Hence, the correct answer is

C. 3.5 V

Q1.98.2 MCQ 1998 1M Ans: B ● ○ ○

In the nodal method, we first apply Kirchhoff's Current Law (KCL) at the selected nodes. The branch currents appearing in the KCL equations are then expressed in terms of node voltages using Ohm's law.

Thus,

Nodal Method = KCL + Ohm's Law

Hence, the correct answer is

B. KCL and Ohm's law

Q1.97.1 MCQ 1997 1M Ans: A ● ○ ○

The voltage V is measured across the same two nodes across which the ideal 10 V source is connected.

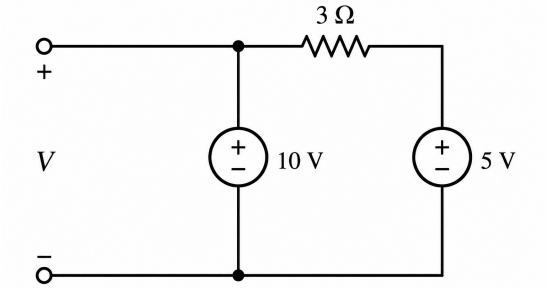


Fig. Solution Q1.97.1

Hence, the correct answer is

A. 10 V

Q1.97.2 MCQ 1997 1M Ans: D ● ● ○

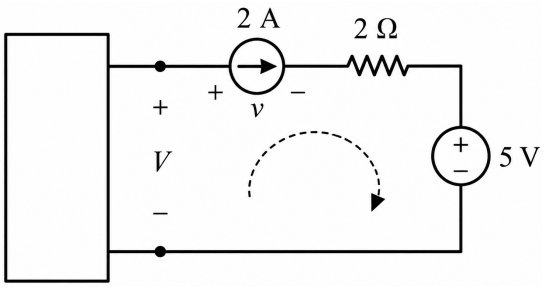


Fig. Solution Q1.97.2

Applying KVL around the loop,

$$-V + v + (2)(2) + 5 = 0 \quad \Rightarrow \quad V = 9 + v$$

The voltage v across the ideal current source is not specified and cannot be determined.
Since v is unknown,V cannot be evaluated uniquely.
Hence, the correct answer is

D. None of these

Mistakes to be avoided

- Many students forget to take voltage drop across current source and evaluate wrong answer.

Q1.97.3 MCQ 1997 1M Ans: A ● ○ ○

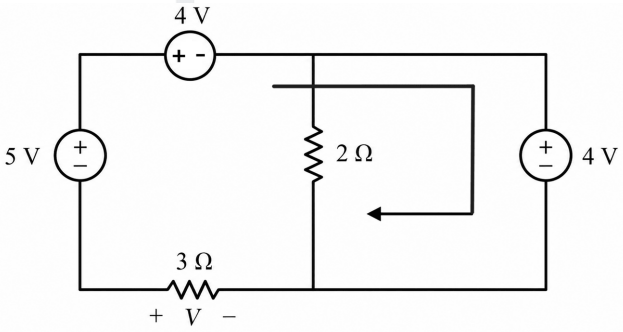


Fig. Solution Q1.97.3

Applying KVL around the loop shown,

$$-V - 5 + 4 + 4 = 0$$

$$\Rightarrow V = 3 \text{ V}$$

Hence, the correct answer is

A. 3 V

Q1.97.4 MCQ 1997 1M Ans: B ● ○ ○

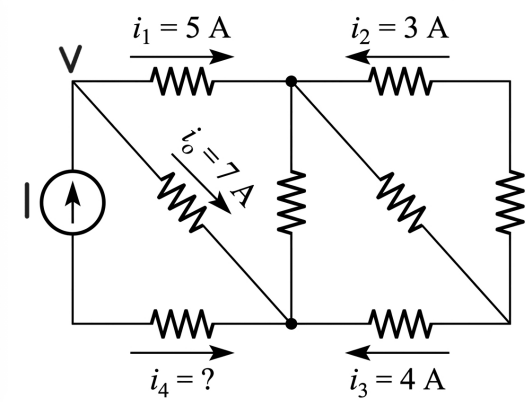


Fig. Solution Q1.97.4

Applying KCL at node V,

$$I = i_1 + i_0 \Rightarrow I = 12 \text{ A}$$

Since,

$$i_4 = -I \Rightarrow i_4 = -12 \text{ A}$$

Hence, the correct answer is

B. -12 A

Q1.93.1 MCQ 1993 2M Ans: A ● ○ ○

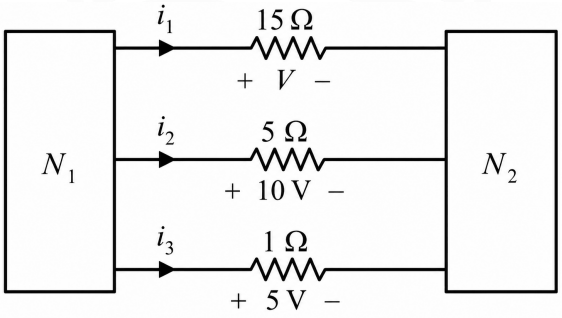


Fig. Solution Q1.93.1

From the given voltages,

$$i_2 = \frac{10}{5} = 2 \text{ A}, \qquad i_3 = \frac{5}{1} = 5 \text{ A}$$

Applying KCL at network N_1,

$$i_1 + i_2 + i_3 = 0$$

$$i_1 = -(2 + 5) \Rightarrow i_1 = -7 \text{ A}$$

Therefore, the voltage across the 15Ω resistor is

$$V = 15\, i_1 = -105 \text{ V}$$

Hence, the correct answer is

A. -105 V

Q1.93.2 MCQ | 1993 | 2M | Ans: D | ● ○ ○

Consider a linear resistive network excited only by ideal voltage sources.

If every resistance is multiplied by a factor k (here k=2), then all branch currents become $I' = \frac{I}{k}$

The voltage across any resistor is therefore

$$V' = I'R' = \left(\frac{I}{k}\right)(kR) = IR = V$$

Hence, the voltage across each resistor remains unchanged.

Hence, the correct answer is

D. not changed

PrepFusion

SOLUTIONS

11

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits — Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at $t = 0^+$, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 48	**MCQ:** 34	**MSQ:** 0	**NAT:** 14
1M: 15	**2M:** 33	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–6M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ | 2026 | 1M | Ans: A | ● ● ● | CLICK HERE FOR VIDEO SOLUTION

Given,

$$C\frac{dV}{dt} = I(t) - m(t), \quad \text{where; } I(t) = q\delta(t)$$

The circuit equivalent to the above equation can be represented as;

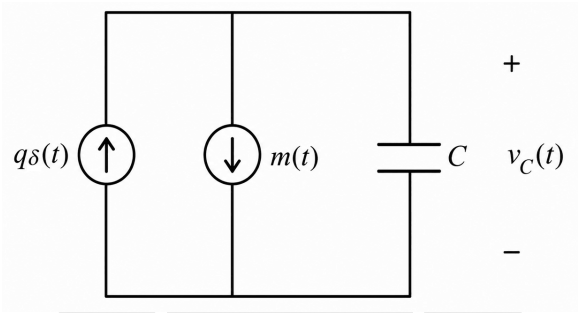


Fig. Solution Q2.26.1

Substituting $I(t) = q\delta(t)$,

$$C\frac{dV}{dt} = q\delta(t) - m(t).$$

Integrating both sides from 0^- to 0^+,

$$\int_{0^-}^{0^+} C\,dV = q\int_{0^-}^{0^+} \delta(t)\,dt - \int_{0^-}^{0^+} m(t)\,dt$$

$$CV(0^+) - CV(0^-) = q - \int_{0^-}^{0^+} m(t)\,dt.$$

Hence,

$$V(0^+) = V_0 + \frac{q}{C} - \frac{1}{C}\int_{0^-}^{0^+} m(t)\,dt \qquad \cdots(1)$$

Now consider the equivalent representation,

$$C\frac{dV}{dt} = -m(t)$$

Integrating from 0^- to 0^+,

$$CV(0^+) - CV(0^-) = -\int_{0^-}^{0^+} m(t)\,dt$$

$$V(0^+) = V(0^-) - \frac{1}{C}\int_{0^-}^{0^+} m(t)\,dt \qquad \cdots(2)$$

Comparing equations (1) and (2),

$$V(0^-) = V_0 + \frac{q}{C}.$$

Thus, the impulse current is absorbed into the initial capacitor voltage, and the equivalent equation becomes

$$C\frac{dV}{dt} = -m(t), \qquad V(0^-) = V_0 + \frac{q}{C}.$$

Hence, the correct answer is

$$\boxed{\textbf{A. } C\frac{dV}{dt} = -m(t), \qquad V(0^-) = V_0 + \frac{q}{C}}$$

Q2.24.1 NAT 2024 1M Ans: 0.75 ● ○ ○

At $t = 0$, the switch is closed.

For an R-L circuit, the time constant is, $\tau = \dfrac{L}{R_{TH}}$.

After closing the switch and deactivating the independent current source (open-circuit), the network seen by the inductor is;

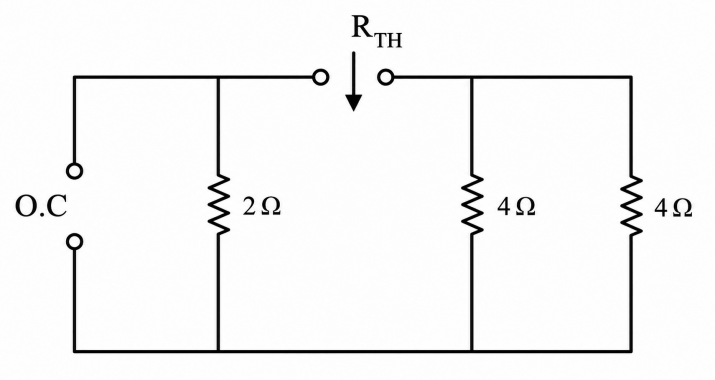


Fig. Solution Q2.24.1

The Thevenin resistance across the inductor is

$$R_{TH} = 2 + (4 \parallel 4)$$

$$R_{TH} = 2 + \frac{4 \times 4}{4 + 4} = 2 + 2 = 4\ \Omega$$

Hence,

$$\tau = \frac{L}{R_{TH}} = \frac{3}{4} = 0.75\ \text{s}$$

Therefore,

$$\boxed{0.75}$$

Key Points

- The **time constant** (τ) is the time required for a first-order response to complete 63.2% of its total change (or reduce to 36.8% of its initial value).

Q2.23.1 NAT 2023 1M Ans: 4 ● ● ○

For $t < 0$, the switch has been closed for a long time. Hence, the inductor behaves as a short circuit.

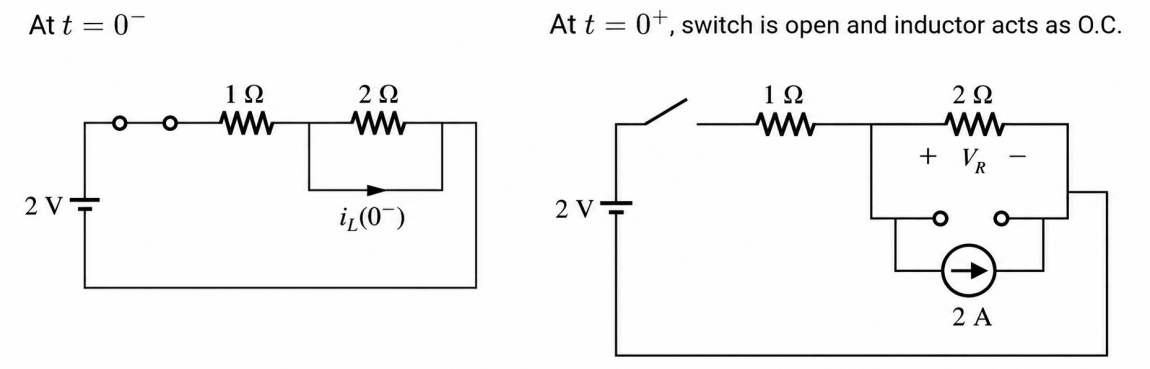


Fig. Solution Q2.23.1

At $t = 0^-$; $2\ \Omega \rightarrow$ short circuit.

$$\Rightarrow i_L(0^-) = \frac{2}{1} = 2\ \text{A}.$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 2\ \text{A}.$$

At $t = 0$, the switch is opened and the source is disconnected. The inductor behaves as 2A current source. Hence, the current through the resistor at $t = 0^+$ is 2A.
Therefore,

$$V_R(0^+) = -2 \times 2 = -4\ \text{V}.$$

Since the response is purely decaying, the maximum magnitude occurs at $t = 0^+$.

$$|V_R|_{\max} = 4\ \text{V}.$$

Hence,

$$\boxed{4}$$

Mistakes to be avoided

- Read question carefully, its asks magnitude only.
- Do not forget to round off the final result as stated in the question.

Q2.23.2 MCQ | 2023 | 2M | Ans: B | ● ● ○

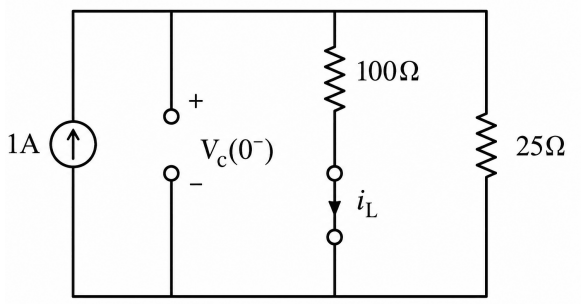


Fig. Solution Q2.23.2

For $t = 0^-$, S_1 is closed and S_2 is open for a long time.

Hence, the capacitor behaves as an open circuit and the inductor behaves as a short circuit.
The inductor current is obtained by current division,

$$i_L(0^-) = \frac{1 \times 25}{100 + 25} = 0.2 \text{ A}$$

The capacitor voltage is

$$v_C(0^-) = (0.2)(100) = 20 \text{ V}$$

At $t = 0^+$, S_1 is opened and S_2 is closed.

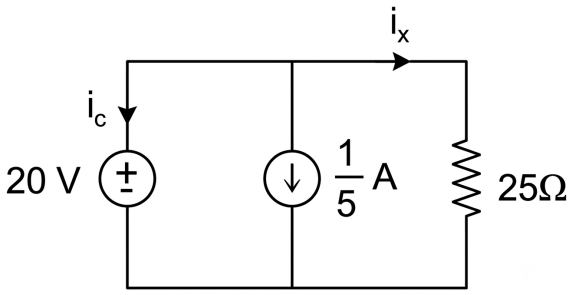


Fig. Solution Q2.23.2

Using the continuity conditions,

$$v_C(0^+) = 20 \text{ V}, \qquad i_L(0^+) = 0.2 \text{ A}$$

The capacitor is therefore equivalent to a 20 V source and the inductor to a 0.2 A current source.
The current through the 25Ω resistor is

$$i_x = \frac{20}{25} = 0.8 \text{ A}$$

Applying KCL,

$$-i_C = i_x + 0.2 = 0.8 + 0.2 = 1$$

$$i_C(0^+) = -1 \text{ A}$$

Hence, the correct answer is

B. -1

Q2.21.1 NAT 2021 2M Ans: 1.44 ● ● ●

Method Laplace Transform

Applying Laplace transform;

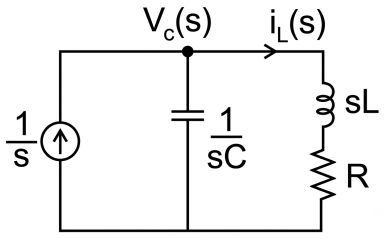


Fig. Solution Q2.21.1

$$\frac{1}{s} = \frac{V_c(s)}{\frac{1}{Cs}} + \frac{V_c(s)}{R + Ls}$$

$$\frac{1}{s} = V_c(s)\left[Cs + \frac{1}{R + Ls}\right] \Rightarrow V_c(s) = \frac{\frac{1}{s}}{Cs + \frac{1}{R + Ls}}$$

$$V_c(s) = \frac{\frac{1}{s}(R + Ls)}{LCs^2 + RCs + 1}$$

The ammeter current is

$$i_L(s) = \frac{V_c(s)}{R + Ls} = \frac{\frac{1}{s}}{LCs^2 + RCs + 1}$$

$$i_L(s) = \frac{\frac{1}{LC}}{s\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

Given,

$$L = 10 \text{ mH}, \qquad C = 100 \text{ pF}, \qquad R = 5 \times 10^3 \ \Omega$$

Comparing the denominator

$$s^2 + \frac{R}{L}s + \frac{1}{LC}$$

with the standard second-order characteristic equation

$$s^2 + 2\xi\omega_n s + \omega_n^2,$$

we get

$$2\xi\omega_n = \frac{R}{L}, \qquad \omega_n^2 = \frac{1}{LC}.$$

Hence,

$$\omega_n = \frac{1}{\sqrt{LC}}, \qquad \xi = \frac{R}{2L\omega_n} = \frac{R}{2}\sqrt{\frac{C}{L}}.$$

Substituting the given values,

$$\xi = \frac{5 \times 10^3}{2}\sqrt{\frac{100 \times 10^{-12}}{10 \times 10^{-3}}} = 0.25.$$

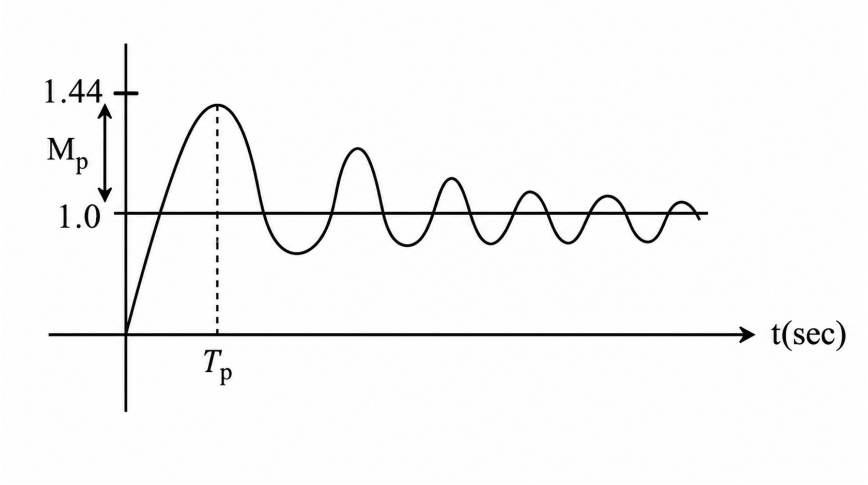


Fig. Solution Q2.21.1

The maximum overshoot is

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}} = 0.44$$

Hence,

$$I_{\max} = I_\infty + M_p = 1 + 0.44 = 1.44 \text{ A}$$

Method Time-Domain Analysis

Applying KCL at the top node,

$$I_s = i_C(t) + i_L(t)$$

Since, $i_C(t) = C\dfrac{dv(t)}{dt}$

and $V_c(t) = L\dfrac{di_L(t)}{dt} + Ri_L(t)$

$$\Rightarrow i_C(t) = LC\frac{d^2i_L}{dt^2} + RC\frac{di_L}{dt}$$

Substituting into KCL,

$$I_s = LC\frac{d^2i_L}{dt^2} + RC\frac{di_L}{dt} + i_L$$

$$\frac{d^2i_L}{dt^2} + \frac{R}{L}\frac{di_L}{dt} + \frac{1}{LC}i_L = \frac{I_s}{LC}$$

Given,

$$I_s = 1 \text{ A}, \quad L = 10 \text{ mH}, \quad C = 100 \text{ pF}, \quad R = 5 \text{ k}\Omega$$

The circuit parameters are

$$\alpha = \frac{R}{2L} = \frac{5000}{2(10\times10^{-3})} = 2.5\times10^5 \text{ rad/s}$$

$$\omega_n = \frac{1}{\sqrt{LC}} = 10^6 \text{ rad/s}$$

Since, $\alpha < \omega_n$ $\Rightarrow$ the response is underdamped.

The damped natural frequency is

$$\omega_d = \sqrt{\omega_n^2 - \alpha^2} = 2.5\times10^5\sqrt{15} \text{ rad/s}$$

Initial conditions are

$$i_L(0) = 0, \qquad \frac{di_L(0)}{dt} = 0$$

The current response is

$$i_L(t) = 1 - e^{-\alpha t}\left[\cos(\omega_d t) + \frac{\alpha}{\omega_d}\sin(\omega_d t)\right]$$

The first maximum occurs at

$$T_p = \frac{\pi}{\omega_d}$$

Hence,

$$i_{L,\max} = 1 + e^{-\alpha\pi/\omega_d}$$

Now,

$$\frac{\alpha}{\omega_d} = \frac{1}{\sqrt{15}}$$

Therefore,

$$i_{L,\max} = 1 + e^{-\pi/\sqrt{15}} = 1 + 0.444 = 1.44 \text{ A}$$

Hence, the required maximum ammeter reading is

$$\boxed{1.44 \text{ A}}$$

Q2.21.2 NAT 2021 2M Ans: 0.139

Step 1: Find the steady-state capacitor voltage

At $t \to \infty$, the capacitor behaves as an open circuit.

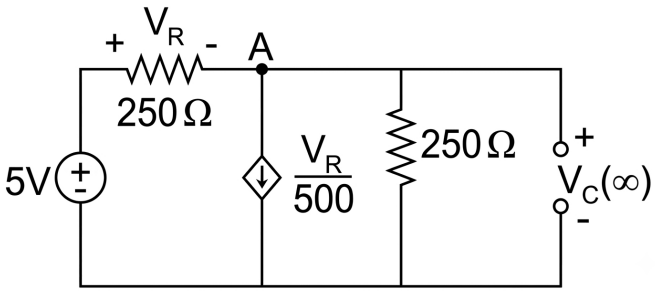


Fig. Solution Q2.21.2

Applying KCL at node A,

$$\frac{V_C(\infty) - 5}{250} + \frac{V_R}{500} + \frac{V_C(\infty)}{250} = 0$$

$$2V_C(\infty) - 10 + V_R + 2V_C(\infty) = 0 \qquad \cdots (i)$$

Now;

$$V_R = -\left(\frac{V_C(\infty) - 5}{250}\right) \times 250$$

$$V_R = -V_C(\infty) + 5 \qquad \cdots (ii)$$

Substituting (ii) into (i),

$$2V_C(\infty) - 10 - V_C(\infty) + 5 + 2V_C(\infty) = 0$$

$$3V_C(\infty) = 5 \Rightarrow V_C(\infty) = \frac{5}{3} \text{ V}$$

Step 2: Find the time constant

Deactivate the independent 5 V source and apply a test voltage V across the capacitor terminals.

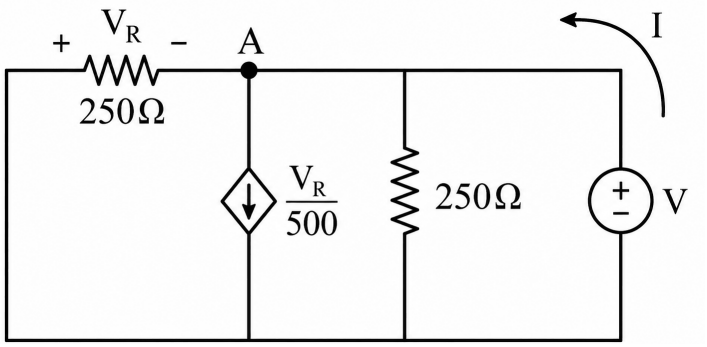


Fig. Solution Q2.21.2

Applying KCL at node A,

$$I = \frac{V}{250} + \frac{V_R}{500} + \frac{V}{250} \qquad \cdots (iii)$$

$$\because V_R = -\left(\frac{V}{250}\right) \times 250 = -V \qquad \cdots (iv)$$

Substituting (iv) into (iii),

$$I = \frac{V}{250} - \frac{V}{500} + \frac{V}{250} = \frac{3V}{500}$$

Hence,

$$R_{th} = \frac{V}{I} = \frac{500}{3}\Omega$$

Therefore,

$$\tau = R_{th}C = \frac{500}{3} \times 0.6 \times 10^{-6} = 10^{-4}\ \text{s}$$

Step 3: Capacitor voltage

The capacitor voltage is

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right] e^{-t/\tau}$$

$$V_C(t) = \frac{5}{3} + \left(-5 - \frac{5}{3}\right) e^{-t/10^{-4}} \Rightarrow V_C(t) = \frac{5}{3} - \frac{20}{3} e^{-t/10^{-4}}$$

At the instant when the capacitor voltage becomes zero,

$$0 = \frac{5}{3} - \frac{20}{3} e^{-t/10^{-4}} \Rightarrow e^{-t/10^{-4}} = \frac{1}{4}$$

$$t = 10^{-4} \ln 4 = 1.386 \times 10^{-4}\ \text{s} = 0.1386\ \text{ms}$$

Hence,

$$\boxed{0.139}$$

Q2.21.3 MCQ | 2021 | 2M | Ans: C | ● ● ○

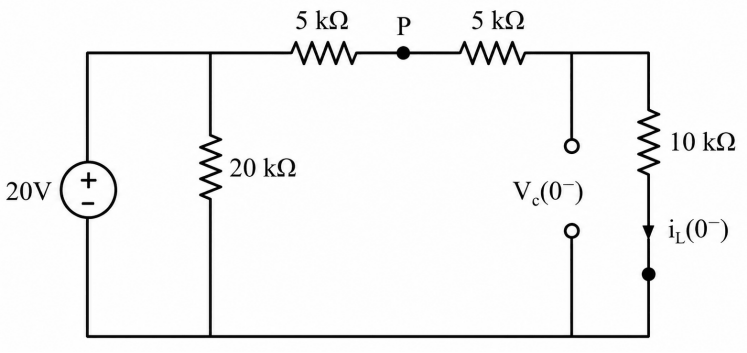


Fig. Solution Q2.21.3

At $t = 0^-$, the switch is at position P for a long time.
Hence,

Capacitor $\rightarrow$ Open circuit, Inductor $\rightarrow$ Short circuit

The inductor current is

$$i_L(0^-) = \frac{20}{20\,\text{k}\Omega} = 1\ \text{mA}$$

Since inductor current is continuous;

$$i_L(0^+) = i_L(0^-) = 1\ \text{mA}$$

Also, the capacitor voltage cannot change instantaneously.
The voltage at the capacitor is obtained by voltage division,

$$V_C(0^-) = 20\left(\frac{10}{5+5+10}\right) = 10\ \text{V}$$

$$\Rightarrow V_C(0^+) = 10\ \text{V}$$

At $t = 0^+$, the switch is moved to position Q.

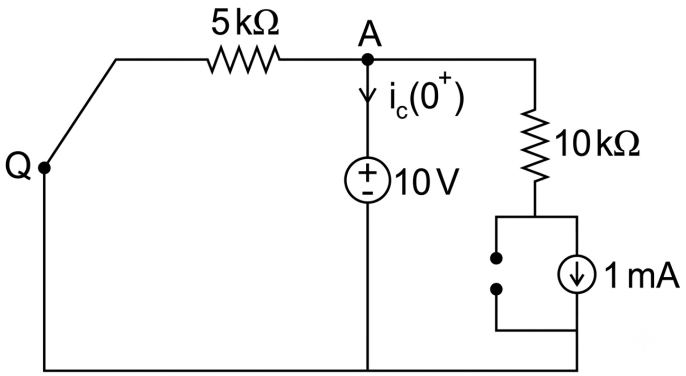


Fig. Solution Q2.21.3

Applying KCL at node A,

$$\frac{10}{5\,\text{k}\Omega} + i_C(0^+) + 1\ \text{mA} = 0$$

$$2 + i_C(0^+) + 1 = 0 \Rightarrow i_C(0^+) = -3\ \text{mA}$$

Using

$$i_C = C\frac{dv_C}{dt}$$

$$1\text{ mF}\left.\frac{dv_C}{dt}\right|_{t=0^+} = -3\text{ mA}$$

$$\left.\frac{dv_C}{dt}\right|_{t=0^+} = -3\text{ V/s}$$

Hence, the correct answer is

C. -3 V/s

Q2.19.1 NAT | 2019 | 2M | Ans: 0.25 | ●●○

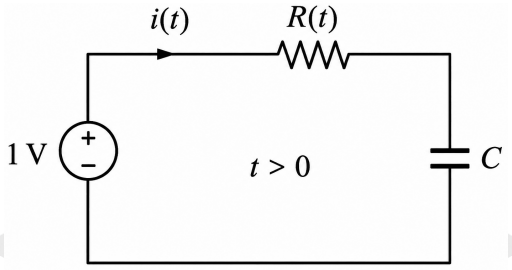


Fig. Solution Q2.19.1

Applying KCL,

$$C\frac{dv(t)}{dt} + \frac{v(t) - V_s}{R(t)} = 0, \qquad \text{where v(t) is the voltage across capacitor}$$

Since, $C = 1, \quad V_s = 1, \quad R(t) = R_0\left(1 - \frac{t}{T}\right)$

and, $T = 3R_0C = 3$ s,

the differential equation becomes

$$\frac{dv(t)}{dt} + \frac{v(t) - 1}{1 - \frac{t}{3}} = 0$$

$$\Rightarrow \frac{dv(t)}{v(t) - 1} = -\frac{dt}{1 - \frac{t}{3}}$$

Integrating both sides,

$$\ln(v(t) - 1) = 3\ln(t - 3) + \ln K, \qquad \text{where } K \text{ is a constant.}$$

Hence,

$$\ln\left(\frac{v(t) - 1}{(t - 3)^3}\right) = \ln K$$

Taking antilog,

$$v(t) = 1 + K(t - 3)^3$$

The capacitor current is

$$i(t) = C\frac{dv(t)}{dt} = \frac{dv(t)}{dt} = 3K(t - 3)^2$$

Using the initial condition, $i(0) = 1$ A,
we get

$$1 = 3K(0 - 3)^2 = 27K$$

$$\Rightarrow K = \frac{1}{27}$$

At $t = \frac{T}{2} = \frac{3}{2} = 1.5$ s,
the current is

$$i\left(\frac{T}{2}\right) = 3\left(\frac{1}{27}\right)(1.5)^2$$

$$= \frac{3}{27} \times \frac{15}{10} \times \frac{15}{10} = \frac{1}{4} = 0.25 \text{ A}$$

Hence,

$$\boxed{0.25}$$

Q2.18.1 NAT | 2018 | 2M | Ans: 0.316 | ● ○ ○

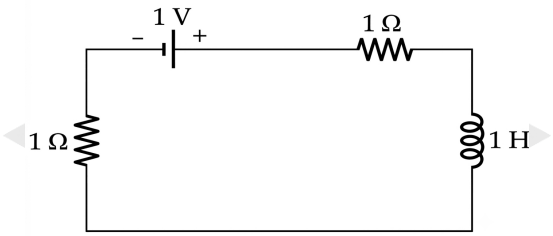


Fig. Solution Q2.18.1

For an RL circuit, the current is

$$i(t) = i(\infty) + [i(0) - i(\infty)]\, e^{-t/\tau}, \qquad \text{where; } \tau = \frac{L}{R}$$

At steady state, the inductor acts as a short circuit.

$$\Rightarrow i(\infty) = \frac{V}{R} = \frac{1}{1+1} = 0.5 \text{ A}$$

Initially, the switch is open, therefore, $i(0^-) = 0$
Since inductor current cannot change instantaneously,

$$i(0^+) = i(0^-) = 0$$

The equivalent resistance seen by the inductor is

$$R = 1 + 1 = 2\ \Omega$$

$$\Rightarrow \tau = \frac{L}{R} = \frac{1}{2} = 0.5 \text{ s}$$

Substituting into the current expression,

$$i(t) = 0.5\left(1 - e^{-2t}\right)$$

At $t = 0.5$ s,

$$i(0.5) = 0.5\left(1 - e^{-1}\right)$$

$$= 0.5(1 - 0.3679) = 0.316 \text{ A}$$

Hence,

$$\boxed{0.316}$$

Q2.17.1 NAT | 2017 | 2M | Ans: 8.16 | ● ● ○

Before switching ($t = 0^-$), the switch is open and the circuit is in steady state. Hence, the inductor behaves as a short circuit.

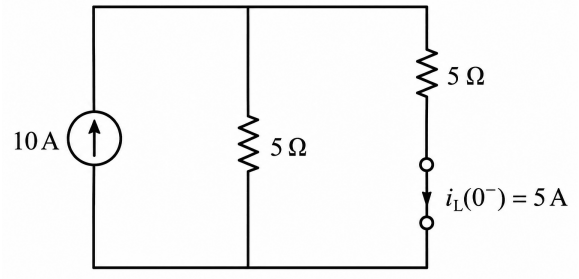


Fig. Solution Q2.17.1

The 10 A current source divides equally between the two 5 Ω branches.
Therefore,

$$i_L(0^-) = \frac{10}{2} = 5 \text{ A}$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 5 \text{ A}$$

After the switch is closed, the middle branch becomes a short circuit.

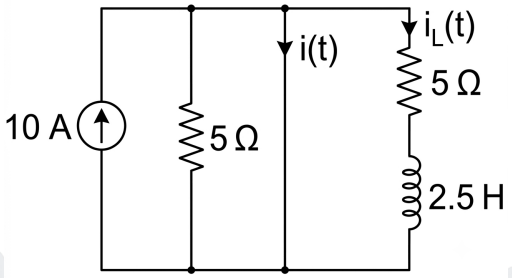


Fig. Solution Q2.17.1

Hence, $i_L(\infty) = 0$
The current through the RL branch is

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]\, e^{-t/\tau}$$

where

$$\tau = \frac{L}{R} = \frac{2.5}{5} = 0.5 \text{ s}$$

$$\Rightarrow i_L(t) = 5e^{-2t}$$

At $t = 0.5$ s,

$$i_L(0.5) = 5e^{-1}$$

By KCL;

$$i(t) = 10 - i_L(t)$$

$$\Rightarrow i(0.5) = 10 - 5e^{-1} = 10 - 5(0.3679) = 8.16 \text{ A}$$

Hence,

$$\boxed{8.16}$$

Q2.17.2 NAT | 2017 | 2M | Ans: 0.34 | ● ● ○

Method Laplace Transform

Applying Laplace transform,

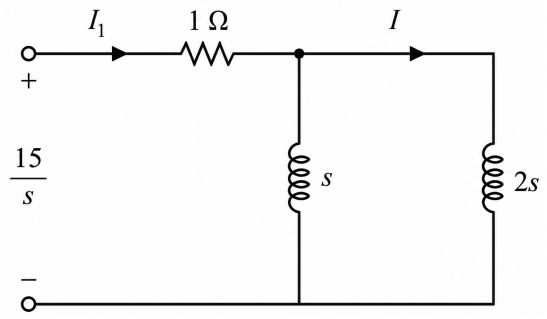


Fig. Solution Q2.17.2

The input current is

$$I_1(s) = \frac{\frac{15}{s}}{1 + (s \parallel 2s)}$$

Now,

$$s \parallel 2s = \frac{s(2s)}{s + 2s} = \frac{2s}{3}$$

Hence,

$$I_1(s) = \frac{\frac{15}{s}}{1 + \frac{2s}{3}} = \frac{45}{s(2s + 3)} = \frac{45}{2s\left(s + \frac{3}{2}\right)}$$

Using current division, the current through the 2 H inductor is

$$I(s) = \frac{s}{s + 2s} I_1(s) = \frac{1}{3} I_1(s)$$

$$\Rightarrow I(s) = \frac{15}{2s\left(s + \frac{3}{2}\right)} = \frac{5}{s} - \frac{5}{s + \frac{3}{2}}$$

Taking inverse Laplace transform,

$$i(t) = 5 - 5e^{-3t/2}$$

Given, $i(t_0) = 2$ A

$$\Rightarrow 5 - 5e^{-3t_0/2} = 2$$

$$e^{-3t_0/2} = \frac{3}{5}$$

$$t_0 = -\frac{2}{3} \ln\left(\frac{3}{5}\right) = 0.34 \text{ s}$$

Hence,

$$\boxed{0.34}$$

Method Time-Domain Analysis

The current through an RL circuit is

$$I(t) = I(\infty) + [I(0) - I(\infty)]\, e^{-t/\tau}$$

At $t < 0$, $\quad I(0^-) = 0$
Since inductor current is continuous, $I(0^+) = 0$
At steady state ($t \to \infty$), both inductors behave as short circuits.

$$I_1 = \frac{15}{1} = 15 \text{ A}$$

The current divides between the parallel inductors as

$$I(\infty) = 15\left(\frac{1}{1+2}\right) = 5 \text{ A}$$

The equivalent inductance is

$$L_{eq} = 1 \parallel 2 = \frac{2}{3} \text{H}$$

The Thevenin resistance seen by the inductors is, $R_{th} = 1\ \Omega$

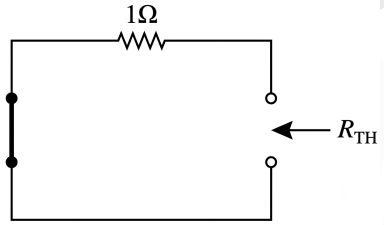


Fig. Solution Q2.17.2

Hence,

$$\tau = \frac{L_{eq}}{R_{th}} = \frac{2}{3} \text{ s}$$

Therefore,

$$I(t) = 5 + (0 - 5)e^{-3t/2} = 5\left(1 - e^{-3t/2}\right)$$

Given, $I(t_0) = 2$ A;

$$5\left(1 - e^{-3t_0/2}\right) = 2 \Rightarrow e^{-3t_0/2} = \frac{3}{5}$$

$$t_0 = -\frac{2}{3}\ln\left(\frac{3}{5}\right) = 0.34 \text{ s}$$

Hence,

$$\boxed{0.34}$$

Q2.16.1 NAT 2016 2M Ans: -1 or 1 or 2 or -2 ● ● ○

Since the direction and branch of the current is not specified in the question;

Case 1: At $t = 0^-$, the circuit is in steady state.
Hence, both capacitors behave as open circuits.

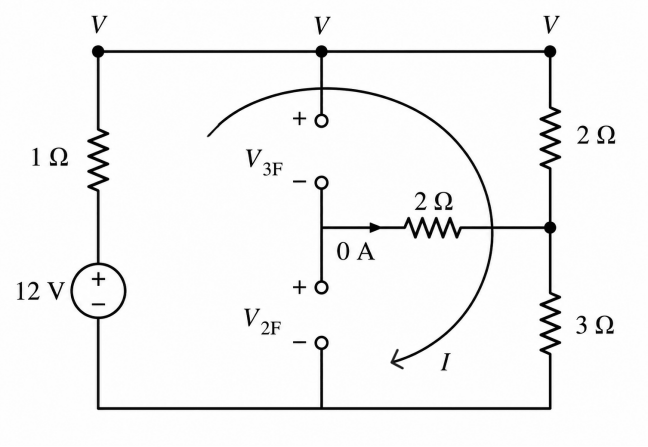


Fig. Solution Q2.16.1

$$\Rightarrow I = \frac{12}{1+2+3} = 2 \text{ A}$$

Hence, the voltage across the bridge network is;

$$V = 2 \times (2+3) = 10 \text{ V}$$

Applying voltage division,

$$V_{3F} = \frac{2}{2+3} \times 10 = 4 \text{ V}, \qquad V_{2F} = \frac{3}{2+3} \times 10 = 6 \text{ V}$$

At $t = 0^+$, each capacitor is replaced by its initial voltage source, and the 3 Ω resistor becomes an open circuit.

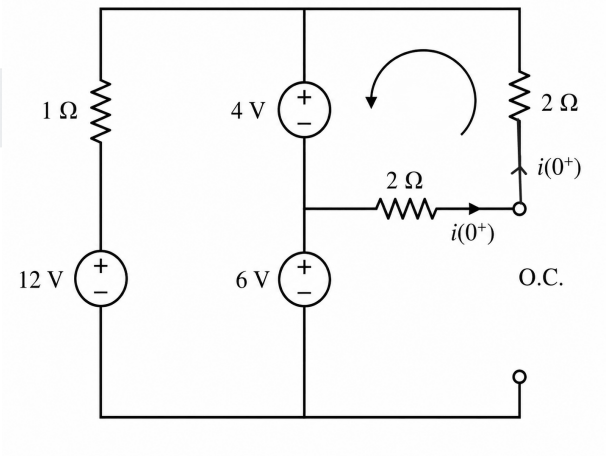


Fig. Solution Q2.16.1

Applying KVL in the loop,

$$2i(0^+) + 2i(0^+) + 4 = 0$$

$$4i(0^+) = -4 \Rightarrow i(0^+) = -1 \text{ A}$$

Case 2: If we assume direction of current to be opposite of what we assumed in Case 1 then another possible answer is

$$\boxed{1A}$$

Case 3: Consider the current in the 2F capacitor branch;

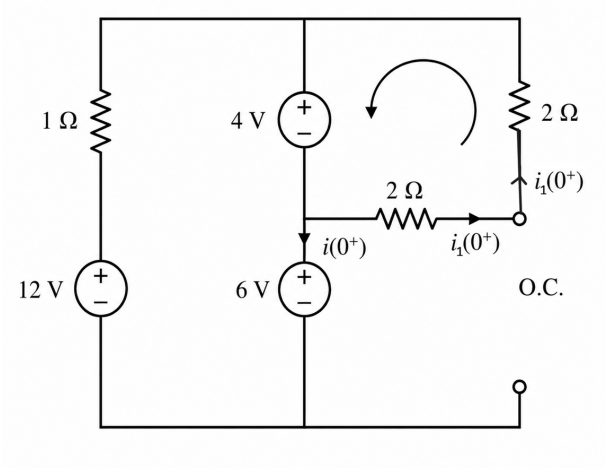


Fig. Solution Q2.16.1

From Case 1, $i_1(0^+) = -1$ A

With the 3 Ω resistor open, the current in the 2 F branch is the current of the outer loop (12 V source, 1 Ω, 4 V and 6 V). Applying KVL,

$$i(0^+) = \frac{12 - 4 - 6}{1} = 2 \text{ A}$$

Case 4: If we assume direction of current to be opposite of what we assumed in Case 3 then another possible answer is

$$\boxed{-2A}$$

Hence the correct answers are;

$$\boxed{-1 \text{ or } 1 \text{ or } -2 \text{ or } 2}$$

Q2.16.2 NAT 2016 2M Ans: 100 ●●○

At $t = 0^-$, the switch has been closed for a long time.

Hence,

Inductor → Short circuit, Capacitor → Open circuit

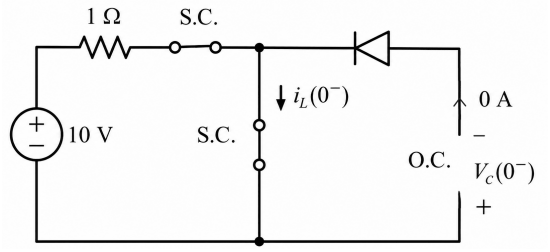


Fig. Solution Q2.16.2

Therefore,

$$i_L(0^-) = \frac{10}{1} = 10 \text{ A}$$

Also,

$$V_L(0^-) = 0, \qquad V_C(0^-) = 0$$

At $t = 0^+$, the switch is opened.

Since the inductor current cannot change instantaneously, $i_L(0^+) = 10$ A

Also, the capacitor voltage is continuous, $V_C(0^+) = 0$

Initially, the diode is forward biased and conducts.

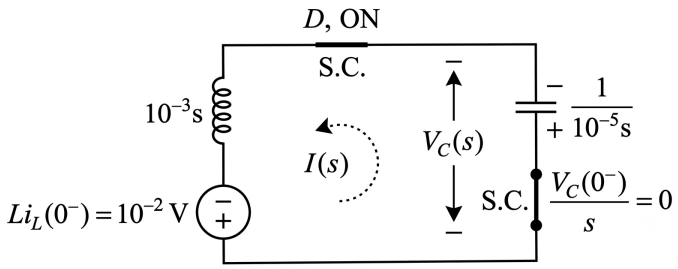


Fig. Solution Q2.16.2

Applying KVL in the Laplace domain,

$$\frac{1}{Cs}I(s) + LsI(s) - Li_L(0^-) = 0$$

$$\frac{1}{10^{-5}s}I(s) + 10^{-3}sI(s) - 10^{-2} = 0$$

$$I(s) = \frac{10^{-2}}{10^{-3}s + \frac{10^5}{s}} = \frac{10s}{s^2 + 10^8}$$

Taking inverse Laplace transform,

$$i(t) = 10\cos(10^4 t)$$

The diode conducts only while the current is positive.
Hence, it turns OFF when

$$10\cos(10^4 t) = 0 \Rightarrow 10^4 t = \frac{\pi}{2}$$

At this instant,

$$V_C(s) = \frac{1}{Cs}I(s) = \frac{1}{10^{-5}s} \times \frac{10s}{s^2 + 10^8} = \frac{10^6}{s^2 + 10^8}$$

Taking inverse Laplace transform,

$$V_C(t) = 100\sin(10^4 t)$$

At $10^4 t = \frac{\pi}{2}$;

$$V_C = 100\sin\frac{\pi}{2} = 100 \text{ V}$$

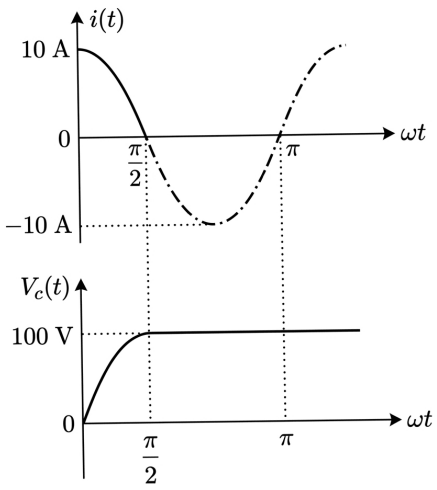


Fig. Solution Q2.16.2

After this instant, diode is reverse biased and isolates the capacitor. Hence, capacitor retains this voltage.
Therefore, the steady-state capacitor voltage is

100

Q2.16.3 MCQ 2016 1M Ans: D

The capacitor voltage is given by

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right] e^{-t/\tau}$$

At $t = 0^-$, the switch is at position 1 for a long time.
Hence, the capacitor behaves as an open circuit.

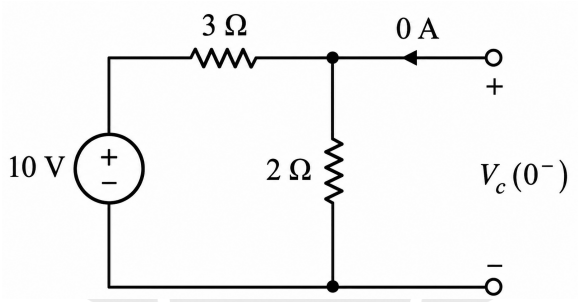


Fig. Solution Q2.16.3

Applying voltage division,

$$V_C(0^-) = \frac{2}{3+2} \times 10 = 4 \text{ V}$$

Since capacitor voltage is continuous, $V_C(0^+) = 4$ V
At $t = 0$, the switch is moved to position 2.

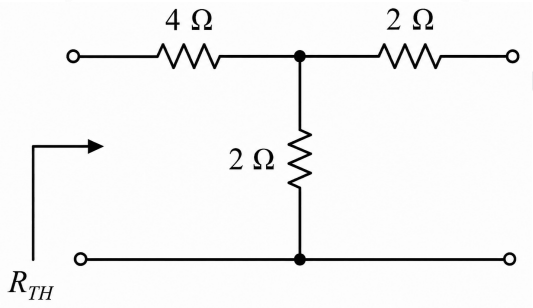


Fig. Solution Q2.16.3

For the RC network, $\tau = R_{\text{th}}C$
To find R_{th}, open-circuit the current source.

$$R_{\text{th}} = 4 + (2 \parallel \infty) = 4 + 2 = 6\ \Omega$$

Hence,

$$\tau = 6 \times 0.1 = 0.6 \text{ s}$$

At $t \to \infty$, the capacitor behaves as an open circuit.

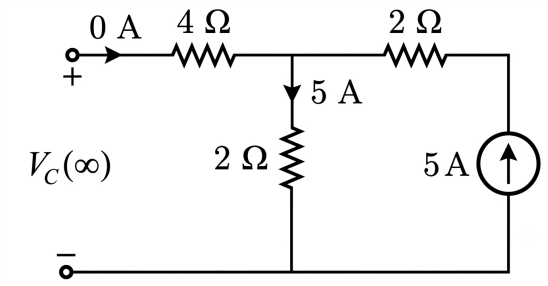


Fig. Solution Q2.16.3

The 5 A current flows through the vertical 2 Ω resistor.
Therefore,

$$V_C(\infty) = 2 \times 5 = 10 \text{ V}$$

$$\Rightarrow V_C(t) = 10 + (4 - 10)e^{-t/0.6}$$

$$V_C(t) = 10 - 6e^{-t/0.6}$$

Hence, the correct answer is

D. $10 - 6e^{-t/0.6}$

Q2.15.1 NAT 2015 2M Ans: 1.5 ●●○

Method Laplace Transform

Applying Laplace transform,

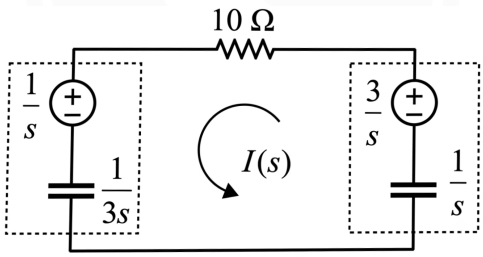


Fig. Solution Q2.15.1

The current in the circuit is

$$I(s) = \frac{\frac{3}{s} - \frac{1}{s}}{10 + \frac{1}{3s} + \frac{1}{s}}$$

$$I(s) = \frac{\frac{2}{s}}{\frac{30s + 4}{3s}} = \frac{6}{30s + 4}$$

$$I(s) = \frac{0.2}{s + \frac{2}{15}}$$

Taking inverse Laplace transform,

$$i(t) = 0.2e^{-2t/15}$$

The power dissipated in the resistor is

$$p(t) = i^2(t)R$$

$$p(t) = (0.2)^2 \times 10\, e^{-4t/15} = 0.4e^{-4t/15}$$

The total energy dissipated is

$$E = \int_0^\infty p(t)\, dt = 0.4 \int_0^\infty e^{-4t/15}\, dt$$

$$E = 0.4\left[-\frac{15}{4}e^{-4t/15}\right]_0^\infty$$

$$E = \frac{0.4 \times 15}{4} = 1.5 \text{ J}$$

Hence,

$$\boxed{1.5}$$

Method Energy Conservation

The initial energy stored in the capacitors is

$$E_i = \frac{1}{2}\left(C_1V_1^2 + C_2V_2^2\right)$$

$$= \frac{1}{2}\left(3 \times 1^2 + 1 \times 3^2\right) = 6 \text{ J}$$

At steady state, both capacitors attain the same voltage.
Using charge conservation,

$$C_1V_1 + C_2V_2 = (C_1 + C_2)V$$

$$3(1) + 1(3) = (3 + 1)V$$

$$V = 1.5 \text{ V}$$

The final energy stored is

$$E_f = \frac{1}{2}(C_1 + C_2)V^2$$

$$= \frac{1}{2}(4)(1.5)^2 = 4.5 \text{ J}$$

Hence, the energy dissipated in the resistor is

$$E = E_i - E_f = 6 - 4.5 = 1.5 \text{ J}$$

Therefore,

$$\boxed{1.5}$$

Q2.15.2 NAT 2015 2M Ans: 2.528 ●●○

The capacitor voltage is given by

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right] e^{-t/\tau}$$

At $t = 0^-$, the switch is open.
Hence, the capacitor is uncharged, $V_C(0^-) = 0$
Since capacitor voltage is continuous, $V_C(0^+) = 0$
At $t = 0$, the switch is closed.
For the RC network, $\tau = R_{th}C$
To find R_{th}, the voltage source is short-circuited.

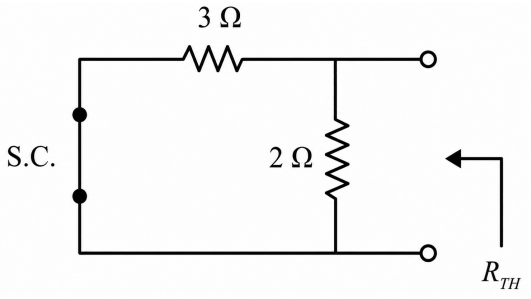


Fig. Solution Q2.15.2

$$R_{\text{th}} = 3 \parallel 2 = \frac{3 \times 2}{3 + 2} = 1.2\ \Omega$$

$$\Rightarrow \tau = 1.2 \times \frac{5}{6} = 1\ \text{s}$$

At steady state, the capacitor behaves as an open circuit.

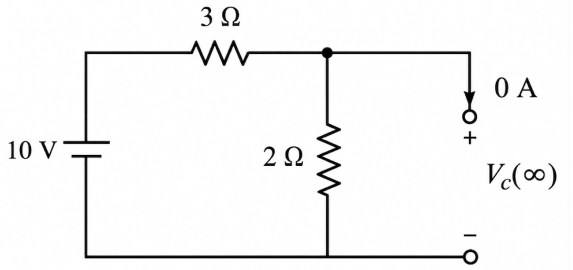


Fig. Solution Q2.15.2

Applying voltage division,

$$V_C(\infty) = \frac{2}{3 + 2} \times 10 = 4\ \text{V}$$

Substituting the values,

$$V_C(t) = 4 + (0 - 4)e^{-t} = 4\left(1 - e^{-t}\right)$$

At $t = 1$ s,

$$V_C(1) = 4\left(1 - e^{-1}\right) = 4\left(1 - \frac{1}{e}\right) = 2.528\ \text{V}$$

Hence,

$$\boxed{2.528}$$

Q2.15.3 MCQ 2015 1M Ans: C ● ○ ○

Method 1

The capacitor voltage is given by, $V_C(t) = V_C(\infty) + [V_C(0^+) - V_C(\infty)]\, e^{-t/\tau}$
At $t = 0^-$, the switch is at position A.

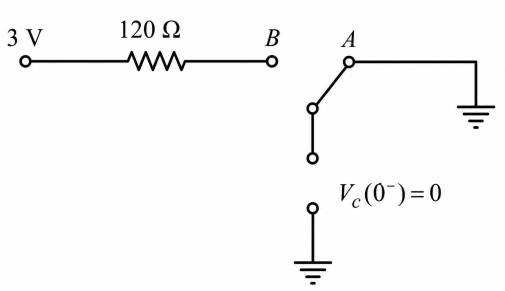


Fig. Solution Q2.15.3

Hence, the capacitor is connected to ground, $V_C(0^-) = 0$
Since capacitor voltage is continuous, $V_C(0^+) = 0$
At $t = 0$, the switch is moved to position B.

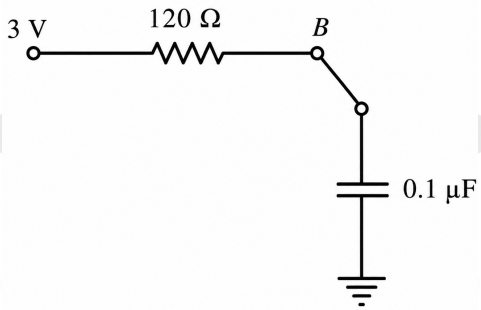


Fig. Solution Q2.15.3

The time constant is

$$\tau = RC = 120 \times 0.1 \times 10^{-6} = 12\ \mu s$$

At steady state, the capacitor behaves as an open circuit, $\Rightarrow V_C(\infty) = 3$ V
Hence,

$$V_C(t) = 3 + (0 - 3)e^{-t/12} = 3\left(1 - e^{-t/12}\right)$$

The capacitor current is

$$i_C(t) = C\frac{dV_C}{dt}$$

$$i_C(t) = 0.1 \times 10^{-6} \cdot \frac{3}{12 \times 10^{-6}} e^{-t/12} = \frac{1}{40} e^{-t/12}\ \text{A}$$

The energy supplied by the 3 V source is

$$E = \int_0^\infty Vi\, dt = \int_0^\infty 3\left(\frac{1}{40} e^{-t/12}\right) dt$$

$$E = \frac{3}{40}\left[-12 \times 10^{-6} e^{-t/12}\right]_0^\infty$$

$$E = \frac{3}{40} \times 12 \times 10^{-6} = 0.9 \times 10^{-6}\ \text{J} = 0.9\ \mu\text{J}$$

Hence, the correct answer is

C. 0.9 μJ

Method 2

Initially, the capacitor is uncharged.
At steady state, $V_C(\infty) = 3$ V
The charge supplied by the source is

$$Q = C\Delta V = 0.1 \times 10^{-6} \times 3 = 0.3 \times 10^{-6} \text{ C}$$

Hence, the energy supplied by the 3 V source is

$$E = VQ = 3 \times 0.3 \times 10^{-6} = 0.9 \times 10^{-6} \text{ J} = 0.9\ \mu\text{J}$$

Hence, the correct answer is

C. $0.9\ \mu$J

Q2.14.1 NAT 2014 2M Ans: 31.25 ● ○ ○

At $t \to \infty$, the inductor behaves as a short circuit.

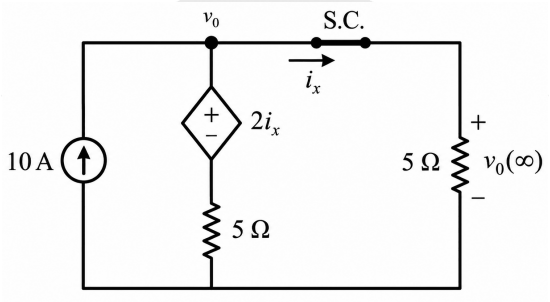


Fig. Solution Q2.14.1

Applying KCL at node v_o,

$$10 = \frac{v_o(\infty) - 2i_x}{5} + \frac{v_o(\infty)}{5}$$

From the circuit,

$$i_x = \frac{v_o(\infty)}{5}$$

Substituting,

$$10 = \frac{v_o(\infty) - \dfrac{2v_o(\infty)}{5}}{5} + \frac{v_o(\infty)}{5}$$

$$10 = \frac{3v_o(\infty)}{25} + \frac{5v_o(\infty)}{25} = \frac{8v_o(\infty)}{25}$$

$$v_o(\infty) = \frac{250}{8} = 31.25 \text{ V}$$

Hence,

31.25

Q2.14.2 MCQ 2014 2M Ans: A ● ● ○

The current through the 2 kΩ resistor is

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right] e^{-t/\tau}$$

Initially, the capacitor is uncharged, $V_C(0^-) = 0$
Since capacitor voltage is continuous, $V_C(0^+) = 0$

At $t = 0^+$, the capacitor behaves as a voltage source of 0 V, i.e., a short circuit.

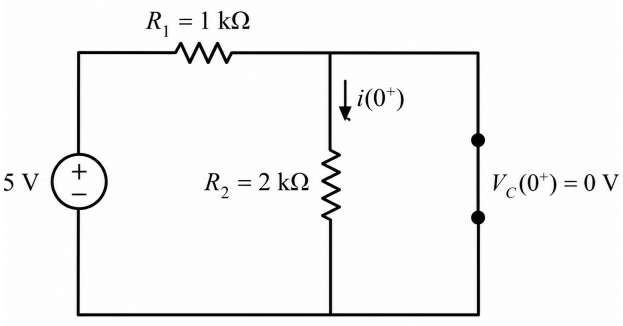


Fig. Solution Q2.14.2

Hence,

$$i(0^+) = \frac{0}{2} = 0\ \text{mA}$$

The time constant is, $\tau = R_{\text{th}}C$

To find R_{th}, the voltage source is short-circuited.

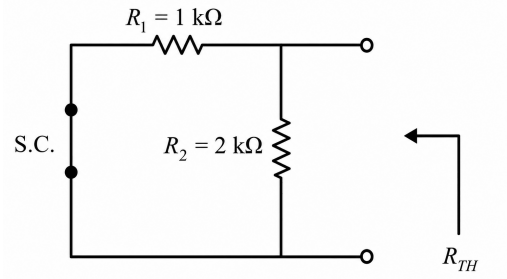


Fig. Solution Q2.14.2

$$R_{\text{th}} = 1 \parallel 2 = \frac{1 \times 2}{1 + 2} = \frac{2}{3}\ \text{k}\Omega$$

Hence,

$$\tau = \frac{2}{3} \times 10^3 \times 10^{-6} = \frac{2}{3}\ \text{ms}$$

At $t \to \infty$, the capacitor behaves as an open circuit.

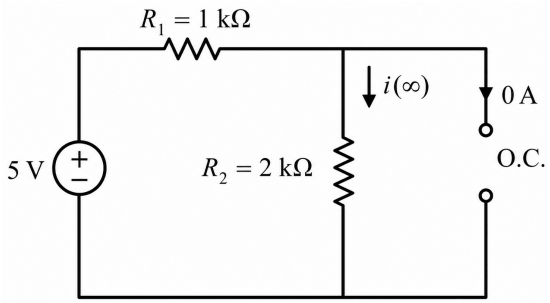


Fig. Solution Q2.14.2

The steady-state current is

$$i(\infty) = \frac{5}{1 + 2} = \frac{5}{3}\ \text{mA}$$

Substituting the values,

$$i(t) = \frac{5}{3} + \left(0 - \frac{5}{3}\right) e^{-t/\tau}$$

$$i(t) = \frac{5}{3}\left(1 - e^{-t/\tau}\right), \qquad \tau = \frac{2}{3}\text{ ms}$$

Hence, the correct answer is

A. $\frac{5}{3}\left(1 - e^{-t/\tau}\right),\ \tau = \frac{2}{3}$ ms

Q2.14.3 MCQ 2014 1M Ans: A ●○○

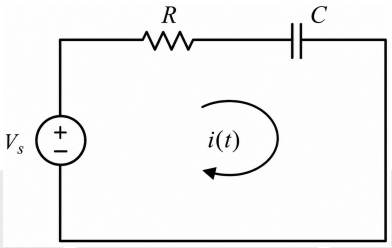


Fig. Solution Q2.14.3

Method 1

Applying KVL,

$$V_s = Ri(t) + V_C(t)$$

Since,

$$V_C(t) = \frac{1}{C}\int_0^t i(u)\,du$$

$$\Rightarrow V_s = Ri(t) + \frac{1}{C}\int_0^t i(u)\,du$$

Differentiating both sides with respect to t,

$$0 = R\frac{di(t)}{dt} + \frac{i(t)}{C} \Rightarrow \frac{di(t)}{dt} = -\frac{1}{RC}i(t)$$

Separating the variables,

$$\frac{di(t)}{i(t)} = -\frac{dt}{RC}$$

Integrating,

$$\int \frac{di(t)}{i(t)} = -\frac{1}{RC}\int dt$$

$$\ln i(t) = -\frac{t}{RC} + K$$

Hence,

$$i(t) = Ae^{-t/(RC)}, \qquad \text{where } A = e^K \text{ is a constant.}$$

Thus, the current is maximum at $t = 0$ and decreases exponentially to zero as $t \to \infty$.

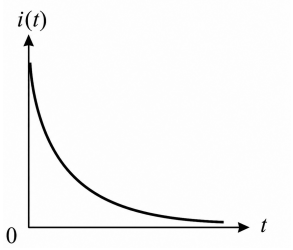


Fig. Solution Q2.14.3

Hence, the correct answer is

A

Method 2

At $t = 0^+$, the capacitor behaves as a short circuit, $\Rightarrow i(0^+) = \frac{V_s}{R}$
At $t \to \infty$, the capacitor behaves as an open circuit, $\Rightarrow i(\infty) = 0$
Hence, the current must decay exponentially from its initial value to its final value.
Therefore,

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right] e^{-t/\tau}$$

Substituting,

$$i(t) = 0 + \left(\frac{V_s}{R} - 0\right) e^{-t/\tau} = \frac{V_s}{R} e^{-t/\tau}$$

Hence, the correct answer is

A

Q2.14.4 NAT 2014 1M Ans: 1.25 ●●○

At $t = 0^-$, the switch is open and the circuit is in steady state.
Hence,

Capacitor → Open circuit, Inductor → Short circuit

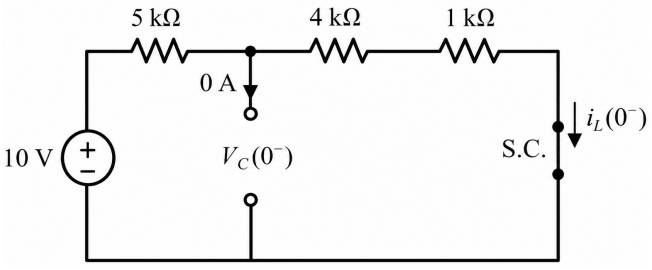


Fig. Solution Q2.14.4

$$\Rightarrow i_L(0^-) = \frac{10}{5+4+1} = 1 \text{ mA} \qquad \text{and} \qquad V_C(0^-) = \frac{5}{5+4+1} \times 10 = 5 \text{ V}$$

Since capacitor voltage and inductor current are continuous,

$$\Rightarrow V_C(0^+) = 5 \text{ V}, \qquad i_L(0^+) = 1 \text{ mA}$$

At $t = 0^+$, the capacitor is replaced by a 5 V source and the inductor by a 1 mA current source.

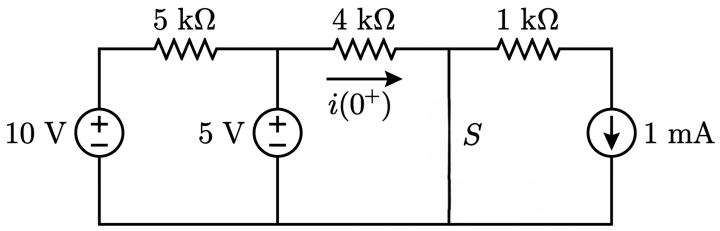


Fig. Solution Q2.14.4

Applying KVL,

$$-5 + 4\,i(0^+) + 0 = 0$$

$$i(0^+) = \frac{5}{4} = 1.25 \text{ mA}$$

Hence,

$$\boxed{1.25}$$

Q2.12.1 MCQ 2012 1M Ans: D

Method Laplace Transform

Applying KVL in the Laplace domain,

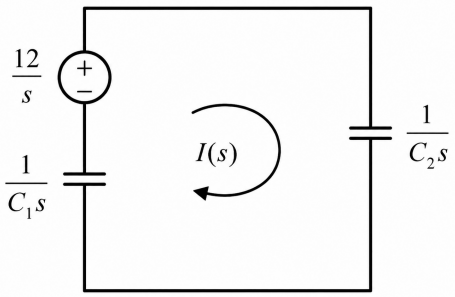


Fig. Solution Q2.12.1

$$\frac{12}{s} = \frac{I(s)}{C_1 s} + \frac{I(s)}{C_2 s} \Rightarrow \frac{I(s)}{s}\left(\frac{1}{C_1} + \frac{1}{C_2}\right) = \frac{12}{s}$$

Let,

$$\frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{C}$$

Hence,

$$I(s) = 12C$$

Taking inverse Laplace transform,

$$i(t) = 12C\,\delta(t)$$

Method Time-Domain Analysis

After the switch is closed,

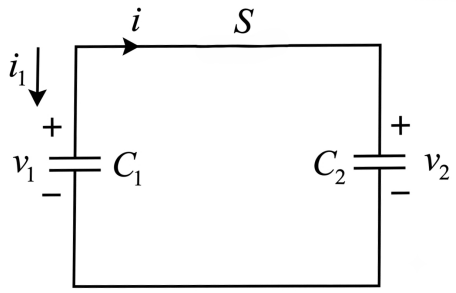


Fig. Solution Q2.12.1

Since the capacitors are connected in parallel, $\Rightarrow v_1(t) = v_2(t)$
The capacitor voltages are

$$v_1(t) = 12 - \frac{1}{C_1}\int_0^t i(\tau)\,d\tau, \qquad v_2(t) = 0 + \frac{1}{C_2}\int_0^t i(\tau)\,d\tau$$

Since,

$$v_1(t) = v_2(t)$$

$$12 - \frac{1}{C_1}\int_0^t i(\tau)\,d\tau = \frac{1}{C_2}\int_0^t i(\tau)\,d\tau$$

$$\left(\frac{1}{C_1} + \frac{1}{C_2}\right)\int_0^t i(\tau)\,d\tau = 12$$

Let,

$$\frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{C}$$

Then,

$$\int_0^t i(\tau)\,d\tau = 12C$$

Differentiating both sides,

$$i(t) = 12C\,\delta(t)$$

Method Conceptual Approach

Immediately after the switch is closed, an ideal charged capacitor is directly connected to an uncharged ideal capacitor through an ideal switch.
Since there is no resistance in the circuit, the charge redistribution takes place instantaneously.
Hence, the capacitor voltage changes as a step function.
Since, $i_C = C\dfrac{dv_C}{dt}$
and the derivative of a step function is an impulse function, $i(t) = 12C\,\delta(t)$
Hence, the correct answer is

D. an impulse function

Key Points

- A capacitor normally cannot change its voltage instantaneously. Therefore, under ordinary conditions,

$$V_C(0^-) = V_C(0^+).$$

However, if an **ideal voltage source** is connected directly across the capacitor, it forces an instantaneous change in the

capacitor voltage. Hence, the continuity property no longer holds, and

$$V_C(0^-) \neq V_C(0^+).$$

Q2.11.1 MCQ | 2011 | 2M | Ans: A | ● ○ ○

Method Transient Analysis

The capacitor current is given by, $i(t) = i(\infty) + [i(0^+) - i(\infty)]\, e^{-t/\tau}$
The initial capacitor voltage is

$$Q = CV$$
$$2.5 \text{ mC} = 50\ \mu\text{F} \times V_C(0^-)$$
$$V_C(0^-) = \frac{2.5 \times 10^{-3}}{50 \times 10^{-6}} = 50 \text{ V}$$

Since capacitor voltage is continuous, $\Rightarrow V_C(0^+) = 50$ V
At $t = 0^+$, the capacitor is replaced by its initial voltage source.

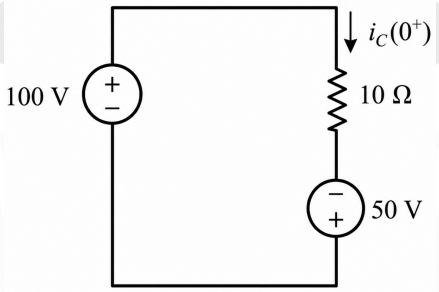


Fig. Solution Q2.11.1

Applying KVL,

$$-100 + 10\, i(0^+) - 50 = 0$$
$$i(0^+) = \frac{100 + 50}{10} = 15 \text{ A}$$

For the RC circuit,

$$\tau = RC = 10 \times 50 \times 10^{-6} = 5 \times 10^{-4} \text{ s}$$

At steady state, the capacitor behaves as an open circuit, $\Rightarrow i(\infty) = 0$
Hence,

$$i(t) = 15e^{-t/(5\times10^{-4})} = 15e^{-2\times10^3 t} \text{ A}$$

Method Option Elimination

At $t = 0^+$, the capacitor is replaced by its initial voltage source of 50 V.
Applying KVL,

$$-100 + 10\, i(0^+) - 50 = 0$$
$$i(0^+) = 15 \text{ A}$$

From the options,

$$\text{(A)} \Rightarrow i(0^+) = 15 \text{ A}, \quad \text{(B)} \Rightarrow i(0^+) = 5 \text{ A}, \quad \text{(C)} \Rightarrow i(0^+) = 10 \text{ A}, \quad \text{(D)} \Rightarrow i(0^+) = -5 \text{ A}$$

Matching with options, we get option A.
Hence, the correct answer is

A. $15e^{-2\times10^3 t}$ A

Q2.10.1 MCQ | 2010 | 2M | Ans: A | ● ● ○

The current through the 10 Ω resistor is, $i(t) = i(\infty) + [i(0^+) - i(\infty)]\, e^{-t/\tau}$

At $t = 0^-$, the switch is open and the circuit is in steady state.

Hence, the inductor behaves as a short circuit.

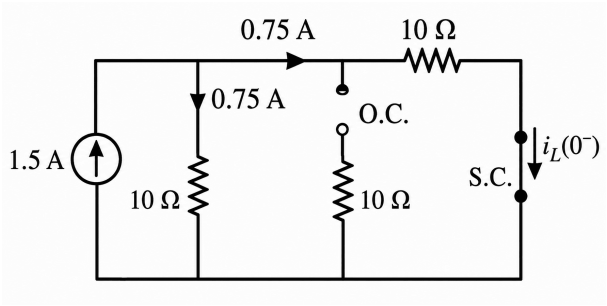


Fig. Solution Q2.10.1

The source current divides equally between the two 10 Ω branches.

$$\Rightarrow i_L(0^-) = \frac{10}{10+10} \times 1.5 = 0.75 \text{ A}$$

Since inductor current is continuous, $i_L(0^+) = i_L(0^-) = 0.75$ A

At $t = 0$, the switch is closed.

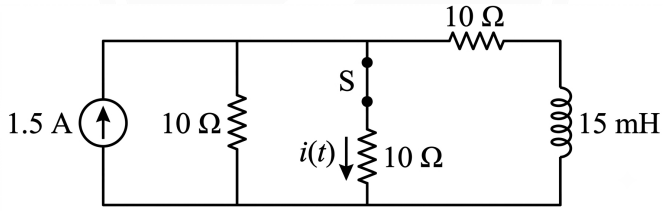


Fig. Solution Q2.10.1

For the RL network, $\tau = \dfrac{L}{R_{\text{th}}}$

To find R_{th}, open-circuit the current source.

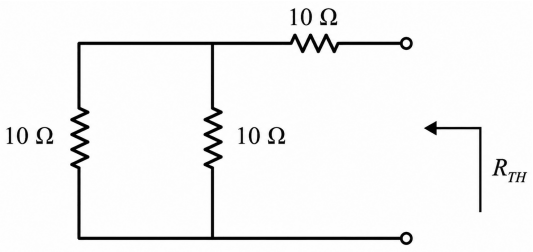


Fig. Solution Q2.10.1

$$R_{\text{th}} = 10 + (10 \parallel 10) = 10 + 5 = 15\ \Omega$$

Hence,

$$\tau = \frac{15 \times 10^{-3}}{15} = 10^{-3} \text{ s}$$

At $t = 0^+$, the inductor is replaced by a current source of 0.75 A.

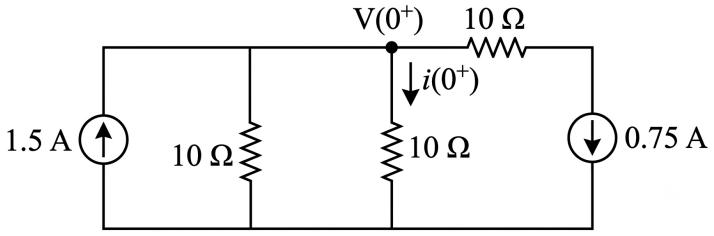


Fig. Solution Q2.10.1

Applying KCL at the top node,

$$\frac{V(0^+)}{10} + \frac{V(0^+)}{10} - 1.5 + 0.75 = 0$$

$$\frac{V(0^+)}{5} = 0.75 \Rightarrow V(0^+) = 3.75 \text{ V}$$

Hence,

$$i(0^+) = \frac{3.75}{10} = 0.375 \text{ A}$$

At $t \to \infty$, the inductor behaves as a short circuit.

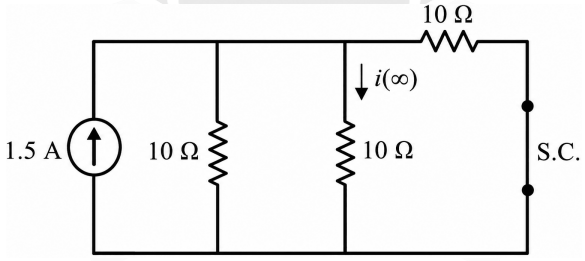


Fig. Solution Q2.10.1

Using current division,

$$i(\infty) = \frac{10}{10+5} \times 1.5 = 0.5 \text{ A}$$

Therefore,

$$i(t) = 0.5 + (0.375 - 0.5)e^{-t/10^{-3}}$$

$$i(t) = 0.5 - 0.125e^{-1000t}$$

Hence, the correct answer is

A. $0.5 - 0.125e^{-1000t}$

Q2.09.1 MCQ 2009 2M Ans: B ● ● ○

The capacitor network can be simplified as

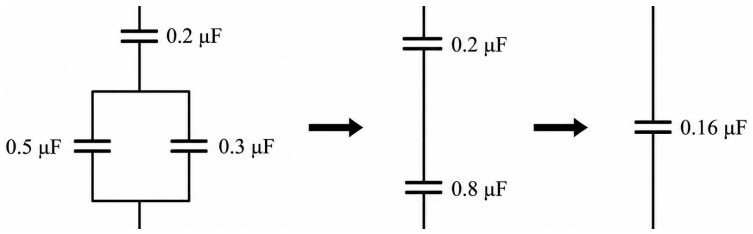


Fig. Solution Q2.09.1

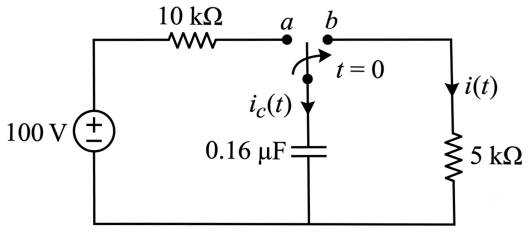


Fig. Solution Q2.09.1

The capacitor current is given by, $i(t) = i(\infty) + [i(0^+) - i(\infty)]\, e^{-t/\tau}$
At $t = 0^-$, the switch is at position a for a long time.
Hence, the capacitor behaves as an open circuit.

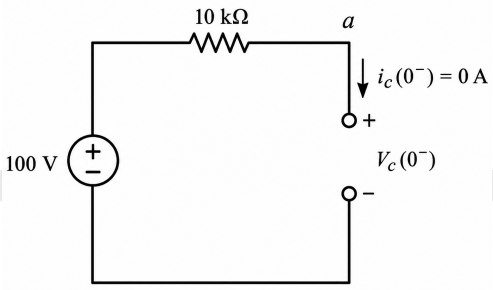


Fig. Solution Q2.09.1

Therefore, $V_C(0^-) = 100$ V
Since capacitor voltage is continuous, $V_C(0^+) = 100$ V
At $t = 0$, the switch is moved to position b.
The time constant is , $\tau = RC = 5 \times 10^3 \times 0.16 \times 10^{-6} = 0.8 \times 10^{-3}$ s
At $t = 0^+$;

$$i(0^+) = \frac{100}{5k} = 20 \text{ mA}$$

At $t \rightarrow \infty$, the capacitor behaves as an open circuit.

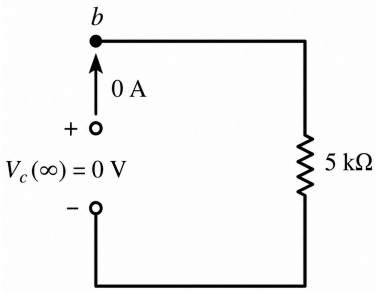


Fig. Solution Q2.09.1

Hence, $i(\infty) = 0$
Substituting,

$$i(t) = 20e^{-t/(0.8\times10^{-3})} = 20e^{-1250t}\, u(t) \text{ mA}$$

Hence, the correct answer is

B. $20e^{-1250t}u(t)$ mA

Q2.09.2 MCQ 2009 2M Ans: A ●●○

The given equation is

$$L\frac{di}{dt} + Ri(t) = V_0\left(1 + Be^{-Rt/L}\sin t\right)u(t)$$

The applied voltage consists of two components:

$$\text{DC source} = V_0 u(t)$$

$$\text{Transient source} = V_0 B e^{-Rt/L}\sin t\, u(t)$$

At steady state ($t \to \infty$), $e^{-Rt/L} \to 0$
Hence, the transient source becomes zero and only the DC source remains.
Also, the inductor behaves as a short circuit.

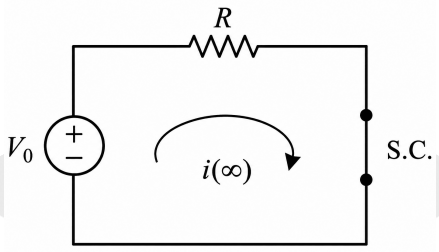


Fig. Solution Q2.09.2

Therefore,

$$i(\infty) = \frac{V_0}{R}$$

Hence, the correct answer is

A. $\dfrac{V_0}{R}$

Q2.08.1 MCQ 2008 2M Ans: D ●●○

Given,

$$i_L(0^-) = 0, \qquad V_C(0^-) = 0$$

Applying Laplace transform,

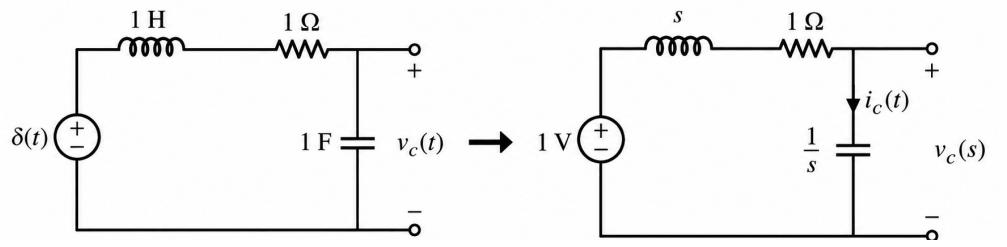


Fig. Solution Q2.08.1

The capacitor voltage is obtained using voltage division,

$$V_C(s) = \frac{\frac{1}{s}}{s + 1 + \frac{1}{s}} \Rightarrow V_C(s) = \frac{1}{s^2 + s + 1}$$

$$\Rightarrow V_C(s) = \frac{1}{\left(s+\frac{1}{2}\right)^2+\frac{3}{4}} = \frac{\frac{\sqrt{3}}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} \times \frac{2}{\sqrt{3}}$$

$$V_C(t) = \frac{2}{\sqrt{3}} e^{-t/2} \sin\left(\frac{\sqrt{3}}{2}t\right), \qquad \text{[Taking inverse Laplace transform]}$$

Hence, the correct answer is

$$\boxed{\textbf{D. } \frac{2}{\sqrt{3}} e^{-t/2} \sin\left(\frac{\sqrt{3}}{2}t\right)}$$

Q2.08.2 MCQ 2008 2M Ans: B ● ● ○

From Q2.08.1,

$$V_C(t) = \frac{2}{\sqrt{3}} e^{-t/2} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

The capacitor current is

$$i_C(t) = C\frac{dV_C(t)}{dt}$$

Since $C = 1$ F,

$$i_C(t) = \frac{2}{\sqrt{3}}\left[-\frac{1}{2}e^{-t/2}\sin\left(\frac{\sqrt{3}}{2}t\right) + e^{-t/2}\frac{\sqrt{3}}{2}\cos\left(\frac{\sqrt{3}}{2}t\right)\right]$$

$$i_C(t) = e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

Since the circuit is series,

$$i_R(t) = i_C(t)$$

Also,

$$V_R(t) = Ri_R(t) = 1 \times i_R(t)$$

Therefore,

$$V_R(t) = e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$$

Hence, the correct answer is

$$\boxed{\textbf{B. } e^{-t/2}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]}$$

Q2.08.3 MCQ 2008 2M Ans: C ● ● ●

Given, $Q(0^-) = 0$ C,

$$V_C(0^-) = \frac{Q(0^-)}{C} \Rightarrow V_C(0^-) = 0 \text{ V}$$

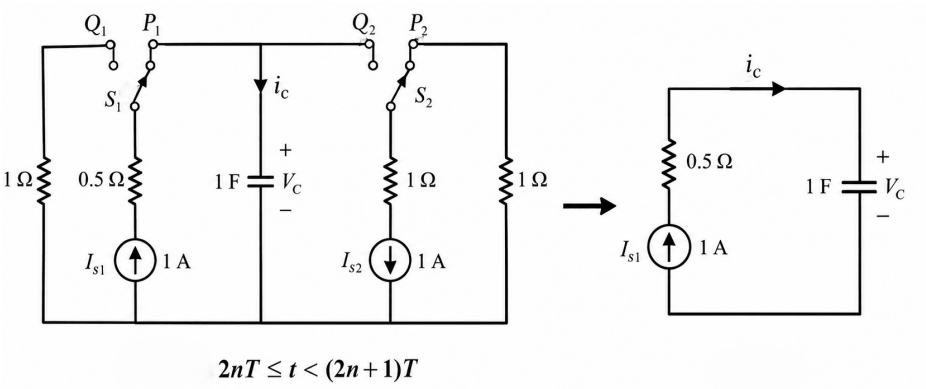
From the property of capacitor,

$$V_C(0^-) = V_C(0^+) = 0 \text{ V}$$

The switching sequence is

n	t	Position of S_1	Position of S_2
0	$0 \le t < T$	$S_1 \rightarrow P_1$	$S_2 \rightarrow P_2$
0	$T \le t < 2T$	$S_1 \rightarrow Q_1$	$S_2 \rightarrow Q_2$
1	$2T \le t < 3T$	$S_1 \rightarrow P_1$	$S_2 \rightarrow P_2$
1	$3T \le t < 4T$	$S_1 \rightarrow Q_1$	$S_2 \rightarrow Q_2$
2	$4T \le t < 5T$	$S_1 \rightarrow P_1$	$S_2 \rightarrow P_2$
2	$5T \le t < 6T$	$S_1 \rightarrow Q_1$	$S_2 \rightarrow Q_2$

Case 1:
When S_1 is connected to P_1 and S_2 is connected to P_2,



$2nT \le t < (2n+1)T$

Fig. Solution Q2.08.3

For $n = 0,\ 0 \le t < T$

$$i_C(t) = 1\ \text{A} = u(t)$$

The capacitor voltage is

$$V_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_C(\tau)\,d\tau = V_C(0^-) + \frac{1}{C}\int_{0}^{t} i_C(\tau)\,d\tau$$

$$\because V_C(0) = 0, \qquad C = 1\ \text{F}$$

$$V_C(t) = 0 + \int_0^t 1\,dt = t$$

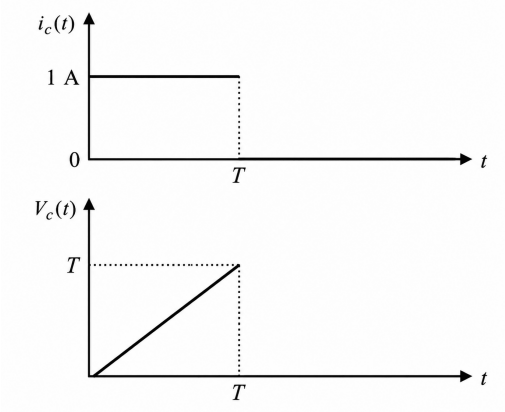


Fig. Solution Q2.08.3

This graph indicates the linear charging of the capacitor as a constant current flows through the capacitor.
Case 2:

Similarly, when S_1 is connected to Q_1 and S_2 is connected to Q_2,

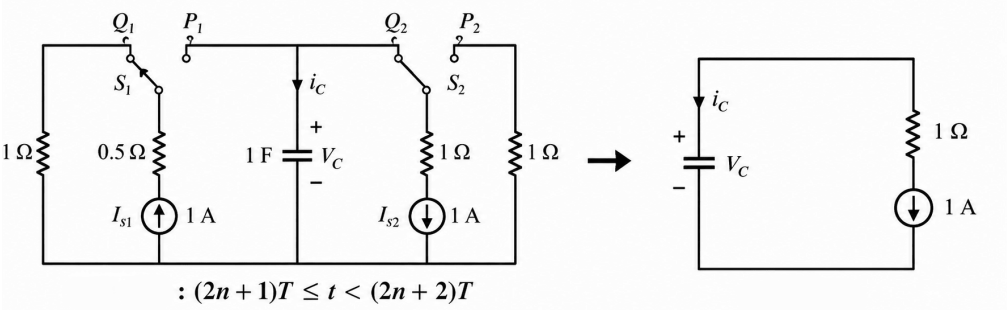


Fig. Solution Q2.08.3

For $n = 0, T \le t < 2T$

$$i_C(t) = -1 \text{ A} = -u(t)$$

The capacitor voltage is

$$V_C(t) = \frac{1}{C}\int_{-\infty}^{0} i_C(t)\,dt + \frac{1}{C}\int_{0}^{T} i_C(t)\,dt + \frac{1}{C}\int_{T}^{t} i_C(t)\,dt$$

Since $C = 1$ F,

$$V_C(t) = 0 + T + \int_{T}^{t} (-1)\,dt V_C(t) = 0 + T - \int_{T}^{t} 1\,dt$$

$$V_C(t) = T - (t - T) = 2T - t$$

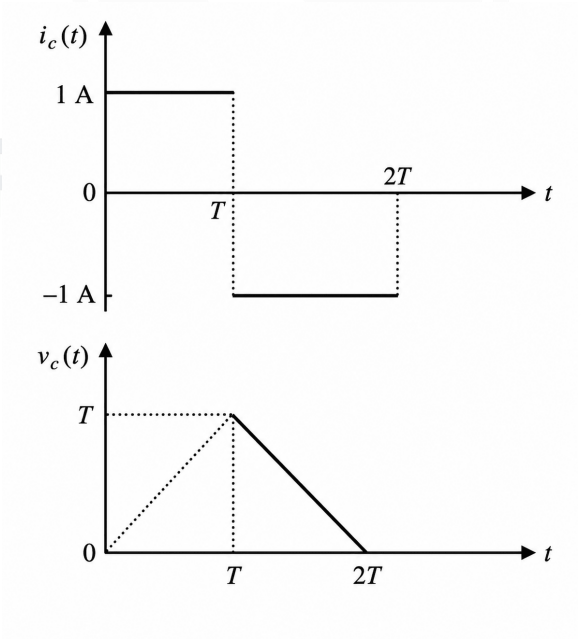


Fig. Solution Q2.08.3

This graph indicates the linear discharging of the capacitor as a constant current flows through the capacitor. Combining both the cases together,

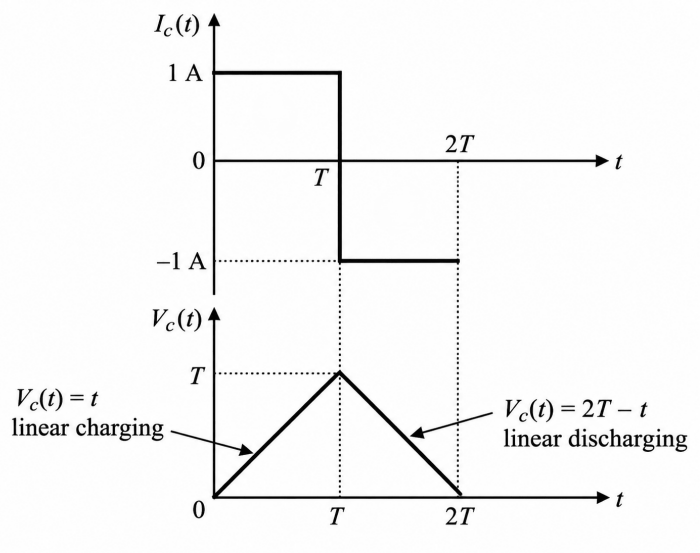


Fig. Solution Q2.08.3

So;

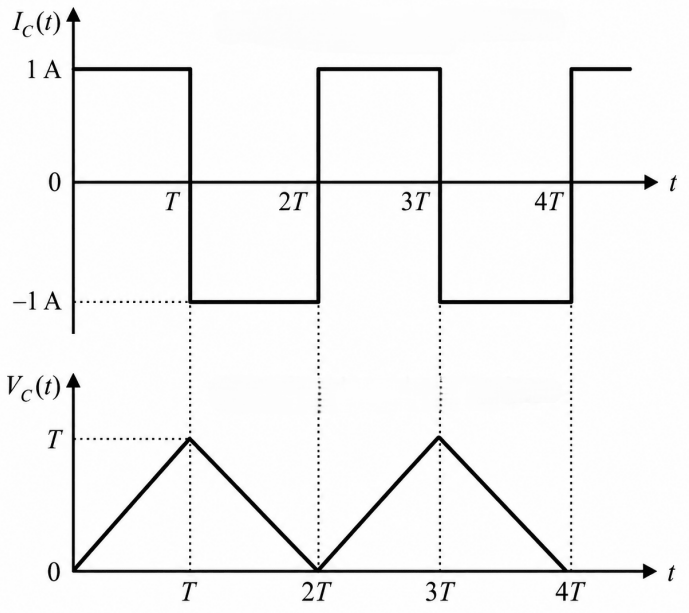


Fig. Solution Q2.08.3

Using unit step functions,

$$V_C(t) = tu(t) - (t-T)u(t-T) - (t-T)u(t-T) + (t-2T)u(t-2T) + \ldots$$

$$= tu(t) - 2(t-T)u(t-T) + 2(t-2T)u(t-2T)$$

$$-(t-3T)u(t-3T) - (t-3T)u(t-3T) + \cdots$$

$$V_C(t) = tu(t) + 2\sum_{n=1}^{\infty}(-1)^n(t-nT)\,u(t-nT)$$

Hence, the correct answer is

C. $tu(t) + 2\sum_{n=1}^{\infty}(-1)^n(t-nT)\,u(t-nT)$

Q2.08.4 MCQ | 2008 | 1M | Ans: B | ● ● ○

The current through the inductor is given by, $i(t) = i(\infty) + [i(0^+) - i(\infty)]\, e^{-t/\tau}$

At $t = 0^-$, the switch is open and the circuit is in steady state.

Hence, the inductor behaves as a short circuit.

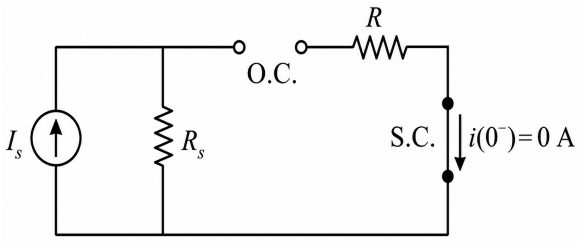


Fig. Solution Q2.08.4

Therefore, $i(0^-) = 0 \implies i(0^-) = i(0^+) = 0$

For the RL network, $\tau = \dfrac{L}{R_{\text{th}}}$

where, R_{th} is the Thevenin resistance seen by the inductor when the current source is open-circuited.

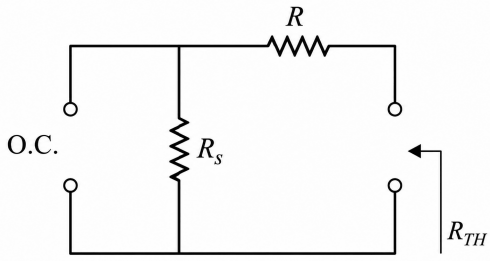


Fig. Solution Q2.08.4

From the figure, $R_{\text{th}} = R + R_s$

Hence,

$$\tau = \frac{L}{R + R_s}$$

At $t = \infty$, the inductor behaves as a short circuit.

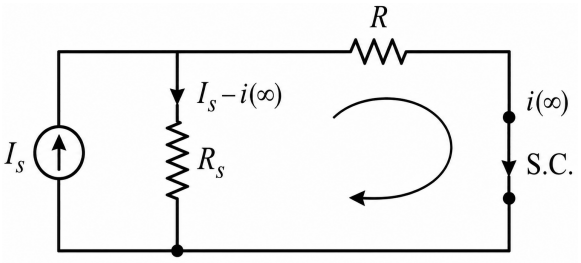


Fig. Solution Q2.08.4

Applying KVL,

$$-[I_s - i(\infty)]\, R_s + Ri(\infty) = 0$$

$$i(\infty)(R + R_s) = I_s R_s$$

$$i(\infty) = \frac{I_s R_s}{R + R_s}$$

Substituting the values,

$$i(t) = \frac{I_s R_s}{R + R_s}\left[1 - e^{-(R+R_s)t/L}\right]$$

$$\frac{di(t)}{dt} = \frac{I_s R_s}{R + R_s}\left[e^{-(R+R_s)t/L} \times \frac{R + R_s}{L}\right], \qquad \text{[Differentiating]}$$

At $t = 0^+$,

$$\left.\frac{di(t)}{dt}\right|_{0^+} = \frac{I_s R_s}{L}$$

Hence, the correct answer is

$$\boxed{\textbf{B. } \frac{R_s I_s}{L}}$$

Q2.07.1 MCQ 2007 2M Ans: A ●○○

The current through the capacitor is given by, $i_C(t) = i_C(\infty) + [i_C(0^+) - i_C(\infty)]\, e^{-t/\tau}$

Given, $V_C(0^-) = 0\ \text{V} \Rightarrow V_C(0^-) = V_C(0^+) = 0\ \text{V}$

At $t \geq 0$, for the RC network, $\tau = R_{\text{th}} C$

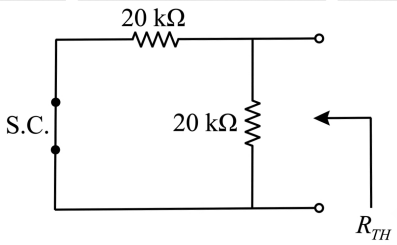


Fig. Solution Q2.07.1

$$R_{\text{th}} = 20 \parallel 20 = 10\ \text{k}\Omega$$

$$\Rightarrow \tau = (10\ \text{k}\Omega)(4\ \mu\text{F}) = 40 \times 10^{-3} = \frac{1}{25}\ \text{s}$$

At $t = 0^+$, the capacitor is replaced with short circuit as it has no initial charge;

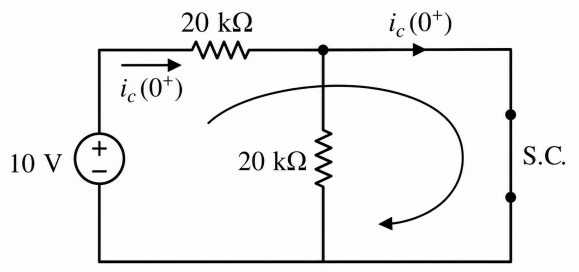


Fig. Solution Q2.07.1

$$i_c(0^+) = \frac{10}{20k} \Rightarrow i_C(0^+) = 0.5\ \text{mA}$$

At $t = \infty$, the capacitor behaves as an open circuit.

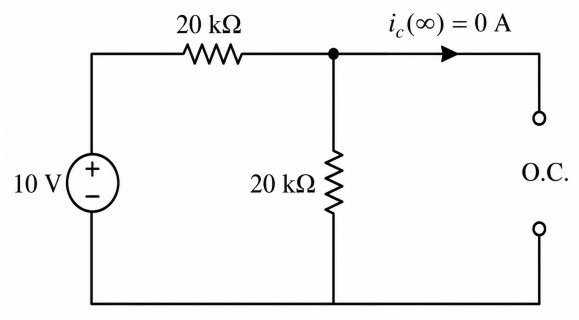


Fig. Solution Q2.07.1

Hence, $i_C(\infty) = 0$ A

Substituting the values,

$$i_C(t) = 0 + (0.5 - 0)e^{-t/(40\times10^{-3})}$$

$$i_C(t) = 0.5e^{-25t} \text{ mA}$$

Hence, the correct answer is

A. $0.5e^{-25t}$ mA

Q2.06.1 MCQ 2006 1M Ans: C

At $t = 0^+$, all the capacitors are initially uncharged.

Hence, both capacitors behave as short circuits.

Thus, the circuit reduces to;

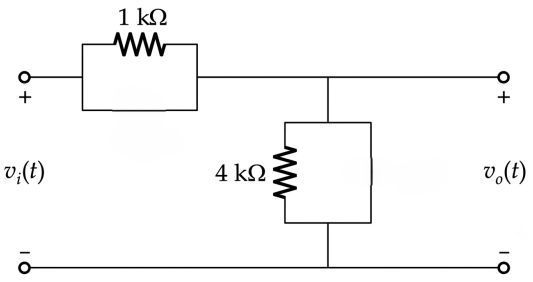


Fig. Solution Q2.06.1

The 1 μF and 4 μF capacitor charges instantaneously through the short-circuit path.

Applying voltage division,

$$v_o(0^+) = 10\left(\frac{4}{1+4}\right) = 8 \text{ V}$$

The circuit reaches steady state instantly;

Hence,

$$v_o(t) = 8u(t) \text{ V}$$

Therefore, the correct answer is

C. $8u(t)$ Volts

Q2.06.2 MCQ 2006 1M Ans: A ● ○ ○

The Laplace-domain equivalent of an inductor with initial current $i(0^-)$ is

$$V(s) = sLI(s) - Li(0^-)$$

or equivalently, an impedance sL in series with a voltage source of magnitude $Li(0^-)$.

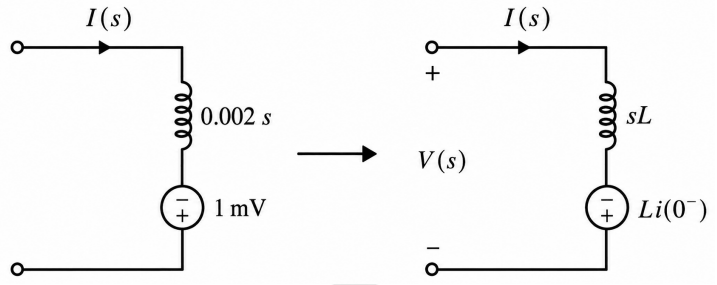


Fig. Solution Q2.06.2

Comparing the given circuit with the standard Laplace equivalent, $sL = 0.002s$
Hence,

$$L = 0.002 \text{ H} = 2 \text{ mH}$$

Also, the series voltage source is

$$Li(0^-) = 1 \text{ mV} = 10^{-3} \text{ V}$$

Therefore,

$$i(0^-) = \frac{10^{-3}}{2 \times 10^{-3}} = 0.5 \text{ A}$$

Hence, the correct answer is

A. 0.5 A

Q2.05.1 MCQ 2005 2M Ans: B ● ○ ○

Given,

$$R = 1 \text{ k}\Omega, \qquad C = 0.1 \ \mu\text{F}$$

The time constant of the RC circuit is;

$$\tau = RC = 1000 \times 0.1 \times 10^{-6} = 100 \ \mu\text{s}$$

The pulse duration is, $T_p = 2$ s
The settling time is

$$t_s = 5\tau = 500 \ \mu\text{s}$$

Since, $T_p \gg t_s$;
the circuit reaches steady state well before the end of the pulse.
Hence, just before $t = 2$ s, the capacitor behaves as an open circuit and is charged to

$$V_C(2^-) = 3 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$V_C(2^+) = V_C(2^-) = 3 \text{ V}$$

At $t = 2^+$, the input voltage falls from 3 V to 0 V, which is equivalent to short-circuiting the input source.

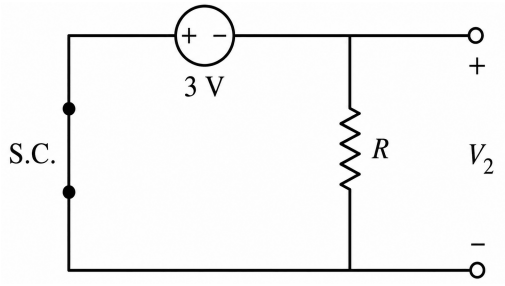


Fig. Solution Q2.05.1

Applying KVL,

$$0 + 3 + V_2 = 0$$

$$\Rightarrow V_2 = -3 \text{ V}$$

Hence, the correct answer is

B. -3 V

Q2.04.1 MCQ 2004 2M Ans: B ● ○ ○

Given,

$$i_L(0^-) = 1 \text{ A}, \qquad v_C(0^-) = -1 \text{ V}$$

The Laplace-domain equivalent of the inductor is an impedance sL in series with a voltage source $Li(0^-)$.

The Laplace-domain equivalent of the capacitor is an impedance $\frac{1}{Cs}$ in series with a voltage source $\frac{v_C(0^-)}{s}$.

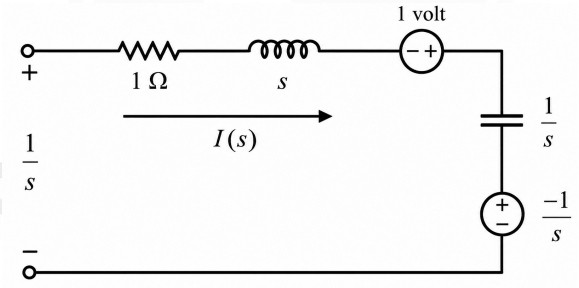


Fig. Solution Q2.04.1

Applying KVL,

$$-\frac{1}{s} + I(s)\left(1 + s + \frac{1}{s}\right) - 1 - \frac{1}{s} = 0$$

$$I(s) = \frac{1 + \frac{2}{s}}{1 + s + \frac{1}{s}}$$

Multiplying the numerator and denominator by s,

$$I(s) = \frac{s + 2}{s^2 + s + 1}$$

Hence, the correct answer is

B. $\frac{s + 2}{s^2 + s + 1}$

Q2.04.2 MCQ | 2004 | 1M | Ans: C | ● ○ ○

Method Transient Analysis

For an RL circuit, $i(t) = i(\infty) + [i(0) - i(\infty)]\, e^{-t/\tau}$
where;

$$\tau = \frac{L}{R} = \frac{1}{2} = 0.5 \text{ s}$$

Before switching, $v_i(t) = 0$
Hence, the circuit is source free and $i(0^-) = 0$
Since the current through an inductor cannot change instantaneously,

$$i(0^+) = i(0^-) = 0$$

At steady state ($t = \infty$), the inductor behaves as a short circuit.

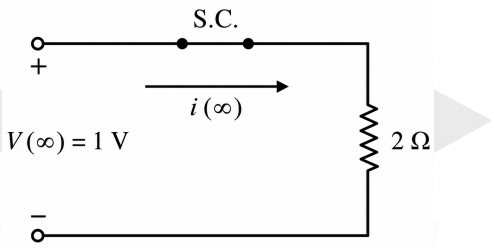


Fig. Solution Q2.04.2

Hence,

$$i(\infty) = \frac{1}{2} = 0.5 \text{ A}$$

Therefore,

$$i(t) = \frac{1}{2}\left(1 - e^{-2t}\right) \text{ A}$$

At $t = \tau = \frac{1}{2}$ s,

$$i\left(\frac{1}{2}\right) = \frac{1}{2}\left(1 - e^{-1}\right) = 0.316 \approx 0.31 \text{ A}$$

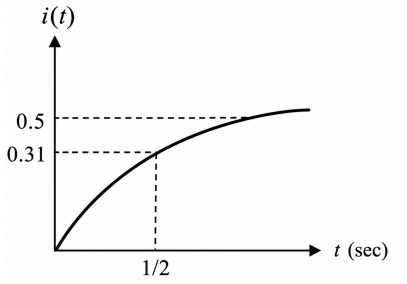


Fig. Solution Q2.04.2

Hence, the waveform corresponding to these values is Option (C).

Method Direct Analysis

The time constant is

$$\tau = \frac{L}{R} = \frac{1}{2} = 0.5 \text{ s}$$

At one time constant,

$$i(\tau) = 63\% \text{ of the steady-state value.}$$

The steady-state current is, $i(\infty) = \frac{1}{2} = 0.5$ A
Therefore,

$$i(\tau) = 0.63 \times 0.5 = 0.315 \approx 0.31 \text{ A}$$

Thus, the required graph must pass through

$$(t, i) = (0.5 \text{ s}, 0.31 \text{ A})$$

and finally settle at 0.5 A.
Hence, the correct answer is

C

Q2.03.1 MCQ 2003 2M Ans: A ●●○

Before switching, the switch is in position 1 for a long time, circuit is in steady state;

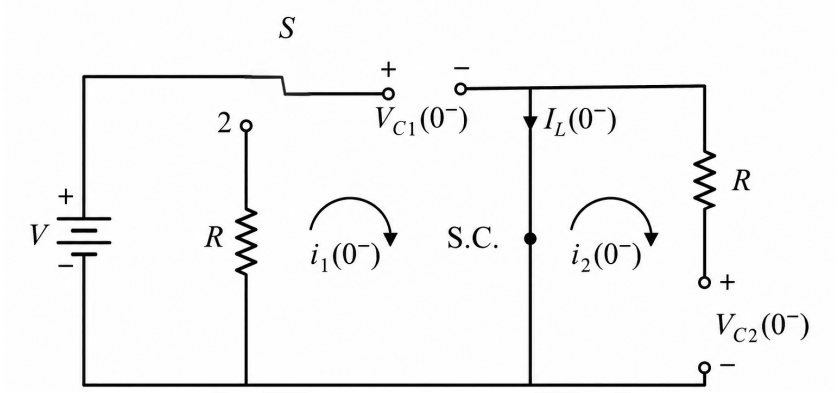


Fig. Solution Q2.03.1

From the circuit,

$$V_{C_1}(0^-) = V, \quad V_{C_2}(0^-) = 0, \quad i_L(0^-) = 0$$

Since capacitor voltage and inductor current cannot change instantaneously,

$$V_{C_1}(0^+) = V, \quad V_{C_2}(0^+) = 0, \quad i_L(0^+) = 0$$

At $t = 0^+$,
Capacitor C_1 is replaced by a voltage source of value V.
Inductor is replaced by a current source of value 0 A, i.e., an open circuit.
Capacitor C_2 is replaced by a voltage source of value 0 V, i.e., a short circuit.

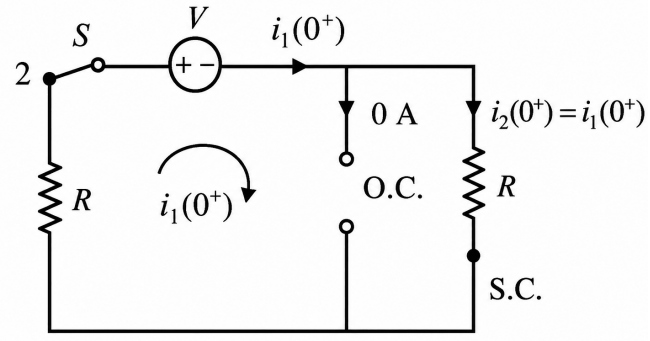


Fig. Solution Q2.03.1

Applying KVL,

$$Ri_1(0^+) + V + Ri_1(0^+) = 0 \Rightarrow 2Ri_1(0^+) = -V$$

$$i_1(0^+) = -\frac{V}{2R}$$

Hence, the correct answer is

A. $-\frac{V}{2R}$

Q2.03.2 MCQ 2003 2M Ans: C ●●○

From the previous question,

$$V_{C_1}(0^-) = V, \qquad V_{C_2}(0^-) = 0, \qquad i_L(0^-) = 0$$

After the switch is moved to position 2, the circuit becomes

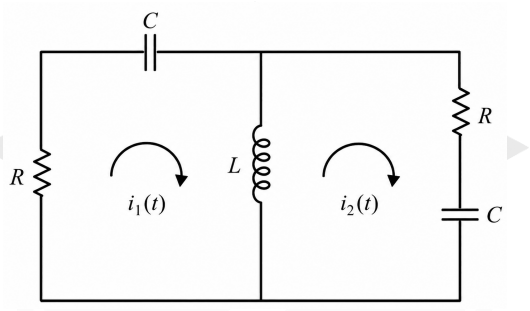


Fig. Solution Q2.03.2

In the Laplace domain,

Capacitor C_1 is replaced by an impedance $\frac{1}{Cs}$ in series with a voltage source $\frac{V}{s}$.

Capacitor C_2 is replaced by an impedance $\frac{1}{Cs}$ since $V_{C_2}(0^-) = 0$.

Inductor is replaced by an impedance Ls since $i_L(0^-) = 0$.

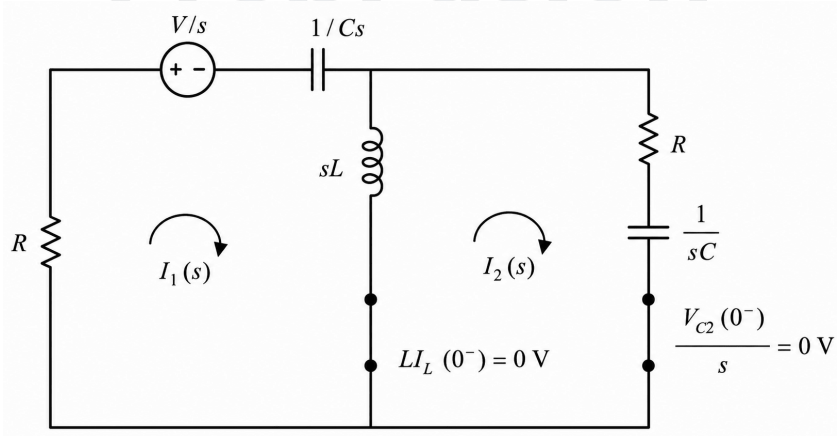


Fig. Solution Q2.03.2

Applying KVL in Loop 1,

$$I_1(s)R + \frac{I_1(s)}{Cs} + [I_1(s) - I_2(s)]\,Ls = -\frac{V}{s}$$

$$I_1(s)\left(R + Ls + \frac{1}{Cs}\right) - Ls\,I_2(s) = -\frac{V}{s}$$

Applying KVL in Loop 2,

$$[I_2(s) - I_1(s)]\, Ls + I_2(s)\left(R + \frac{1}{Cs}\right) = 0$$

$$-Ls\, I_1(s) + I_2(s)\left(R + Ls + \frac{1}{Cs}\right) = 0$$

Hence,

$$\begin{bmatrix} R + Ls + \frac{1}{Cs} & -Ls \\ -Ls & R + Ls + \frac{1}{Cs} \end{bmatrix} \begin{Bmatrix} I_1(s) \\ I_2(s) \end{Bmatrix} = \begin{Bmatrix} -\frac{V}{s} \\ 0 \end{Bmatrix}$$

Therefore, the correct answer is

C

Q2.02.1 MCQ 2002 1M Ans: C

Before switching, the switch is closed for a long time.

Hence, the circuit is in steady state and the inductor behaves as a short circuit.

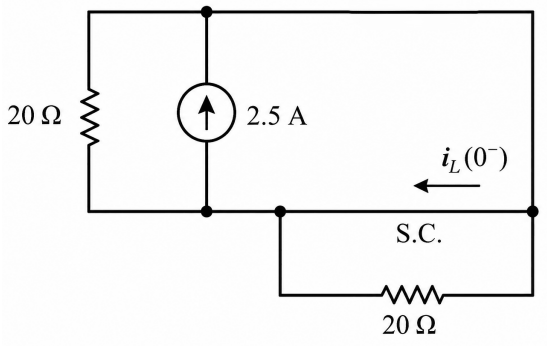


Fig. Solution Q2.02.1

Therefore, $i_L(0^-) = 2.5$ A

Since the current through an inductor cannot change instantaneously, $i_L(0^+) = i_L(0^-) = 2.5$ A

At $t = 0^+$, the switch is opened and the inductor is replaced by a current source of value 2.5 A.

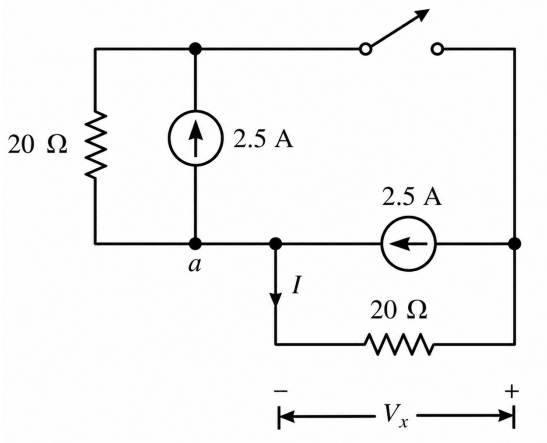


Fig. Solution Q2.02.1

The 2.5 A current flows through the 20 Ω resistor.

Hence, $I = 2.5$ A
By Ohm's law,

$$V_x = -IR = -2.5 \times 20 = -50 \text{ V}$$

Hence, the correct answer is

C. -50 V

Mistakes to be avoided

- Do not mark option D, as the switch is open so there is no return path for 2.5A source. This is why both 2.5A current flows in their own parallel 20ohm resistor.

Q2.97.1 MCQ 1997 2M Ans: B ● ○ ○

Initially, $V_C(0^-) = 6$ V

At steady state ($t = \infty$), the capacitor behaves as an open circuit.
Hence, there is no current through the resistor and the capacitor charges to the source voltage.

$$\Rightarrow V_C(\infty) = 10 \text{ V}$$

The energy supplied to the capacitor is

$$\Delta W_C = \frac{1}{2} C \left[V_C^2(\infty) - V_C^2(0^-)\right]$$

$$\Delta W_C = \frac{1}{2} \times 2 \left(10^2 - 6^2\right) = 64 \text{ J}$$

The energy supplied by the source is

$$W_S = V \int_0^\infty i(t)\, dt = VQ$$

where

$$Q = C\left[V_C(\infty) - V_C(0^-)\right] = 2(10 - 6) = 8 \text{ C}$$

Hence,

$$W_S = 10 \times 8 = 80 \text{ J}$$

Therefore, the energy absorbed by the resistor is

$$W_R = W_S - \Delta W_C = 80 - 64 = 16 \text{ J}$$

Hence, the correct answer is

B. 16 J

Q2.96.1 MCQ 1996 2M Ans: B ● ○ ○

At steady state,

Inductor $\rightarrow$ Short circuit, Capacitor $\rightarrow$ Open circuit

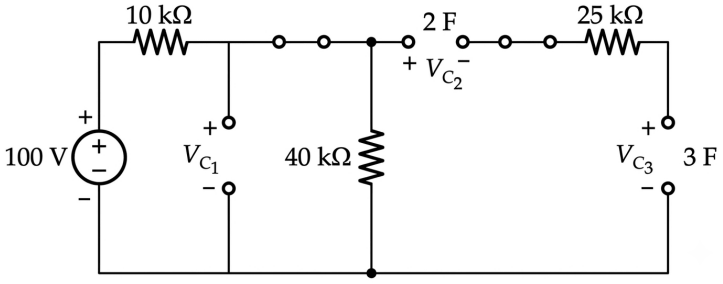


Fig. Solution Q2.96.1

The 10 kΩ and 40 kΩ resistors form a voltage divider across the 100 V source.
Hence,

$$V_{C1} = 100 \times \frac{40}{10 + 40} = 80 \text{ V}$$

Then using capacitive voltage division,

$$V_{C2} = 80 \times \frac{3}{2 + 3} = 48 \text{ V}$$

$$V_{C3} = 80 \times \frac{2}{2 + 3} = 32 \text{ V}$$

Therefore, the correct answer is

B. 80 V, 48 V, 32 V

Q2.95.1 MCQ 1995 1M Ans: B ● ○ ○

When a DC voltage source V is connected to a series RC circuit, at steady state the capacitor behaves as an open circuit.
$\Rightarrow V_C = V$

The energy stored in the capacitor is, $W_C = \frac{1}{2}CV^2$
The total energy supplied by the source is

$$W_S = VQ = V(CV) = CV^2$$

$$\Rightarrow \frac{W_C}{W_S} = \frac{\frac{1}{2}CV^2}{CV^2} = \frac{1}{2} = 0.5$$

Hence, the correct answer is

B. 0.500

Q2.94.1 MCQ 1994 1M Ans: B ● ○ ○

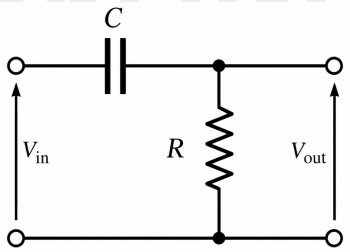


Fig. Solution Q2.94.1

For an RC differentiating circuit,

$$V_o = RC\frac{dv_i}{dt}$$

Given, $v_i(t) = 100t$
Hence,

$$\frac{dv_i}{dt} = 100 \text{ V/s}$$

Therefore,

$$V_o = (5 \times 10^3)(4 \times 10^{-6})(100) = 2 \text{ V}$$

Since the slope of the ramp input is constant, the output voltage is constant.

Hence, the maximum output voltage is

$$\boxed{2\text{ V}}$$

Therefore, the correct answer is

$$\boxed{\textbf{B. } 2.0\text{ V}}$$

Q2.92.1 MCQ 1992 2M Ans: C ● ● ○

The charging time constant is obtained when the switch is at position A.

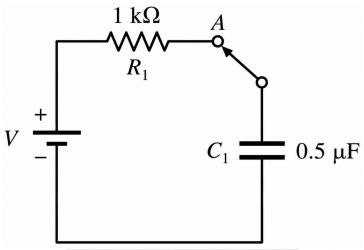


Fig. Solution Q2.92.1

Hence,

$$\tau_c = R_1 C = (1\text{ k}\Omega)(0.5\ \mu\text{F}) = 0.5 \times 10^{-3}\text{ s}$$

When the switch is moved to position B, the circuit has no independent source.

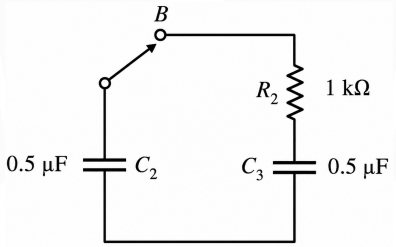


Fig. Solution Q2.92.1

The two capacitors are connected in series.
Hence, the equivalent capacitance is

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{0.5 \times 0.5}{0.5 + 0.5} = 0.25\ \mu\text{F}$$

The discharging time constant is

$$\tau_d = R_2 C_{\text{eq}} = (1\text{ k}\Omega)(0.25\ \mu\text{F}) = 0.25 \times 10^{-3}\text{ s}$$

Hence, the correct answer is

$$\boxed{\textbf{C. } 1, 3}$$

Q2.92.2 MCQ 1992 2M Ans: B ● ● ○

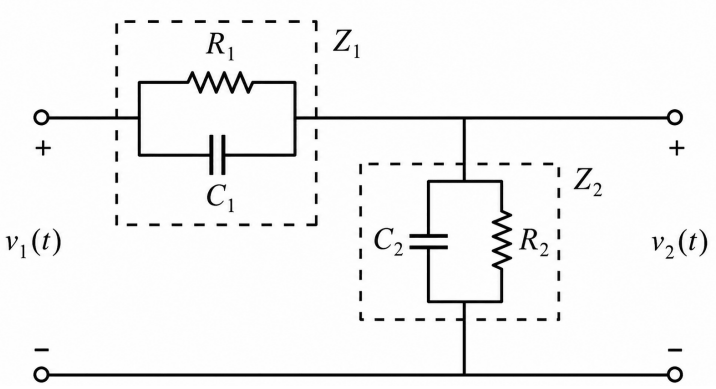


Fig. Solution Q2.92.2

Method Using Impulse Response

The impedance of the left branch is

$$Z_1(s) = R_1 \parallel \frac{1}{C_1 s} = \frac{R_1\left(\frac{1}{C_1 s}\right)}{R_1 + \frac{1}{C_1 s}} = \frac{R_1}{R_1 C_1 s + 1}$$

Given, $R_1C_1 = R_2C_2$

$$\Rightarrow Z_1(s) = \frac{R_1}{R_2 C_2 s + 1}$$

Similarly,

$$Z_2(s) = R_2 \parallel \frac{1}{C_2 s} = \frac{R_2}{R_2 C_2 s + 1}$$

Applying voltage division,

$$\frac{V_2(s)}{V_1(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

Substituting $Z_1(s)$ and $Z_2(s)$,

$$\frac{V_2(s)}{V_1(s)} = \frac{\frac{R_2}{R_2 C_2 s + 1}}{\frac{R_1}{R_2 C_2 s + 1} + \frac{R_2}{R_2 C_2 s + 1}} = \frac{R_2}{R_1 + R_2}$$

Therefore,

$$V_2(s) = \frac{R_2}{R_1 + R_2} V_1(s)$$

Given, $v_1(t) = \delta(t) \Rightarrow V_1(s) = 1$

Therefore,

$$V_2(s) = \frac{R_2}{R_1 + R_2}$$

$$v_2(t) = \frac{R_2}{R_1 + R_2}\delta(t) \qquad \text{[Taking inverse Laplace transform]}$$

Method Using Step Response

For an LTI circuit,

$$h(t) = \frac{d}{dt}s(t)$$

i.e., the impulse response is the derivative of the step response.

Since both branches have the same time constant.
Hence, the frequency-dependent terms cancel while applying voltage division.
Therefore,

$$\frac{V_2(s)}{V_1(s)} = \frac{Z_2}{Z_1 + Z_2} = \frac{R_2}{R_1 + R_2},$$

which is independent of s.
Thus, the circuit behaves as an ideal resistive voltage divider with a constant gain.

$$v_2(t) = \frac{R_2}{R_1 + R_2} u(t)$$

Differentiating both sides,

$$\frac{d}{dt} v_2(t) = \frac{R_2}{R_1 + R_2} \frac{d}{dt} u(t)$$

Since,

$$\frac{d}{dt} u(t) = \delta(t),$$

the impulse response is

$$h(t) = \frac{R_2}{R_1 + R_2} \delta(t)$$

Hence, the correct answer is

B. $\dfrac{R_2}{R_1 + R_2} \delta(t)$

Q2.89.1 MCQ 1989 2M Ans: C ●●○

Given, $C = \frac{1}{2}$ F
The impedance of the capacitor in the Laplace domain is

$$Z_C(s) = \frac{1}{Cs} = \frac{2}{s}$$

$$\Rightarrow I_C(s) = \frac{V_C(s)}{Z_C(s)} = \frac{s(s+1)}{2(s^3 + s^2 + s + 1)}$$

By the Initial Value Theorem,

$$i_C(0^+) = \lim_{s \to \infty} sI_C(s)$$

Substituting $I_C(s)$,

$$i_C(0^+) = \lim_{s \to \infty} \frac{s^2(s+1)}{2(s^3 + s^2 + s + 1)}$$

Dividing the numerator and denominator by s^3,

$$i_C(0^+) = \lim_{s \to \infty} \frac{1 + \frac{1}{s}}{2\left(1 + \frac{1}{s} + \frac{1}{s^2} + \frac{1}{s^3}\right)} = \frac{1}{2} \text{ A}$$

Hence, the correct answer is

C. $\frac{1}{2}$ A

Key Points

- **Initial Value Theorem (IVT)** $f(0^+) = \lim_{s\to\infty} sF(s)$

  Conditions:

  - $f(t)$ must not contain an impulse (or its derivatives) at $t = 0$.
  - The function $f(t)$ must be causal
  - The Laplace transform $F(s)$ must be a strictly proper or biproper rational function, meaning the highest power of s in the numerator must not exceed the highest power of s in the denominator.

- **Final Value Theorem (FVT)**; $f(\infty) = \lim_{s\to 0} sF(s)$

  Conditions:

  - All poles of $sF(s)$ must lie in the left-half plane (LHP), except at most one simple pole at the origin.
  - $sF(s)$ must not have poles in the right-half plane (RHP).
  - $sF(s)$ must not have repeated poles at the origin.
  - $sF(s)$ must not have poles on the $j\omega$-axis (except a simple pole at $s = 0$).
  - The final value $f(\infty)$ must exist (system must be stable).

Q2.89.2 MCQ 1989 2M Ans: B ● ○ ○

Under steady-state condition,

Inductor → Short circuit, Capacitor → Open circuit

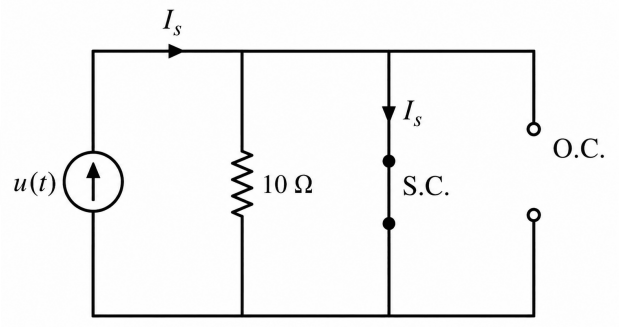


Fig. Solution Q2.89.2

Therefore, the entire source current flows through the inductor.
Hence, the correct answer is

B. the inductor

Q2.87.1 MCQ 1987 2M Ans: A ● ○ ○

For an inductor,

$$i_L(t) = \frac{1}{L}\int v_L(t)\,dt$$

Thus, the current through an inductor is the integration of the applied voltage.
A square wave, when integrated, produces a triangular waveform.
Given, $L = 1$ mH
The applied voltage is +1 V for 0.5 ms, followed by -1 V for the next 0.5 ms.

The change in current during the positive half-cycle is

$$\Delta i = \frac{V}{L}\Delta t = \frac{1}{1 \times 10^{-3}}\left(0.5 \times 10^{-3}\right) = 0.5 \text{ A}$$

Similarly, during the negative half-cycle, $\Delta i = -0.5$ A
Hence, the current increases linearly to 0.5 A and then decreases linearly back to zero, repeating periodically.

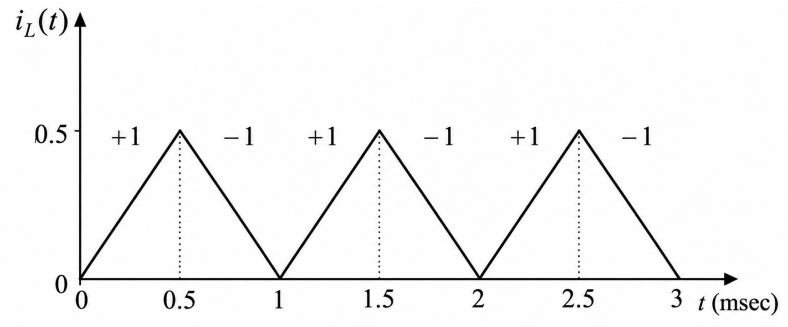


Fig. Solution Q2.87.1

Hence, the correct answer is

A. Triangular wave with peak amplitude of 0.5 A

PrepFusion

SOLUTIONS

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and Conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Unit Snapshot*

Stream: EC	**Questions:** 39	**MCQ:** 24	**MSQ:** 1	**NAT:** 14
1M: 17	**2M:** 22	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–3M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Q3.24.1 NAT | 2024 | 2M | Ans: 2.5 | ● ● ○

For maximum power transfer, $R_L = R_{th}$

Deactivate the independent 50 V source by replacing it with a short circuit and apply a test voltage V across the load terminals.

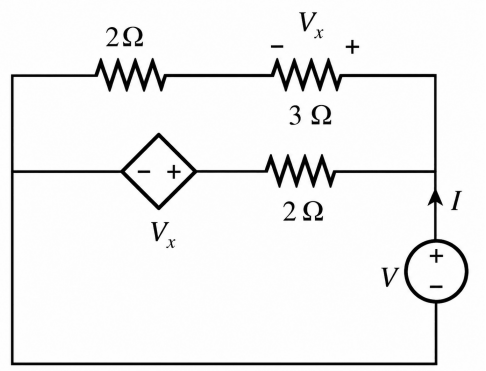


Fig. Solution Q3.24.1

Let the current drawn from the test source be I.

Applying KCL at the load terminal,

$$I = \frac{V - V_x}{2} + \frac{V}{5}$$

$$\because V_x = \frac{3}{2+3}V = \frac{3V}{5}$$

$$\Rightarrow I = \frac{V - \frac{3V}{5}}{2} + \frac{V}{5} = \frac{V}{5} + \frac{V}{5} = \frac{2V}{5}$$

Hence,

$$R_{th} = \frac{V}{I} = \frac{V}{\frac{2V}{5}} = \frac{5}{2} = 2.5\ \Omega$$

Therefore, for maximum power transfer,

$$R_L = R_{th} = 2.5\ \Omega$$

Hence,

$$\boxed{2.5\ \Omega}$$

Q3.22.1 MCQ | 2022 | 1M | Ans: B | ● ○ ○

Method Nodal Analysis

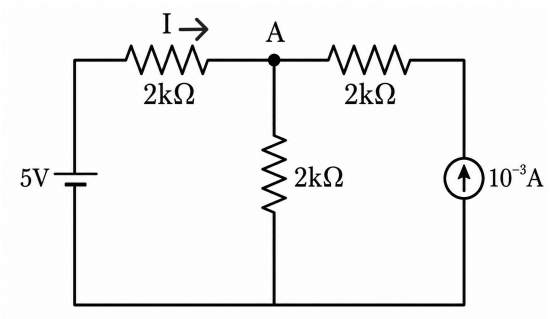


Fig. Solution Q3.22.1

Let the voltage at node A be V_A.
Applying KCL at node A,

$$\frac{V_A}{2\,\text{k}\Omega} + \frac{V_A - 5}{2\,\text{k}\Omega} = 10^{-3}$$

$$2V_A - 5 = 2 \times 10^3 \times 10^{-3}$$

$$2V_A = 7 \Rightarrow V_A = 3.5\ \text{V}$$

The required current is

$$I = \frac{5 - V_A}{2\,\text{k}\Omega} = \frac{5 - 3.5}{2 \times 10^3} = 0.75 \times 10^{-3}\ \text{A}$$

Method Superposition Theorem

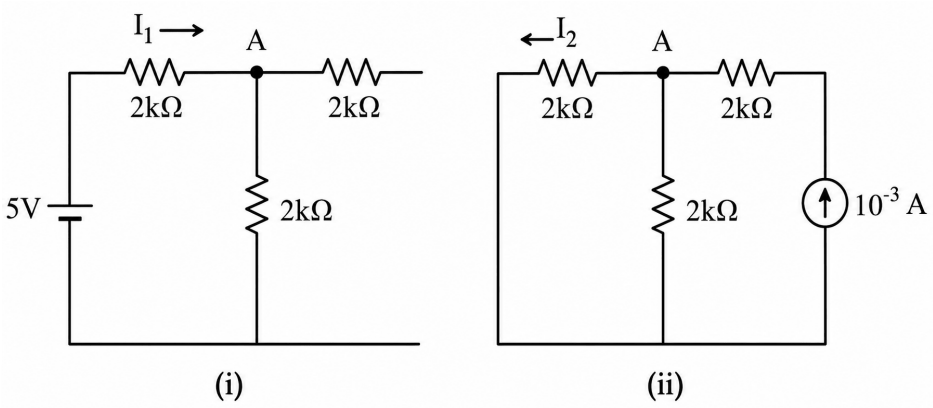


Fig. Solution Q3.22.1

Case 1: Only the 5 V source is active
The current source is opened.

$$I_1 = \frac{5}{2\,\text{k}\Omega + 2\,\text{k}\Omega} = \frac{5}{4 \times 10^3} = 1.25 \times 10^{-3}\ \text{A}$$

Case 2: Only the 10^{-3} A current source is active
The voltage source is short-circuited.
The 10^{-3} A current divides equally between the two identical 2 kΩ branches.

$$I_2 = \frac{10^{-3}}{2} = 0.5 \times 10^{-3}\ \text{A}$$

The direction of I_2 is opposite to the reference direction of I.

$$\Rightarrow I = I_1 - I_2$$

$$= 1.25 \times 10^{-3} - 0.5 \times 10^{-3} = 0.75 \times 10^{-3}\ \text{A}$$

Hence, the correct answer is

B. 0.75×10^{-3} A

Q3.20.1 MCQ | 2020 | 1M | Ans: B | ● ● ○

Transform the left-hand current source into its equivalent voltage source.

$$1\text{ A} \parallel 1\ \Omega \implies 1\text{ V in series with } 1\ \Omega$$

The two current sources are now in parallel $\Rightarrow I_N = 1 + 2 = 3$ A
The equivalent parallel resistance is,

$$R_N = 3 \parallel 2 = \frac{3 \times 2}{3 + 2} = \frac{6}{5}\ \Omega$$

Converting this Norton equivalent into its Thevenin equivalent,

$$V_{th} = I_N R_N = 3 \times \frac{6}{5} = \frac{18}{5} = 3.6\text{ V}$$

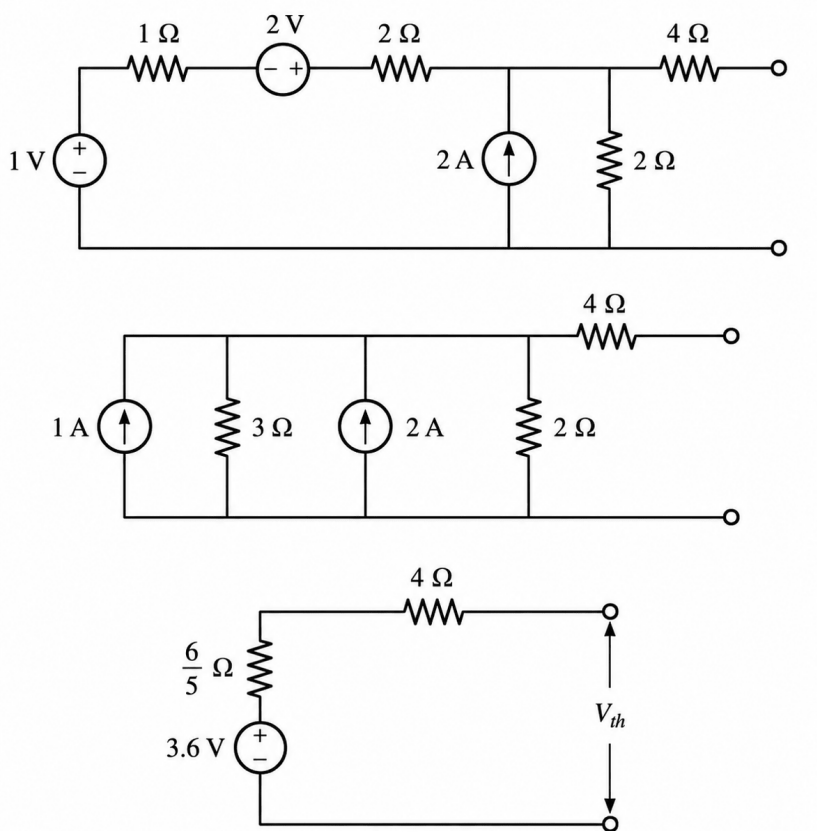


Fig. Solution Q3.20.1

$$\Rightarrow V_{th} = 3.6\text{ V}$$

Hence, the correct answer is

B. 3.6 V

Q3.19.1 MCQ | 2019 | 1M | Ans: B | ● ○ ○

A passive linear bilateral network satisfies the Reciprocity Theorem. It states that if a voltage source V_1 applied at one port produces a current I_1 at another port, then interchanging the position of the excitation and the response gives

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

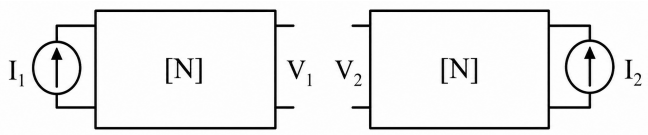


Fig. Solution Q3.19.1

From the given circuit in question,

$$V_1 = 5\text{ V}, \qquad I_1 = 1\text{ A}$$

Applying the Reciprocity Theorem,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2} \Rightarrow \frac{5}{1} = \frac{5}{I_2}$$

$$I_2 = 1\text{ A}$$

Hence, the correct answer is

B. 1 A

Key Points

- **Reciprocity Theorem** is applicable only to **linear, bilateral (reciprocal), passive (resistive)** networks.
- The theorem is valid only when the network contains **a single independent source**. Dependent sources may be present if the network remains linear and bilateral.
- If multiple independent sources are present, apply the **Superposition Theorem** first and then use Reciprocity to each source individually.

Q3.18.1 NAT | 2018 | 2M | Ans: 8.00 | ● ● ○

By symmetry of the network, [B and C] and [D and E] are at the same potential since they are equidistant from the source terminals through identical resistive paths. Hence, the resistors connected between these equipotential points carry zero current and can be removed.

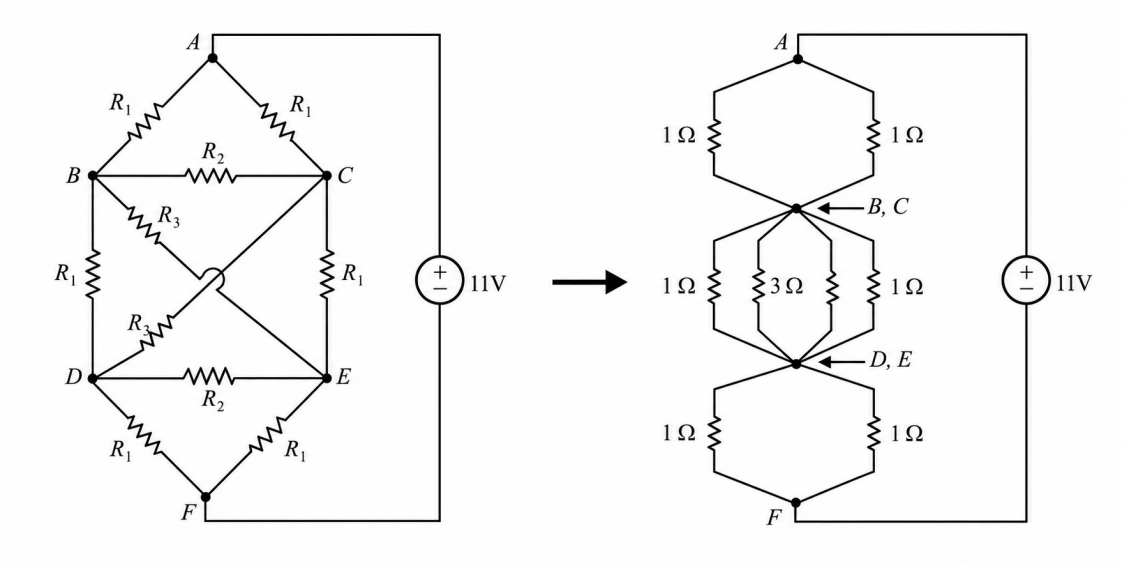


Fig. Solution Q3.18.1

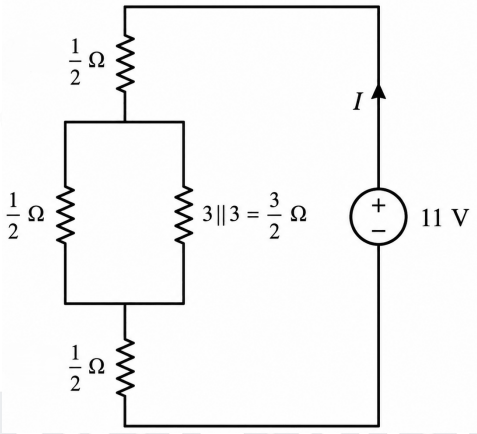


Fig. Solution Q3.18.1

Hence,

$$R_{eq} = \frac{1}{2} + \frac{3}{8} + \frac{1}{2} = \frac{11}{8}\ \Omega$$

Therefore, the current supplied by the source is

$$I = \frac{V}{R_{eq}} = \frac{11}{11/8} = 8\ \text{A}$$

Hence,

$$\boxed{8.00}$$

Q3.17.1 NAT 2017 2M Ans: -1.00 ●●○

To find the Thevenin resistance, deactivate the independent 10 V source and apply a 1 A test current source across terminals P and Q.

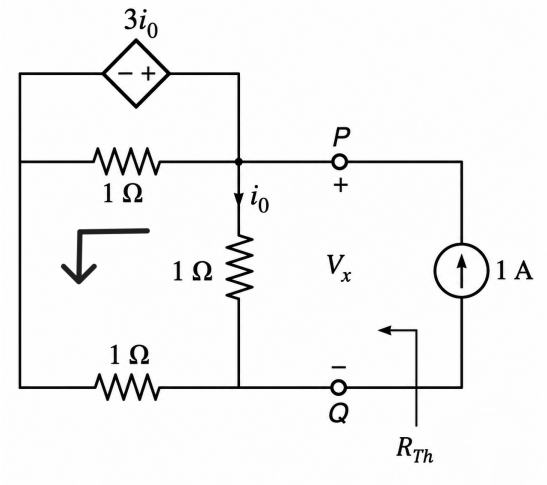


Fig. Solution Q3.17.1

Let the voltage across the test source be V_x.
The current through the 1 Ω resistor is, $i_o = V_x$
Applying KVL in the left loop,

$$V_x = 3i_o + (1 - i_o) = 2i_o + 1$$

$$\Rightarrow 2i_o + 1 = i_o$$

$$i_o = -1 \text{ A} \Rightarrow V_x = -1 \text{ V}$$

The Thevenin resistance is

$$R_{\text{Th}} = \frac{V_x}{1} = -1 \ \Omega$$

Hence,

$$\boxed{-1.00}$$

Q3.16.1 MCQ | 2016 | 1M | Ans: B | ● ● ○

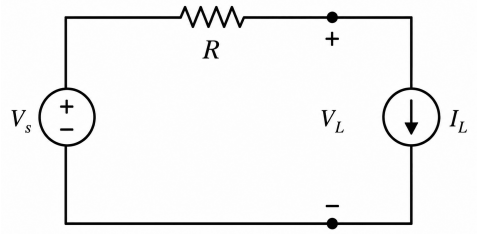


Fig. Solution Q3.16.1

Applying KVL to the circuit,

$$V_L = V_s - I_L R$$

where V_L is the voltage across the constant current load.
The power absorbed by the load is

$$P_L = V_L I_L = (V_s - I_L R) I_L$$

$$P_L = V_s I_L - I_L^2 R$$

For maximum power transfer,

$$\frac{dP_L}{dI_L} = 0$$

$$\Rightarrow \frac{dP_L}{dI_L} = V_s - 2I_L R$$

$$V_s - 2I_L R = 0$$

Hence,

$$I_L = \frac{V_s}{2R}$$

Therefore, the correct answer is

B. $\frac{V_s}{2R}$

Q3.16.2 NAT 2016 2M Ans: 0.80 ●●○

For maximum power transfer, $R = R_{\text{Th}}$

And the maximum power is, $P_{\max} = \frac{V_{\text{Th}}^2}{4R_{\text{Th}}}$

Step 1: V_{Th}

Open-circuit the load R.

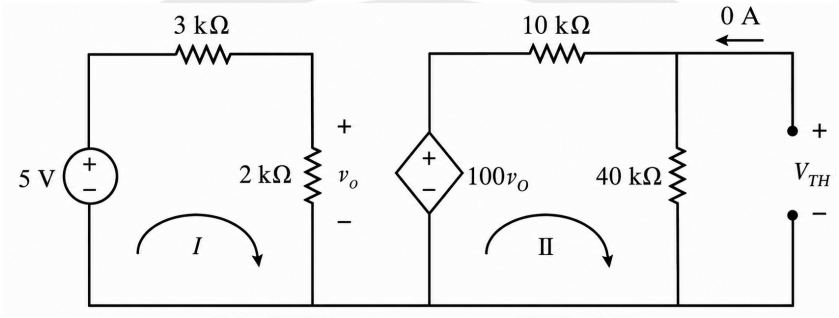


Fig. Solution Q3.16.2

Using voltage division in loop I;

$$v_o = \frac{2}{3+2} \times 5 = 2 \text{ V}$$

Hence, the dependent voltage source is, $100v_o = 100 \times 2 = 200$ V

Applying voltage division in loop II,

$$V_{\text{Th}} = \frac{40}{10+40} \times 200 = 160 \text{ V}$$

Step 2: R_{Th}

Deactivate the independent source by replacing the 5 V source with a short circuit.

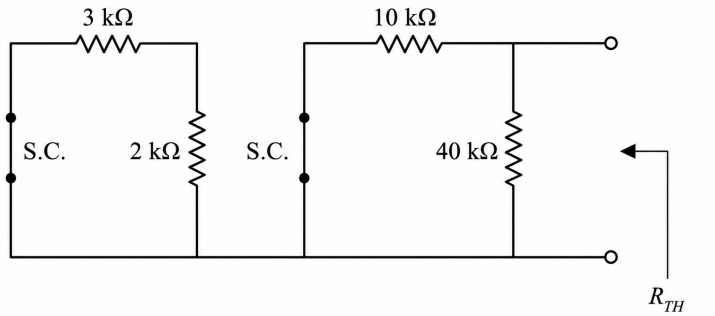


Fig. Solution Q3.16.2

Therefore,

$$R_{Th} = 10\ \text{k}\Omega \parallel 40\ \text{k}\Omega = \frac{10 \times 40}{10 + 40} = 8\ \text{k}\Omega$$

Substituting in the maximum power formula,

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{160^2}{4 \times 8000} = 0.8\ \text{W}$$

Hence,

$$\boxed{0.80}$$

Q3.16.3 NAT | 2016 | 2M | Ans: -1.00 | ● ● ○

Method Nodal Analysis

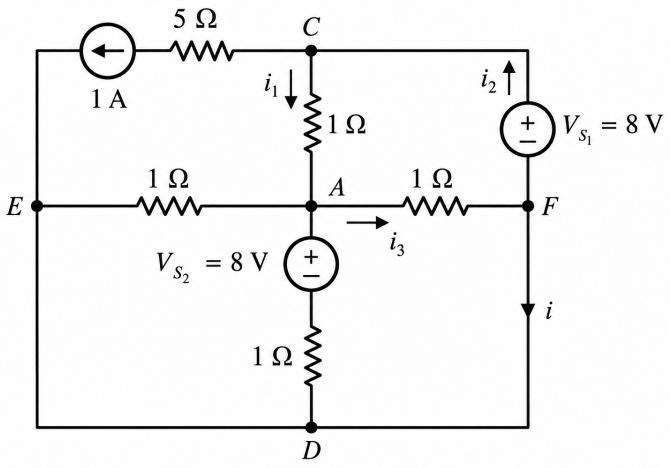


Fig. Solution Q3.16.3

Taking node D as reference $\Rightarrow V_D = V_E = V_F = 0$
Let the voltage at node A be V_A.
Applying KCL at node A,

$$\frac{V_A - V_F}{1} + \frac{V_A - V_E}{1} + \frac{V_A - V_C}{1} + \frac{V_A - 8 - V_D}{1} = 0$$

Substituting the node voltages,

$$\frac{V_A}{1} + \frac{V_A}{1} + \frac{V_A - 8}{1} + \frac{V_A - 8}{1} = 0$$

$$4V_A = 16 \Rightarrow V_A = 4\ \text{V}$$

Hence,

$$i_1 = \frac{8 - 4}{1} = 4\ \text{A}$$

$$i_3 = \frac{V_A - V_F}{1} = \frac{4 - 0}{1} = 4\ \text{A}$$

Applying KCL at node C,

$$-i_2 + i_1 = 1$$

$$i_2 = 4 - 1 = 3\ \text{A}$$

Applying KCL at node F,

$$i = -i_3 + i_2$$

$$i = -4 + 3 = -1 \text{ A}$$

Method Thevenin's Theorem

Calculation of R_{Th}

Deactivate all independent sources.

Replace the voltage sources by short circuits and the current source by an open circuit.

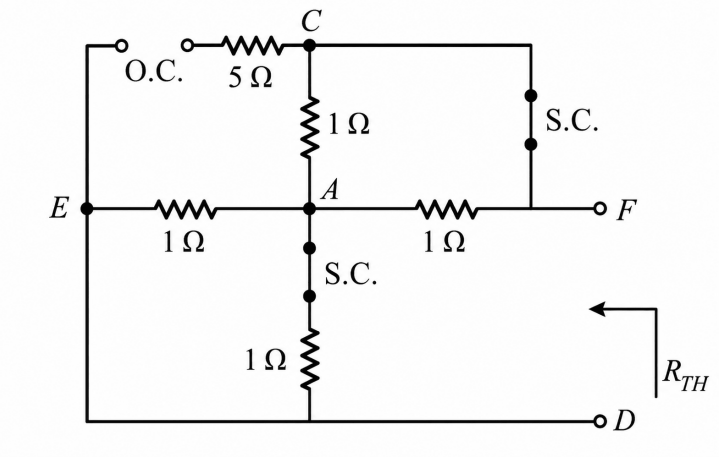


Fig. Solution Q3.16.3

The equivalent resistance is

$$R_{\text{Th}} = 1 \parallel 1 + (1 \parallel 1) = 1\ \Omega$$

Calculation of V_{Th}

Open-circuit the output terminals.

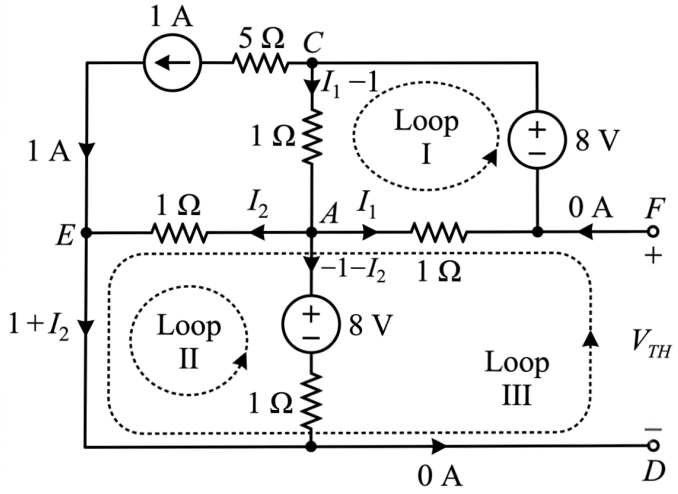


Fig. Solution Q3.16.3

Applying KVL in Loop I,

$$-8 + I_1 - 1 + I_1 = 0$$

$$I_1 = \frac{9}{2} \text{ A}$$

Applying KVL in Loop II,

$$I_2 + (1 + I_2) - 8 = 0$$

$$2I_2 = 7 \Rightarrow I_2 = \frac{7}{2}\text{ A}$$

Applying KVL in Loop III,

$$-V_{\text{Th}} - I_1 + I_2 = 0$$

$$V_{\text{Th}} = I_2 - I_1 = \frac{7}{2} - \frac{9}{2} = -1\text{ V}$$

The Thevenin equivalent circuit is

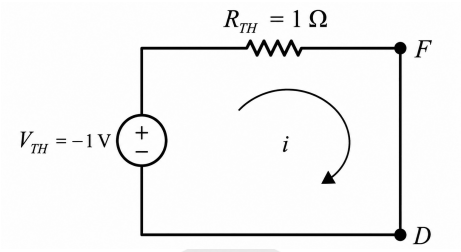


Fig. Solution Q3.16.3

Hence,

$$i = \frac{V_{\text{Th}}}{R_{\text{Th}}} = \frac{-1}{1} = -1\text{ A}$$

Therefore,

$$\boxed{-1.00}$$

Q3.16.4 MCQ | 2016 | 2M | Ans: D

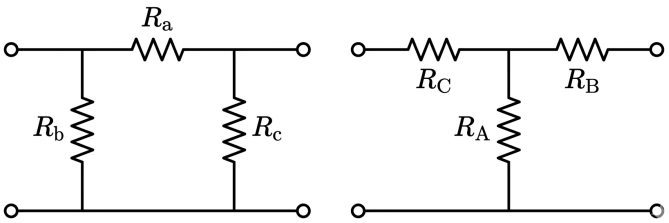


Fig. Solution Q3.16.4

For a delta-to-star transformation,

$$R_A = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_B = \frac{R_c R_a}{R_a + R_b + R_c}, \quad R_C = \frac{R_a R_b}{R_a + R_b + R_c}$$

For a star-to-delta transformation,

$$R_a = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A}, \quad R_b = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}, \quad R_c = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

Therefore, on simplifying the given network;

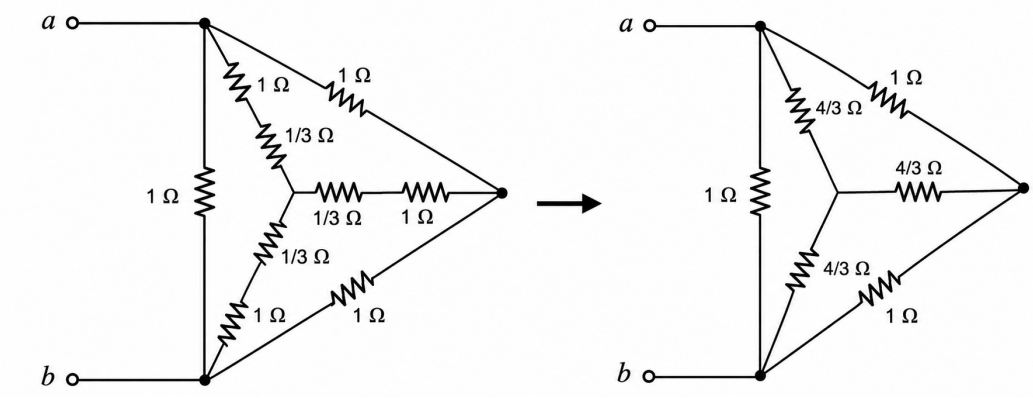


Fig. Solution Q3.16.4

Using star to delta conversion;

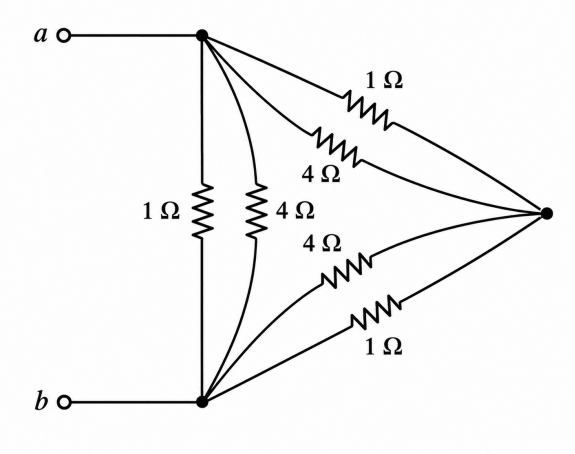


Fig. Solution Q3.16.4

$$R_{ab} = (1 \parallel 4) \parallel [(1 \parallel 4) + (1 \parallel 4)] = 0.8 \parallel 1.6 = \frac{8}{15}\ \Omega$$

Hence, the correct answer is

D. $\frac{8}{15}\ \Omega$

Q3.15.1 NAT 2015 1M Ans: 10 ●○○

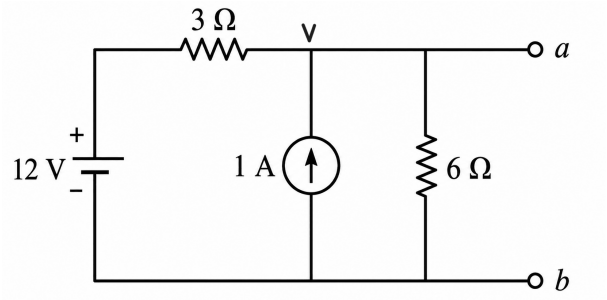


Fig. Solution Q3.15.1

Also, $V_{ab} = V_{th}$

Applying KCL at node V,

$$-0 - 1 + \frac{V_{th}}{6} + \frac{V_{th} - 12}{3} = 0$$

Multiplying throughout by 6,

$$-6 + V_{th} + 2V_{th} - 24 = 0$$

$$3V_{th} = 30 \Rightarrow V_{th} = 10 \text{ V}$$

Hence, the Thevenin equivalent voltage across terminals $a - b$ is

$$\boxed{10}$$

Q3.15.2 NAT 2015 2M Ans: 1.33 ●●○

To determine the Norton resistance, apply a test voltage source V_{dc} across terminals $a - b$ and assume the current drawn from the source is I_{dc}.

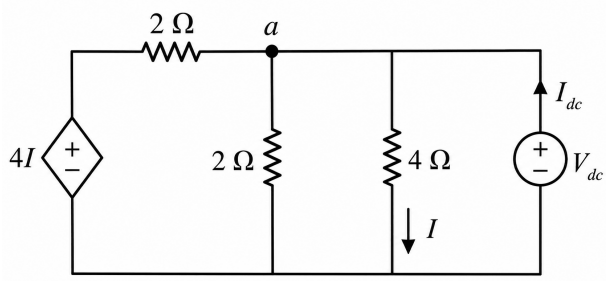


Fig. Solution Q3.15.2

The controlling current through the 4 Ω resistor is, $I = \frac{V_{dc}}{4}$

Hence, the dependent voltage source is

$$4I = 4\left(\frac{V_{dc}}{4}\right) = V_{dc}$$

Applying KCL at node a,

$$\frac{V_{dc} - 4I}{2} + \frac{V_{dc}}{2} + \frac{V_{dc}}{4} - I_{dc} = 0$$

Substituting $4I = V_{dc}$,

$$\frac{V_{dc} - V_{dc}}{2} + \frac{V_{dc}}{2} + \frac{V_{dc}}{4} = I_{dc}$$

$$\frac{3V_{dc}}{4} = I_{dc}$$

Therefore,

$$R_N = R_{Th} = \frac{V_{dc}}{I_{dc}} = \frac{V_{dc}}{\frac{3V_{dc}}{4}} = \frac{4}{3}\ \Omega = 1.33\ \Omega$$

Hence,

$$\boxed{1.33}$$

Key Points

- Norton Equivalent Resistance and Thevenin Equivalent resistance are same.

Q3.15.3 NAT | 2015 | 1M | Ans: 100 | ● ● ○

Since the circuit is symmetric about the line perpendicular to the line joining terminals a and b, the corresponding nodes on either side of the line of symmetry are at the same potential.

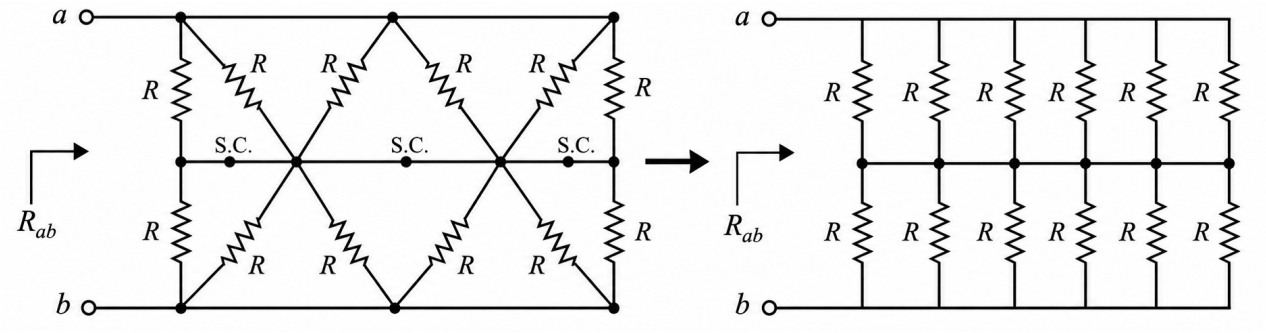


Fig. Solution Q3.15.3

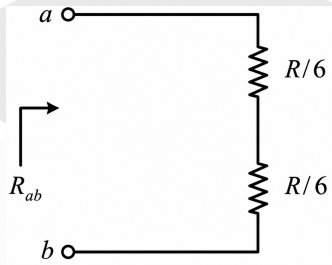


Fig. Solution Q3.15.3

Hence,

$$R_{ab} = \frac{R}{6} + \frac{R}{6} = \frac{2R}{6} = \frac{R}{3}$$

Given, $R = 300\ \Omega$

$$\Rightarrow R_{ab} = \frac{300}{3} = 100\ \Omega$$

Hence,

$$\boxed{100}$$

Q3.14.1 MCQ | 2014 | 1M | Ans: D | ● ○ ○

According to Norton's theorem, any linear bilateral network can be replaced by an equivalent current source in parallel with an equivalent resistance (or impedance in AC circuits). Hence, the correct answer is

D. in parallel with a current source

Q3.14.2 MCQ | 2014 | 1M | Ans: C | ● ○ ○

For maximum power transfer in AC circuits, the load impedance must be the complex conjugate of the source (or Thevenin) impedance. Hence, $Z_2 = Z_1^*$, where $(^*)$ denotes the complex conjugate. Therefore, the correct answer is

C. $Z_2 = Z_1^*$

Q3.14.3 NAT | 2014 | 2M | Ans: 2.62 | ● ● ○

The 2R resistance can be break down into two R in series. Since the ladder network is infinite, removing the first section leaves the equivalent resistance unchanged.

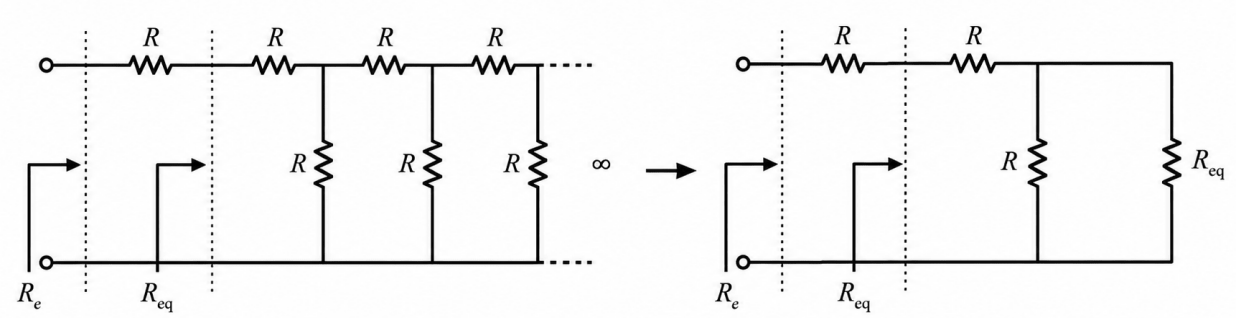


Fig. Solution Q3.14.3

Let the equivalent resistance of the remaining ladder be R_{eq}.
Then,

$$R_{eq} = R + (R \parallel R_{eq})$$

$$R_{eq} = R + \frac{RR_{eq}}{R + R_{eq}}$$

Multiplying both sides by $(R + R_{eq})$,

$$R_{eq}(R + R_{eq}) = R(R + R_{eq}) + RR_{eq}$$

$$RR_{eq} + R_{eq}^2 = R^2 + 2RR_{eq}$$

$$R_{eq}^2 - RR_{eq} - R^2 = 0$$

Solving,

$$R_{eq} = \frac{R \pm \sqrt{R^2 + 4R^2}}{2} = \frac{R(1 \pm \sqrt{5})}{2}$$

Since resistance cannot be negative,

$$R_{eq} = \frac{1 + \sqrt{5}}{2} R = 1.618R$$

The required equivalent resistance is

$$R_e = R + R_{eq} = R + 1.618R = 2.618R$$

Hence,

$$\frac{R_e}{R} = 2.618 \approx 2.62$$

Therefore,

$$\boxed{2.62}$$

Q3.14.4 NAT | 2014 | 1M | Ans: 2.80 | ● ○ ○

Method Mesh Analysis

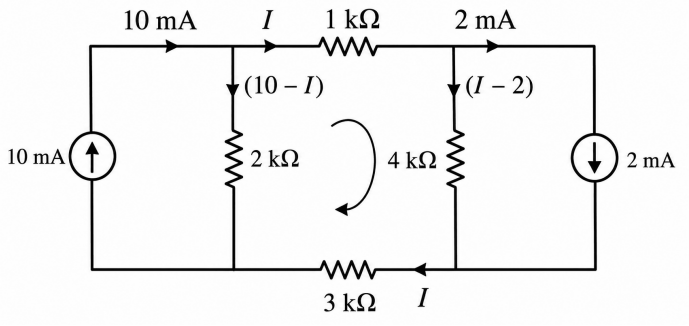


Fig. Solution Q3.14.4

Let the current through R_2 be I.
Applying KVL to the loop,

$$I(1) + (I-2)(4) + 3I - 2(10-I) = 0$$

$$10I - 8 - 20 = 0 \Rightarrow I = 2.8 \text{ mA}$$

Method Superposition Theorem

Case 1: Only the 10 mA current source is active
The 2 mA current source is opened.

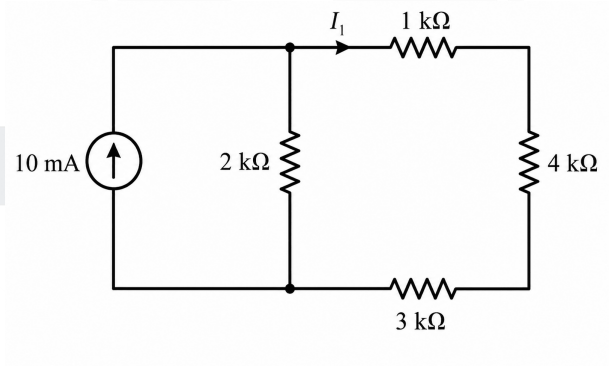


Fig. Solution Q3.14.4

Using current division,

$$I_1 = \frac{2}{8+2} \times 10 = 2 \text{ mA}$$

Case 2: Only the 2 mA current source is active
The 10 mA current source is opened.

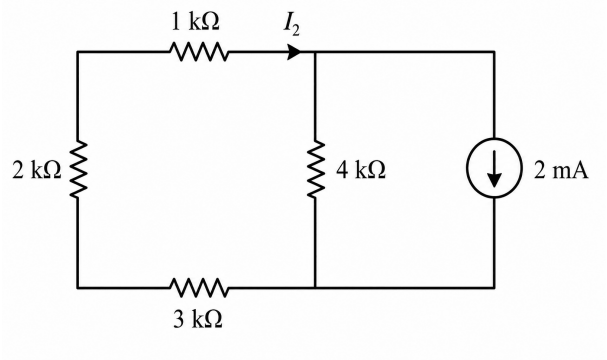


Fig. Solution Q3.14.4

Again, by current division,

$$I_2 = \frac{4}{6+4} \times 2 = 0.8 \text{ mA}$$

By superposition,

$$I = I_1 + I_2 = 2 + 0.8 = 2.8 \text{ mA}$$

Hence,

$$\boxed{2.80}$$

Q3.14.5 NAT 2014 1M Ans: 0.50 ● ○ ○

Method Nodal Analysis

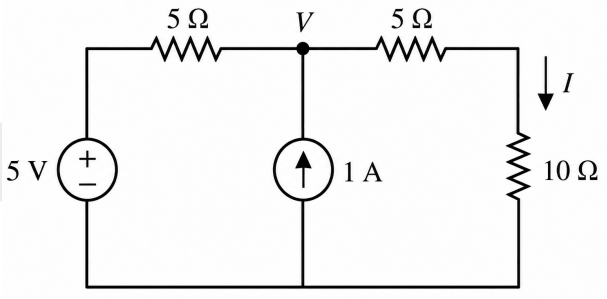


Fig. Solution Q3.14.5

Applying KCL at the node V,

$$\frac{V-5}{5} + \frac{V}{15} - 1 = 0$$

Multiplying throughout by 15,

$$3(V-5) + V - 15 = 0 \Rightarrow V = \frac{15}{2} \text{ V}$$

The current through the 10 Ω resistor is

$$I = \frac{V}{5+10} = \frac{15/2}{15} = \frac{1}{2} \text{ A}$$

Method Superposition Theorem

Case 1: Only the 5 V source is active

The 1 A current source is opened.

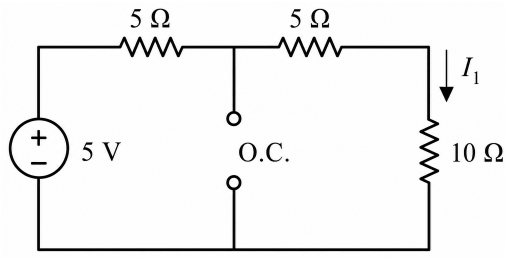


Fig. Solution Q3.14.5

The current through the 10 Ω resistor is

$$I_1 = \frac{5}{5+5+10} = \frac{5}{20} = \frac{1}{4}\text{ A}$$

Case 2: Only the 1 A current source is active

The 5 V source is short-circuited.

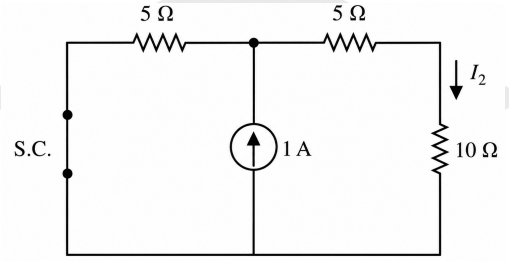


Fig. Solution Q3.14.5

Using current division,

$$I_2 = \frac{5}{5+5+10} \times 1 = \frac{5}{20} = \frac{1}{4}\text{ A}$$

By superposition,

$$I = I_1 + I_2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}\text{ A}$$

Hence,

$$\boxed{0.50}$$

Q3.14.6 NAT 2014 2M Ans: 29.09 ●●○

The given Y-network is

$$R_a = 10\ \Omega, \qquad R_b = 11\ \Omega, \qquad R_c = 10\ \Omega$$

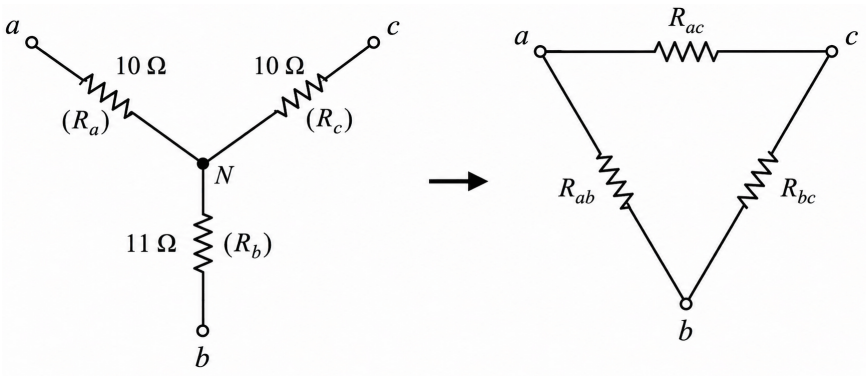


Fig. Solution Q3.14.6

For a Y-to-Δ transformation,

$$R_{ab} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_c}, \quad R_{bc} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_a}, \quad R_{ac} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_b}$$

Since the numerator is same in all three expressions, the smallest Δ resistance corresponds to largest denominator.

$$\because R_b > R_a = R_c \Rightarrow R_{ac} < R_{ab} = R_{bc}$$

Now,

$$R_{ac} = \frac{10 \times 11 + 11 \times 10 + 10 \times 10}{11} = \frac{320}{11} = 29.09\ \Omega$$

Hence, the lowest resistance in the equivalent Δ-network is

$$\boxed{29.09}$$

Q3.14.7 NAT 2014 2M Ans: 10 ● ○ ○

The given Y-network has

$$R_a = 5\ \Omega, \quad R_b = 3\ \Omega, \quad R_c = 7.5\ \Omega$$

For a Y-to-Δ transformation,

$$R_1 = \frac{R_aR_b + R_bR_c + R_cR_a}{R_c}$$

Substituting the given values,

$$R_1 = \frac{5 \times 3 + 3 \times 7.5 + 7.5 \times 5}{7.5}$$

$$= \frac{15 + 22.5 + 37.5}{7.5} = \frac{75}{7.5} = 10\ \Omega$$

Hence,

$$\boxed{10}$$

Q3.13.1 MCQ 2013 1M Ans: B ● ○ ○

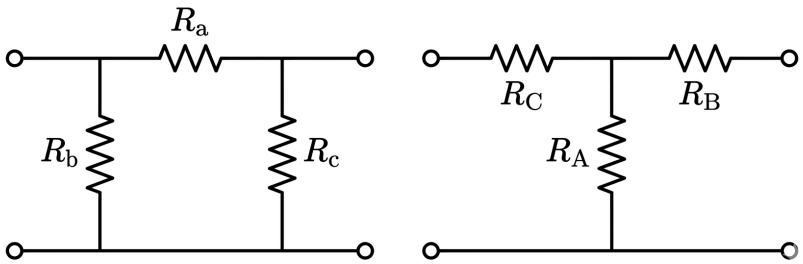


Fig. Solution Q3.13.1

For a delta-to-star transformation,

$$R_A = \frac{R_bR_c}{R_a + R_b + R_c}, \quad R_B = \frac{R_cR_a}{R_a + R_b + R_c}, \quad R_C = \frac{R_aR_b}{R_a + R_b + R_c}$$

If all delta resistances are scaled by a factor k,

$$R'_a = kR_a, \quad R'_b = kR_b, \quad R'_c = kR_c$$

Then,

$$R'_A = \frac{R'_b R'_c}{R'_a + R'_b + R'_c} = \frac{(kR_b)(kR_c)}{k(R_a + R_b + R_c)}$$

$$R'_A = k\frac{R_b R_c}{R_a + R_b + R_c} = kR_A$$

Similarly,

$$R'_B = kR_B, \qquad R'_C = kR_C$$

Therefore,

B. k

Q3.12.1 MCQ 2012 2M Ans: A ●●○

For a one-port network, the current entering one terminal is equal to the current leaving the other terminal.

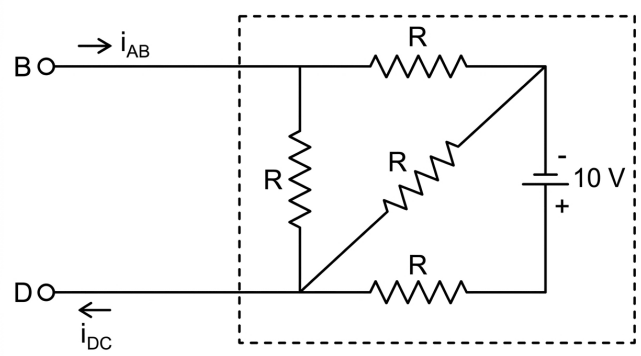


Fig. Solution Q3.12.1

Hence,

$$i_{AB} = i_{DC}$$

Therefore, if a current i flows from V_A to V_B, the same current i must flow through the branch connecting V_C and V_D.

Method 1 (Using Source Transformation)

Transform the 2 A current source in parallel with 1Ω into its equivalent Voltage source:

$$V = IR = (2)(1) = 2 \text{ V}$$

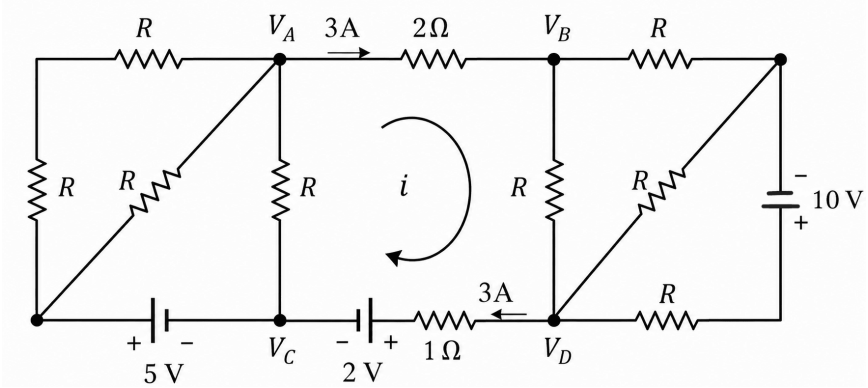


Fig. Solution Q3.12.1

Given,

$$V_A - V_B = 6 \text{ V}$$

Current through the 2Ω resistor is

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3 \text{ A}$$

By the above one-port property, the same current flows through the branch between C and D.

$$i_{DC} = 3 \text{ A}$$

in series with 1Ω.
Hence,

$$V_{DC} = (3)(1) + 2 = 5 \text{ V}$$

Therefore,

$$V_{CD} = V_C - V_D = -5 \text{ V}$$

Method 2 (Without Source Transformation)

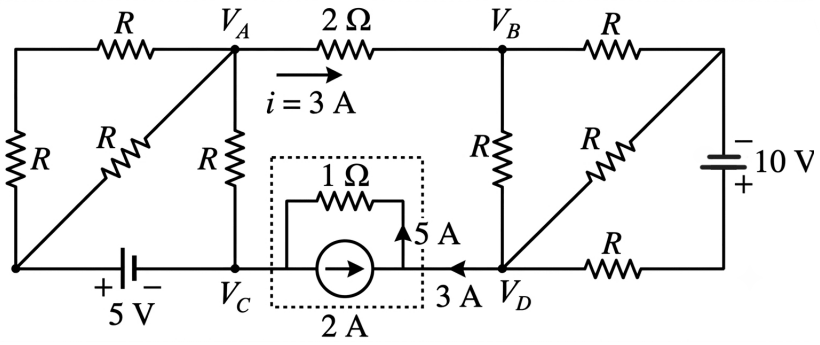


Fig. Solution Q3.12.1

Given,

$$V_A - V_B = 6 \text{ V}$$

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3 \text{ A}$$

The current source supplies

$$2 \text{ A}$$

from C to D. Hence the current through the parallel 1Ω resistor is

$$i_R = 3 + 2 = 5 \text{ A}$$

Thus,

$$V_{DC} = (5)(1) = 5 \text{ V}$$

Therefore,

$$V_{CD} = V_C - V_D = -5 \text{ V}$$

Hence, the correct answer is

A. -5 V

Q3.12.2 MCQ | 2012 | 1M | Ans: A ● ● ○

Apply a test voltage source V across terminals 1 and 2.

Let the current drawn from the source be I.

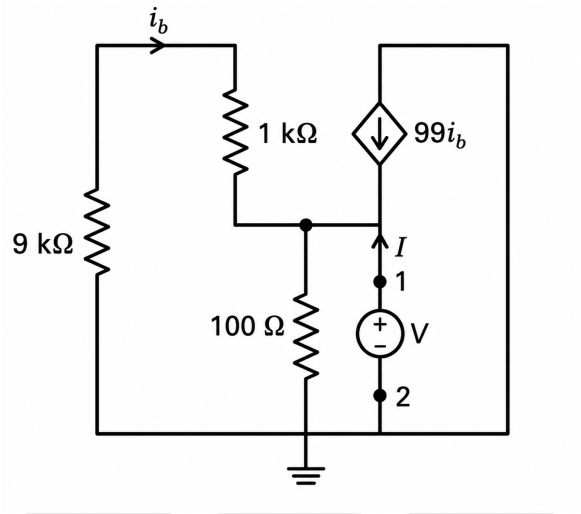


Fig. Solution Q3.12.2

Then,

$$Z_{12} = \frac{V}{I}$$

The 1 kΩ and 9 kΩ resistors are in series, hence

$$i_b = \frac{0 - V}{1\text{k} + 9\text{k}}$$

$$i_b = -\frac{V}{10^4} \quad (1)$$

Applying KCL at node 1,

$$-i_b + \frac{V}{100} - 99i_b - I = 0$$

$$V = 10^4 i_b + 100I \quad (2)$$

Substituting (1) into (2),

$$V = 10^4\left(-\frac{V}{10^4}\right) + 100I$$

$$2V = 100I \Rightarrow \frac{V}{I} = 50\ \Omega$$

Therefore,

$$\boxed{Z_{12} = 50\ \Omega}$$

Hence, the correct answer is

$$\boxed{\textbf{A. } 50\ \Omega}$$

Q3.11.1 MCQ 2011 1M Ans: C ●○○

For maximum power transfer, $R_L = R_{\text{Th}}$

Hence, we first determine the Thevenin resistance seen from the load terminals.

Deactivate all the independent sources.

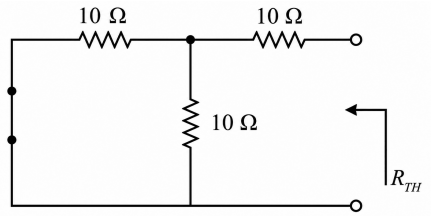


Fig. Solution Q3.11.1

The equivalent resistance is

$$R_{\text{Th}} = (10 \parallel 10) + 10$$

$$\Rightarrow R_{\text{Th}} = 5 + 10 = 15\ \Omega$$

Hence,

$$R_L = R_{\text{Th}} = 15\ \Omega$$

Therefore, the correct answer is

C. 15 Ω

Q3.10.1 MCQ 2010 2M Ans: A ●●○

Method Nodal Analysis

Assume node C as the reference node.

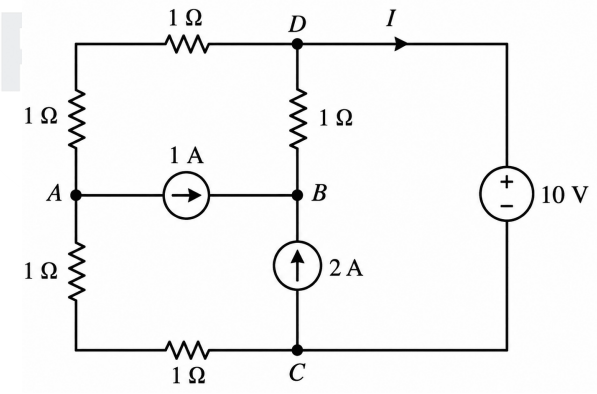


Fig. Solution Q3.10.1

Hence, $V_C = 0$

Applying KCL at node A,

$$\frac{V_A - 10}{2} + \frac{V_A}{2} + 1 = 0$$

$$V_A = 4\ \text{V}$$

Applying KCL at node B,

$$-1 - 2 + \frac{V_B - 10}{1} = 0$$

$$V_B = 13 \text{ V}$$

Applying KCL at node D,

$$I = \frac{10 - V_A}{2} + \frac{10 - V_B}{1}$$

$$= \frac{10 - 4}{2} + 10 - 13 = 3 - 3 = 0 \text{ A}$$

Thus, the current entering the positive terminal of the voltage source is 0 A
Hence, the power supplied by the 10 V source is

$$P = VI = 10(0) = 0 \text{ W}$$

Method Superposition Theorem

Case 1: Only the 1 A current source is active

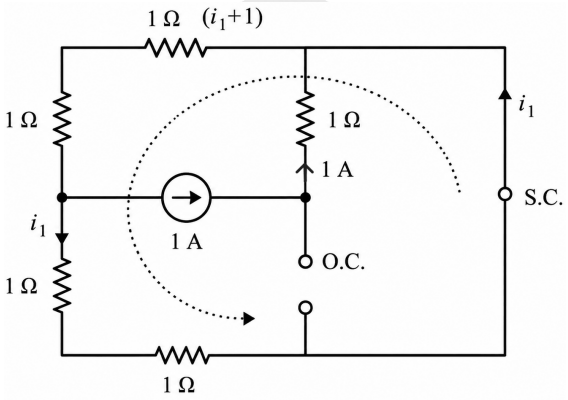


Fig. Solution Q3.10.1

Applying KVL in the loop,

$$(i_1 + 1) + (i_1 + 1) + 2i_1 = 0$$

$$4i_1 = -2 \Rightarrow i_1 = -\frac{1}{2} \text{ A}$$

Case 2: Only the 2 A current source is active

The 1 A current source is opened and the voltage source is short-circuited.

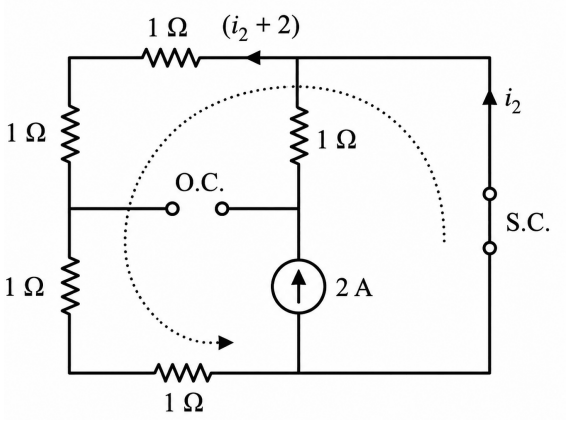


Fig. Solution Q3.10.1

Applying KVL in the outer loop,

$$(i_2 + 2) + (i_2 + 2) + (i_2 + 2) + (i_2 + 2) = 0$$

$$4i_2 + 8 = 0 \Rightarrow i_2 = -2 \text{ A}$$

Case 3: Only the 10 V source is active

Both current sources are opened.

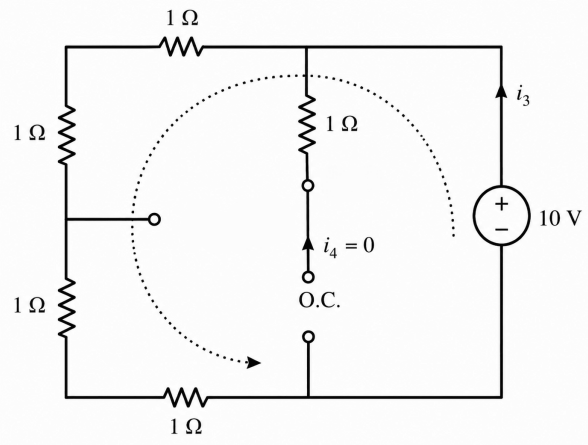


Fig. Solution Q3.10.1

Applying KVL,

$$i_3 = \frac{10}{4} = 2.5 \text{ A}$$

Hence,

$$i = i_1 + i_2 + i_3 = -\frac{1}{2} - 2 + 2.5 = 0$$

Therefore,

$$P = Vi = 10 \times 0 = 0 \text{ W}$$

Hence, the correct answer is

A. 0 W

Q3.09.1 MCQ 2009 2M Ans: C ●●○

For maximum power transfer, $R_L = R_{\text{Th}}$

Since the circuit contains a dependent source, apply a test voltage source V_{dc} across terminals $a - b$ and let the current drawn be I_{dc}.

Deactivate the independent 100 V source by replacing it with a short circuit.

The circuit can be redrawn as

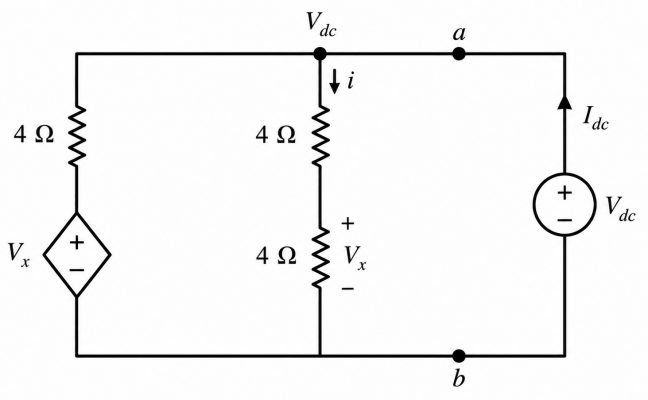


Fig. Solution Q3.09.1

Applying KCL at node a,

$$\frac{V_{dc} - V_x}{4} + \frac{V_{dc}}{8} - I_{dc} = 0$$

$$2(V_{dc} - V_x) + V_{dc} = 8I_{dc}$$

$$3V_{dc} - 2V_x = 8I_{dc} \qquad \cdots (1)$$

From the figure,

$$i = \frac{V_{dc}}{8} = \frac{V_x}{4}$$

$$\Rightarrow V_x = \frac{V_{dc}}{2} \qquad \cdots (2)$$

Substituting (2) into (1),

$$3V_{dc} - V_{dc} = 8I_{dc}$$

$$R_{\text{Th}} = \frac{V_{dc}}{I_{dc}} = 4\ \Omega$$

Hence,

$$R_L = R_{\text{Th}} = 4\ \Omega$$

Therefore, the correct answer is

C. 4 Ω

Q3.08.1 MCQ 2008 2M Ans: A ● ● ○

At steady state (DC condition), $\omega = 0$

Hence, $X_L = \omega L = 0 \Rightarrow$ The inductor behaves as a short circuit.

Also, $X_C = \dfrac{1}{\omega C} = \infty \Rightarrow$ The capacitor behaves as an open circuit.

The steady-state equivalent circuit is shown below.

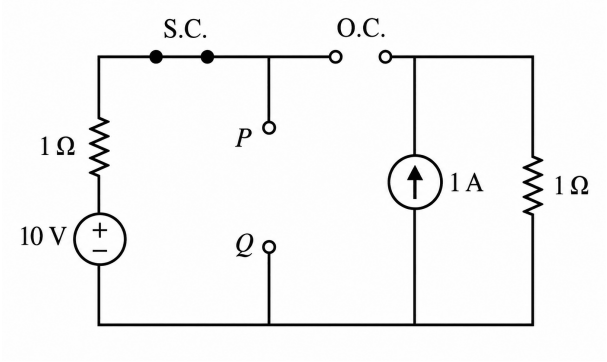


Fig. Solution Q3.08.1

To find the Thevenin impedance, deactivate all the independent sources.
The resulting circuit is

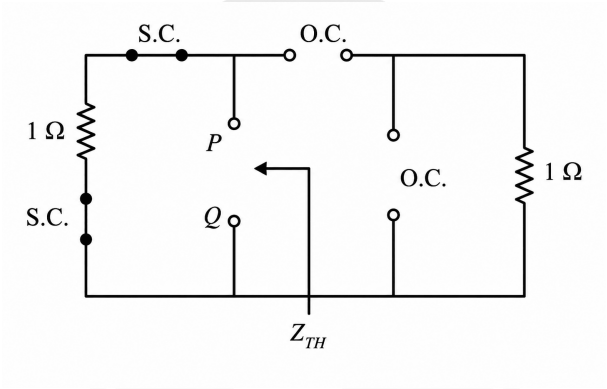


Fig. Solution Q3.08.1

Looking into terminals P and Q,

$$Z_{TH} = 1\ \Omega$$

Hence, the correct answer is

A. 1 Ω

Mistakes to be avoided

- Since there is no AC source present in the circuit ⇒ no need to use Laplace.

Q3.07.1 MCQ | 2007 | 2M | Ans: D | ● ● ○

Calculation of V_{TH}:

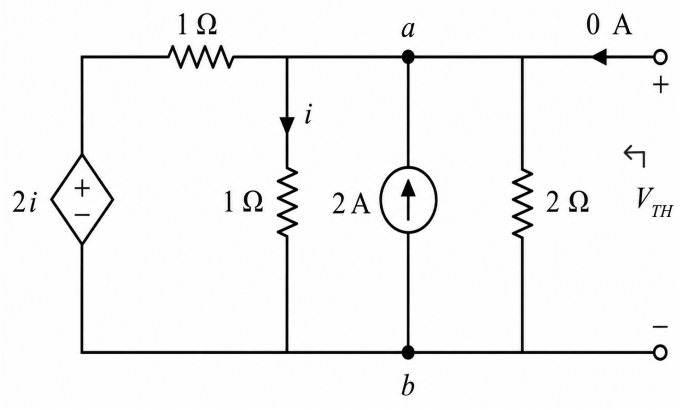


Fig. Solution Q3.07.1

Applying KCL at node a,

$$-2 + \frac{V_{TH}}{2} + \frac{V_{TH}}{1} + \frac{V_{TH} - 2i}{1} = 0$$

where,

$$i = \frac{V_{TH}}{1} = V_{TH}$$

Substituting,

$$-2 + \frac{V_{TH}}{2} + V_{TH} + V_{TH} - 2V_{TH} = 0$$

$$2 = \frac{V_{TH}}{2} + V_{TH} - V_{TH} = \frac{V_{TH}}{2}$$

$$V_{TH} = 4 \text{ V}$$

Calculation of R_{TH}:

Since a dependent source is present, apply a test voltage source V_{dc} across terminals $X - Y$ and let the current entering the network be I_{dc}.

Deactivate all the independent sources.

The modified circuit is shown below.

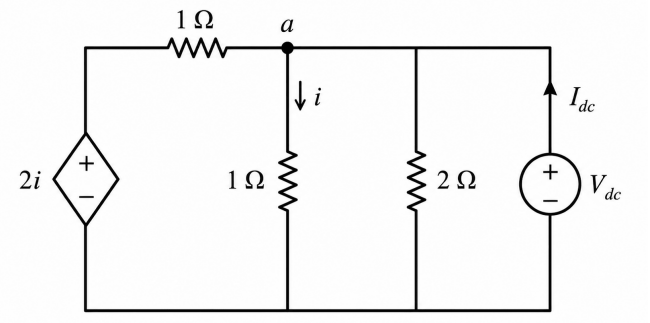


Fig. Solution Q3.07.1

Applying KCL at node a,

$$-I_{dc} + \frac{V_{dc}}{2} + \frac{V_{dc}}{1} + \frac{V_{dc} - 2i}{1} = 0$$

where,

$$i = \frac{V_{dc}}{1} = V_{dc}$$

Substituting,

$$-I_{dc} + \frac{V_{dc}}{2} + V_{dc} + V_{dc} - 2V_{dc} = 0$$

$$I_{dc} = \frac{V_{dc}}{2}$$

Hence,

$$R_{TH} = \frac{V_{dc}}{I_{dc}} = 2\ \Omega$$

Hence, the correct answer is

D. 4 V, 2 Ω

Q3.05.1 MCQ 2005 2M Ans: B

Calculation of V_{TH}:

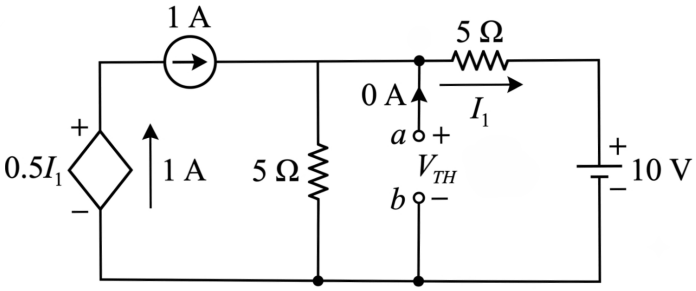


Fig. Solution Q3.05.1

Applying KCL at node a,

$$-1 + \frac{V_{TH}}{5} + \frac{V_{TH} - 10}{5} = 0$$

$$1 = \frac{2V_{TH}}{5} - 2$$

$$V_{TH} = \frac{15}{2} = 7.5\ \text{V}$$

Calculation of R_{TH}:

Since a dependent source is present, apply a test voltage source V_{dc} across terminals $a - b$ and let the current entering the network be I_{dc}.

Deactivate all the independent sources.

The modified circuit is shown below.

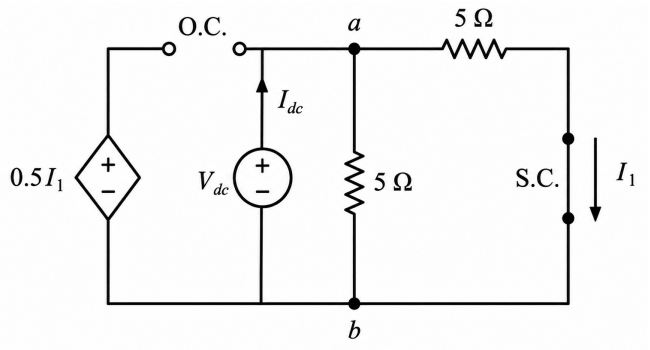


Fig. Solution Q3.05.1

Applying KCL at node a,

$$-I_{dc} + \frac{V_{dc}}{5} + I_1 = 0$$

$$\because I_1 = \frac{V_{dc}}{5}$$

$$\Rightarrow -I_{dc} + \frac{V_{dc}}{5} + \frac{V_{dc}}{5} = 0$$

$$I_{dc} = \frac{2V_{dc}}{5}$$

Hence,

$$R_{TH} = \frac{V_{dc}}{I_{dc}} = 2.5\ \Omega$$

Hence, the correct answer is

B. 7.5 V, 2.5 Ω

Q3.05.2 MCQ 2005 1M Ans: C ● ○ ○

Using the Maximum Power Transfer Theorem, $R_L = R_{TH}$

The Thevenin equivalent resistance seen by the load is, $R_{TH} = 100\ \Omega$

The maximum power transferred to the load is

$$P_{\max} = \frac{V_{TH}^2}{4R_{TH}}$$

Here,

$$V_{TH} = 10\ \text{V}, \qquad R_{TH} = 100\ \Omega$$

Therefore,

$$P_{\max} = \frac{10^2}{4 \times 100} = \frac{100}{400} = 0.25\ \text{W}$$

Hence, the correct answer is

C. 0.25 W

Mistakes to be avoided

- Maximum Power Transfer Theorem ($R_L = R_{th}$) is applicable when the *load resistance is variable*. Do not apply this if load resistance is not variable.

Q3.03.1 MCQ 2003 2M Ans: A ● ● ○

Method Current Distribution Method

Let a current i enter one corner of the cube and leave the diagonally opposite corner.

By symmetry, the current divides equally among the three branches connected to the source node.

$$\text{Current in each branch} = \frac{i}{3}$$

The current distribution is shown below.

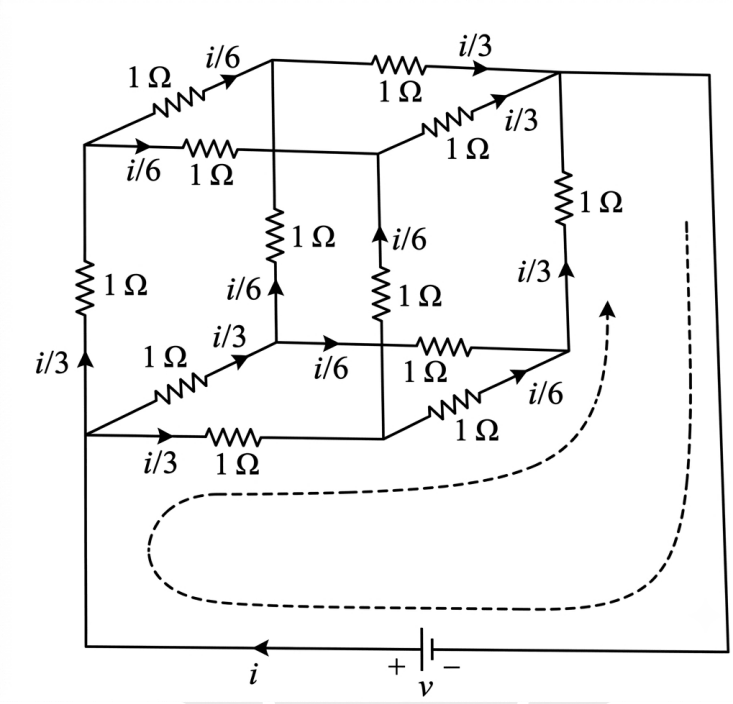


Fig. Solution Q3.03.1

Applying KVL along any path from one corner to the opposite corner,

$$v = \frac{i}{3} \times 1 + \frac{i}{6} \times 1 + \frac{i}{3} \times 1$$

$$v = \frac{5i}{6}$$

Therefore,

$$R_{eq} = \frac{v}{i} = \frac{5}{6}\,\Omega$$

Method Symmetry Method

Let brach resistance of each branch be $R = 1\Omega$.
From vertices E, D and B to terminal A, the path resistance is R.
Also, from these vertices to terminal H, the path resistance is $R + R$.
Hence, E, D and B are at the same potential (equipotential).
Similarly, G, F, C are also equipotential.

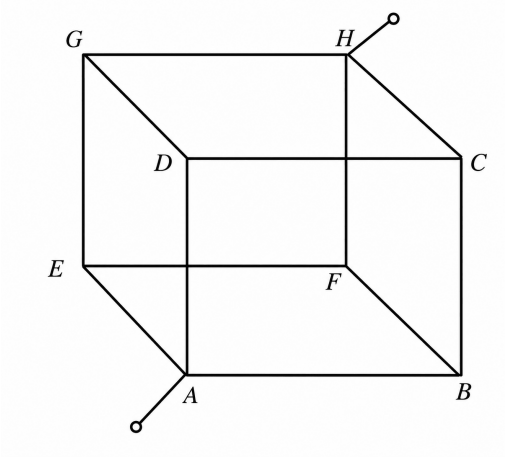


Fig. Solution Q3.03.1

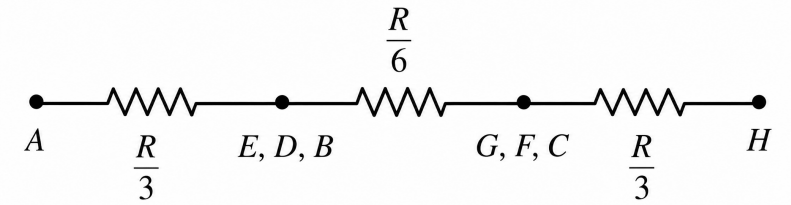


Fig. Solution Q3.03.1

Thus,

$$R_{eq} = \frac{1}{3} + \frac{1}{6} + \frac{1}{3} = \frac{5}{6}\,\Omega$$

Hence, the correct answer is

A. $\frac{5}{6}\,\Omega$

Key Points

- For a cube with each branch having resistance R:

Terminals	R_{eq}
Across an edge	$\frac{7}{12}R$
Across a face diagonal	$\frac{3}{4}R$
Across a body (space) diagonal	$\frac{5}{6}R$

Q3.02.1 MCQ 2002 2M Ans: A ●○○

Using the Maximum Power Transfer Theorem, $R_L = R_{TH}$

Since a dependent source is present, apply a test voltage source V_{dc} across terminals $a - b$ and let the current entering the network be I_{dc}.

Deactivate the independent source.

The modified circuit is shown below.

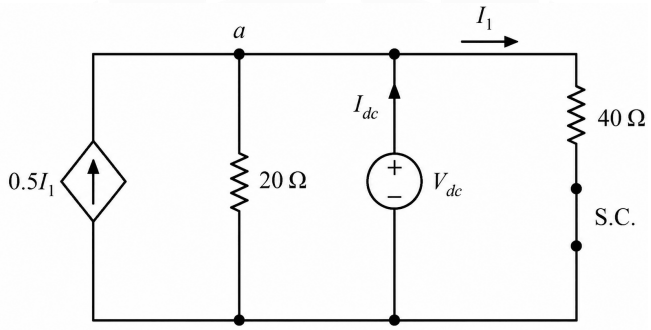


Fig. Solution Q3.02.1

Applying KCL at node a,

$$-I_{dc} - 0.5I_1 + I_1 + \frac{V_{dc}}{20} = 0$$

where,

$$I_1 = \frac{V_{dc}}{40}$$

Substituting,

$$I_{dc} = \frac{V_{dc}}{80} + \frac{V_{dc}}{20} = \frac{5V_{dc}}{80}$$

Hence,

$$R_{TH} = \frac{V_{dc}}{I_{dc}} = \frac{80}{5} = 16\ \Omega$$

Hence, the correct answer is

A. 16 Ω

Q3.01.1 MCQ 2001 2M Ans: D ●●○

Using source transformation, convert the 8 A current source in parallel with the 10 Ω resistor into its equivalent voltage source.

$$V = IR = 8 \times 10 = 80 \text{ V}$$

The transformed circuit is shown below.

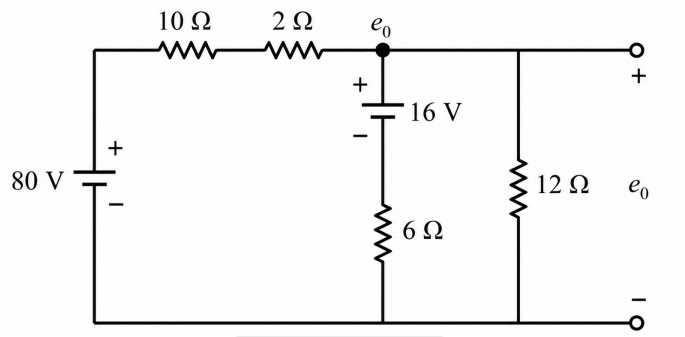


Fig. Solution Q3.01.1

Applying KCL at node e_o,

$$\frac{e_o - 80}{12} + \frac{e_o}{12} + \frac{e_o - 16}{6} = 0$$

Multiplying throughout by 12,

$$e_o - 80 + e_o + 2(e_o - 16) = 0$$

$$4e_o = 112$$

$$e_o = \frac{112}{4} = 28 \text{ V}$$

Hence, the correct answer is

D. 28 V

Q3.00.1 MCQ 2000 2M Ans: C ●●○

The given network is linear, bilateral and consists only of passive elements.

Hence, the **Reciprocity Theorem** can be applied.

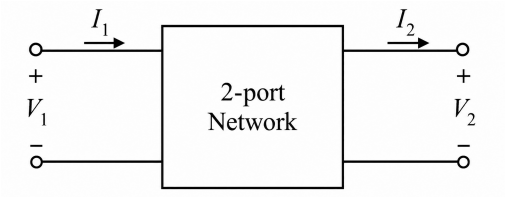


Fig. Solution Q3.00.1

According to the Reciprocity Theorem,

$$\frac{V_1}{V_2} = \frac{I_1}{I_2}$$

From Fig. (a),

$$V_1 = 10 \text{ V}, \qquad I_1 = 2 \text{ A}$$

From Fig. (b),

$$V_2 = -20 \text{ V}$$

(The polarity of the 20 V source is opposite to that of the 10 V source.)
Therefore,

$$\frac{10}{-20} = \frac{2}{I_2}$$
$$I_2 = -4 \text{ A}$$

Hence, the correct answer is

C. -4 A

Q3.99.1 MCQ 1999 2M Ans: C ● ○ ○

Using the Maximum Power Transfer Theorem, $R = R_{TH}$
Hence, find the Thevenin resistance seen from the load terminals.
Deactivate all the independent sources.
The modified circuit is shown below.

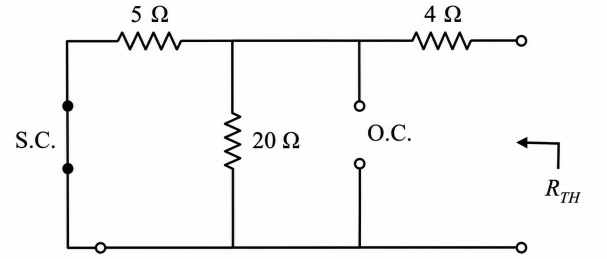


Fig. Solution Q3.99.1

Looking into the load terminals,

$$R_{TH} = (5 \parallel 20) + 4$$

Now,

$$5 \parallel 20 = \frac{5 \times 20}{5 + 20} = 4\ \Omega$$

Therefore,

$$R_{TH} = 4 + 4 = 8\ \Omega$$
$$\Rightarrow R = R_{TH} = 8\ \Omega$$

Hence, the correct answer is

C. 8 Ω

Q3.98.1 MCQ 1998 1M Ans: A ● ○ ○

The Superposition Theorem is applicable only to **linear networks**.
It can be applied to networks containing

- independent sources,
- linear dependent sources,
- linear passive elements (R, L and C),
- linear transformers.

However, it is **not applicable** to networks containing non-linear elements since the principle of linearity is violated.
Hence, the correct answer is

A. Non-linear elements

Q3.95.1 MSQ | 1995 | 1M | Ans: A,D | ● ● ○

Method Superposition Theorem

(i) Let the voltage source V alone be active.

Replace the current source by its internal resistance, i.e., an open circuit.

The current through resistor R is I_1.

Power dissipated in R is

$$P_1 = I_1^2 R = 4 \text{ W}$$

(ii) Let the current source I alone be active.

Replace the voltage source by its internal resistance, i.e., a short circuit.

The current through resistor R is I_2.

Power dissipated in R is

$$P_2 = I_2^2 R = 9 \text{ W}$$

(iii) When both sources are active

According to the Superposition Theorem,

$$I = I_1 \pm I_2$$

Therefore,

$$P = (I_1 \pm I_2)^2 R = I_1^2 R + I_2^2 R \pm 2 I_1 I_2 R$$

where,

$$I_1 I_2 R = \sqrt{I_1^2 R \times I_2^2 R} = \sqrt{P_1 P_2} = \sqrt{4 \times 9} = 6 \text{ W}$$

$$\Rightarrow P = 4 + 9 \pm 2(6)$$

Thus,

$$P_{\max} = 25 \text{ W}, \qquad P_{\min} = 1 \text{ W}$$

Method Direct Formula

The maximum and minimum power are given by

$$P_{\max} = \left(\sqrt{P_1} + \sqrt{P_2}\right)^2$$

$$P_{\max} = (\sqrt{4} + \sqrt{9})^2 = (2 + 3)^2 = 25 \text{ W}$$

Similarly,

$$P_{\min} = \left(\sqrt{P_1} - \sqrt{P_2}\right)^2$$

$$P_{\min} = (\sqrt{4} - \sqrt{9})^2 = (2 - 3)^2 = 1 \text{ W}$$

Hence, the correct answers are

A. 1 W and **D.** 25 W

Key Points

- **Superposition Theorem** is applicable only for **linear responses** such as voltage and current, because these quantities vary linearly with the sources.
- It **cannot be applied directly to power**, since power is a nonlinear quantity:

$$P = VI, \qquad P = I^2 R, \qquad P = \frac{V^2}{R}$$

- To determine power in a circuit with multiple independent sources, first use superposition to find the **total voltage or**

total current, and then compute the power using the complete response.

Q3.95.2 MCQ 1995 1M Ans: A ● ● ○

Using the Maximum Power Transfer Theorem, $R = R_{TH}$

Deactivate the independent voltage source by replacing it with a short circuit.

The modified circuit is shown below.

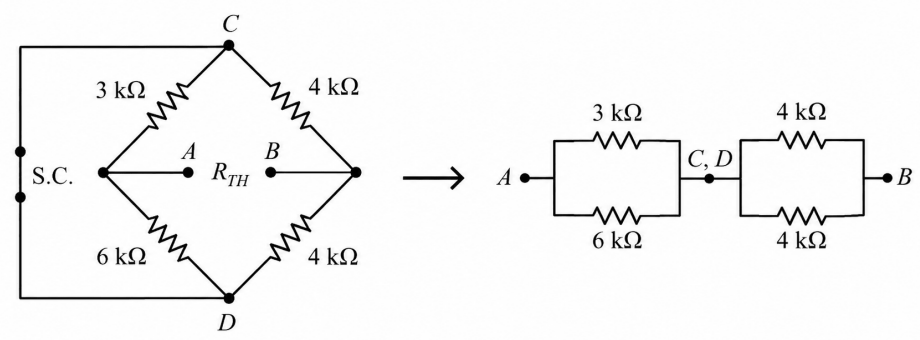


Fig. Solution Q3.95.2

The short circuit makes nodes C and D equipotential.

Hence,

$$R_{TH} = (3 \parallel 6) + (4 \parallel 4)$$

Now,

$$3 \parallel 6 = \frac{3 \times 6}{3 + 6} = 2\ \text{k}\Omega, \qquad 4 \parallel 4 = \frac{4 \times 4}{4 + 4} = 2\ \text{k}\Omega$$

Therefore,

$$R_{TH} = 2 + 2 = 4\ \text{k}\Omega$$

Hence,

$$R = R_{TH} = 4\ \text{k}\Omega$$

Hence, the correct answer is

A. 4.00 kΩ

Q3.93.1 NAT 1993 2M Ans: 50 ● ● ○

When S_1 is closed, the ideal ammeter measures the short-circuit current across the switch branch.

Hence, $I_{SC} = 5$ A

When S_1 is open, the ideal voltmeter measures the open-circuit voltage across the switch branch $\Rightarrow V_{OC} = V_{M_2}$

So, to find V_{M_2} we find thevenin resistance, because we know short circuit current.

The Thevenin resistance seen across terminals of S_1 is obtained by deactivating the independent voltage source.

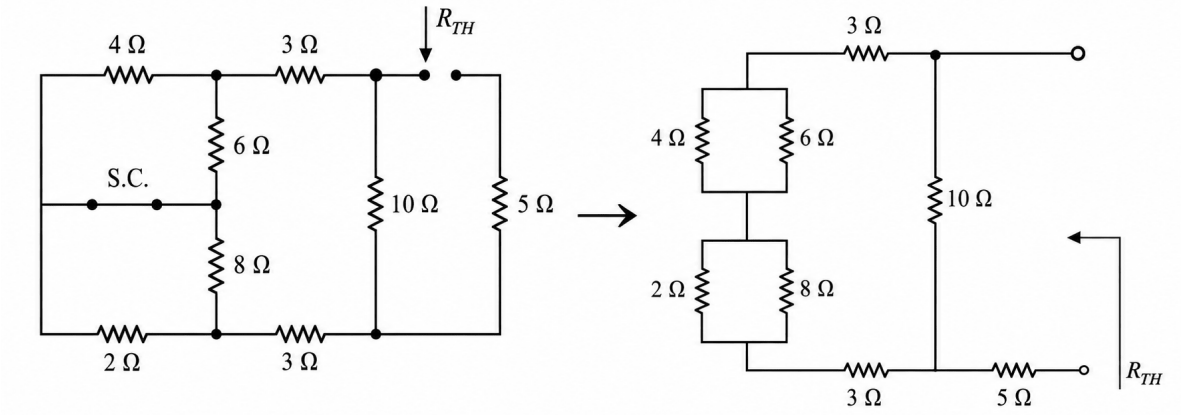


Fig. Solution Q3.93.1

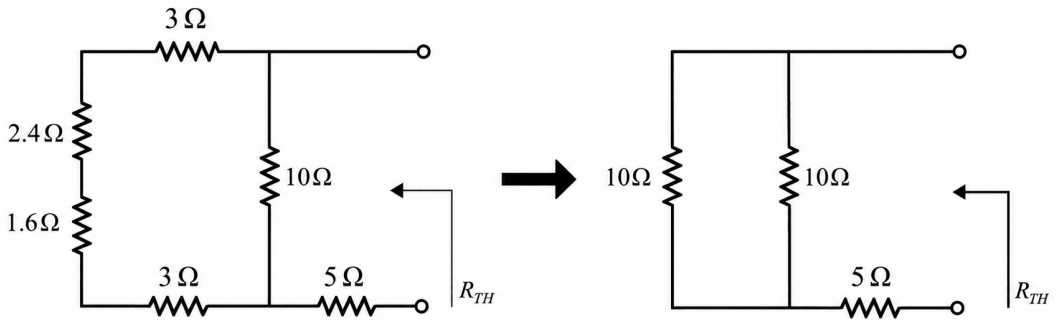


Fig. Solution Q3.93.1

$$R_{TH} = 10 \parallel 10 + 5 = 5 + 5 = 10\ \Omega$$

Therefore,

$$V_{OC} = I_{SC} R_{TH} = 5 \times 10 = 50\ \text{V}$$

Hence, the ideal voltmeter reading is

$$\boxed{50\ \text{V}}$$

SOLUTIONS

IV

2nd Order Dynamics and Complex Impedance

Topics

1. Source-Free 2nd Order RLC Circuits
2. Initial Conditions and Sourced RLC Circuits
3. LC Oscillators
4. Equivalent Impedance of AC Circuits

Unit Snapshot*

Stream: EC	**Questions:** 2	**MCQ:** 1	**MSQ:** 0	**NAT:** 0
1M: 1	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~0M	**Range:** 0M(–)–0M(–)

**See preface for how Avg Weightage and Range are calculated.*

Q4.94.1 Descriptive | 1994 | 1M | Ans: False | ● ● ○

A driving-point function (such as an input impedance $Z(s)$ or input admittance $Y(s)$) of a passive network must be physically realizable using actual passive components (R, L, and C).

Thus the degrees of the numerator and denominator polynomials cannot differ by more than 1. The mathematical and physical reasons for this constraint are detailed below:

1. High-Frequency Behavior ($s \to \infty$) At extremely high frequencies, any physically realizable passive network must simplify to look like either a single resistor, a single inductor, or a single capacitor.

Let n be the highest degree of the numerator polynomial $N(s)$, and m be the highest degree of the denominator polynomial $D(s)$. As $s \to \infty$, the behavior of $Z(s)$ is dominated by its highest-order terms:

$$Z(s) \approx H \cdot s^{(n-m)}$$

Depending on the difference between n and m, three physical cases can exist for a passive element:

Case 1 ($n - m = 1$)**:** $Z(s) \approx H \cdot s$, which represents a pure **inductor** ($Z(s) = sL$).

Case 2 ($n - m = 0$)**:** $Z(s) \approx H$, which represents a pure **resistor** ($Z(s) = R$).

Case 3 ($m - n = 1$)**:** $Z(s) \approx \frac{H}{s}$, which represents a pure **capacitor** ($Z(s) = \frac{1}{sC}$).

For the given input impedance function:

$$Z(s) = \frac{5}{s^2 + 4}$$

Here, $n = 0$ and $m = 2$. The degree difference is $m - n = 2$. As $s \to \infty$:

$$Z(s) \approx \frac{5}{s^2}$$

There is no individual passive component or combination of components whose basic impedance decreases proportionally to $\frac{1}{s^2}$ at high frequencies.

2. Multiplicity of Poles at Infinity For a network function to be realizable, any poles or zeros located on the imaginary axis ($j\omega$-axis), including the point at infinity, must be **simple** (i.e., they must have a multiplicity/order of exactly 1).

If we examine the corresponding admittance function $Y(s)$:

$$Y(s) = \frac{1}{Z(s)} = \frac{s^2 + 4}{5} = \frac{1}{5}s^2 + \frac{4}{5}$$

Taking the limit as $s \to \infty$, the admittance becomes:

$$\lim_{s \to \infty} Y(s) \propto s^2$$

This indicates that $Y(s)$ has a **pole at infinity of order 2**.

Therefore, because the difference in degrees is 2, the function cannot represent the input impedance of a passive network.

False

Q4.92.1 MCQ | 1992 | 2M | Ans: C | ● ● ○

To determine which networks have identical driving-point functions, we need to find the driving-point function for each network.

1. Network N_1

The network N_1 consists of three parallel branches and the total driving-point admittance $Y_1(s)$ can be computed by summing the admittances of the parallel branches:

$$Y_1(s) = \frac{1}{Z_{b1}} + \frac{1}{Z_{b2}} + \frac{1}{Z_{b3}}$$

$$Y_1(s) = \frac{1}{2s + 1} + \frac{s}{2s + 1} + s$$

$$Y_1(s) = \frac{1+s}{2s+1} + s = \frac{1+s+s(2s+1)}{2s+1} = \frac{2s^2+2s+1}{2s+1}$$

2. Network N_2

The network N_2 consists of two parallel branches. Finding the total driving-point admittance $Y_2(s)$:

$$Y_2(s) = \frac{1}{2s+1} + \frac{s}{2s+1} = \frac{s+1}{2s+1}$$

3. Network N_3

The network N_3 consists of two parallel branches. Finding the total driving-point admittance $Y_3(s)$:

$$Y_3(s) = \frac{1}{Z_{b1}} + \frac{1}{Z_{b2}} = \frac{1}{1+\frac{s}{s+1}} + s = \frac{s+1}{2s+1} + s$$

$$Y_3(s) = \frac{s+1+s(2s+1)}{2s+1} = \frac{2s^2+2s+1}{2s+1}$$

4. Network N_4

The network N_4 consists of two parallel branches. Finding the total driving-point admittance $Y_4(s)$:

$$Y_4(s) = \frac{1}{2s+1} + s = \frac{1+s(2s+1)}{2s+1} = \frac{2s^2+s+1}{2s+1}$$

On comparing all driving point functions networks N_1 and N_3 have identical driving-point functions.

$\boxed{(c) N1 \text{ and } N3}$

PrepFusion

SOLUTIONS

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Introduction to Power in AC Circuits
6. Active, Reactive and Apparent Power in Complex Networks
7. Series RLC Resonance Circuit
8. Quality Factor and Stored Energy in Series Resonance
9. Parallel RLC Resonance Circuit
10. Application of Network Theorems in AC Circuits
11. Maximum Power Transfer in AC Circuits

Unit Snapshot*

Stream: EC	**Questions:** 84	**MCQ:** 58	**MSQ:** 1	**NAT:** 24
1M: 39	**2M:** 38	**Descriptive:** 7	**Avg Weightage** ~3M	**Range:** 1M(2023)–5M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 NAT 2026 1M Ans: 71.50 to 72.50 ● ○ ○

For the given circuit, we need to find the power dissipated in the 1 kΩ resistor. Since the components are connected in parallel, the voltage across the resistor is $v_i(t) = 12\sin(\omega t)$.

We know that the power dissipated in a resistor is given by:

$$P = \frac{V_{rms}^2}{R}$$

Here, the voltage is sinusoidal, hence $V_{rms} = \frac{12}{\sqrt{2}}$. Substituting these values:

$$P = \frac{144/2}{1000} = 0.072\,\text{W}$$

Final Result: $P = 72\,\text{mW}$

$$71.50\,to\,72.50$$

Q5.26.2 MCQ 2026 2M Ans: B ● ● ○

The currents I_C and I_L as shown in the figure are given by:

$$I_C = \frac{V}{R_C + \frac{1}{j\omega C}} = \frac{V}{R_C - \frac{j}{\omega C}} = \frac{V}{\sqrt{R_C^2 + \left(\frac{1}{\omega C}\right)^2}} \angle \tan^{-1}\left(\frac{1}{\omega C R_C}\right)$$

$$I_L = \frac{V}{R_L + j\omega L} = \frac{V}{\sqrt{R_L^2 + (\omega L)^2}} \angle -\tan^{-1}\left(\frac{\omega L}{R_L}\right)$$

The phase angles θ_1 and θ_2 can be defined as:

$$\theta_1 = \tan^{-1}\left(\frac{1}{\omega C R_C}\right)$$

$$\theta_2 = \tan^{-1}\left(\frac{\omega L}{R_L}\right)$$

Given the condition that the sum of the phase angles is 90°:

$$\theta_1 + \theta_2 = 90^\circ$$

Taking the tangent of both sides:

$$\tan(\theta_1 + \theta_2) = \tan 90^\circ$$

Using the trigonometric identity $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$, the expression becomes infinite when the denominator is zero:

$$\frac{\frac{1}{\omega C R_C} + \frac{\omega L}{R_L}}{1 - \left(\frac{1}{\omega C R_C} \cdot \frac{\omega L}{R_L}\right)} \to \infty$$

Equating the denominator to zero:

$$1 - \frac{1}{R_C R_L} \cdot \frac{\omega L}{\omega C} = 0$$

$$1 = \frac{1}{R_C R_L} \cdot \frac{L}{C}$$

$$R_C R_L = \frac{L}{C}$$

For the specific circuit parameters provided in the derivation:

$$\boxed{R_C R_L = 1}$$

Since the frequency is fixed at 159.15 kHz, we do not explore any other solutions by further simplifying the expressions.

$$\boxed{(b) 1}$$

Q5.25.1 MCQ 2025 1M Ans: A ● ○ ○

The voltage source is in parallel to each component - R, L and C. The impedances can be computed from the admittances as folows :

$$X_C = \frac{1}{j0.25} = -j4\,\Omega$$

$$X_L = \frac{1}{-j0.1} = j10\,\Omega$$

$$R = \frac{1}{0.2} = 5\,\Omega$$

Source voltage = $1\angle 90^\circ = j1$ V Now we can compute the current through each element.

$$i_C = \frac{j1}{-j4} = -0.25A = 0.25\angle 180^\circ \text{ A}$$

$$i_L = \frac{j1}{j10} = 0.1A = 0.1\angle 0^\circ \text{ A}$$

$$i_R = \frac{j1}{5} = j0.2A = 0.2\angle 90^\circ \text{ A}$$

$$\boxed{(a) i_C = 0.25\angle 180^\circ \text{ A}, i_L = 0.1\angle 0^\circ \text{ A}, i_R = 0.2\angle 90^\circ \text{ A}}$$

Q5.25.2 MCQ 2025 2M Ans: A ● ○ ○

Given data from the circuit parameters: $\omega = 200$ rad/s

$$I_{M2} = 5\text{ A}, \quad V_{M1} = 25\text{ V}$$

The circuit between the source and the meters is linear and time-invariant, so in sinusoidal steady state every voltage and current at its output is a sinusoid at the source frequency. Hence the load is analysed at $\omega = 200$ rad/s.

The total impedance of the load Z_L is calculated as:

$$Z_L = \frac{V_L}{I_L} = \frac{25\text{ V}}{5\text{ A}} = 5\,\Omega$$

For a series RLC load configuration, the impedance expression is given by:

$$Z_L = \sqrt{R^2 + (X_L - X_C)^2}$$

The inductive reactance X_L is:

$$X_L = \omega L = 200 \times 1 = 200\,\Omega$$

Substituting the known values into the impedance equation:

$$5 = \sqrt{5^2 + (X_L - X_C)^2}$$

Squaring both sides of the equation:

$$5^2 = 5^2 + (X_L - X_C)^2$$

$$(X_L - X_C)^2 = 0$$

$$X_C = X_L = 200\,\Omega$$

The capacitive reactance X_C is given by the formula:

$$X_C = \frac{1}{\omega C} = \frac{1}{200C} = 200\,\Omega$$

Solving for the capacitance C:

$$C = \frac{1}{4 \times 10^4}\,\text{F} = 25\,\mu\text{F}$$

$$\boxed{(a) 25\,\mu\text{F}}$$

Q5.23.1 MCQ | 2023 | 1M | Ans: D | ● ○ ○

Given : $\omega_c = 10^6 rad/s$ and Q = 1000. By inspecting the options, we can see that for all cases $L = 1\,\mu\text{H}\, and\, C = 1\,\mu\text{F}$, which aligns with the fact that

$$\omega_c = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6} \times 10^{-6}}} = 10^6 rad/s$$

. Hence we need to calculate the value of R. As per the definition of quality factor for a series RLC circuit,

$$Q = \frac{\omega_c L}{R}$$

Substituting, $Q = 10^6 \times 10^{-6}/R \implies 1000 = 1/R \implies R = 0.001\Omega$

$$\boxed{(d) R = 0.001\,\Omega, L = 1\,\mu\text{H}, C = 1\,\mu\text{F}}$$

Q5.22.1 MCQ | 2022 | 2M | Ans: A | ● ○ ○

We can determine the values of R_1 and R_2 from the locus diagram of the impedance $Z(j\omega)$ as plotted as ω. In the locus diagram, we can see that at $\omega = 0$ rad/s, $-Im(Z) = 0\text{k}\Omega$ and $Re(Z) = 5\text{k}\Omega$ and at $\omega = \infty$ rad/s, $-Im(Z) = 0\text{k}\Omega$ and $Re(Z) = 2\text{k}\Omega$.

1. At $\omega = 0$ rad/s: At $\omega = 0$ rad/s, capacitive impedance $X_C = \infty\text{k}\Omega$ (inversely proportional to ω), i.e. it appears as if the capacitive branch is open circuited.

$\therefore Z(jw) = R_1 + R_2 = 5\text{k}\Omega$.

1. At $\omega = \infty$ rad/s: At $\omega = \infty$ rad/s, capacitive impedance $X_C = 0\text{k}\Omega$, i.e. it appears as if the capacitive branch is short circuited.

$$\therefore Z(jw) = R_1 = 2\text{k}\Omega$$

$$\implies R_2 = 3\,\text{k}\Omega$$

$$\boxed{(a) R_1 = 2\text{k}\Omega, R_2 = 3\text{k}\Omega}$$

Q5.22.2 MCQ | 2022 | 2M | Ans: B | ● ● ○

From the given waveform of the steady-state current $I_{(t)}$, we can deduce the mathematical expression for the current. The peak amplitude of the current is:

$$I_{\max} = \frac{1}{\sqrt{2}}\,\text{A}$$

The waveform passes through zero going positive at a delayed time, reaching its peak at $t = \frac{\pi}{4}$. Thus, the phase difference is $\frac{\pi}{4}$ lagging. The time-domain expression for the current is:

$$I_{(t)} = \frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)\,\text{A}$$

Given that the input voltage is $V(t) = \sin(t)$ V, the current lags the voltage by $\frac{\pi}{4}$. A lagging current signifies that the passive network contains inductive reactance. Hence, the load impedance can be represented as:

$$Z = j\omega L$$

The total circuit impedance seen by the source includes the series component, yielding:

$$I(t) = \frac{V_{(t)}}{1+Z} = \frac{\sin(t)}{Z+1}\,\text{A} \quad \text{— (i)}$$

From the waveform, the maximum value of the current is:

$$|I(t)|_{\max} = \frac{1}{\sqrt{2}}\,\text{A}$$

Using the magnitude of the transfer function from equation (i):

$$|I(t)|_{\max} = \frac{1}{|Z+1|} = \frac{1}{|j\omega L+1|} = \frac{1}{\sqrt{2}}$$

Equating the denominators:

$$|j\omega L + 1| = \sqrt{2}$$

$$\sqrt{1+\omega^2 L^2} = \sqrt{2}$$

Squaring both sides:

$$1 + \omega^2 L^2 = 2$$

Here $\omega = 1$ rad/s:

$$\therefore\ L^2 = 1$$

$$\implies L = 1\,\text{H}$$

Key Points

- A sinusoidal current waveform of the form $I_m \sin(\omega t - \phi)$ indicates a phase lag of ϕ radians with respect to a standard reference sine wave $\sin(\omega t)$.
- In an AC circuit, an inductive load causes the steady-state current to lag behind the applied voltage.

Mistakes to be avoided

- Do not confuse the peak value $I_{\max} = \frac{1}{\sqrt{2}}$ A with the RMS value when calculating the magnitude equation.
- Ensure that the phase delay from the time axis shifts properly into a negative phase angle in the sine argument.

$\boxed{(b) \textit{an inductor of 1 H}}$

Q5.20.1 NAT | 2020 | 1M | Ans: 2.80 to 2.85 | ● ○ ○

From the given information we can say $\omega = 5$ rad/s. The current and voltage equations can be written in phasor form as :

$$V = 200\angle 0^\circ$$

$$I = 10\angle -45^\circ$$

We know $V = I.Z \implies Z = \frac{V}{I}$

$$\therefore Z = \frac{200\angle 0^\circ}{10\angle -45^\circ} = 20\angle 45^\circ$$

We can thus make two conclusions :

$$tan^{-1}\frac{\omega L}{R} = \frac{\pi}{4} \text{ and } \sqrt{(\omega L)^2 + R^2} = 20$$

From the first equation it follows that $\omega L = R$ (as $tan(\frac{\pi}{4} = 1)$.

$$\therefore \sqrt{(\omega L)^2 + (\omega L)^2} = 20 \implies \omega L = \sqrt{200} \implies L = 2.83\,\text{H}$$

$$\boxed{2.80 to 2.85}$$

Q5.20.2 MCQ | 2020 | 2M | Ans: C | ● ○ ○

By applying the nodal analysis to find the net current, the total current I can be expressed as the sum of individual component currents:

$$I = [I_1 + I_2]$$

$$I = -\left[\frac{120\angle -90^\circ}{80 - j35} + \frac{120\angle -30^\circ}{80 - j35}\right]$$

(A negative sign is added because the voltage applied would result in a current flowing in a direction opposite to the defined direction:)

$$I = -\frac{120\angle -90^\circ + 120\angle -30^\circ}{80 - j35}$$

Converting to polar form and evaluating the expression:

$$I = 2.38\angle 143.7^\circ\,\text{A}$$

$$\boxed{(c) 2.38\angle 143.63^\circ\,\text{A}}$$

Q5.19.1 MCQ | 2019 | 2M | Ans: C | ● ○ ○

To solve the problem we can first convert the capacitors arranged in a Y to that in Δ. When arranged in Δ, the capacitors would be of value C/3 (C being the original value). Hence we have a Δ-connected load, whose each arm consists of a resistance R and capacitance C/3 in parallel. Let each arm impedance be Z.

$$Z = R||(3)/j\omega C$$

Here $\omega = 1000$ rad/s, $R = 1\ k\,\Omega$ and $C = 1\ \mu$F

$$\therefore Z = 900 - j300\Omega$$

The impedance seen by the voltage source would $Z_{eq} = Z||2Z = 600 - j200\Omega$ The voltage in phasor form is $V = 2\angle 0^\circ$

$$\therefore I = \frac{2\angle 0^\circ}{600 - j200} = 0.003 + j0.001\,\text{A} = 3 + j1\,\text{mA}$$

. The time domain equation would be $i(t) = 3sin(1000t) + cos(1000t)$ mA.

Key Points

- The same problem can be solved by converting the load to an equivalent Y form, but the approach of converting to Δ form is more convenient and simple. Hence the other approach is not recommended.

Mistakes to be avoided

- When converting a network from Y to Δ the arm impedances get multiplied by 3. However since capacitive impedance is inversely proportional to the capacitance value, the capacitance needs to be divided by 3.

$$\boxed{(c) 3sin(1000t) + cos(1000t)}$$

Q5.18.1 MCQ | 2018 | 2M | Ans: C | ● ○ ○

This problem can be solved by using voltage division between the two resistances and the capacitance.
Given : $\omega = 5rad/s, C = 1\mu\text{F}, R = 100\,\text{k}\Omega$

$$\text{Capacitive Impedance, } X_C = \frac{1}{j\omega C} = -j200\,\text{k}\Omega$$

Voltage in phasor form, $V = 5\angle 0°$ Therefore, voltage across capacitance :

$$V_C = V\frac{X_C}{X_C + R + R} = 5 \times \frac{-j200}{-j200 + 200} = 2.5 - j2.5 = 2.5\sqrt{2}\angle -45°$$

The time domain equation would be

$$v(t) = 2.5\sqrt{2}sin(5t - \frac{\pi}{4})v = 2.5\sqrt{2}sin(5t - 0.25\pi)V$$

$$\boxed{(c) 2.5\sqrt{2}sin(5t - 0.25\pi)}$$

Q5.17.1 NAT | 2017 | 2M | Ans: 2.55 to 2.65 | ● ○ ○

This problem can be solved by using voltage division between two impedances. Let V_1 be across impedance Z_1 and V_2 be across impedance Z_2. Here $\omega = 3$ rad/s The two impedances can be calculated as :

$$Z_1 = R + j\omega L = 4 + j(3)(1)\Omega \implies |Z_1| = 5\Omega$$

$$Z_2 = R + 1/(j\omega C) = 5 + 1/(j(3)(1/36))\Omega \implies |Z_2| = 13\Omega$$

Now amplitude of $V_1 = 100 \times \frac{|Z_1|}{|Z_1+Z_2|}$ and amplitude of $V_2 = 100 \times \frac{|Z_2|}{|Z_1+Z_2|}$

$$\therefore \frac{\text{amplitude of } V_2}{\text{amplitude of } V_1} = \frac{100|Z_2|/|Z_1 + Z_2|}{100|Z_1|/|Z_1 + Z_2|} = \frac{|Z_2|}{|Z_1|} = \frac{13}{5} = 2.6$$

$$\boxed{2.55\ to\ 2.65}$$

Q5.17.2 NAT | 2017 | 1M | Ans: 0.9 to 1.1 | ● ● ○

We know $\angle V_1 = \angle I + \angle Z$

$$V_1 = I.Z_1 \text{ and } V_2 = I.Z_2 \implies \angle(V_1 - V_2) = \angle(Z_1 - Z_2)$$

Hence for a phase difference of 45° between the voltages V_1 and V_2, the impedances Z_1 and Z_2 must have a phase angle difference of 45° between them.

$$Z_1 = 1\angle 0^\circ\ \Omega$$

The impedance Z_2 is:

$$Z_2 = 1 + j\omega$$

Since the phase angle of Z_1 is 0°, the phase angle of Z_2 must be 45° to maintain the required phase difference:

$$\angle Z_2 = \tan^{-1}\left(\frac{\omega}{1}\right) = 45^\circ$$

$$\frac{\omega}{1} = \tan(45^\circ) = 1$$

$$\omega = 1\ \text{rad/s}$$

Mistakes to be avoided

- Do not confuse the absolute phase angles of the voltages with the relative phase difference between them.

$$\boxed{0.9\ to\ 1.1}$$

Q5.17.3 NAT 2017 1M Ans: 0.19 to 0.21 ● ● ○

We know $\angle V_1 = \angle I + \angle Z$ Thus, For the current to be in phase with the voltage the angle of impedance must be zero.

$$\text{Impedance, } Z = 5 + j\omega 5 + \frac{1}{j\omega 5} = 5 + j(5\omega - \frac{1}{5\omega})$$

For the phase angle to be zero the imaginary part must be zero, or $5\omega = \frac{1}{5\omega} \implies \omega = 1/5 = 0.2rad/s$ Then we can use voltage division in order to find the required quantity. Let V_1 be across resistance Z_1 and V_2 be across capacitance Z_2.

$$Z_1 = 5\Omega \text{ and } Z_2 = \frac{1}{j(0.2)(5)} = -j\Omega$$

Therefore we can write :

$$\frac{\text{amplitude of voltage across the capacitor}}{\text{amplitude of voltage across the resistor}} = \frac{|Z_2|}{|Z_1|} = \frac{1}{5} = 0.2$$

(Also refer to solutions for Q5.17.1 and Q5.17.2)

$$\boxed{0.19\ to\ 0.21}$$

Q5.16.1 NAT 2016 1M Ans: 0.30 to 0.34 ● ○ ○

At the resonant frequency of a parallel RLC network, the net reactive current becomes zero because the inductive and capacitive currents cancel each other out. Thus, the resistor carries the entire source current:

$$I_R = I$$

The current through the inductor is computed using the quality factor Q:

$$I_L = QI\angle -90^\circ\ \text{A}$$

The quality factor Q for a parallel RLC circuit is defined as:

$$Q = \omega_0 CR$$

Expressing ω_0 as $\frac{1}{\sqrt{LC}}$, the quality factor formula becomes:

$$Q = R\sqrt{\frac{C}{L}}$$

Substituting the given component parameters ($R = 10\,\Omega$, $C = 10 \times 10^{-6}$ F, and $L = 10 \times 10^{-3}$ H):

$$Q = 10\sqrt{\frac{10 \times 10^{-6}}{10 \times 10^{-3}}}$$

$$Q = 10\sqrt{10^{-3}} = 10 \times \frac{1}{10\sqrt{10}}$$

$$Q = \frac{1}{\sqrt{10}} \approx 0.3163$$

Therefore, the ratio of the magnitude of the inductor current to the resistor current at resonance is:

$$\frac{|I_L|}{|I_R|} = Q = 0.3163$$

Key Points

- The quality factor Q of a parallel resonant circuit measures the current magnification across the reactive components relative to the input current: $Q = \frac{|I_L|}{I} = \frac{|I_C|}{I}$.
- In terms of component values, the parallel resonance quality factor is defined mathematically as $Q = R\sqrt{\frac{C}{L}}$, which is inversely proportional to the series circuit variant.

Mistakes to be avoided

- Do not mistakenly use the series resonance quality factor formula ($Q = \frac{1}{R}\sqrt{\frac{L}{C}}$) for a parallel circuit configuration.

PrepFusion

0.30 *to* 0.34

Q5.16.2 MCQ 2016 1M Ans: B ●●○

The input has two frequencies, so we use superposition and treat each component separately.

Component at $\omega = 200$ rad/s: We can redraw the network by plugging in the impedances in place of the respective inductances and capacitances as follows.

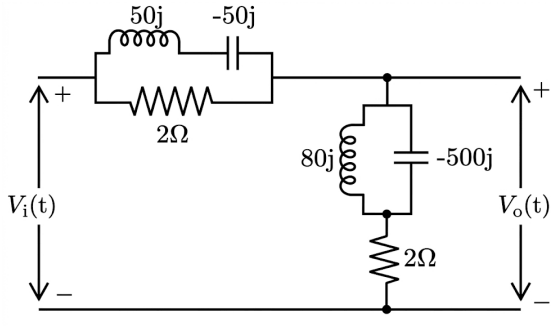


Fig. Solution Q5.16.2

The network can be then simplified as :

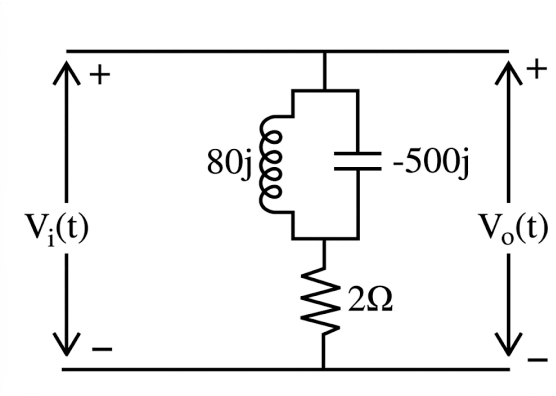


Fig. Solution Q5.16.2

The series LC ($j50\,\Omega$ and $-j50\,\Omega$) is at series resonance, so it shorts the $2\,\Omega$ resistor and there is no drop across the series element. Hence the $2\cos(200t)$ component appears unchanged at the output.

Component at $\omega = 500$ rad/s: in the shunt branch $X_L = 500 \times 0.4 = 200\,\Omega$ and $X_C = \dfrac{1}{500 \times 10 \times 10^{-6}} = 200\,\Omega$, so the parallel LC is at resonance and its impedance is infinite. The shunt branch is open, no current flows through the series element and there is no drop across it, so the $4\sin(500t)$ component also appears unchanged at the output.

By superposition, $v_o(t) = v_i(t)$

$$\therefore v_o(t) = 2\cos(200t) + 4\sin(500t)$$

$$\boxed{(b) 2\cos(200t) + 4\sin(500t)}$$

Q5.15.1 NAT 2015 1M Ans: 24 to 26 ● ○ ○

For the condition of resonance,

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-10}}} = 10^5 rad/s$$

At resonance the entire voltage appears across the resistor.

$$\therefore V_R = 10V$$

Also at resonance, the magnitude of the voltage across the capacitance $|V_C| = Q.|V_R|$, where Q is the quality factor. For a series RLC circuit, $Q = \frac{1}{R}\sqrt{\frac{L}{C}} = 2.5$

$$\therefore |V_C| = 2.5|V_R| = 2.5 \times 10 = 25V$$

$$\boxed{24 \text{ to } 26}$$

Q5.15.2 NAT 2015 2M Ans: 1.6 to 1.7 ● ● ○

According to the Maximum Power Transfer Theorem, the maximum power is transferred to the load R_L when $R_L = |Z_{th}|$ where Z_{th} is the Thevenin impedance when viewed from load terminals. We can compute the Thevenin impedance as follows :

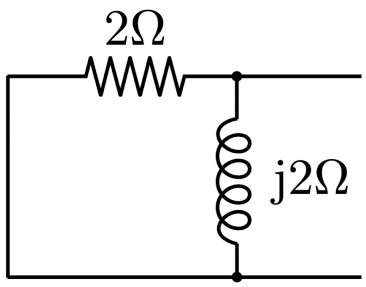


Fig. Solution Q5.15.2

$$Z_{th} = 2||j2 \implies Z_{th} = 1 + j1\Omega$$
$$\therefore R_L = |Z_{th}| = \sqrt{2}\,\Omega$$

Next, we calculate the Thevenin's voltage.

$$V_{th} = 4 \times \frac{j2}{2 + j2} \implies V_{th} = 2\sqrt{2}\angle 45°V$$

Thus the Thevenin equivalent circuit looks like :

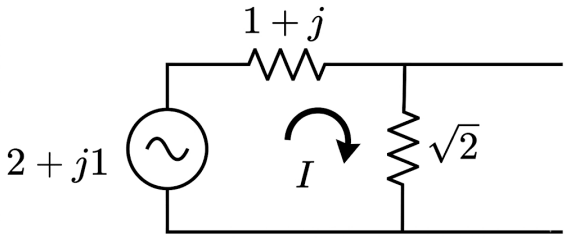


Fig. Solution Q5.15.2

Current $I = \frac{V_{th}}{Z_{th}+R_L} = \frac{2+j2}{\sqrt{2}+1+j1} \implies |I| = 1.082A$ Therefore, Maximum power transferred to load resistance (in Watts) is :

$$|I|^2 R_L = 1.65\,\text{W}$$

Mistakes to be avoided

- Do not use the formula $P_{max} = V_{th}^2/4|Z_{th}$ directly. Instead draw the complete Thevenin's equivalent circuit and then attempt to find the maximum power transferred.

1.6 *to* 1.7

Q5.15.3 NAT 2015 1M Ans: 100 ●○○

For a series RLC circuit, the relationship between the RMS source voltage V_s and the RMS voltage drops across the resistor (V_R), inductor (V_L), and capacitor (V_C) is given by the phasor relationship:

$$V_s^2 = V_R^2 + (V_L - V_C)^2$$

Substituting the given values into the equation ($V_s^2 = 10000$, $V_R^2 = 6400$, and $V_L = 40\,\text{V}$):

$$10000 = 6400 + (40 - V_C)^2$$

Subtracting 6400 from both sides:

$$3600 = (40 - V_C)^2$$

Taking the square root of both sides:

$$\pm 60 = 40 - V_C$$

This yields two possible mathematical solutions for V_C: 1. $40 - V_C = 60 \implies V_C = -20\,\text{V}$ 2. $40 - V_C = -60 \implies V_C = 100\,\text{V}$

Since the RMS voltage magnitude must be a positive quantity, we select the physically meaningful result:

$$V_C = 40 - (-60) = 100\,\text{V}$$

Key Points

- The voltage across the resistor is in phase with the current, while the inductive voltage leads the current by 90° and the capacitive voltage lags by 90°.
- Because the inductor and capacitor voltages are 180° out of phase, they directly subtract from one another along the imaginary axis, leading to the term $(V_L - V_C)^2$.

100

Q5.15.4 NAT | 2015 | 1M | Ans: 4.9 to 5.1 | ● ○ ○

In the problem, we are asked to find the average value of the voltage V_{ab}. Since, we are dealing with the average value, we can open the capacitor and short the inductor.

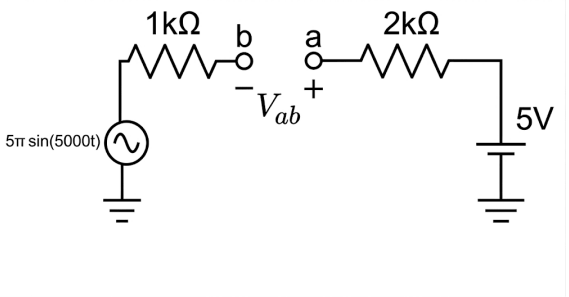


Fig. Solution Q5.15.4

Next we can check the effect of the two sources one by one (Superposition Theorem). For the source $5\pi\, sin(5000t)V$, keeping 5V source shorted, the average voltage contributed to V_{ab} is 0. For the source 5V, keeping $5\pi\, sin(5000t)V$source shorted, the average voltage contributed to V_{ab} is 5 V. Hence, $V_{ab} = 5V$.

Q5.15.5 NAT | 2015 | 1M | Ans: 0.49 to 0.51 | ● ○ ○

At very high frequencies we can treat the capacitive impedance to be zero, i.e., capacitors to be shorted. Thus the network becomes

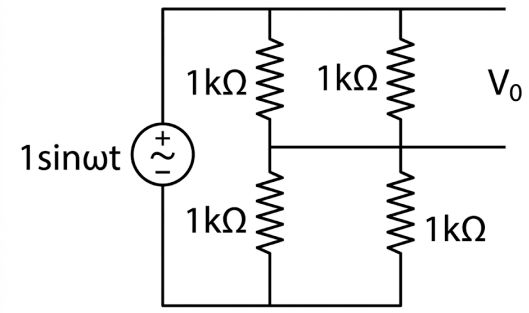


Fig. Solution Q5.15.5

This can be simplified as :

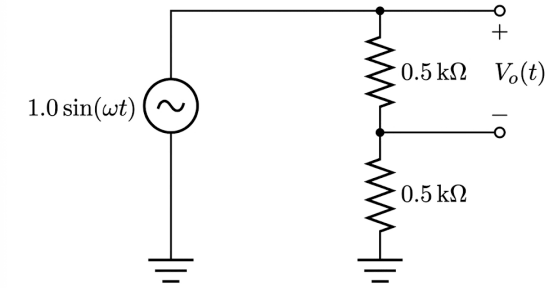


Fig. Solution Q5.15.5

Thus the output voltage $v_o(t) = v_i(t)/2 = 0.5\,sin(\omega t)V$ Hence the peak amplitude of $v_o(t) = 0.5V$.

$$\boxed{0.49\,to\,0.51}$$

Q5.15.6 NAT 2015 2M Ans: 20

At the given input frequency $\omega = 5000$ rad/s, we analyze the equivalent network impedance. The parallel combination impedance block Z is formulated as follows:

$$Z = \frac{(5j - jX_c) \cdot 5j}{(10 - X_c)j} = \frac{(5 - X_c) \cdot 5j}{10 - X_c}$$

The total equivalent network impedance Z_{eq} looking into the circuit includes the remaining series elements:

$$Z_{eq} = Z + 50 + 5j$$

For the steady-state network current to drop to zero ($I = 0$), the input driving-point impedance must approach infinity:

$$Z_{eq} = Z + 50 + 5j \rightarrow \infty$$

An infinite impedance condition implies the denominator of the rational expression becoming zero:

$$10 - X_c = 0$$

$$X_c = 10\,\Omega$$

The capacitive reactance formula is given by:

$$X_c = \frac{1}{\omega C}$$

Substituting the known values ($\omega = 5000\,\text{rad/s}$ and $X_c = 10\,\Omega$):

$$\frac{1}{5000 \times C} = 10$$

Solving directly for the capacitance C:

$$C = \frac{1}{50000}\,\text{F} = 20 \times 10^{-6}\,\text{F} = 20\,\mu\text{F}$$

$$\boxed{20}$$

Q5.14.1 MCQ | 2014 | 1M | Ans: C | ● ○ ○

According to maximum power transfer theorem, between two cascaded sections of an electrical network, the relationship between the output impedance Z_1 of the first section to the input impedance Z_2 of the second section is $Z_2 = Z_1^*$.

$$\boxed{(c)\, Z_2 = Z_1^*}$$

Q5.14.2 MCQ | 2014 | 2M | Ans: B | ● ○ ○

The relationship between real power P, reactive power Q, and complex apparent power S can be described using the power factor angle θ:

$$P = S\cos(\theta) \quad \text{and} \quad Q = S\sin(\theta)$$

By convention, the reactive power sign is defined such that for a leading power factor $Q < 0$, and for a lagging power factor $Q > 0$.

The reactive power component Q can be calculated from the real power P as:

$$Q = P\tan(\theta)$$

Given a leading power factor load with $P = 10\,\text{kW}$ and $\cos(\theta) = 0.8$:

$$Q = -10\,\text{kW} \times \tan\left(\cos^{-1}(0.8)\right)$$

$$Q = -10\,\text{kW} \times 0.75 = -7.5\,\text{kVAR} \quad \text{[leading power factor]}$$

Therefore, the complex apparent power for the first load S_1 is written as:

$$S_1 = 10 - j7.5\,\text{kVA}$$

The second load is given as lagging with a complex apparent power S_2:

$$S_2 = 8 + j6\,\text{kVA}$$

The total complex apparent power S drawn by the combined system is the vector sum of the individual loads:

$$S = S_1 + S_2$$

$$S = (10 - j7.5) + (8 + j6)$$

$$S = (10 + 8) + j(-7.5 + 6)$$

$$S = 18 - j1.5\,\text{kVA}$$

Key Points

- A leading power factor signifies a capacitive load configuration, resulting in a negative reactive power component (supplying VARs).
- A lagging power factor signifies an inductive load configuration, resulting in a positive reactive power component (absorbing VARs).

Mistakes to be avoided

- Do not directly add the magnitudes of apparent power ($|S_1| + |S_2| \neq |S|$); vector components must be isolated and summed independently.
- Pay close attention to sign conventions: a leading power factor must be treated with a negative imaginary component, otherwise the total net reactive power will be incorrectly calculated.

$$\boxed{(b)(18 - j1.5)kVA}$$

Q5.14.3 NAT | 2014 | 2M | Ans: 0.45 to 0.47 | ● ● ○

The resonant frequency ω_r of the given series RLC circuit is calculated as:

$$\omega_r = \frac{1}{\sqrt{LC}} = 1 \text{ rad/s}$$

Since the circuit operates at a leading power factor, it exhibits predominantly capacitive behavior. Therefore, the operating angular frequency must be lower than the resonant frequency: $\omega < \omega_r$

At resonance, the net reactive components cancel each other out, meaning the magnitude of the circuit current I_r depends solely on the resistive part of the network:

$$I_r = \frac{V}{R} = \frac{V}{1} = V$$

We seek the operating frequency ω at which the current magnitude reduces to exactly half of its peak resonant value ($I = \frac{I_r}{2}$):

$$\left| \frac{V}{1 + \left(1 \times \omega - \frac{1}{\omega}\right) j} \right| = \left| \frac{V}{2} \right|$$

Equating the magnitudes of the denominators:

$$1 + \left(\omega - \frac{1}{\omega}\right)^2 = 4$$

$$\left(\omega - \frac{1}{\omega}\right)^2 = 3$$

Taking the square root on both sides yields:

$$\omega - \frac{1}{\omega} = \pm\sqrt{3}$$

Rearranging this into a standard quadratic equation form:

$$\omega^2 \mp \sqrt{3}\omega - 1 = 0$$

Solving the quadratic equation using the standard algebraic formula:

$$\omega = \frac{\pm\sqrt{3} \pm \sqrt{3+4}}{2} = \frac{\pm\sqrt{3} \pm \sqrt{7}}{2}$$

Since the angular frequency ω must be a positive real value and satisfy the leading power factor constraint ($\omega < \omega_r$), we select the appropriate root:

$$\omega = \frac{-\sqrt{3} + \sqrt{7}}{2} \approx 0.456 \text{ rad/s}$$

$$\boxed{0.456}$$

Key Points

- A series RLC network transitions behavior based on frequency: it is capacitive (leading power factor) when $\omega < \omega_r$, purely resistive at resonance when $\omega = \omega_r$, and inductive (lagging power factor) when $\omega > \omega_r$.
- At resonance, circuit impedance drops to its absolute minimum value ($Z = R$), driving the current magnitude to its peak value.

$$\boxed{0.45 \text{ to } 0.47}$$

Q5.14.4 MCQ 2014 2M Ans: D ●●○

By applying Kirchhoff's Current Law (KCL) nodal analysis at the output node, we establish the steady-state current summation equation for the circuit branch components:

$$-4 + \frac{V_2 + 10}{-j3} + \frac{V_2}{6} + \frac{V_2}{j6} = 0$$

Grouping all terms associated with the unknown node voltage variable V_2 onto the left side and moving the constant scalar components to the right side:

$$V_2\left(\frac{1}{-j3} + \frac{1}{6} + \frac{1}{j6}\right) = \frac{10}{j3} + 4$$

Multiplying the numerator and denominator fractions by the common complex factor to clear denominators:

$$V_2\left(\frac{-2}{j} + 1 + \frac{1}{j}\right) = \frac{20}{j} + 24$$

Using the fundamental complex algebraic identity $\frac{1}{j} = -j$ to simplify the imaginary phase values:

$$V_2(j2 + 1 - j) = -j20 + 24$$

$$V_2(1 + j1) = 24 - j20$$

Isolating the node voltage variable V_2 by dividing by the complex coefficient factor:

$$V_2 = \frac{24 - j20}{1 + j1}$$

Multiplying by the complex conjugate of the denominator $(1 - j1)$ to rationalize and evaluate the expression:

$$V_2 = \frac{(24 - j20)(1 - j1)}{(1 + j1)(1 - j1)} = \frac{24 - j24 - j20 - 20}{2}$$

$$V_2 = \frac{4 - j44}{2} = 2 - j22 \text{ V}$$

$$\boxed{(d) 2 - j22V}$$

Q5.14.5 NAT | 2014 | 2M | Ans: 1.9 to 2.1 | ● ● ○

To find the resonance condition where the driving-point input impedance $Z_{bb'}$ becomes purely resistive, we look at the complex expression of the network's total impedance:

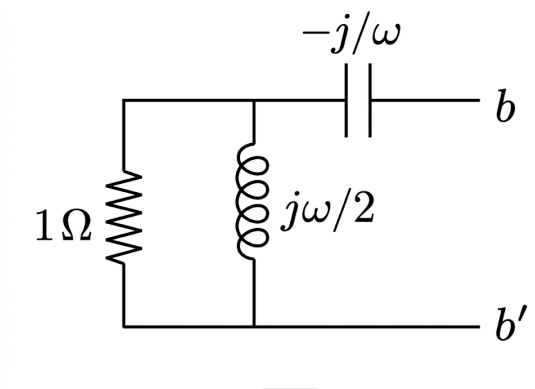


Fig. Solution Q5.14.5

$$Z_{bb'} = \frac{1 \times j\frac{\omega}{2}}{1 + j\frac{\omega}{2}} - \frac{j}{\omega}$$

To simplify, we multiply the numerator and denominator of the first term by the complex conjugate of its denominator, which is $\left(1 - j\frac{\omega}{2}\right)$:

$$Z_{bb'} = \frac{j\frac{\omega}{2}\left(1 - j\frac{\omega}{2}\right)}{1 + \frac{\omega^2}{4}} - \frac{j}{\omega}$$

Expanding the numerator components yields:

$$Z_{bb'} = \frac{j\frac{\omega}{2} + \frac{\omega^2}{4}}{1 + \frac{\omega^2}{4}} - \frac{j}{\omega}$$

Grouping the real and imaginary components independently to isolate the total network impedance parameters:

$$Z_{bb'} = \frac{\frac{\omega^2}{4}}{1 + \frac{\omega^2}{4}} + j\left[\frac{\frac{\omega}{2}}{1 + \frac{\omega^2}{4}} - \frac{1}{\omega}\right]$$

For the input impedance to be purely resistive, the net reactive parameter must cancel out completely, forcing the imaginary part of the function to zero:

$$\text{Img}(Z_{bb'}) = 0$$

$$\frac{\frac{\omega}{2}}{1 + \frac{\omega^2}{4}} = \frac{1}{\omega}$$

Cross-multiplying to solve the linear frequency relationship:

$$\frac{\omega^2}{2} = 1 + \frac{\omega^2}{4}$$

Subtracting $\frac{\omega^2}{4}$ from both sides of the equation:

$$\frac{\omega^2}{4} = 1$$

$$\omega^2 = 4$$

Taking the positive square root for a physically valid angular frequency solution:

$$\omega = 2\,\text{rad/s}$$

$$\boxed{1.9\,to\,2.1}$$

Q5.14.6 MCQ 2014 2M Ans: B ●●○

The equivalent capacitance C_{eq} of the parallel combination shown in the circuit network is calculated as:

$$C_{eq} = \frac{C}{2} \parallel C = \frac{C}{2} + C = \frac{3C}{2}$$

The transfer function $G(s)$ of this is derived using the standard voltage divider rule across the equivalent capacitive impedance element:

$$G(s) = \frac{\frac{1}{\frac{3}{2}sC}}{R + \frac{1}{\frac{3}{2}sC}} = \frac{\frac{2}{3sC}}{R + \frac{2}{3sC}} = \frac{2}{3sRC + 2}$$

Substituting $s = j\omega$ to evaluate the frequency response behavior of the transfer function $G(j\omega)$:

$$G(j\omega) = \frac{0.5 \times 2}{2 + j3\omega RC}$$

$|A(\omega)|$ is given as 0.25 and input voltage magnitude $|V_{in}| = 1$:

$$|A(\omega)| = 1 \times |G(j\omega)| = \frac{1}{\sqrt{4 + 9\omega^2R^2C^2}} = 0.25$$

Squaring both sides of the equation to clear the radical sign:

$$\frac{1}{4 + 9\omega^2R^2C^2} = \frac{1}{16}$$

$$4 + 9\omega^2R^2C^2 = 16$$

$$9\omega^2R^2C^2 = 12$$

$$\omega RC = \sqrt{\frac{12}{9}} = \frac{2}{\sqrt{3}}$$

Rearranging the terms we get ω:

$$\omega = \frac{2}{\sqrt{3}RC}$$

$$\boxed{(b)\,\frac{2}{\sqrt{3}RC}}$$

Q5.13.1 MCQ 2013 1M Ans: C ●○○

According to maximum power transfer theorem, the purely resistive load should have a value equal to the magnitude of the internal impedance of the source in order to have maximum power transferred to it.

$$R_L = |Z_{in}| \implies R_L = \sqrt{4^2 + 3^2} = 5\Omega$$

$$\boxed{(c)5}$$

Q5.13.2 MCQ 2013 2M Ans: C ● ● ○

To determine the open-circuit Thevenin voltage V_{th} across the load terminals where R_l is connected, we analyze the network by looking into from the load terminals (opening R_L) and thus output current is zero ($i_2 = 0\,\text{A}$).

Due to this thedependant voltage source $j40I_2$ contributes 0V and appears shorted, and the source voltage appears across the resistor ($3\,\Omega$) and the inductor ($4j\,\Omega$). The voltage drop V_l across the reactive inductive component is given by :

$$V_l = v_s \times \frac{4j}{3+4j}$$

$$V_l = 80\angle 90^\circ\,\text{V}$$

Because the output loop terminal is completely open ($i_2 = 0\,\text{A}$), there is zero potential drop across the series $j6 + 5\,\Omega$. Therefore,

$$V_{th} = 10V_l$$

Substituting the evaluated phasor value of V_l into the dependent balance relation:

$$V_{th} = 10 \times (80\angle 90^\circ) = 800\angle 90^\circ\,\text{V}$$

$$\boxed{(c) 800\angle 90^\circ}$$

Q5.13.3 MCQ 2013 2M Ans: B ● ○ ○

Case 1 : It is given that $V_{WX1} = 100\text{V}$. Therefore from the turns ratio, the secondary voltage would be $100 \times 1.25 = 125\,\text{V}$. Owing to the attenuation factor of 0.8, $V_{YZ1} = 0.8 \times 125 = 100\,\text{V}$. **Case 2 :** It is given that $V_{YZ2} = 100\text{V}$. Owing to the attenuation factor of 0.8, the secondary voltage would be $100/0.8 = 125\,\text{V}$. Considering the turns ratio, $V_{WX2} = 125/1.25 = 100\,\text{V}$. Thus, we can conclude

$$V_{YZ1}/V_{WX1} = 100/100 \text{ and } V_{WX2}/V_{YZ2} = 100/100$$

$$\boxed{(b) 100/100 \textit{ and } 100/100}$$

Q5.12.1 MCQ 2012 1M Ans: B ● ○ ○

It is given that $i(t) = 5cos(100\pi t + 100)$ A.

$$\therefore I_{rms} = \frac{5}{\sqrt{2}}\,\text{A}$$

Average power delivered $P = I_{rms}^2 R$, where R = 4Ω Substituting, we get $P = 50W$

$$\boxed{(b) 50W}$$

Q5.12.2 MCQ 2012 1M Ans: C ● ● ○

A ground node reference is established at the common junction point of the voltage and current sources, setting its potential to 0 V.

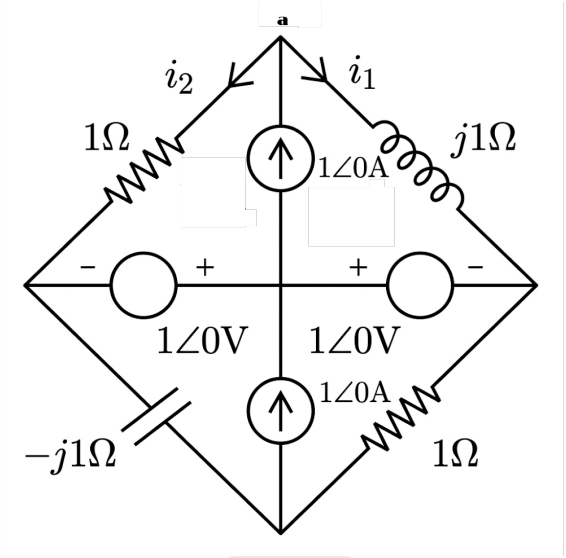


Fig. Solution Q5.12.2

Let the unknown node potential across the $1\angle 0°$ A source be V.
Applying KVL along the meshes in the network :

$$1 = -V + ji_1$$

$$1 = -V + i_2$$

$$\implies ji_1 - i_2 = 0$$

Applying KCL at node 'a' we get:

$$1 = i_1 + i_2$$

Combining the results we get :

$$ji_1 + i_1 = 1$$

$$\therefore i_1 = \frac{1}{1+j}\text{ A}$$

$$\boxed{(c)\ \frac{1}{1+j}A}$$

Q5.12.3 MCQ 2012 2M Ans: A ●●○

First, we need to convert circuit B to its Thevenin equivalent form.

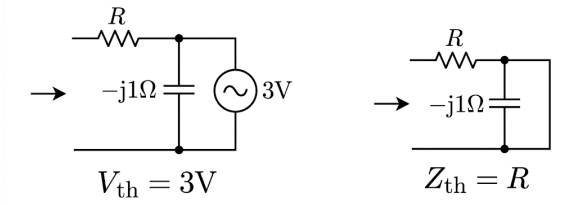


Fig. Solution Q5.12.3

The simplified circuit now looks like :

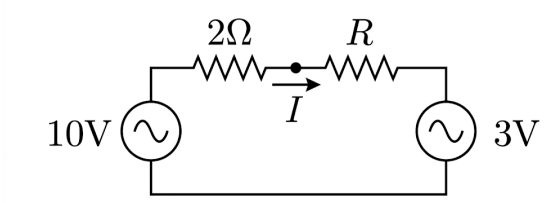


Fig. Solution Q5.12.3

The circuit current I as a function of load resistance R is expressed as:

$$I = \frac{10 - 3}{2 + R} = \frac{7}{2 + R}$$

The power transferred from circuit A to circuit B is given by the relation:

$$P = I^2R + 3I$$

Substituting the expression for current I:

$$P = I(IR + 3)$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R}{2 + R} + 3\right]$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R + 3(2 + R)}{2 + R}\right]$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R + 6 + 3R}{2 + R}\right]$$

$$P = \frac{7(10R + 6)}{(2 + R)^2} = \frac{70R + 42}{(2 + R)^2}$$

To find the condition for maximum power transfer, we set the first derivative of power with respect to R to zero ($\frac{dP}{dR} = 0$):

$$\frac{dP}{dR} = \frac{(2 + R)^2 \cdot (70) - (70R + 42) \cdot 2(2 + R)}{(2 + R)^4} = 0$$

Factoring out $(2 + R)$ from the numerator (where $2 + R \neq 0$):

$$(2 + R)\left[70(2 + R) - 2(70R + 42)\right] = 0$$

$$140 + 70R - 140R - 84 = 0$$

$$56 - 70R = 0$$

$$R = \frac{56}{70} = 0.8\,\Omega$$

Since a negative solution ($R = -2\,\Omega$) is physically non-realizable for a passive resistor, it is ignored.

Key Points

- **Maximum Power Transfer Theorem (Modified):** When a circuit contains independent or dependent sources within the load network, the traditional condition $R_L = R_{th}$ changes. Maximization must be derived from fundamental calculus principles using $\frac{dP}{dR_L} = 0$.

Mistakes to be avoided

- Do not apply $R = R_{th} = 2\,\Omega$ here, because circuit B contains an active terminal characteristic ($3I$ component) rather than a purely passive linear resistor.

$$\boxed{(a)0.8\Omega}$$

Q5.11.1 MCQ 2011 1M Ans: A ●○○

To find the Norton's equivalent current, we need to short-circuit the terminals P and Q. Let the node'a' have voltage be V.

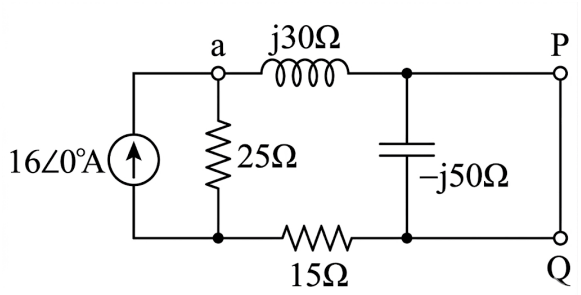


Fig. Solution Q5.11.1

Applying Kirchhoff's Current Law (KCL) at node 'a':

$$16 = \frac{V}{25} + \frac{V}{j30 + 15}$$

Solving for voltage V:

$$16 = V\left(\frac{1}{25} + \frac{1}{15 + j30}\right)$$

$$V = \frac{16 \times 25 \times (15 + j30)}{25 + (15 + j30)}$$

$$V = \frac{400 \times (15 + j30)}{40 + j30}$$

$$V = \frac{6000 + j12000}{40 + j30} = 240 + j120\,\text{V}$$

Now, Norton's equivalent current flowing through the short-circuited path is the current passing through the branch containing the $j30\,\Omega$ inductor and $15\,\Omega$ resistor:

$$I = \frac{V}{15 + j30}$$

Substituting the value of V:

$$I = \frac{240 + j120}{15 + j30} = 6.4 - j4.8\,\text{A}$$

$$\boxed{(a)6.4 - j4.8}$$

Q5.11.2 MCQ | 2011 | 1M | Ans: A | ● ● ○

For any sinusoidal input passing through an LTI (Linear Time-Invariant) system with a transfer function $G(s)$, the steady-state output response $V_o(j\omega)$ is given by:

$$V_o(j\omega) = |V_i(j\omega)| \cdot |G(j\omega)| \angle \left(\angle V_i(j\omega) + \angle G(j\omega) \right)$$

First, let us determine the transfer function $G(s)$ of the system using s-domain analysis. The parallel combination in the circuit is given by:

$$\frac{1}{Cs} \parallel R = \frac{\frac{1}{Cs} \cdot R}{\frac{1}{Cs} + R} = \frac{R}{RCs + 1}$$

Using the voltage divider rule, the transfer function $G(s) = \frac{V_o(s)}{V_i(s)}$ is expressed as:

$$G(s) = \frac{\frac{1}{Cs} \parallel R}{R + \frac{1}{Cs} + \left(\frac{1}{Cs} \parallel R \right)}$$

$$G(s) = \frac{\frac{R}{RCs+1}}{R + \frac{1}{Cs} + \frac{R}{RCs+1}}$$

Multiplying the numerator and the denominator by $(RCs + 1)$:

$$G(s) = \frac{R}{\left(R + \frac{1}{Cs} \right)(RCs + 1) + R}$$

$$G(s) = \frac{R}{R^2Cs + R + R + \frac{1}{Cs} + R}$$

Multiplying the numerator and the denominator by Cs to eliminate the fraction:

$$G(s) = \frac{RCs}{R^2C^2s^2 + 3RCs + 1}$$

To find the frequency response, substitute $s = j\omega$:

$$G(j\omega) = \frac{jRC\omega}{(j\omega)^2R^2C^2 + 3jRC\omega + 1} = \frac{jRC\omega}{-\omega^2R^2C^2 + 3jRC\omega + 1}$$

Given the operating frequency is $\omega = \frac{1}{RC}$. Substituting ω into the frequency response equation:

$$G(j\omega)|_{\omega=\frac{1}{RC}} = \frac{j \cdot (1)}{-\left(\frac{1}{RC} \right)^2 R^2C^2 + 3j \cdot (1) + 1}$$

$$G(j\omega)|_{\omega=\frac{1}{RC}} = \frac{j}{-1 + 3j + 1} = \frac{j}{3j} = \frac{1}{3}$$

Evaluating the magnitude and phase of the transfer function at this frequency:

$$|G(j\omega)| = \frac{1}{3}, \quad \angle G(j\omega) = 0^\circ$$

Thus, we can conclude that the amplitude of the output voltage is scaled to $\frac{1}{3}$rd of the input amplitude, and the phase difference between the output and input is zero. Assuming an input voltage of $V_p \cos(\omega t)$, the time-domain output voltage $V_o(t)$ is:

$$V_o(t) = \frac{V_p}{3} \cos\left(\frac{t}{RC} \right)$$

$$\boxed{(a)(V_p/3)\cos(t/RC)}$$

Q5.11.3 MCQ 2011 2M Ans: B ● ● ○

When the lower delta network consisting of three balanced 6 Ω resistors is converted to an equivalent star network, each leg of the star network has a resistance R given by:

$$R = \frac{6 \times 6}{6 + 6 + 6} = 2\,\Omega$$

The network now looks as shown below :

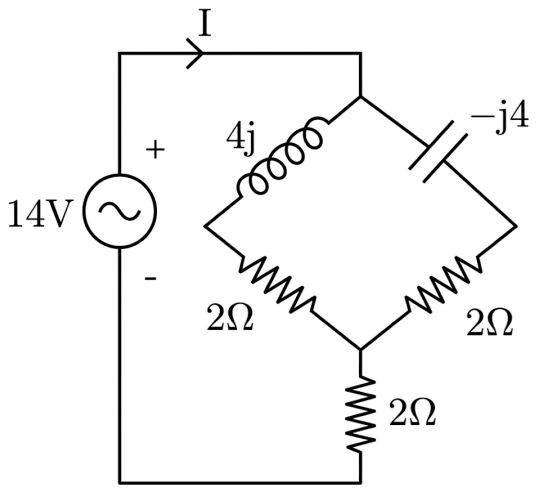


Fig. Solution Q5.11.3

The total equivalent impedance Z seen by the 14 V AC source is:

$$Z = 2 + \frac{(2 + 4j)(2 - 4j)}{(2 + 4j) + (2 - 4j)}$$

Simplifying the parallel combination in the numerator and denominator:

$$Z = 2 + \frac{4 - (4j)^2}{4} = 2 + \frac{4 + 16}{4} = 2 + \frac{20}{4} = 7\,\Omega$$

Using Ohm's law, the total source current I is calculated as:

$$I = \frac{V_s}{Z} = \frac{14\angle 0^\circ\ \text{V}}{7\,\Omega} = 2\angle 0^\circ\ \text{A}$$

$$\boxed{(b) 2.0\angle 0^\circ\ \text{A}}$$

Q5.10.1 MCQ 2010 2M Ans: A ● ○ ○

Based on the given information, for the AC equivalent circuit, the circuit parameters are:

- Source voltage: $V_s = 20\angle 0^\circ$ V
- Series inductive reactance: $Z_L = 20j\,\Omega$
- Parallel combination consists of a resistor $R = 1\,\Omega$ and a capacitive reactance $Z_C = -20j\,\Omega$.

First, let us find the equivalent impedance of the parallel combination of the 1 Ω resistor and the $-20j$ Ω capacitor:

$$Z_p = 1 \parallel (-20j) = \frac{1 \times (-20j)}{1 - 20j} = \frac{-20j}{1 - 20j}\,\Omega$$

Using the voltage division rule, the voltage V across the $1\,\Omega$ resistor is given by:

$$V = V_s \times \frac{Z_p}{Z_L + Z_p}$$

Substituting the given values into the voltage divider formula:

$$V = 20\angle 0^\circ \times \frac{\frac{-20j}{1-20j}}{20j + \frac{-20j}{1-20j}}$$

Multiplying the numerator and the denominator by $(1 - 20j)$:

$$V = 20\angle 0^\circ \times \frac{-20j}{20j(1-20j) - 20j}$$

$$V = 20\angle 0^\circ \times \frac{-20j}{20j + 400 - 20j}$$

$$V = 20\angle 0^\circ \times \frac{-20j}{400} = 20\angle 0^\circ \times \left(-\frac{j}{20}\right) = -j1\,\text{V}$$

Now, the current I flowing through the $1\,\Omega$ resistor is calculated using Ohm's law:

$$I = \frac{V}{R} = \frac{-j1\,\text{V}}{1\,\Omega} = -j1\,\text{A}$$

$$\boxed{(a) - j1A}$$

Q5.10.2 MCQ 2010 1M Ans: D ● ○ ○

Each of the statements can be expplained as follows:

Option A : The bandwidth (BW) of a parallel RLC circuit is defined by the range of frequencies over which the power delivered to the circuit is at least half of its maximum value. The formula is:

$$\frac{1}{RC}$$

Since R is in the denominator, as the resistance R increases, the bandwidth BW must decrease. This statement is **correct**.

Option B : Looking at the formula BW $= \frac{1}{RC}$, we can see that the inductance (L) does not appear in the expression for bandwidth. This statement is **correct**.

Option C : At resonance, the inductive reactance $(X_L = \omega L)$ and capacitive reactance $(X_C = \frac{1}{\omega C})$ are equal in magnitude but opposite in phase. The total admittance is given by $Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$. At resonance, the imaginary part becomes zero, leaving only $Y = \frac{1}{R}$. Since $Z = \frac{1}{Y} = R$, the impedance is purely resistive (a real number). This statement is **correct**.

Option D : As established in the previous case, at resonance, the total admittance Y reaches its minimum value because the reactive currents cancel out, leaving only the resistive current. Since impedance Z is the inverse of admittance $(Z = 1/Y)$, when admittance is at its minimum, impedance must be at its maximum.This statement is **incorrect**.

$$\boxed{(d)\textit{At resonance, the magnitude of input impedance attains its minimum value}}$$

Q5.09.1 MCQ 2009 2M Ans: B ●●○

The reactive power consumed by the load can be found using the following methods.

Method 1

The current flowing through the network is $I = \frac{20}{1+2j+7+4j} = 1.6 - j1.2A = 2\angle -36.87°$ Reactive power consumed is given by $Q = |I|^2 X$ For the load impedance, reactance $X = 4$

$$\therefore Q = 2^2 \times 4 = 16VAR$$

Method 2

The current flowing through the network is $I = \frac{20}{1+2j+7+4j} = 1.6 - j1.2A = 2\angle -36.87°$ The voltage across the load (by voltage division) is given by

$$V_L = 20 \times \frac{7 + j4}{1 + j2 + 7 + j4} = 16 - j2V = 16.12\angle -7.12°V$$

Power consumed by the load $S_L = V_L I^*$

$$\therefore S_L = 16.12\angle -7.12° \times 2\angle 36.87° = 28 + j16VA$$

Hence $Q = 16VAR$

$$\boxed{(b)16VAR}$$

Q5.07.1 MCQ 2007 2M Ans: D ●○○

The voltage V_{AB} is essentially the voltage across the current source. It can be computed by calculating the equivalent impedance in series with the current source (or across the terminals of the current source), and multiplying that with the current value, as the entire voltage drops across that impedance. The equivalent impedance as seen from the terminals of the current source is :

$$Z_{eq} = (5 - j3)||(5 + j3) = 3.4\Omega$$

$$\therefore V_{AB} = 5\angle 30° \times 3.4 = 17\angle 30°V$$

Key Points

- We can find out the voltage V_{AB} using nodal analysis at node A, however it is not recommended as the approach using equivalent impedance is significantly shorter and simpler.

$$\boxed{(d)17\angle 30°}$$

Q5.07.2 MCQ 2007 2M Ans: D ●○○

We know that for a series RLC circuit the bandwidth is given by $B = \frac{R}{L}$ For circuit 1 :

$$B_1 = \frac{R}{L_1}$$

For circuit 2 :

$$B_2 = \frac{R}{L_2} = \frac{R}{L_1/4} = 4B_1$$

$\therefore \frac{B_1}{B_2} = \frac{1}{4}$

$$\boxed{(d)\frac{1}{4}}$$

Q5.07.3 MCQ 2007 1M Ans: C ● ○ ○

To determine the type of filter the circuit behaves as, we investigate its response at $\omega = 0$ and at $\omega = \infty$. **Case 1 : at** $\omega = 0$ We know that capacitive impedance is inversely proportional to ω. Therefore, if $\omega = 0$, then capacitive impedance is infinity (or very large), that is, we can treat the capacitor as open. Hence the network looks like this :

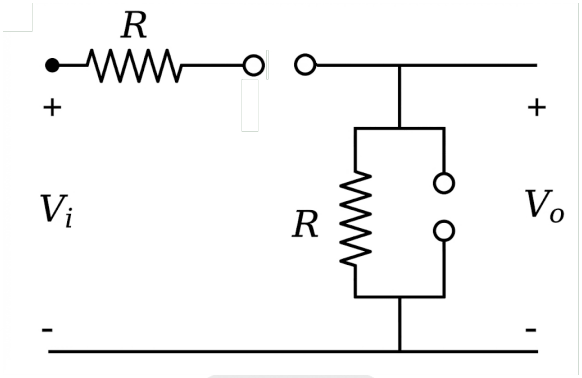


Fig. Solution Q5.07.3

Here $V_o = 0V$ **Case 2 : at** $\omega = \infty$

If $\omega = \infty$, then capacitive impedance is 0 (or very small), that is, we can treat the capacitor as shorted. Hence the network looks like this :

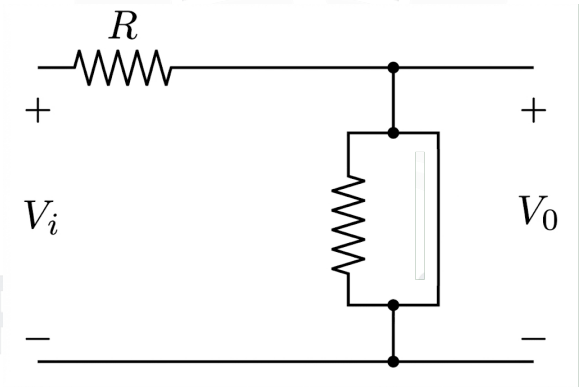


Fig. Solution Q5.07.3

Here $V_o = 0V$

$$\boxed{(c) A \text{ band-pass filter}}$$

Q5.07.4 MCQ 2007 1M Ans: D ● ○ ○

According to maximum power transfer theorem, for maximum power to be transferred to the load we need to have $Z_L = Z_S^*$

$$\therefore Z_L = Z_L = R_S - jX_S$$

$$\boxed{(d) Z_L = R_S - jX_S}$$

Q5.06.1 MCQ 2006 2M Ans: A ● ● ○

Given the driving-point impedance relationship:

$$Z_2(s) = R_{neg} + Z_1(s)$$

Expressing $Z_1(s)$ in terms of its real and imaginary parts:

$$Z_2(s) = R_{neg} + \text{Re}\, Z_1(s) + j\text{Im}\, Z_1(s)$$

For the network function $Z_2(s)$ to be a Positive Real Function, its real part must be non-negative in the right-half of the s-plane ($\text{Re}\, s \geq 0$). Specifically, on the imaginary axis ($s = j\omega$):

$$\text{Re}\, Z_2(s) \geq 0$$

Substituting the expression for the real part:

$$R_{neg} + \text{Re}\, Z_1(s) \geq 0$$

$$\text{Re}\, Z_1(s) \geq -R_{neg}$$

Since R_{neg} represents a negative resistance value ($R_{neg} < 0$), the quantity $-R_{neg}$ is a positive value. Thus, we can write:

$$\text{Re}\, Z_1(s) \geq |R_{neg}|$$

Evaluating this condition on the real frequency axis ($s = j\omega$) for all frequencies:

$$|R_{neg}| \leq \text{Re}\, Z_1(j\omega) \quad \text{for all } \omega$$

$$\boxed{(a)|R_{neg}| \leq ReZ_1(j\omega), \forall\omega}$$

Q5.05.1 MCQ 2005 2M Ans: A ● ○ ○

We can perform a nodal analysis at the node at which current i_1 enters. Writing KCL at that node (i_1 and the 5 A source current enter it, the $10\angle 60^\circ$ A source current leaves it),

$$i_1 + 5 - 10\angle 60^\circ = 0 \implies i_1 = 10\angle 60^\circ - 5$$

$$i_1 = 5 + j\frac{10\sqrt{3}}{2} - 5 = j\frac{10\sqrt{3}}{2}A$$

This gives the instantaneous value of the current.

Key Points

- For this problem, we can use other methods like superposition or finding the Thevenin's or Norton's equivalent. However, since current in two branches are known, to find the current in the third branch, writing KCL is the most effective solution.

$$\boxed{(a)\frac{10\sqrt{3}}{2}\angle 90^\circ Amps}$$

Q5.05.2 MCQ 2005 1M Ans: B ● ○ ○

The resonant frequency of an RLC circuit is given by :

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

Substituting,

$$f_c = \frac{1}{2\pi\sqrt{1/400 \times 10^{-6}}} = \frac{1}{\pi} \times 10^4 Hz$$

$$\boxed{(b)\ \frac{1}{\pi} \times 10^4 Hz}$$

Q5.04.1 MCQ 2004 2M Ans: D ● ○ ○

Let us analyze the statements given in the question regarding resonance conditions.

Analysis of Series $R-L-C$ Circuit: At the resonant frequency (ω_o), the inductive reactance (X_l) and capacitive reactance (X_c) cancel each other out:

$$X_l = X_c \implies \omega_o L = \frac{1}{\omega_o C}$$

Consequently, the total impedance (Z_{series}) of a series $R-L-C$ circuit reaches its minimum value and is purely resistive:

$$Z_{series} = R + j\left(\omega_o L - \frac{1}{\omega_o C}\right) = R$$

Analysis of Parallel $R-L-C$ Circuit: For a parallel $R-L-C$ circuit, the quality factor (Q) is defined as the ratio of the sheet resistance to the inductive reactance at resonance:

$$Q = \frac{R}{X_l} = \frac{R}{\omega_o L}$$

We know that conductance (G) is the reciprocal of resistance:

$$G = \frac{1}{R}$$

Substituting $R = \frac{1}{G}$ into the quality factor equation:

$$Q = \frac{1}{G\omega_o L}$$

From this expression, it is evident that the quality factor is inversely proportional to the conductance:

$$Q \propto \frac{1}{G}$$

Therefore, increasing the conductance G will result in a decrease in the quality factor Q.

Based on the evaluation of the network behaviors at resonance, both statements s_1 and s_2 are false.

Key Points

- In a series resonant circuit, impedance is minimum ($Z = R$), and current is maximum.
- In a parallel resonant circuit, admittance is minimum ($Y = G$), and impedance is maximum ($Z = R$).

- The quality factor formulas are opposite for series and parallel topologies:

$$Q_{series} = \frac{\omega_o L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$Q_{parallel} = \frac{R}{\omega_o L} = R\sqrt{\frac{C}{L}} = \frac{1}{G}\sqrt{\frac{C}{L}}$$

Mistakes to be avoided

- Do not apply the series quality factor formula ($Q = \frac{\omega L}{R}$) to a parallel $R - L - C$ network configuration.

$$\boxed{(d) \text{both } S_1 \text{ and } S_2 \text{ are FALSE.}}$$

Q5.04.2 MCQ 2004 2M Ans: B ● ○ ○

For the given transfer function, the characteristic equation is obtained from the denominator polynomial:

$$s^2 + 20s + 10^6 = 0$$

Comparing this with the standard second-order characteristic equation of a series resonant circuit:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

By comparing the coefficients of matching powers of s, we get:

$$\frac{R}{L} = 20 = Bandwidth$$

$$\omega_o^2 = \frac{1}{LC} = 10^6 \implies \omega_o = 10^3 \text{ rad/s} = Centre\ Frequency$$

The quality factor Q for a series second-order network is defined as:

$$Q = \frac{Centre\ Frequency}{Bandwidth}$$

Substituting the derived values into the quality factor expression:

$$Q = \frac{10^3}{20} = 50$$

$$\boxed{(b) 50}$$

Q5.04.3 MCQ 2004 1M Ans: A ● ● ○

The transfer function H(s) of the system is given by :

$$H(s) = \frac{1}{sRC + 1}$$

The frequency response would be :

$$H(\omega) = \frac{1}{j\omega RC + 1} = \frac{1}{\sqrt{(\omega RC)^2 + 1}} \angle - tan^{-1}(\omega RC)$$

The output of an LTI system with sinusoidal input is given by $V_o(s) = |V_i(\omega)||H(\omega)|\angle(\angle V_i(\omega) + \angle H(\omega))$ Here it is given that RC = 1 ms, $\omega = 10^3 rad/s$ and $|V_i| = \sqrt{2}$ Therefore,

$$H(\omega) = \frac{1}{j10^3(10^{-3}) + 1} = \frac{1}{1+j} = \frac{1}{\sqrt{2}}\angle -45^\circ$$

$$\therefore V_o(s) = |V_i(\omega)|\frac{1}{\sqrt{2}}\angle(\angle V_i(\omega) - 45^\circ)$$

In time domain we can write,

$$V_o(t) = sin(10^3 t - 45^\circ)$$

$$\boxed{(a) sin(10^3 t - 45^\circ)}$$

Q5.04.4 MCQ 2004 1M Ans: A ●●○

Given the input frequency $\omega = 2$ rad/s and a parallel AC circuit consisting of a resistor, an inductor, and a capacitor, we need to find the current i(t).

Let the input voltage be $v(t) = \sin(2t)$. The total input current i(t) entering the parallel network is the sum of the currents passing through each branch:

$$i(t) = \frac{v(t)}{R} + \frac{v(t)}{j\omega L} + v(t) \times j\omega C$$

From the circuit parameters provided at $\omega = 2$ rad/s:

- Resistance: $R = \frac{1}{3}\,\Omega$
- Inductive Reactance: $X_L = j\omega L = j2\omega = \frac{j}{2}\,\Omega$
- Capacitive Reactance: $X_C = \frac{1}{j\omega C} = -\frac{j}{6}\,\Omega$

Substituting the individual sinusoidal branch currents by converting phasor admittances back to time-domain expressions (where a $-j$ phase shifts the sine by -90° and $+j$ shifts it by $+90^\circ$):

$$i(t) = 3\sin(2t) + 2\sin(2t - 90^\circ) + 6\sin(2t + 90^\circ)$$

Using standard trigonometric identities ($\sin(2t - 90^\circ) = -\cos(2t)$ and $\sin(2t + 90^\circ) = \cos(2t)$):

$$i(t) = 3\sin(2t) - 2\cos(2t) + 6\cos(2t)$$

$$i(t) = 3\sin(2t) + 4\cos(2t)$$

To combine these terms into a single sinusoidal expression of the form $A\sin(2t + \theta)$:

$$A = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) \approx 53.13^\circ$$

$$\boxed{(a) 5sin(2t + 53.1^\circ)}$$

Q5.03.1 MCQ 2003 1M Ans: C ● ○ ○

The source impedance is given by

$$Z_S = R_S + j\omega L_S = 1 + j1\Omega$$

According to Maximum power transfer theorem, the maximum power is transferred to the load if the load impedance is the complex conjugate of the Thevenin impedance looking into the network from the load terminals. In this case, the load impedance should be the complex conjugate of the source impedance that is

$$Z_L = 1 - j1\Omega$$

Thus it includes a resistance of 1Ω and a capacitive impedance of 1Ω. $\frac{-j}{\omega C} = -j1 \implies C = 1F$ (given$\omega = 1rad/sec$)

$$\boxed{(c)\,1\Omega \textit{ resistance in series with 1 F capacitor}}$$

Q5.03.2 MCQ 2003 1M Ans: B ● ○ ○

Given initial parameters for the series $R - L - C$ circuit:

$$\omega_{n1} = 1\,\text{kHz}$$

$$Q_1 = 100$$

The resonant frequency (ω_n) and quality factor (Q) of a series $R - L - C$ circuit are given by the standard relations:

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$Q = \frac{\omega_n L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Let the new resistance, inductance, and capacitance parameters be doubled such that $R_2 = 2R_1$, $L_2 = 2L_1$, and $C_2 = 2C_1$. Calculating the new resonant frequency ω_{n2}:

$$\omega_{n2} = \frac{1}{\sqrt{L_2C_2}} = \frac{1}{\sqrt{(2L_1)(2C_1)}} = \frac{1}{2\sqrt{L_1C_1}} = \frac{\omega_{n1}}{2}$$

Thus, the resonant frequency is halved:

$$\omega_{n2} = \frac{1\,\text{kHz}}{2} = 0.5\,\text{kHz}$$

Calculating the new quality factor Q_2:

$$Q_2 = \frac{1}{R_2}\sqrt{\frac{L_2}{C_2}} = \frac{1}{2R_1}\sqrt{\frac{2L_1}{2C_1}} = \frac{1}{2}\left(\frac{1}{R_1}\sqrt{\frac{L_1}{C_1}}\right) = \frac{Q_1}{2}$$

Thus, the quality factor is also halved:

$$Q_2 = \frac{100}{2} = 50$$

The final value of the new quality factor is:

$$\boxed{(b)50}$$

Q5.01.1 MCQ 2001 2M Ans: A ●○○

Given that $\omega = 10 rad/sec$ Therefore the Thevenin impedance looking into the network from the load terminals is given by :

$$Z_{TH} = 10 + j(10)(1) = 10 + j10\Omega$$

According to Maximum power transfer theorem, the maximum power is transferred to resistive load if it is equal to the magnitude of this calculated impedance.

$$\therefore R = |Z_{TH}| = \sqrt{10^2 + 10^2} = 14.14\Omega$$

$$\boxed{(a) 14.14\Omega}$$

Q5.01.2 MCQ 2001 2M Ans: A ●●○

To find the time-domain output load current response, we first establish the total equivalent input impedance $Z_{in}(s)$ seen by the source $e(t) = E_m \cos(\omega t)$:

$$Z_{in}(s) = R_1 \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{R_1\left(R_2 + \frac{1}{sC_2}\right)}{R_1 + R_2 + \frac{1}{sC_2}}$$

Substituting $s = j\omega$ into the total input current expression $i(t) = \frac{e(t)}{Z_{in}(j\omega)}$:

$$i(t) = \frac{E_m \cos(\omega t)}{R_1 \parallel \left(R_2 + \frac{1}{j\omega C_2}\right)}$$

$$i(t) = \frac{E_m \cos(\omega t)}{\left(\frac{R_1 R_2 + \frac{R_1}{j\omega C_2}}{R_1 + R_2 + \frac{1}{j\omega C_2}}\right)} = \frac{E_m \cos(\omega t)}{\left(R_1 R_2 + \frac{R_1}{j\omega C_2}\right)} \times \left(R_1 + R_2 + \frac{1}{j\omega C_2}\right)$$

Using the current division rule, the fraction of the total current flowing through the load branch containing R_2 and C_2 is denoted as $i_L(t)$:

$$i_L(t) = i(t) \times \frac{R_1}{R_1 + R_2 + \frac{1}{j\omega C_2}}$$

Substituting the expression of $i(t)$ into the current division equation yields:

$$i_L(t) = E_m \cos(\omega t)\left[\frac{R_1}{R_1 R_2 + \frac{R_1}{j\omega C_2}}\right]$$

We can now analyze the limiting cases using a boundary-frequency analysis approach:

Case 1: At low-frequency limit ($\omega = 0$)** As $\omega \to 0$, the reactance of the capacitor approaches infinity ($X_{c2} \to \infty$), acting as an open circuit:

$$i_2(t) = E_m \times 0 = 0$$

Case 2: At high-frequency limit ($\omega = \infty$)** As $\omega \to \infty$, the reactance of the capacitor approaches zero ($X_{c2} \to 0$), acting as a perfect short circuit:

$$i_2(t) = E_m \cos(\omega t)\left[\frac{R_1}{R_1 R_2 + 0}\right] = \frac{E_m}{R_2}\cos(\omega t)$$

Comparing these boundary responses with the options shows that only option A satisfies these exact conditions.

$$\boxed{(a)}$$

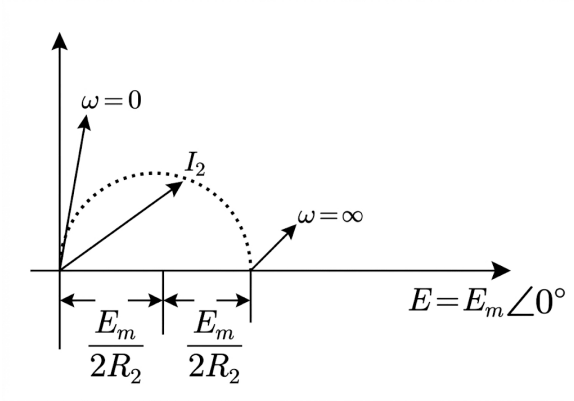


Fig. Solution Q5.01.2

Q5.01.3 NAT 2001 2M Ans: *

Applying KCL at node where E_2 is measured:

$$\frac{E_2 - 10\angle 20°}{j} + \frac{E_2 - 0}{1 - j} + \frac{E_2 - 0}{1 + 2j} = 0$$

$$E_2 \left[-j + \frac{1 + j}{2} + \frac{1 - 2j}{5} \right] = \frac{10\angle 20°}{j}$$

$$E_2 \left[-10j + 5 + 5j + 2 - 4j \right] = \frac{100\angle 20°}{j}$$

$$E_2 (7 - 9j) = \frac{100\angle 20°}{j}$$

Solving for the node phasor voltage E_2:

$$E_2 = \frac{100\angle 20°}{(7 - 9j)j} = 8.77\angle -17.87° \text{ V}$$

Next, evaluating the individual branch current expressions using the calculated node voltage value:

$$I = \frac{10\angle 20° - E_2}{j} = \frac{10\angle 20° - 8.77\angle -17.87°}{j}$$

$$I = 6.2\angle -9.75° \text{ A}$$

For the branch current I_1:

$$I_1 = \frac{E_2}{1 - j} = \frac{8.77\angle -17.87°}{1 - j} = 6.201\angle 27.13° \text{ A}$$

Assuming a voltage divider output configuration or a dependent relation scaled across the branch elements where E_0 is evaluated from the node voltage E_2:

$$E_0 = 0.5 \times \frac{E_2}{1 + 2j} = 0.5 \times \frac{8.77\angle -17.87°}{1 + 2j}$$

$$\boxed{E_2 = 8.77\angle -17.87° \text{ V}, E_0 = 1.96\angle -81.30° \text{ V}, I = 6.2\angle -9.75° \text{ A}, I_1 = 6.201\angle 27.13° \text{ A}}$$

Q5.00.1 MCQ | 2000 | 1M | Ans: D

To check the type of filter the given circuit behaves as, we need to check its response at $\omega = 0 rad/sec$ and $\omega = \infty rad/sec$.
Case 1 : At $\omega = 0 rad/sec$ We know that inductive impedance is proportional to frequency and capacitive impedance is inversely proportional to it. $\therefore$, at $\omega = 0 rad/sec$, inductive impedance is 0 (L appears shorted) and capacitive impedance is ∞ (C appears open).

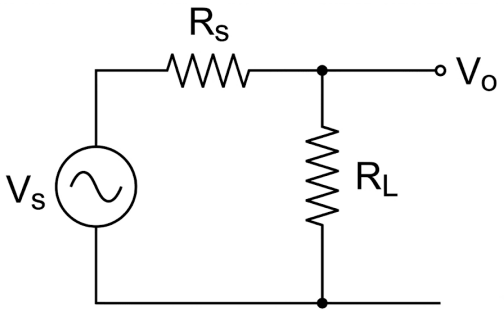


Fig. Solution Q5.00.1

Hence $V_o = \frac{R_L}{R_S+R_L} \times V_s$

Case 2 : At $\omega = \infty rad/sec$

$\therefore$, at $\omega = \infty rad/sec$, inductive impedance is ∞ (L appears open) and capacitive impedance is 0 (C appears shorted).

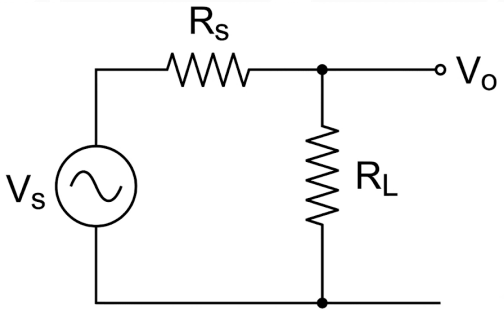


Fig. Solution Q5.00.1

Hence $V_o = \frac{R_L}{R_S+R_L} \times V_s$

From the above analysis we can conclude that the output remains high at both $\omega = 0 rad/sec$ and $\omega = \infty rad/sec$. Analysing the given options, this behaviour is possible only for a band reject filter.

$$\boxed{(d) Band\ reject\ filter}$$

Key Points

- High output at both $\omega = 0 rad/sec$ and $\omega = \infty rad/sec$ can signify either band reject behaviour or all pass behaviour. To differentiate between both we need to check for the placement of poles and zeroes, which would be symmetric in case of an all pass filter.
- Since the options given here do not include all pass filter, we do not explore that area. However, mere observation tells us that this circuit would not have symmetric poles and zeroes.

Q5.00.2 MCQ | 2000 | 2M | Ans: A | ● ● ○

Let us find the transfer function of this network. The network is a simple first order low pass filter with transfer function $H(s) = \frac{1}{sRC+1}$, where $R = 1\,k\Omega$ and $C = 10\mu F$.

$$\therefore, H(s) = \frac{1}{0.01s + 1} = \frac{100}{s + 100}$$

By principle of superposition, we need to investigate the output at two frequencies as per given input, i.e. at dc and at 100 rad/sec. For a dc input of magnitude 3V, the capacitor would eventually charge upto 3V. Hence the output for the 3V input would be 3V itself. Next to investigate for the input 4 sin 100t V, we need to check the frequency reponse for 100 rad/sec frequency.

$$|H(\omega)| = \frac{100}{j100 + 100} = \frac{1}{\sqrt{2}}\angle -45^\circ$$

We know,

$$|V_o(j\omega)| = |H(\omega)||V_i(\omega)| \; and \; \angle V_o(j\omega) = \angle H(\omega) + \angle V_i(\omega)$$

$$\therefore |V_o(j\omega)| = 4 \times \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} \; and \angle V_o(j\omega) = 0^\circ - 45^\circ = -45^\circ$$

Thus in time domain,

$$V_o(t) = \frac{4}{\sqrt{2}} \sin\left(100t - \frac{\pi}{4}\right)$$

Combining both outputs, $V_o(t) = 3 + \frac{4}{\sqrt{2}} \sin\left(100t - \frac{\pi}{4}\right)$ V

$$\boxed{(a) 3 + \frac{4}{\sqrt{2}} \sin\left(100t - \frac{\pi}{4}\right) V}$$

Q5.00.3 NAT | 2000 | 5M | Ans: * | ● ● ○

Impedance across terminal a & b will be maximum when the parallel RLC circuit is in resonance because in a parallel RLC circuit, impedance is maximum at resonance frequency.

The admittance Y_1 of the parallel RLC combination is given by:

$$Y_1 = \frac{1}{j\omega_0 L} + \frac{1}{\frac{1}{j\omega_0 C}} + \frac{1}{R}$$

Substituting the given values from the circuit parameters ($L = 4\,\text{H}$, $C = 4\,\text{F}$):

$$Y_1 = \frac{1}{j\omega_0 \cdot 4} + j\omega_0 \cdot 4 + \frac{1}{R} = \frac{-j}{4\omega_0} + j\omega_0 \cdot 4 + \frac{1}{R}$$

$$Y_1 = \frac{1}{R} + j\left[4\omega_0 - \frac{1}{4\omega_0}\right]$$

At resonance, the imaginary part of admittance must be zero ($\text{Im}(Y_1) = 0$):

$$4\omega_0 = \frac{1}{4\omega_0}$$

$$\omega_0^2 = \frac{1}{16} \implies \omega_0 = \frac{1}{4}\,\text{rad/sec} = 0.25\,\text{rad/sec}$$

(b) Finding Total Impedance At resonance frequency $\omega_0 = \frac{1}{4}$ rad/sec:

$$Y_1 = \frac{1}{R} \implies Z_1 = R = 2\,\Omega$$

The total impedance Z_{th} across the terminals is the sum of the series resistor and the resonant tank circuit impedance:

$$Z_{th} = 2 + 2 = 4\,\Omega$$

Hence, the value is 4. (c) Finding Time-Domain Currents Given input voltage:

$$V_i(t) = V\sin(\omega_0 t) = V\sin(0.25t)\text{ V}$$

At resonance $\omega_0 = \frac{1}{4}$ rad/sec, by voltage division, the voltage across the parallel combination $V_c(t)$ is:

$$V_c(t) = V_i(t) \times \frac{2}{2+2} = \frac{V}{2}\sin(0.25t)\text{ V}$$

Now, calculating individual branch currents in time-domain:

1. Capacitor Current $i_C(t)$:

$$i_C(t) = V_c(t) \times 4j\omega_0 = \frac{V}{2}\sin(0.25t) \times 4 \times \frac{1}{4} \times j$$

$$i_C(t) = \frac{V}{2}\sin(0.25t + 90^\circ) = \frac{V}{2}\cos(0.25t)\text{ A}$$

2. Resistor Current $i_R(t)$:

$$i_R(t) = \frac{V_c(t)}{2} = \frac{V}{4}\sin(0.25t)\text{ A}$$

3. Inductor Current $i_L(t)$:

$$i_L(t) = \frac{V_c(t)}{4j\omega_0} = \frac{-jV\sin(0.25t)}{2 \times 4 \times \frac{1}{4}}$$

$$i_L(t) = -\frac{V}{2}\cos(0.25t)\text{ A}$$

$$\boxed{(a)\omega_0 = 0.25\text{ rad/sec}, (b)Z_{th} = 4\,\Omega, (c)i_L(t) = -\frac{V}{2}\cos(0.25t)\text{ A}}$$

Q5.99.1 MCQ 1999 2M Ans: A ● ○ ○

The Thevenin's equivalent voltage is calculated by voltage division between the 3Ω and j4 Ω elements. This is because the voltage source is in parallel with the branch containing (3+j4) Ω, j2 Ω and -j6 Ω. Thus, the other 2 elements don't play a role in calculation of the Thevenin's voltage.

$$V_{TH} = \frac{j4}{3+j4} \times 100V = 64 + j48V = 16(4 + j3)V = j16(3 - j4)V$$

$$\boxed{(a)j16(3 - j4)}$$

Q5.99.2 NAT 1999 5M Ans: * ● ● ○

Since we have to specifically use Superposition method, we can divide the analysis into 2 cases. **Case 1 :** Here, we turn off the $10\angle 20^\circ A$ current source and observe the effect of the 10 A current source.

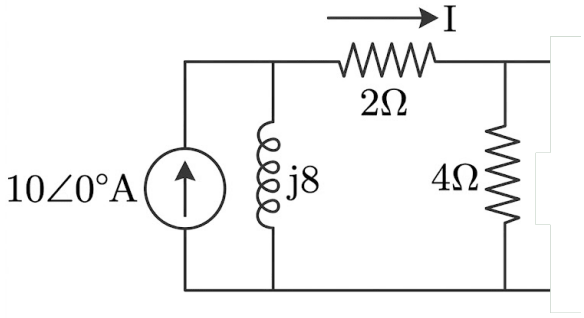


Fig. Solution Q5.99.2

By rule of current division :

$$I_1 = \frac{j8}{j8+2+4} \times 10A = 6.4 + j4.8A$$

(I_1 is the component of I due to 10A source). **Case 2 :** Here, we turn off the 10A current source and observe the effect of the $10\angle 20°A$ current source.

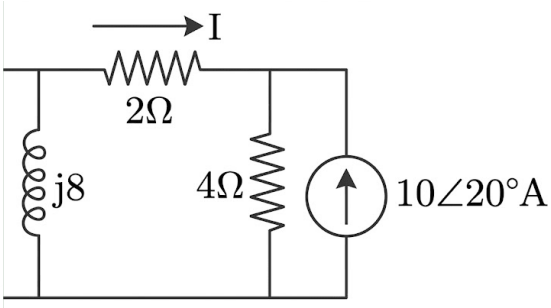


Fig. Solution Q5.99.2

By rule of current division :

$$I_2 = -\frac{4}{j8+2+4} \times 10\angle 20°A = -3.35 + j2.19A$$

(I_2 is the component of I due to $10\angle 20°A$ source).

$$I + I_1 + I_2 = 3.05 + j6.99A = 7.6\angle 66°A$$

$$\boxed{7.6\angle 66°A}$$

Q5.99.3 NAT 1999 5M Ans: * ●●○

Given a coil in series with a capacitance,the resonant frequency is given as $f_0 = 50\,\text{Hz}$ with a capacitor $C_1 = 10\,\mu\text{F}$. The frequency is said to be the resonant frequency as it is at this frequency when the current is maximum.

$$f_0 = \frac{1}{2\pi\sqrt{LC_1}} \implies 50 = \frac{1}{2\pi\sqrt{L \times 10 \times 10^{-6}}}$$

$$\sqrt{L \times 10^{-5}} = \frac{1}{100\pi}$$

$$L \times 10^{-5} = \frac{1}{10000\pi^2} \implies L = \frac{10}{10\pi^2} = \frac{1}{\pi^2} \approx 1.0132\,\text{H}$$

Using the quality factor $Q = 10$:

$$Q = \frac{\omega_0 L}{R} \implies R = \frac{2\pi f_0 L}{Q}$$

$$R = \frac{2\pi \times 50 \times 1.0132}{10} = 10\pi \times 1.0132 \approx 31.83\,\Omega$$

Next, we add a capacitance C_2 in parallel with the RLC circuit in order to have the maximum current at 100Hz (resonant frequency). For the the combined input admittance $Y(j\omega)$ is expressed as:

$$Y(j\omega) = \frac{1}{R + j\omega L + \frac{1}{j\omega C_1}} + j\omega C_2$$

$$Y(j\omega) = \frac{j\omega C_1}{1 - \omega^2 LC_1 + j\omega C_1 R} + j\omega C_2$$

Multiplying the numerator and denominator of the first term by its complex conjugate to segregate the real and imaginary components:

$$Y(j\omega) = \frac{j\omega C_1\left[(1-\omega^2 LC_1) - j\omega C_1 R\right]}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2} + j\omega C_2$$

$$Y(j\omega) = \frac{\omega^2 C_1^2 R}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2} + j\left[\frac{\omega C_1(1-\omega^2 LC_1)}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2} + \omega C_2\right]$$

For the composite circuit to operate at parallel resonance, the net input admittance must be purely conductive. This requires the total imaginary component (susceptance) to vanish:

$$\text{Im}\left[Y(j\omega)\right] = 0$$

$$\frac{\omega C_1(1-\omega^2 LC_1)}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2} + \omega C_2 = 0$$

Factoring out the non-zero operating frequency $\omega \neq 0$:

$$\frac{C_1(1-\omega^2 LC_1)}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2} + C_2 = 0$$

$$C_2 = \frac{C_1(\omega^2 LC_1 - 1)}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2}$$

The resonant frequency is $f = 100\,\text{Hz}$. Thus, the angular frequency ω is:

$$\omega = 2\pi f = 2\pi \times 100 = 200\pi \approx 628.32\,\text{rad/s}$$

The values of L,R, and C_1 are also known. Substituting these values back into the expression for C_2:

$$C_2 = \frac{C_1(\omega^2 LC_1 - 1)}{(1-\omega^2 LC_1)^2 + (\omega C_1 R)^2}$$

$$\therefore\ C_2 = 3.32\,\mu\text{F}$$

At this frequency, the imaginary part of impedance is 0. Thus $Y(j\omega) = Re[Y(j\omega)] = \frac{\omega^2 C_1^2 R}{(1-\omega^2 LC_1)^2+(\omega C_1 R)^2}$ Substituting values of ω, R, L and C_1, $Y(j\omega) = 1.39 \times 10^{-4} S$. The equivalent impedance is the inverse of this value.

$$Z(j\omega) = \frac{1}{Y(j\omega)} = 7193.8\Omega$$

The voltage applied is 100V RMS. Thus the current in the circuit is :

$$I_{RMS} = \frac{V_{RMS}}{Z(j\omega)} = \frac{100}{7193.8} = 13.9mA$$

$$\boxed{C_2 = 3.32\,\mu\text{F}, I_{RMS} = 13.9mA}$$

Q5.98.1 MCQ | 1998 | 1M | Ans: B | ● ○ ○

At ideal parallel resonance, the reactive branch currents cancel:

$$I_L = -I_C, \qquad I_R = I_s = 1\,\text{mA}.$$

The resistor current is in phase with the voltage, while each reactive current is in quadrature with it. Hence

$$|I_R + I_L| = \sqrt{(1\,\text{mA})^2 + |I_L|^2} > 1\,\text{mA},$$

and, similarly,

$$|I_R + I_C| = \sqrt{(1\,\text{mA})^2 + |I_C|^2} > 1\,\text{mA}.$$

Thus (A) and (C) are false, while both (B) and (D) are true for the options as printed. The printed question does not specify a unique correct choice.

Q5.98.2 NAT | 1998 | 3M | Ans: * | ● ● ○

The total admittance Y of the parallel combination of elements is given by:

$$Y = \frac{1}{Z} = \frac{1}{R + j\omega L} + \frac{1}{R + \frac{1}{j\omega C}}$$

$$Y = \frac{R - j\omega L}{R^2 + (\omega L)^2} + \frac{R + \frac{j}{\omega C}}{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

Separating the real (conductance) and imaginary (susceptance) parts:

$$Y = \left[\frac{R}{R^2 + (\omega L)^2} + \frac{R}{R^2 + \left(\frac{1}{\omega C}\right)^2}\right] + j\left[\frac{\frac{1}{\omega C}}{R^2 + \left(\frac{1}{\omega C}\right)^2} - \frac{\omega L}{R^2 + (\omega L)^2}\right]$$

At the resonance frequency, the imaginary part of the admittance must be zero, i.e., $\text{Im}(Y) = 0$:

$$\frac{\omega L}{R^2 + (\omega L)^2} = \frac{\frac{1}{\omega C}}{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

Simplifying the right-hand side denominator expression:

$$\frac{\omega L}{R^2 + (\omega L)^2} = \frac{1}{\omega C\left(R^2 + \left(\frac{1}{\omega C}\right)^2\right)}$$

$$\omega^2 LC \times R^2 + \omega^2 LC \times \frac{1}{\omega^2 C^2} = R^2 + (\omega L)^2$$

$$\omega^2 LCR^2 + \frac{L}{C} = R^2 + \omega^2 L^2$$

$$\omega^2(LCR^2 - L^2) = R^2 - \frac{L}{C}$$

$$\omega^2 LC\left(R^2 - \frac{L}{C}\right) = R^2 - \frac{L}{C}$$

If $R^2 \neq \frac{L}{C}$, the term $\left(R^2 - \frac{L}{C}\right)$ cancels out from both sides, yielding the standard resonance frequency:

$$\omega = \frac{1}{\sqrt{LC}}$$

The input impedance Z_{in} at resonance under this specific condition becomes:

$$Z_{in} = \frac{R^2 + \frac{L}{C}}{2R}$$

However, if we choose the condition where $L = CR^2$ (or $R^2 = \frac{L}{C}$), then the resonance equation $\omega^2 \times 0 = 0$ is satisfied for all values of ω. Substituting $L = CR^2$ into the impedance expression:

$$Z_{in} = \frac{R^2 + R^2}{2R} = R$$

Under this condition, the circuit impedance becomes entirely resistive and independent of frequency. Thus, the circuit resonates at an infinite number of frequencies.

Key Points

- The same problem can be solved by finding the equivalent impedance of the circuit and setting the imaginary part to zero, however, since this involves a parallel combination of network elements calculating the admittance is easier.
- When the condition $R = \sqrt{\frac{L}{C}}$ is met for both branches, the circuit enters a condition of universal resonance, meaning it behaves purely resistively at all operating frequencies.

$Resonant\ frequency = \frac{1}{\sqrt{LC}}$,
$Impedance\ at\ resonance = \frac{R^2 + \frac{L}{C}}{2R}$,
and impedance becomes entirely resistive and independent of frequency, and resonates at an infinite number of frequencies when $L = CR^2$.

Q5.98.3 NAT 1998 3M Ans: 5 ●○○

According to maximum power transfer theorem, to transfer maximum power to a resistive load, the value of resistance must be equal to the magnitude of the Thevenin impedance looking into the circuit from the load terminals which in this case is the internal impedance of the source.

$$R_L = \sqrt{R_S^2 + X_S^2} = \sqrt{3^2 + 4^2} = 5\Omega$$

$\boxed{5}$

Q5.97.1 MCQ 1997 1M Ans: B ●●○

The given circuit is a balanced Wheatstone bridge. We can check the ratio of the two arm resistances : $2/2 = 4/4 = 1$. Hence no current flows through the j3||-j4 impedance,i.e., we can ignore it for the rest of the analysis. Thus the equivalent impedance, Z_{eq} is the parallel combination of (2+2) and (4+4) Ω.

$$Z_{eq} = 4||8 = \frac{8}{3}\Omega$$

$\boxed{(b)(\frac{8}{3})\Omega}$

Q5.97.2 NAT 1997 5M Ans: * ●●○

(a) Equivalent impedance Z_{ab} at limiting frequencies: We know that inductive reactance is directly proportional to and capacitive reactance is inversely proportional to frequency. Therefore, At $\omega = 0$ (DC condition): The inductive reactance is $X_l = j\omega L = 0 \Rightarrow$ Short circuit. The capacitive reactance is $X_c = \frac{1}{j\omega C} = \infty \Rightarrow$ Open circuit.

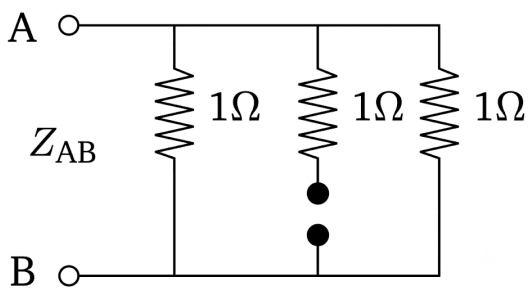


Fig. Solution Q5.97.2

The circuit simplifies to the parallel combination of two 1 Ω resistors :

$$Z_{AB} = 1 \parallel 1 = 0.5\,\Omega$$

At $\omega = \infty$ (High-frequency condition): The inductive reactance is $X_l = j\omega L = \infty \Rightarrow$ Open circuit. The capacitive reactance is $X_c = \frac{1}{j\omega C} = 0 \Rightarrow$ Short circuit.

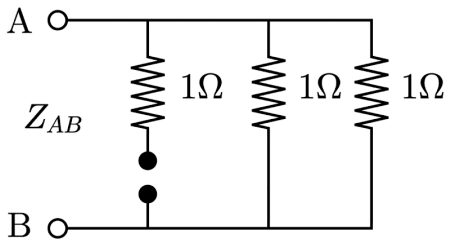


Fig. Solution Q5.97.2

The circuit simplifies to the parallel combination of two 1 Ω resistors :

$$Z_{AB} = 1 \parallel 1 = 0.5\,\Omega$$

(b) Equivalent Admittance across AB and Resonant Frequency:
The equivalent admittance Y_{ab} of the parallel network is:

$$Y_{AB} = \frac{1}{1} + \frac{1}{1 + \frac{4}{j\omega}} + \frac{1}{1 + 4j\omega}$$

$$Y_{AB} = 1 + \frac{1 + \frac{j4}{\omega}}{1 + \frac{16}{\omega^2}} + \frac{1 - 4j\omega}{1 + 16\omega^2}$$

Grouping the real and imaginary parts together:

$$Y_{AB} = \left[1 + \frac{1}{1 + \frac{16}{\omega^2}} + \frac{1}{1 + 16\omega^2}\right] + j\left[\frac{\frac{4}{\omega}}{1 + \frac{16}{\omega^2}} - \frac{4\omega}{1 + 16\omega^2}\right]$$

For the resonance frequency ω_0, the imaginary part of the admittance must be zero ($\mathrm{Im}(Y_{ab}) = 0$):

$$\frac{\frac{4}{\omega_0}}{1 + \frac{16}{\omega_0^2}} - \frac{4\omega_0}{1 + 16\omega_0^2} = 0$$

$$\frac{4}{\omega_0\left(1 + \frac{16}{\omega_0^2}\right)} = \frac{4\omega_0}{1 + 16\omega_0^2}$$

Simplifying the equation:

$$1 + 16\omega_0^2 = \omega_0^2\left(1 + \frac{16}{\omega_0^2}\right)$$

$$1 + 16\omega_0^2 = \omega_0^2 + 16$$

$$15\omega_0^2 = 15$$

$$\omega_0 = 1\ \text{rad/sec}$$

Given the input source current is:

$$i_s(t) = I\sin(\omega_0 t)$$

From the given network at the resonant frequency $\omega = \omega_0 = 1$ rad/sec, the total admittance value is:

$$Y_{AB} = 1 + \left[\frac{1}{1+\frac{16}{1}} + \frac{1}{1+16}\right] + j[0]$$

$$Y_{AB} = 1 + \frac{1}{17} + \frac{1}{17} = \frac{19}{17}\,\text{S}$$

The total current expression from the source is:

$$i_s(t) = 10\sin(t) \times \left[1 + \frac{1}{17} + \frac{1}{17}\right]$$

$$i_s(t) = 10\sin(t) \times \frac{19}{17} = \frac{190}{17}\sin(t)\,\text{A}$$

Comparing this with $i_s(t) = I\sin(\omega_0 t)$, the amplitude I is:

$$\boxed{I = \frac{190}{17}\,\text{A}}$$

The current in the capacitive branch $i_c(t)$ can be computed as:

$$i_2(t) = \frac{10\sin(\omega_0 t)}{1+\frac{4}{j\omega_0}} = \frac{10\sin(t)}{1-4j}$$

$$i_2(t) = \frac{10}{17}\sin(t) \times (1+4j)$$

$$i_2(t) = \frac{10}{\sqrt{17}}\sin(t+76^\circ)\,\text{A}$$

$$\boxed{(a) Z_{AB} = 0.5\Omega\, at\, \omega = 0\, and\, \infty rad/s,\ (b) I = \frac{190}{17}\sin(t)\,\text{A}, \omega_0 = 1\,\text{rad/sec}, i_2(t) = \frac{10}{\sqrt{17}}\sin(t+76^\circ)\,\text{A}}$$

Q5.97.3 NAT 1997 5M Ans: * ●●○

(a): State-Space Representation

Let the state variables be defined as the inductor currents and the capacitor voltage:

$$\frac{di_{l1}(t)}{dt} = \dot{X}_1, \quad \frac{dV_{c2}(t)}{dt} = \dot{X}_2, \quad \frac{di_{l3}(t)}{dt} = \dot{X}_3$$

Given circuit parameters: $R_1 = 2\,\Omega$, $R_2 = 4\,\Omega$, $R_3 = 4\,\Omega$, $L_1 = 1\,\text{H}$, $C_2 = 1\,\text{F}$, $L_3 = 1\,\text{H}$.
Applying Kirchhoff's Current Law (KCL) at Node A:

$$\frac{e(t) - L_1\frac{di_{l1}(t)}{dt}}{2} = i_{l1}(t) + \frac{L_1\frac{di_{l1}(t)}{dt} - V_{c2}(t)}{4}$$

Substitute the values $L_1 = 1\,\text{H}$ and clear the fractions by multiplying by 4:

$$2e(t) - 2\dot{X}_1 = 4X_1 + \dot{X}_1 - X_2$$

$$2e(t) = 4X_1 + 3\dot{X}_1 - X_2$$

Rearranging to solve for $\dot{X}_1$:

$$\dot{X}_1 = -\frac{4}{3}X_1 + \frac{1}{3}X_2 + \frac{2}{3}e(t) \quad \text{— (1)}$$

Applying KCL at Node B:

$$\frac{V_{c2}(t) - L_1\frac{di_{l1}(t)}{dt}}{4} + C_2\frac{dV_{c2}(t)}{dt} + i_{l3}(t) = 0$$

Substituting values $L_1 = 1\,\text{H}$ and $C_2 = 1\,\text{F}$, then multiplying by 4:

$$X_2 - \dot{X}_1 + 4\dot{X}_2 + 4X_3 = 0$$

Substitute equation (1) into this expression:

$$X_2 - \left(-\frac{4}{3}X_1 + \frac{1}{3}X_2 + \frac{2}{3}e(t)\right) + 4\dot{X}_2 + 4X_3 = 0$$

$$\frac{4}{3}X_1 + \frac{2}{3}X_2 - \frac{2}{3}e(t) + 4\dot{X}_2 + 4X_3 = 0$$

Solving for $\dot{X}_2$:

$$4\dot{X}_2 = -\frac{4}{3}X_1 - \frac{2}{3}X_2 - 4X_3 + \frac{2}{3}e(t)$$

$$\dot{X}_2 = -\frac{1}{3}X_1 - \frac{1}{6}X_2 - X_3 + \frac{1}{6}e(t) \quad \text{— (2)}$$

From the loop containing the capacitor C_2, resistor R_3, and inductor L_3:

$$V_{c2}(t) - R_3 i_{l3}(t) - L_3\frac{di_{l3}(t)}{dt} = 0$$

$$X_2 - 4X_3 - \dot{X}_3 = 0$$

Solving for $\dot{X}_3$:

$$\dot{X}_3 = 0X_1 + X_2 - 4X_3 \quad \text{— (3)}$$

Combining equations (1), (2), and (3) into standard matrix state-space form $\dot{X} = AX + Bu$:

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}-\frac{4}{3} & \frac{1}{3} & 0\\ -\frac{1}{3} & -\frac{1}{6} & -1\\ 0 & 1 & -4\end{bmatrix}\begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix} + \begin{bmatrix}\frac{2}{3}\\ \frac{1}{6}\\ 0\end{bmatrix}e(t)$$

(b): Energy Dissipation Calculation

Given initial states at $t = 0$:

$$e(t) = 0 \quad (\text{for } t \geq 0)$$

$$i_{l1}(0) = 0\,\text{A}, \quad V_{c2}(0) = 0\,\text{V}, \quad i_{l3}(0) = 1\,\text{A}$$

The initial total energy stored in the circuit elements at $t = 0$ is solely due to the inductor L_3:

$$E(0) = \frac{1}{2}L_1 i_{l1}^2(0) + \frac{1}{2}C_2 V_{c2}^2(0) + \frac{1}{2}L_3 i_{l3}^2(0)$$

$$E(0) = 0 + 0 + \frac{1}{2} \times 1 \times (1)^2 = 0.5\,\text{J}$$

As $t \to \infty$, because there is no sustaining independent source ($e(t) = 0$) and the circuit contains dissipative resistive elements ($2\,\Omega$, $4\,\Omega$, $4\,\Omega$), all stored energy will eventually be completely dissipated. Thus, the final energy stored is:

$$E(\infty) = 0\,\text{J}$$

Using the conservation of energy principle, the total energy dissipated in the resistors over the interval $(0, \infty)$ is:

$$\Delta E = E(0) - E(\infty) = 0.5 - 0 = 0.50\,\text{J}$$

$$\boxed{\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = \begin{bmatrix}-\frac{4}{3} & \frac{1}{3} & 0\\ -\frac{1}{3} & -\frac{1}{6} & -1\\ 0 & 1 & -4\end{bmatrix}\begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix} + \begin{bmatrix}\frac{2}{3}\\ \frac{1}{6}\\ 0\end{bmatrix} e(t),\ \text{Energy dissipated} = 0.50J}$$

Q5.96.1 MCQ 1996 1M Ans: B ● ○ ○

The ammeter A_1 gives the vector sum of the currents carried by the parallel branches, or read by ammeters A_2 and A_3.

We can see that A_2 is connected to a branch containing a pure inductor and A_3 is connected to a branch containing a pure resistor. Assuming ideal ammeters, we can safely say that the two branch currents are in quadrature (that is 90° apart). This is because current through resistor is in phase with the voltage across it, while current through inductor lags behind the voltage across it by 90°. Hence the magnitude of the current value read by A_1 is

$$A_1 = \sqrt{A_2^2 + A_3^2} = \sqrt{3^2 + 4^2} = 5A$$

$$\boxed{(b) 5A}$$

Q5.95.1 MCQ 1995 1M Ans: C ● ○ ○

The RMS value of current i(t) is given by

$$I_{RMS} = \sqrt{I_o^2 + I_1^2/2 + I_3^2/2}$$

Substituting,

$$I_{RMS} = \sqrt{3^2 + 4^2/2 + 4^2/2} = 5A$$

Power dissipated in a network is given by

$$P_{diss} = I_{RMS}^2 R = 5^2 \times 10 = 250W$$

$$\boxed{(c) 5A, 250W \text{ respectively}}$$

Q5.95.2 Descriptive 1995 1M Ans: 100,0.001 ● ○ ○

At resonant frequency, the imaginary part of impedance becomes 0, and the impedance equals the resistance present in the circuit.

$R = 100\Omega$ from given information that circuit has an impedance of (100+j0)Ω at its resonant angular frequency

We know that for a series RLC circuit, quality factor is given by

$$Q = \frac{\omega_0 L}{R}$$

Here, $\omega_0 = 10^7 rad/s$, Q = 100 and $R = 100\Omega$ Therefore, $L = \frac{QR}{\omega_0} = \frac{100\times100}{10^7} = 0.001H$

$$\boxed{R = 100\Omega, L = 0.001H}$$

Q5.95.3 NAT 1995 2M Ans: *

The two inductors in series can be either additively or subtractively coupled. We know that mutual inductance and coefficient of coupling are related as

$$M = k\sqrt{L_1 L_2} = 0.1\sqrt{2 \times 2} = 0.2H$$

Additive coupling :

$$L_{eq} = L_1 + L_2 + 2M = 2 + 2 + 2 \times 0.2 = 4.4H$$

Subtractive coupling :

$$L_{eq} = L_1 + L_2 - 2M = 2 + 2 - 2 \times 0.2 = 3.6H$$

$$\boxed{4.4H, 3.6H}$$

Q5.94.1 MCQ 1994 1M Ans: D

Method 1

From the given data capacitive impedance $X_C = -j20\Omega$ and inductive reactance $X_L = j20\Omega$. Since it is a series RLC circuit and $X_L = -X_C$, the circuit is at resonance.

$$V_R = 200V_{rms}$$

Hence $V_C = -jQV_R$ Here $Q = |X_L|/R = 20/10 = 2$

$$\therefore, V_C = -j2 \times 200V_{rms} = 400\angle -90^\circ V_{rms}$$

Method 2

From the given data capacitive impedance $X_C = -j20\Omega$ and inductive reactance $X_L = j20\Omega$. Since it is a series RLC circuit and $X)L = -X_C$, the circuit is at resonance, and the circuit impedance equals the resistance.

$$V_R = 200V_{rms}$$

$$I = 200/10 = 20A$$

= Thus $V_C = IX_C$

$$V_C = 20 \times -j20 = -j400V_{rms} = 400\angle -90^\circ V_{rms}$$

$$\boxed{(d)400\angle -90^\circ}$$

Q5.93.1 MCQ 1993 1M Ans: A

The two inductances are additively coupled. Hence net inductance $L_{eq} = L_1 + L_2 + 2M$ To have the circuit in resonance $X_L = -X_C \implies X_L = j12\Omega\ X_{Leq} = 12\Omega \implies \omega(L_1 + L_2 + 2M) = 12$ Since $X_{L1} = 2\Omega$ and $X_{L2} = 8\Omega$, we can conclude that $\omega M = 1\Omega$. Now $M = k\sqrt{L_1 L_2}$

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{\omega M}{\omega\sqrt{L_1 L_2}} = \frac{\omega M}{\sqrt{\omega L_1 \times \omega L_2}} = \frac{1}{\sqrt{2 \times 8}} = 0.25$$

$$\boxed{(a)0.25}$$

Q5.93.2 MCQ 1993 2M Ans: C ● ○ ○

Let the number of turns in primary winding be N. In order to apply Maximum Power Transfer theorem, we need to refer 2Ω resistance to the primary. $R_{L1} = 2 \times (\frac{N}{40})^2$ By Maximum Power Transfer theorem, $R_{L1} = 8\Omega \implies 2 \times (\frac{N}{40})^2 = 8$

$$\therefore, N = \sqrt{8 \times 1600/2} = 80$$

$$\boxed{(c) 80}$$

Q5.93.3 MCQ 1993 1M Ans: C ● ○ ○

The ammeter A_3 gives the vector sum of the currents carried by the parallel branches, or read by ammeters A_1 and A_2.

We can see that A_2 is connected to a branch containing a pure capacitor and A_1 is connected to a branch containing a pure resistor. Assuming ideal ammeters, we can safely say that the two branch currents are in quadrature (that is 90° apart). This is because current through resistor is in phase with the voltage across it, while current through capacitor leads the voltage across it by 90°. Hence the magnitude of the current value read by A_3 is

$$A_1 = \sqrt{A_1^2 + A_2^2} = \sqrt{5^2 + 12^2} = 13A$$

$$\boxed{(c) 13A}$$

Q5.92.1 MCQ 1992 2M Ans: B ● ○ ○

In a pure resistor, the voltage V_r and current I_r are always in phase with each other. For the given RLC network configuration, it is observed that the total circuit current is leading the supply voltage.

From the phasor diagram analysis of this system, the magnitude of the capacitive voltage phasor is greater than the inductive voltage phasor:

$$V_c > V_l$$

This phase relationship indicates that the circuit is operating with net capacitive properties:

$$\therefore X_c > X_l$$

Substituting the reactances in terms of the operating frequency ω:

$$\frac{1}{\omega C} > \omega L$$

$$\omega^2 < \frac{1}{LC}$$

$$\omega < \frac{1}{\sqrt{LC}}$$

Since the resonant frequency is defined as $\omega_0 = \frac{1}{\sqrt{LC}}$, this operating frequency condition satisfies:

$$\omega < \omega_0$$

Thus, the circuit operates at a frequency less than the resonant frequency.

$$\boxed{(b) \textit{Lesser than the resonance frequency}}$$

Q5.91.1 MCQ 1991 1M Ans: A ● ○ ○

Consider a standard series $R - L - C$ circuit connected to an AC voltage source V. The current alternating in the loop is limited by the total impedance of the branch components:

$$i = \frac{V}{R + j\omega L + \frac{1}{j\omega C}} = \frac{V}{R + Z_l + Z_c} = \frac{V}{Z_{\text{total}}}$$

At the resonant frequency (ω_0), the inductive reactance and capacitive reactance cancel each other out precisely ($\omega_0 L = \frac{1}{\omega_0 C}$). Under this condition, the total input impedance simplifies to:

$$Z_{\text{total}} = R$$

Because the net reactive impedance is zero, the circuit impedance drops to its minimum possible value, which is purely resistive. Consequently, the steady-state current magnitude reaches its maximum peak value at resonance:

$$i_{\max} = \frac{V}{R}$$

Therefore, the current is maximum at the resonance frequency for a series RLC circuit.

$\boxed{(a)\textit{Equal to the resonant frequency}}$

Q5.90.1 MCQ 1990 1M Ans: B ● ○ ○

From the figure, we see that there is subtractive coulping between the two inductors in series.

$$\therefore L_{eq} = L_1 + L_2 - 2M = 2 + 2 - 2 \times 1 = 2H$$

The capacitor has value of 2F. Therefore, the resonant frequency in Hz is

$$f_c = \frac{1}{2\pi\sqrt{L_{eq}C}} = \frac{1}{2\pi\sqrt{2 \times 2}} = \frac{1}{4\pi} Hz$$

$\boxed{(b)\frac{1}{4\pi} Hz}$

Q5.89.1 MCQ 1989 2M Ans: A ● ● ○

Case 1: Only Source 1 is active (Source 2 is short-circuited) The power dissipated in resistor R is given as $P_1 = 1\,\text{W}$.

$$P_1 = |I_1|^2 \times R$$

$$1 = |I_1|^2 \times 1 \implies |I_1| = 1\,\text{A}$$

Looking at the polarity of Source 1, the reference direction for the current phasor I_1 flowing downward through R can be written as:

$$I_1 = -|I_1| = -1\,\text{A}$$

Case 2: Only Source 2 is active (Source 1 is short-circuited) The power dissipated in resistor R is given as $P_2 = 4\,\text{W}$.

$$P_2 = |I_2|^2 \times R$$

$$4 = |I_2|^2 \times 1 \implies |I_2| = 2\,\text{A}$$

Looking at the polarity of Source 2, the reference direction for the current phasor I_2 flowing downward through R can be written as:

$$I_2 = +|I_2| = 2\,\text{A}$$

Case 3: Both sources are active simultaneously Since both sources operate at the same frequency ω, we can algebraically combine the parallel current contributions using superposition:

$$I_{\text{total}} = I_1 + I_2$$

$$I_{\text{total}} = -1 + 2 = 1\,\text{A}$$

The total power P_{total} dissipated in resistor R is:

$$P_{\text{total}} = |I_{\text{total}}|^2 \times R$$

$$P_{\text{total}} = (1)^2 \times 1 = 1\,\text{W}$$

$$\boxed{(a)1W}$$

Q5.89.2 MCQ 1989 1M Ans: A ●○○

To match a complex load impedance $Z_l = R_l + jX_l$ to a source with an internal impedance $Z_s = R_s + jX_s$ using an ideal transformer, we look at how the transformer transforms magnitudes.

An ideal transformer with a primary-to-secondary turns ratio of $n = \frac{N_1}{N_2}$ alters the secondary load impedance when viewed from its primary side. The reflected load impedance Z_l' connected across the primary terminals is given by:

$$Z_l' = n^2 Z_l$$

Taking the absolute magnitude of both sides:

$$|Z_l'| = n^2 |Z_l|$$

For maximum power transfer matching between a source and its reflected load network when transforming through an ideal transformer, the magnitude of the reflected load impedance must equal the magnitude of the internal source impedance ($|Z_l'| = |Z_s|$):

$$|Z_s| = n^2 |Z_l|$$

Solving for the turns ratio n^2:

$$n^2 = \frac{|Z_s|}{|Z_l|}$$

Inverting the relationship depending on the reference conversion direction matching primary-to-secondary configurations:

$$n = \sqrt{\left|\frac{Z_l}{Z_s}\right|}$$

$$\boxed{(a)\sqrt{|Z_L/Z_S|}}$$

Key Points

- An ideal transformer acts as an impedance matching device changing the impedance scale by the square of its turns ratio (n^2).
- It can change only the magnitude of a complex load, not its phase angle, because it lacks reactive components.

Mistakes to be avoided

- Do not confuse simple transformer impedance transformation with conjugate impedance matching ($Z_l = Z_s^*$). An ideal transformer alone cannot cancel out the reactive component unless the transformation matches the overall absolute magnitudes.

Q5.89.3 MSQ | 1989 | 2M | Ans: A,D | ● ● ○

To determine how to increase the half-power bandwidth of the given resonant circuit, we evaluate the circuit's parallel resonance characteristics and its quality factor relationships.

The half-power bandwidth (BW) of a resonant circuit is fundamentally related to its resonant frequency (ω_0) and its quality factor (Q_0):

$$\text{BW} = \frac{\omega_0}{Q_0}$$

For an ideal parallel resonant tank circuit containing a capacitance C in parallel with an equivalent resistance R_{eq}, the quality factor is expressed as:

$$Q_0 = \omega_0 R_{\text{eq}} C$$

Substituting Q_0 back into the bandwidth equation shows that bandwidth depends inversely on the total parallel equivalent resistance:

$$\text{BW} = \frac{\omega_0}{\omega_0 R_{\text{eq}} C} = \frac{1}{R_{\text{eq}} C}$$

Thus, to **increase** the half-power bandwidth (BW), the total equivalent parallel resistance R_{eq} must be **decreased**. NOTE : R_{eq} is the effective resistance as seen by the capacitance C.

Let us analyze how changing R_1 and R_2 impacts the equivalent parallel damping:

1. **Effect of R_2:** Resistor R_2 is connected directly in parallel with the capacitive branch. A direct decrease in the value of R_2 will lower the overall parallel combination resistance R_{eq}.

$$R_{\text{eq}} \approx R_{\text{series-converted}} \parallel R_2$$

Decreasing R_2 directly causes R_{eq} to decrease, which consequently increases the bandwidth. Thus, **Option D is correct**.

2. **Effect of R_1:** The inductive branch contains a series resistance R_1. To find its parallel equivalence near resonance, we perform a series-to-parallel transformation of the practical inductor branch (R_1 in series with L). The equivalent parallel resistance R_{p1} contributed by this branch is:

$$R_{p1} = R_1 \left(1 + Q_{\text{coil}}^2\right) = R_1 \left(1 + \frac{\omega^2 L^2}{R_1^2}\right) = R_1 + \frac{\omega^2 L^2}{R_1}$$

Near resonance, assuming a reasonably high coil quality factor ($Q_{\text{coil}} \gg 1$):

$$R_{p1} \approx \frac{\omega_0^2 L^2}{R_1}$$

The overall equivalent parallel resistance R_{eq} of the entire network becomes:

$$R_{\text{eq}} = R_{p1} \parallel R_2 = \left(\frac{\omega_0^2 L^2}{R_1}\right) \parallel R_2$$

From this expression, if we **increase** R_1, the value of the equivalent parallel resistance R_{p1} drops. A decrease in R_{p1} lowers the total parallel combination resistance R_{eq}. Since a lower R_{eq} increases the bandwidth:

$$\text{BW} \uparrow \implies R_{\text{eq}} \downarrow \implies R_{p1} \downarrow \implies R_1 \uparrow$$

Therefore, increasing R_1 also serves to increase the bandwidth. Thus, **Option A is correct**.

$$\boxed{(a) increasing\ R_1\ ,\ (b) decreasing\ R_2}$$

Q5.88.1 MCQ 1988 1M Ans: B ● ○ ○

According to Maximum Power Transfer Theorem, the maximum power is transferred to load impedance when it equals the complex conjugate of the Thevenin Impedance of the circuit looking from the load terminals. Here the Thevenin impedance is the source impedance.

$$\therefore Z_L = Z_S^*$$

$$\boxed{(b) \text{Complex conjugate of } Z_S^*}$$

Q5.87.1 NAT 1987 2M Ans: 0 ● ● ○

The problem involves calculating the current through a specific branch (the 1 F capacitor). We first convert the network components to reactances,then performing a Delta-to-Star (Δ-Y) transformation, and analyzing the simplified bridge network.

Step 1: Conversion of Circuit Elements to Reactances The source frequency from the voltage expression $2\sin(100t)$ is:

$$\omega = 100 \text{ rad/s}$$

The impedances of the elements are calculated as follows: For the 1 F capacitor:

$$Z_{C1} = \frac{-j}{\omega C} = \frac{-j}{100 \times 1} = -0.01j\ \Omega$$

For the 0.5 F capacitor:

$$Z_{C2} = \frac{-j}{\omega C} = \frac{-j}{0.5 \times 100} = -0.02j\ \Omega$$

For the 1 H inductor:

$$Z_L = j\omega L = j(100 \times 1) = j100\ \Omega$$

Step 2: Delta to Star (Δ-Y) Transformation

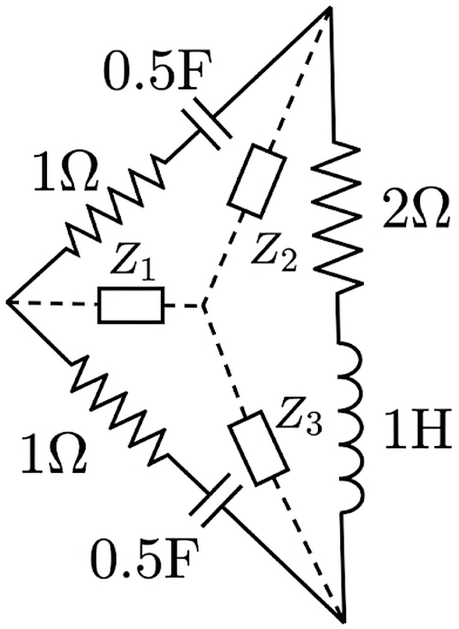


Fig. Solution Q5.87.1

By converting the outer delta network comprising the reactances into an equivalent star network, the calculated star impedances are: For Z_1:

$$Z_1 = \frac{(1-0.02j)(1-0.02j)}{1-0.02j+1-0.02j+2+j100} = \frac{0.9996-0.04j}{4+j99.96} \approx 0\,\Omega$$

For Z_2 and Z_3:

$$Z_2 = Z_3 = \frac{(1-0.02j)(2+j100)}{4+j99.96} = \frac{4+j99.96}{4+j99.96} = 1\,\Omega$$

Step 3: Analysis of the Simplified Circuit

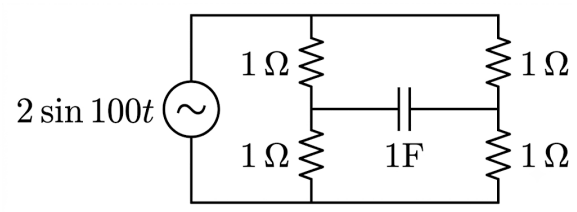


Fig. Solution Q5.87.1

Replacing the delta section with its star equivalent transforms the circuit configuration into a balanced Wheatstone bridge network consisting of four 1 Ω bridge arms and a central branch containing the 1 F capacitor(ignoring Z_1).

For a balanced Wheatstone bridge:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \implies \frac{1\,\Omega}{1\,\Omega} = \frac{1\,\Omega}{1\,\Omega}$$

Because the bridge is perfectly balanced, the potential difference across the central branch is zero. Consequently, no current flows through the central 1 F capacitor:

$$\boxed{0}$$

SOLUTIONS

VI

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

Unit Snapshot*

Stream: EC	**Questions:** 41	**MCQ:** 28	**MSQ:** 2	**NAT:** 11
1M: 15	**2M:** 24	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2020)–4M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 2M Ans: D ● ○ ○

We need to find out the ABCD parameters of the corresponding network. This can be done using the following approaches.

Method 1

Calculating ABCD parameters from mesh equations:

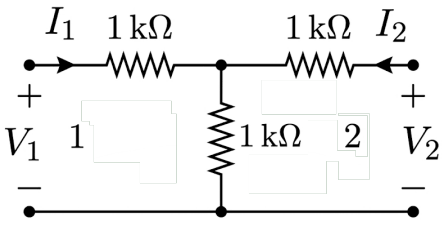


Fig. Solution Q6.26.1

Applying Kirchhoff's Voltage Law (KVL) in the respective loops of the two-port network:

KVL in mesh 1:

$$V_1 = 2\text{k}I_1 + 1\text{k}I_2 \quad \text{— (1)}$$

KVL in mesh 2:

$$V_2 = 2\text{k}I_2 + 1\text{k}I_1 \quad \text{— (2)}$$

The general form of the ABCD parameters is defined as:

$$V_1 = AV_2 - BI_2 \quad \text{— (3)}$$

$$I_1 = CV_2 - DI_2 \quad \text{— (4)}$$

From equation (2), we can express I_1 in terms of V_2 and I_2:

$$1\text{k}I_1 = V_2 - 2\text{k}I_2$$

$$I_1 = \frac{1}{10^3}V_2 - \frac{2 \times 10^3}{10^3}I_2$$

$$I_1 = 10^{-3}V_2 - 2I_2 \quad \text{— (5)}$$

Comparing equation (5) with the standard form in equation (4):

$$\therefore C = 10^{-3}, \quad D = 2$$

Substituting equation (5) into equation (1):

$$V_1 = 2\text{k}(10^{-3}V_2 - 2I_2) + 1\text{k}I_2$$

$$V_1 = (2 \times 10^3 \times 10^{-3})V_2 - (4 \times 10^3)I_2 + 1\text{k}I_2$$

$$V_1 = 2V_2 - 3\text{k}I_2 \quad \text{— (6)}$$

Comparing equation (6) with the standard form in equation (3):

$$\therefore A = 2, \quad B = 3 \times 10^3$$

Method 2

Calculating ABCD parameters by opening and shorting of respective ports: The general form of ABCD parameters is:

$$V_1 = AV_2 - BI_2 \quad \text{— (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{— (2)}$$

The parameters are defined under specific port conditions:

$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}, \quad B = -\left.\frac{V_1}{I_2}\right|_{V_2=0}, \quad C = \left.\frac{I_1}{V_2}\right|_{I_2=0}, \quad D = -\left.\frac{I_1}{I_2}\right|_{V_2=0}$$

We analyze the circuit for two conditions:

- Keeping Port 2 open ($I_2 = 0$)
- Keeping Port 2 shorted ($V_2 = 0$)

1. Port 2 Open ($I_2 = 0$):

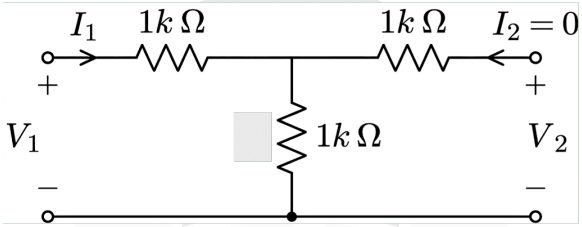


Fig. Solution Q6.26.1

$$V_2 = I_1(1\,\text{k})$$

$$I_1 = \frac{V_1}{1\,\text{k} + 1\,\text{k}} = \frac{V_1}{2\,\text{k}}$$

Substituting I_1 in the V_2 equation:

$$V_2 = \frac{V_1}{2\,\text{k}}(1\,\text{k}) = \frac{V_1}{2} \implies A = \left.\frac{V_1}{V_2}\right|_{I_2=0} = 2$$

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0} = \frac{1}{1\,\text{k}} = 10^{-3}$$

2. Port 2 Shorted ($V_2 = 0$):

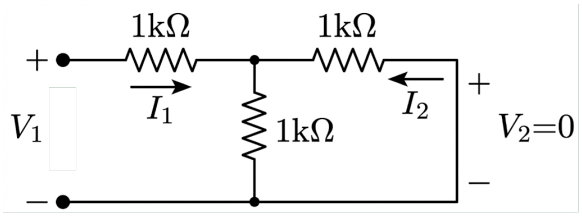


Fig. Solution Q6.26.1

When port 2 is shorted, the shunt 1 kΩ resistor is in parallel with the output 1 kΩ resistor. The total resistance seen at port 1 is:

$$R_{in} = 1\,\text{k} + (1\,\text{k} \parallel 1\,\text{k}) = 1\,\text{k} + 0.5\,\text{k} = 1.5\,\text{k}\Omega$$

$$V_1 = I_1(1.5\,\text{k})$$

By current division, the output current I_2 (defined entering the port) is:

$$I_2 = -\frac{I_1}{2} \implies D = -\frac{I_1}{I_2}\bigg|_{V_2=0} = 2$$

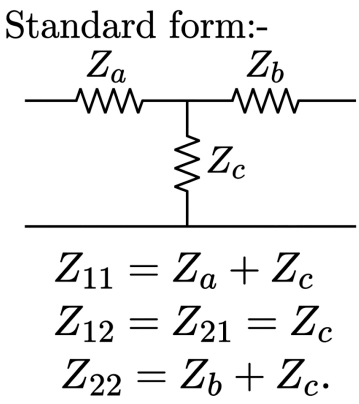
Substituting $I_1 = -2I_2$ into the V_1 equation:

$$V_1 = -2I_2(1.5\,\text{k}) = -3\,\text{k}I_2$$

$$B = -\frac{V_1}{I_2}\bigg|_{V_2=0} = 3\,\text{k}$$

Method 3

Transforming Z parameters to ABCD parameters:



$$Z_{11} = Z_a + Z_c$$
$$Z_{12} = Z_{21} = Z_c$$
$$Z_{22} = Z_b + Z_c.$$

Fig. Solution Q6.26.1

The given Z parameters for the network are:

$$Z = \begin{bmatrix} 2\times10^3 & 1\times10^3 \\ 1\times10^3 & 2\times10^3 \end{bmatrix}$$

From the Z-parameter definition, we have:

$$V_1 = 2\times10^3 I_1 + 10^3 I_2 \quad \text{— (1)}$$

$$V_2 = 10^3 I_1 + 2\times10^3 I_2 \quad \text{— (2)}$$

From equation (2), we can express I_1 in terms of V_2 and I_2:

$$10^3 I_1 = V_2 - 2\times10^3 I_2$$

$$I_1 = \frac{1}{10^3}V_2 - \frac{2\times10^3}{10^3}I_2$$

$$I_1 = 10^{-3}V_2 - 2I_2 \quad \text{— (3)}$$

Comparing this with the ABCD parameter equation $I_1 = CV_2 - DI_2$, we get:

$$C = 10^{-3}\,\Omega^{-1}$$

$$D = 2$$

Substituting I_1 from equation (3) into equation (1):

$$V_1 = 2 \times 10^3 (10^{-3} V_2 - 2I_2) + 10^3 I_2$$

$$V_1 = (2 \times 10^3 \times 10^{-3}) V_2 - (2 \times 10^3 \times 2) I_2 + 10^3 I_2$$

$$V_1 = 2V_2 - 4 \times 10^3 I_2 + 10^3 I_2$$

$$V_1 = 2V_2 - 3 \times 10^3 I_2 \quad \text{— (4)}$$

Comparing this with the ABCD parameter equation $V_1 = AV_2 - BI_2$, we get:

$$\therefore A = 2$$

$$B = 3 \times 10^3\ \Omega$$

Key Points

- **Unit Check:** A and D are dimensionless, B has units of Ohms (Ω), and C has units of Siemens (S).

Mistakes to be avoided

- Sign Convention : Ensure you take the direction of the port currents and their corresponding signs in the ABCD parameter equation correctly. For example, if port-2 current enters the network, we will have the equations used in the solution above. However, if the current leaves the network, the '-' signs will be replaced by '+' signs (as in transmission lines).

$$\boxed{(d) A = 2, B = 3 \times 10^3\ \Omega, C = 10^{-3}\ \Omega^{-1}, D = 2}$$

Q6.25.1 MCQ | 2025 | 2M | Ans: A | ● ○ ○

Method 1

The network can be simplified into a standard form using Y − Δ transformation (star-delta transformation). Applying Y − Δ transformation to the 2 Ω delta network:

$$R_Y = \frac{R_\Delta \times R_\Delta}{\sum R_\Delta} = \frac{2 \times 2}{2 + 2 + 2} = \frac{4}{6} = \frac{2}{3}\ \Omega$$

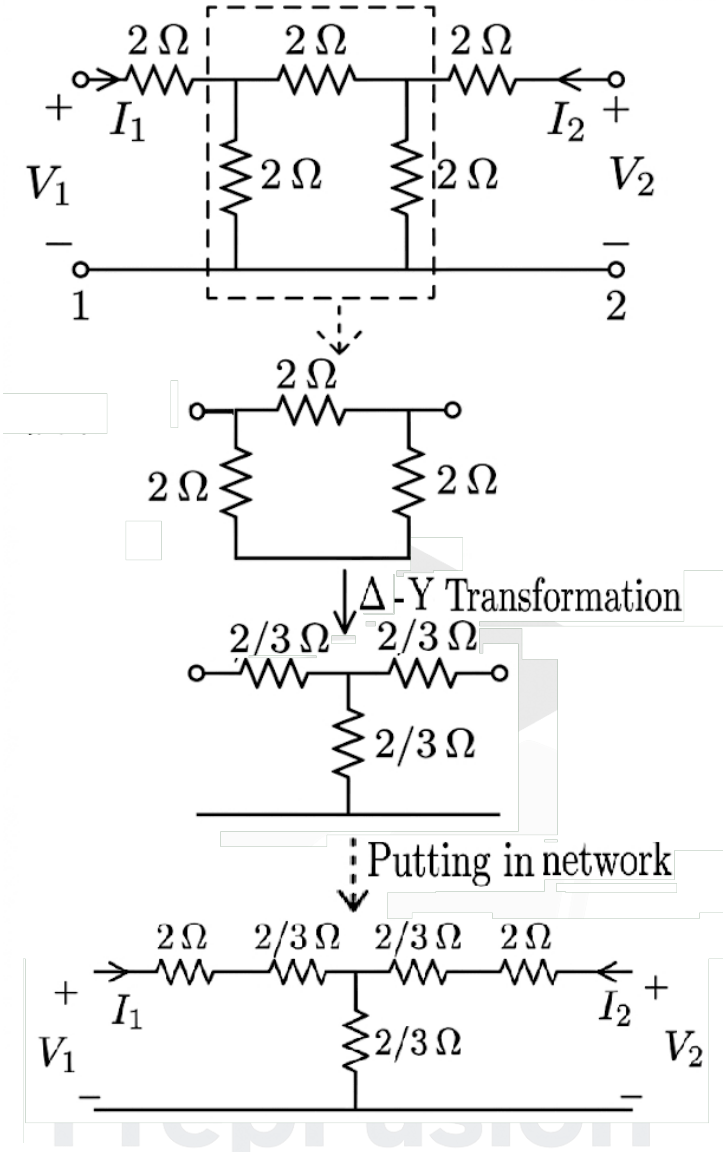


Fig. Solution Q6.25.1

Putting this back into the original network:

- Total series resistance at port 1: $R_1 = 2 + 2/3 = 8/3\,\Omega$.
- Total series resistance at port 2: $R_2 = 2 + 2/3 = 8/3\,\Omega$.
- Common shunt resistance: $R_3 = 2/3\,\Omega$.

The simplified network is now in a standard T-network form. For a T-network, the Z-parameters are (Ref. Q6.26.1, Method 3):

$$Z_{11} = R_1 + R_3 = \frac{8}{3} + \frac{2}{3} = \frac{10}{3}\,\Omega$$

$$Z_{22} = R_2 + R_3 = \frac{8}{3} + \frac{2}{3} = \frac{10}{3}\,\Omega$$

$$Z_{12} = Z_{21} = R_3 = \frac{2}{3}\,\Omega$$

The Z-parameter matrix is:

$$Z = \begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$$

Method 2

The Z-parameters are defined by the following equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad — (1)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad — (2)$$

The individual parameters are calculated as:

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}, \quad Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}, \quad Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}, \quad Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$$

We will solve the network under two conditions:

a) Keeping Port 1 open ($I_1 = 0$)

b) Keeping Port 2 open ($I_2 = 0$)

a) Port 1 Open ($I_1 = 0$):

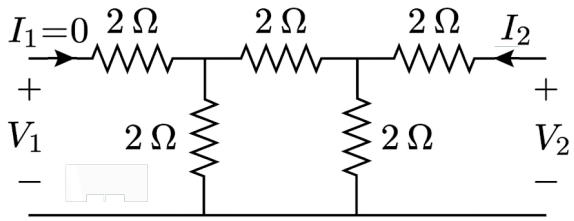


Fig. Solution Q6.25.1

With Port 1 open, the current I_2 flows through the network. The equivalent resistance seen from Port 2 is:

$$V_2 = I_2 \times (2 + 2 \parallel 4)$$

$$\Longrightarrow V_2 = I_2 \times (2 + 4/3)$$

$$\Longrightarrow V_2 = I_2 \times 10/3$$

$$\Longrightarrow Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} = \frac{10}{3}\,\Omega$$

Using the current division rule, the current through the shunt resistor at Port 1 is determined, and V_1 is:

$$V_1 = 2 \times I_2 \times \frac{2}{2+4}$$

$$\Longrightarrow V_1 = I_2 \times 2/3$$

$$\Longrightarrow Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = \frac{2}{3}\,\Omega$$

b) Port 2 Open ($I_2 = 0$):

With Port 2 open, the current I_1 enters Port 1. The equivalent resistance seen from Port 1 is:

$$V_1 = I_1 \times (2 + 2 \parallel 4)$$

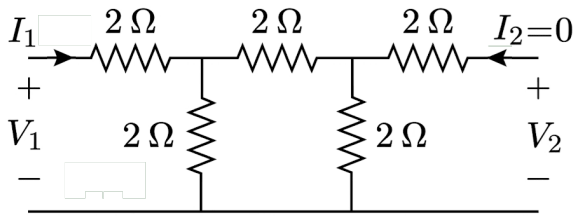


Fig. Solution Q6.25.1

$$\implies V_1 = I_1 \times 10/3$$

$$\implies Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = \frac{10}{3}\,\Omega$$

The voltage V_2 is the voltage across the shunt resistor at Port 2:

$$V_2 = 2 \times I_1 \times \frac{2}{2+4}$$

$$\implies V_2 = I_1 \times 2/3$$

$$\implies Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = \frac{2}{3}\,\Omega$$

Key Points

- **T-Network Z-parameters:** For a T-network with series arms Z_a, Z_b and shunt arm Z_c, the parameters are $Z_{11} = Z_a + Z_c$, $Z_{22} = Z_b + Z_c$, and $Z_{12} = Z_{21} = Z_c$.

$$(a)\begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$$

Q6.24.1 NAT 2024 2M Ans: 1.5 to 1.5 ●○○

The Y parameters relate port current and voltages in the following manner :

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = Y \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$$

$\therefore$ Port-2 must be shorted for V_2 to be 0.

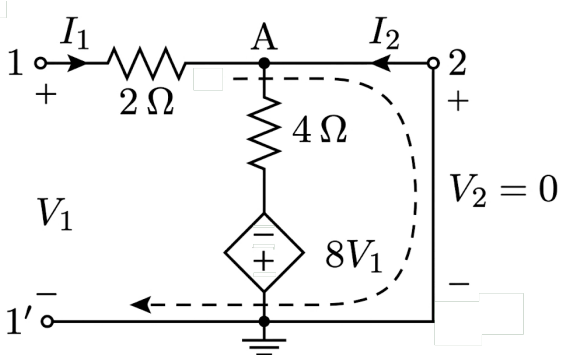


Fig. Solution Q6.24.1

Writing KVL equation for outer loop:

$$V_1 = 2I_1 \implies I_1 = 0.5V_1$$

Writing KCL equation at node A:

$$-I_1 + \frac{V_A + 8V_1}{4} - I_2 = 0$$

But $V_A = 0$.

$$\therefore -I_1 + 2V_1 - I_2 = 0$$

$$\implies -0.5V_1 + 2V_1 - I_2 = 0$$

$$\implies I_2 = 1.5V_1$$

$$\implies Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0} = 1.5S$$

1.5 to 1.5

Q6.23.1 NAT 2023 1M Ans: 80 to 80

The Y-parameters of the network are defined by the equations:

$$I_1 = y_{11}V_1 + y_{12}V_2$$

$$I_2 = y_{21}V_1 + y_{22}V_2$$

For Maximum Power Transfer, we need to make the load impedance, Z_L equal to the Thevenin Impedance, Z_{Th}, as seen from Port-2. To find Z_{Th}, we will analyze the network from Port-2. The Thevenin IMpedance can be calculated in the following ways :

Method 1

From the given equations we can write :

$$100\,I_1 = 2V_1 - V_2 \quad \text{— (1)}$$

$$100\,I_2 = -V_1 + \frac{4}{3}V_2 \quad \text{— (2)}$$

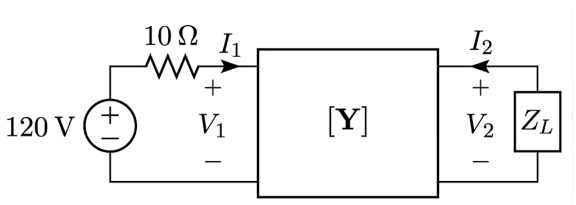


Fig. Solution Q6.23.1

Applying Kirchhoff's voltage law (KVL) to Port 1:

$$V_1 = 120 - 10\,I_1 \quad \text{— (3)}$$

We determine the Thevenin impedance by finding the open-circuit voltage (V_{oc}) and short-circuit current (I_{sc}) at Port-2:

$$Z_{Th} = \left.\frac{V_{oc}}{I_{sc}}\right|_{\text{Port-2}}$$

Thus we check for two conditions:

a) Port 2 open (To find V_{oc})

b) Port 2 shorted (To find I_{sc})

a) Port 2 Open: From equation (2):

$$0 = -V_1 + \frac{4}{3}V_2 \implies V_1 = \frac{4}{3}V_2$$

Now substitute V_1 into equation (1):

$$100\,I_1 = 2\left(\frac{4}{3}V_2\right) - V_2$$

$$\implies 100\,I_1 = \frac{8}{3}V_2 - V_2$$

$$\implies 100\,I_1 = \frac{5}{3}V_2 \quad \text{— (4)}$$

From Port-1 KVL (equation 3), substitute I_1 from equation (4) and V_1 from earlier:

$$V_1 = 120 - 10\left(\frac{5\,V_2}{3\times 100}\right) \implies \frac{4}{3}V_2 = 120 - 10\left(\frac{V_2}{60}\right)$$

$$\implies \frac{4}{3}V_2 + \frac{1}{6}V_2 = 120 \implies V_2\left(\frac{8+1}{6}\right) = 120$$

$$\implies V_2\left(\frac{9}{6}\right) = 120 \implies V_2\left(\frac{3}{2}\right) = 120$$

$$\implies V_2 = \frac{2\times 120}{3} = 80\,\text{V}$$

So, open-circuit voltage at Port-2 is:

$$V_{oc} = 80\,\text{V}$$

b) Port 2 Shorted:. From equations (1) and (2):

$$100\,I_1 = 2V_1 \implies V_1 = 50\,I_1$$

From equation (3) (Port 1 KVL):

$$V_1 = 120 - 10\,I_1$$

Equating the two expressions for V_1:

$$50\,I_1 = 120 - 10\,I_1 \implies 60\,I_1 = 120 \implies I_1 = 2\,\text{A}$$

Then, Port-1 voltage is:

$$V_1 = 50(2) = 100\,\text{V}$$

Now find I_2 using equation (2) with $V_2 = 0$:

$$100\,I_2 = -V_1 \implies 100\,I_2 = -100$$

$$\implies I_2 = -1\,\text{A}$$

The short-circuit current flowing out of Port-2 is:

$$I_{sc} = -I_2 = 1\,\text{A}$$

Finally, the Thevenin impedance Z_{Th} at Port-2 is:

$$Z_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{80\,\text{V}}{1\,\text{A}} = 80\,\Omega$$

For maximum power transfer to the load:

$$Z_L = Z_{Th} = 80\,\Omega$$

Therefore, the required load impedance is $80\,\Omega$.

Method 2

The admittance parameters (Y-parameters) for the two-port network are provided as follows:

$$[Y] = \begin{bmatrix} \frac{2}{100} & -\frac{1}{100} \\ -\frac{1}{100} & \frac{4}{300} \end{bmatrix}$$

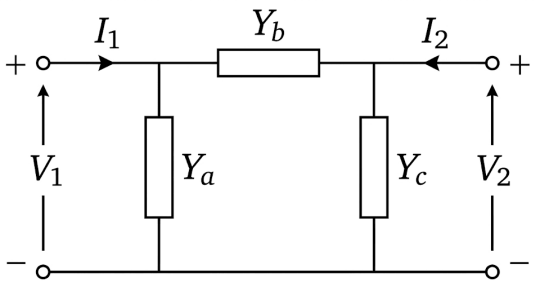


Fig. Solution Q6.23.1

For the above π-model representation of a two-port network, the relationship between the admittance elements (Y_a, Y_b, Y_c) and the Y-parameters is given by:

- $Y_{12} = Y_{21} = -Y_b \implies -\frac{1}{100} = -Y_b \implies Y_b = \frac{1}{100}\,\text{S}$
- $Y_{11} = Y_a + Y_b \implies \frac{2}{100} = Y_a + \frac{1}{100} \implies Y_a = \frac{1}{100}\,\text{S}$
- $Y_{22} = Y_b + Y_c \implies \frac{4}{300} = \frac{1}{100} + Y_c \implies Y_c = \frac{4}{300} - \frac{3}{300} = \frac{1}{300}\,\text{S}$

Converting these admittances to impedances $(Z = 1/Y)$:

$$Z_a = 100\,\Omega, \quad Z_b = 100\,\Omega, \quad Z_c = 300\,\Omega$$

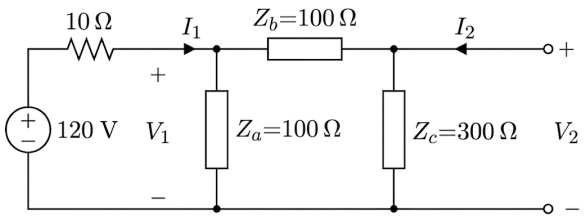


Fig. Solution Q6.23.1

To find the Thevenin impedance (Z_{Th}) looking into Port 2, we de-activate the independent voltage source (replace it with a short circuit) and also perform a delta-to-star conversion on the network formed by Z_a, Z_b, and Z_c:

- $R_1 = \frac{100\times100}{100+100+300} = \frac{10000}{500} = 20\,\Omega$
- $R_2 = \frac{100\times300}{500} = 60\,\Omega$
- $R_3 = \frac{100\times300}{500} = 60\,\Omega$

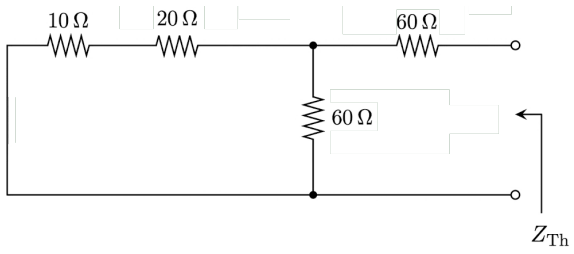


Fig. Solution Q6.23.1

The equivalent impedance Z_{Th} seen from the load terminals is:

$$Z_{Th} = 60 + [(20 + 10) \parallel 60]$$

$$Z_{Th} = 60 + \left(\frac{30 \times 60}{30 + 60}\right) = 60 + \frac{1800}{90} = 60 + 20 = 80\,\Omega$$

According to the Maximum Power Transfer Theorem, the load impedance Z_L must equal the Thevenin impedance of the source network:

$$Z_L = Z_{Th} = 80\,\Omega$$

The required value for maximum power transfer is $80\,\Omega$.

Key Points

- **Y-parameters** for a π-network are defined as $Y_{11} = Y_a + Y_b$, $Y_{12} = Y_{21} = -Y_b$, and $Y_{22} = Y_c + Y_b$.
- **Maximum Power Transfer Theorem:** For maximum power delivery in a resistive circuit, the load resistance must equal the Thevenin resistance ($R_L = R_{Th}$) of the source network.
- **Efficiency:** Under the condition of maximum power transfer, the power efficiency of the circuit is exactly 50%.

Mistakes to be avoided

- **Unit Conversion:** Ensure impedances (Z) are converted to admittances ($Y = 1/Z$) before applying standard π-model formulas.
- **Negative Sign in Transfer Admittance:** Do not forget the negative sign when equating the series element Y_b to the transfer parameters Y_{12} and Y_{21}.

80 to 80

Q6.23.2 MCQ 2023 2M Ans: A ●●○

Note: The dependent current source in the question figure should be $h_{21}I_1$ instead of $h_{21}V_1$ to comply with standard h-parameter relations.

The standard two-port network relationships using hybrid (h) parameters are defined by the following equations:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

To achieve maximum voltage gain ($\frac{V_2}{V_1}$) in an amplifier network, the ideal characteristics of the h-parameters must satisfy specific conditions:

- **Input Impedance (h_{11}):** Should be minimum (ideally zero) so that there is negligible voltage drop on the input side. This ensures that the maximum input signal is effectively transferred to the amplifier stage.

- **Reverse Voltage Gain (h_{12}):** Should be zero so that the output voltage does not feed back and corrupt or degrade the input signal. The absence of reverse feedback is crucial for maintaining higher stable voltage gain.
- **Forward Current Gain (h_{21}):** Should be very high because it represents the forward amplification factor. Higher h_{21} ensures more current is driven into the load, providing stronger amplification from input to output.
- **Output Admittance (h_{22}):** Should be zero because it represents parallel output admittance. Zero output admittance implies infinite output resistance ($R_o = \infty$), which maximizes the output voltage drop across the connected load.

Evaluating these required conditions together confirms that option A is the correct choice.

$$\boxed{(a)\, h_{11} = 0, h_{12} = 0, h_{21} = \text{ very high and } h_{22} = 0}$$

Q6.22.1 NAT 2022 2M Ans: 3.9 to 4.1 ●●○

From figure (a) $V_1 = 10\,\text{V}$. For this problem, we need to draw the Thevenin equivalent circuit as seen from Port 2. From plot in figure (b), the open-circuit voltage is:

$$V_{oc} = 5\,\text{V} = V_{th}$$

The slope of plot in figure (b) is calculated as:

$$\text{Slope} = \frac{4-5}{4-0}\frac{\text{V}}{\text{mA}} = -0.25\,\text{k}\Omega = -250\,\Omega$$

The equation of the V_2 vs $-I_2$ plot is:

$$V_2 - 5 = -250(-I_2 - 0)$$

$$V_2 = 5 + 250I_2$$

When $V_2 = 0$, the short-circuit current is derived from:

$$I_2 = -\frac{5}{250} = -20\,\text{mA}$$

Since $I_{sc} = -I_2$:

$$I_{sc} = 20\,\text{mA}$$

The Thevenin impedance looking in from Port 2 is:

$$Z_{th} = \frac{V_{oc}}{I_{sc}} = \frac{5\,\text{V}}{20\,\text{mA}} = 250\,\Omega$$

We can now draw the resultant network with the load:

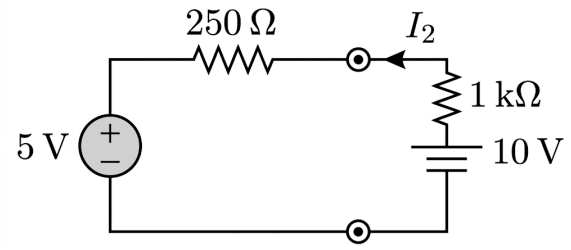


Fig. Solution Q6.22.1

The current I_2 is calculated as:

$$I_2 = \frac{10-5}{250+1000}\,\text{A} = \frac{5}{1250}\,\text{A}$$

$$I_2 = 4\,\text{mA}$$

Mistakes to be avoided

- **Unit Conversion:** Pay close attention to the units on the graph axes (e.g., mA to A) to avoid a scale error in the impedance calculation.
- **Current Direction:** Ensure the sign of I_2 is correctly handled according to the port convention defined in the problem.

3.9 to 4.1

Q6.21.1 MCQ | 2021 | 2M | Ans: C | ● ● ○

The standard equations for Y-parameters are:

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

We can solve this problem in the following methods :

Method 1

Applying Kirchhoff's Current Law (KCL) to the given circuit:

Write KCL at node where V_1 is applied wrt reference:

$$I_1 = \frac{V_1}{1} + \frac{V_1 - V_2}{1} - 3V_2$$

$$I_1 = V_1 + V_1 - V_2 - 3V_2$$

$$I_1 = 2V_1 - 4V_2 \quad \ldots (i)$$

Write KCL at node where V_2 is applied wrt reference:

$$I_2 = \frac{V_2}{1} + \frac{V_2 - V_1}{1}$$

$$I_2 = V_2 + V_2 - V_1$$

$$I_2 = -V_1 + 2V_2 \quad \ldots (ii)$$

Comparing equations (i) and (ii) with standard Y-parameter equations, we get:

$$[Y] = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix} \mho$$

Method 2

From the standard equations for Y-parameters we have :

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

From these equations, the individual parameters are defined as:

$$y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}, \quad y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$

$$y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}, \quad y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

We can consider the shorting of each port one by one in order to find the Y-parameters.

1. Port 1 shorted ($V_1 = 0$)

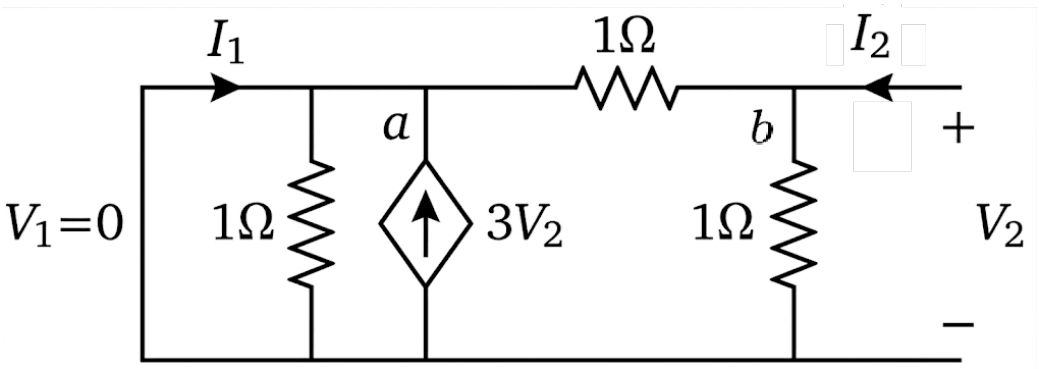


Fig. Solution Q6.21.1

Applying KCL at node a:

$$-I_1 - 3V_2 + \frac{0 - V_2}{1} = 0$$

$$\Rightarrow -4V_2 + I_1 = 0$$

$$\Rightarrow y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0} = -4$$

Applying KCL at node b:

$$-I_2 + \frac{V_2}{1} + \frac{V_2}{1} = 0$$

$$\Rightarrow I_2 = 2V_2$$

$$\Rightarrow y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0} = 2$$

2. Port 2 shorted ($V_2 = 0$) The dependent current source $3V_2$ becomes 0 A (open circuit).

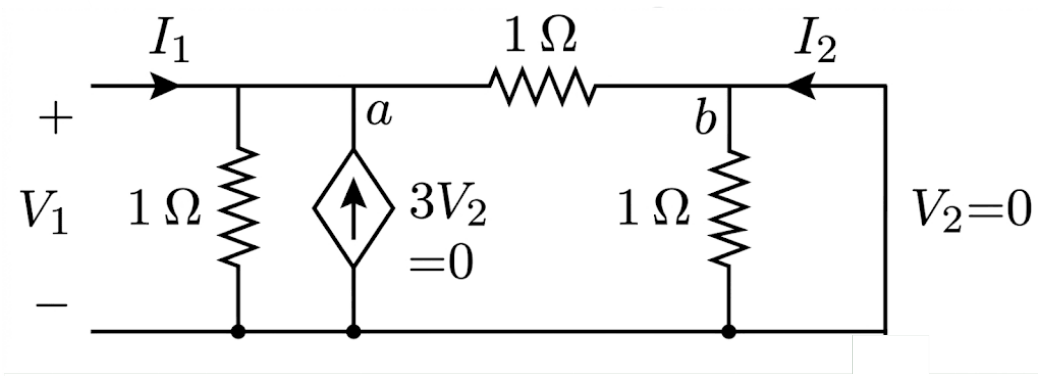


Fig. Solution Q6.21.1

Applying KCL at node a:

$$-I_1 + \frac{V_1}{1} + \frac{V_1 - 0}{1} = 0$$

$$\Rightarrow I_1 = 2V_1$$

$$\Rightarrow y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0} = 2$$

Applying KCL at node b:

$$-I_2 + \frac{0 - V_1}{1} = 0$$

$$\Rightarrow V_1 = -I_2$$

$$\Rightarrow y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0} = -1$$

Therefore, the y-parameters are:

$$y_{11} = 2, y_{12} = -4, y_{21} = -1, y_{22} = 2$$

$$\boxed{(c) y_{11} = 2, y_{12} = -4, y_{21} = -1, y_{22} = 2}$$

Q6.20.1 NAT | 2020 | 1M | Ans: 48 to 48 | ● ● ○

For Maximum Power Transfer, $Z_L = Z_{th}$.
From the given circuit, we can write the following equations:

$$V_1 = 120 - 10I_1 \quad \text{— (1)}$$

$$I_{\text{load}} = -I_2 \quad \text{— (2)}$$

The impedance matrix can be expanded as:

$$V_1 = 40I_1 + 60I_2 \quad \text{— (3)}$$

$$V_2 = 60I_1 + 120I_2 \quad \text{— (4)}$$

where $Z_{th} = \frac{V_{oc}}{I_{sc}}$ looking into the network at Port 2.
Now, $V_{oc} = V_2$ when $I_2 = 0$. From (4):

$$V_2 = 60I_1$$

From (3):

$$V_1 = 40I_1$$

From (1):

$$40I_1 = 120 - 10I_1$$

$$\Rightarrow 50I_1 = 120$$

$$\Rightarrow I_1 = \frac{120}{50} = 2.4\,\text{A}$$

$$\therefore V_2 = 60 \times 2.4 = 144\,\text{V}$$

Next, $I_{sc} = -I_2$ when $V_2 = 0$ (from (2)). From (4):

$$60I_1 = -120I_2$$

$$I_2 = -0.5I_1$$

From (3):

$$V_1 = 40I_1 + 60(-0.5I_1)$$

$$V_1 = 40I_1 - 30I_1$$

$$V_1 = 10I_1$$

From (1):

$$10I_1 = 120 - 10I_1$$

$$\Rightarrow I_1 = \frac{120}{20} = 6\,\text{A}$$

$$I_2 = -3\,\text{A}$$

$$I_{sc} = 3\,\text{A}$$

$$Z_{th} = \frac{144}{3} = 48\,\Omega$$

$$\therefore Z_L = Z_{th} = 48\,\Omega$$

48 to 48

Q6.18.1 NAT | 2018 | 1M | Ans: 4.3 to 5.3 | ● ○ ○

The general form of the ABCD parameters is defined as:

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Thus, the transmission parameter B of a two-port network is defined under short-circuit conditions at terminal 2 ($V_2 = 0$):

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$$

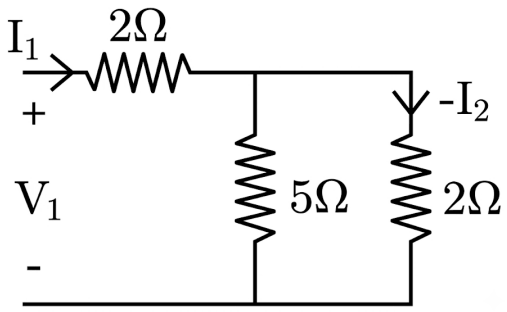


Fig. Solution Q6.18.1

When terminal 2 is short-circuited, the input current I_1 is given by:

$$I_1 = \frac{V_1}{2 + (5 \parallel 2)} = \frac{V_1}{2 + \frac{10}{7}}$$

Using the current division rule, the short-circuit output current $-I_2$ can be expressed in terms of I_1 as:

$$-I_2 = I_1 \times \frac{5}{5+2} = I_1 \times \frac{5}{7}$$

Substituting the value of I_1 into the current equation:

$$-I_2 = \frac{V_1 \times \frac{5}{7}}{2 + \frac{10}{7}} = \frac{5V_1}{24}$$

Rearranging the expression to find the ratio $\frac{V_1}{-I_2}$:

$$\frac{V_1}{-I_2} = \frac{24}{5} = 4.8$$

(You can also refer to solution of Q 6.26.1 for other methods of solving the same question)

4.3 to 5.3

Q6.16.1 MCQ 2016 1M Ans: D ● ○ ○

For a 2-port network to be reciprocal, the ABCD Parameters should be such that $AD - BC = 1$. Now, Determinant(T) = 1
$\therefore$ Determinant(T) = 1

$$\boxed{(d)\text{Determinant}(T) = 1}$$

Q6.16.2 NAT 2016 1M Ans: 2.98 to 3.02 ● ○ ○

Method 1

Z-parameter of T equivalent network is

$$[Z] = \begin{bmatrix} Z_a + Z_c & Z_c \\ Z_c & Z_b + Z_c \end{bmatrix}$$

Comparing it with

$$\begin{bmatrix} 2j\omega & j\omega \\ j\omega & 3 + j2\omega \end{bmatrix}$$

We get $Z_c = j\omega$ and $Z_b + Z_c = 3 + j2\omega$

$$Z_b + j\omega = 3 + j2\omega$$

$$Z_b = 3 + j\omega$$

$$\boxed{R_b = 3\,\Omega}$$

Method 2

From std Z parameter equations we have:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Let us consider Z_{22} where $Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$
$\therefore$ We need to keep Port 1 open.

$$V_2 = I_2(Z_b + Z_c)$$

$$\text{And } Z_{22} = Z_b + Z_c = 3 + 2j\omega$$

Next we consider Z_{12}, where $Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$
$\therefore$ Same condition of keeping Port 1 open holds.

$$V_1 = Z_c I_2$$

$$\therefore Z_{12} = Z_c = j\omega$$

Now, $Z_{22} = Z_b + Z_c$

$$Z_b = Z_{22} - Z_c = 3 + j\omega$$

Comparing with $Z_b(j\omega) = R_b + j\omega$, $R_b = 3\Omega$

$$\boxed{2.98 \text{ to } 3.02}$$

Q6.16.3 MCQ | 2016 | 2M | Ans: A | ●●○

The given circuit configuration presents a parallel arrangement of two resistors with specifically defined port voltages and currents. We need to determine the open-circuit impedance (Z) parameter matrix.

The standard two-port network definitions for Z-parameters are:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

We evaluate the parameters by setting one port current to zero at a time.

Case 1: Port-2 is open-circuited ($I_2 = 0$) When Port-2 is open, the current I_1 entering Port-1 flows through the parallel combination of the $3\,\Omega$ and $6\,\Omega$ resistors.

The equivalent input resistance looking into Port-1 is:

$$R_{eq} = 3 \parallel 6 = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\,\Omega$$

Therefore, the voltage V_1 across Port-1 is:

$$V_1 = I_1(3 \parallel 6) = 2I_1$$

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = 2\,\Omega$$

Now observe the polarity of V_2 relative to V_1. The reference positive terminal of V_1 is at the left node, whereas the reference positive terminal of V_2 is at the right node. Since the two branches are in parallel, the physical voltage across them is identical, but the defined polarity for V_2 is inverted relative to V_1:

$$V_2 = -V_1 = -2I_1$$

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = \frac{-2I_1}{I_1} = -2\,\Omega$$

Case 2: Port-1 is open-circuited ($I_1 = 0$) By network symmetry, when a current I_2 enters Port-2 with Port-1 left open, it sees the same parallel combination of $3\,\Omega$ and $6\,\Omega$:

$$V_2 = I_2(3 \parallel 6) = 2I_2$$

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} = 2\,\Omega$$

Due to the inverted polarity orientation across the parallel node pair, the voltage tracked at Port-1 is:

$$V_1 = -V_2 = -2I_2$$

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = \frac{-2I_2}{I_2} = -2\,\Omega$$

Assembling the calculated coefficients into the open-circuit impedance matrix:

$$[Z] = \begin{bmatrix} 2 & -4.5 \\ -4.5 & 2 \end{bmatrix} \Longrightarrow \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \Omega$$

Mistakes to be avoided

- **Polarity Inversion Blindness:** Always check the explicit + and − sign markings given on the circuit schematic. Inverting the reference terminal signs changes the sign of your transfer parameters (Z_{12}, Z_{21}).

$$\boxed{(a)\ \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}}$$

Q6.15.1 MCQ 2015 1M Ans: * ● ○ ○

Method 1

The given nodal equations for the two-port network are:

$$I_1 = \left(\frac{1}{5} + \frac{1}{10}\right) V_1 - \frac{1}{5} V_2$$

$$I_2 = \left(-\frac{1}{5}\right) V_1 + \left(\frac{1}{5} + \frac{1}{10}\right) V_2$$

Simplifying the coefficients:

$$\frac{1}{5} + \frac{1}{10} = 0.2 + 0.1 = 0.3$$

$$-\frac{1}{5} = -0.2$$

Substituting these values back into the equations, we get:

$$I_1 = 0.3V_1 - 0.2V_2$$

$$I_2 = -0.2V_1 + 0.3V_2$$

Expressing the equations in the standard admittance parameter (Y-parameter) matrix form:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = [Y] \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$[Y] = \begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$$

(If we ignore the negative sign, then only option A will match).

Method 2

For a π-model representation of a two-port network, elements of the admittance matrix is given by: $Y_{11} = Y_a + Y_b$, $Y_{12} = Y_{21} = -Y_b$, and $Y_{22} = Y_c + Y_b$. Here $Y_a = \frac{1}{10}$, $Y_b = \frac{1}{5}$ and $Y_c = \frac{1}{10}$. Hence we can conclude :

- $Y_{12} = Y_{21} = -\frac{1}{5} = -0.2\,\text{S}$
- $Y_{11} = \frac{1}{10} + \frac{1}{5} = \frac{3}{10}\,\text{S}$
- $Y_{22} = \frac{1}{10} + \frac{1}{5} = \frac{3}{10}\,\text{S}$

(If we ignore the negative sign, then only option A will match).

$$\boxed{*(a) \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}}$$

Q6.15.2 MCQ 2015 2M Ans: B ● ● ○

Given the network equations:

$$V_1 = (7 + j4)I_1 + 2I_2 \quad \text{— (1)}$$

$$V_2 = 2I_1 + (7 - j4)I_2 \quad \text{— (2)}$$

From equation (2), we can express I_1 in terms of V_2 and I_2:

$$(7 - j4)I_1 = V_2 - 2I_2$$

$$I_1 = \frac{V_2}{7 - j4} - \frac{2}{7 - j4} I_2$$

Alternatively, rewriting equation (2) directly as presented in the steps to solve for I_1:

$$I_1 = \frac{V_2}{2} - \frac{(7 - j4)}{2} I_2$$

$$I_1 = 0.5V_2 - (3.5 - j2)I_2 \quad \text{— (3)}$$

By substituting equation (3) into equation (1):

$$V_1 = (7 + j4)\,[0.5V_2 - (3.5 - j2)I_2] + 2I_2$$

Multiplying the terms:

$$(7 + j4)(0.5) = 3.5 + j2$$

$$(7 + j4)(3.5 - j2) = 24.5 - j14 + j14 - j^2 8 = 24.5 + 8 = 32.5$$

Substituting these back into the equation for V_1:

$$V_1 = (3.5 + j2)V_2 - 32.5I_2 + 2I_2$$

$$V_1 = (3.5 + j2)V_2 - 30.5I_2$$

The standard transmission parameters (T-parameters or $ABCD$ parameters) are defined by:

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Comparing the derived equations with the standard definition, the transmission matrix $[T]$ is:

$$[T] = \begin{bmatrix} 3.5 + j2 & 30.5 \\ 0.5 & 3.5 - j2 \end{bmatrix}$$

We can solve this problem using some other methods as well. You can refer to the solution for Qs. 6.26.1 for other approaches to this type of problem. The only difference between these two problems is that one deals with real impedances while another with complex ones.

$$\boxed{(b)\ \begin{bmatrix} 3.5 + j2 & 30.5 \\ 0.5 & 3.5 - j2 \end{bmatrix}}$$

Q6.14.1 NAT 2014 2M Ans: 1.24 to 1.26 ● ● ○

The hybrid parameters (h-parameters) of a two-port network are defined by the following system of equations:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

The output admittance parameter h_{22} is defined under the open-circuited input condition ($I_1 = 0$):

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

Method 1

We can simplify the network into a standard π - form if we consider the resistances connected in parallel to each other (voltages on both ends of a resistor being equal implies parallel connection).

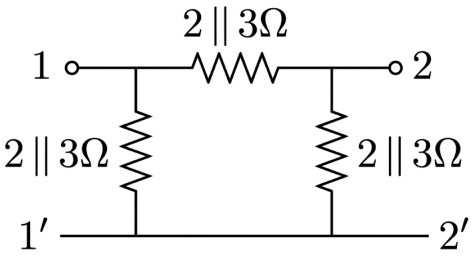


Fig. Solution Q6.14.1

Let $R = 2||3\Omega = \frac{6}{5}\Omega$ For $I_1 = 0$, Port 1 should be open.

$$\frac{V_2}{R} + \frac{V_2}{2R} = I_2$$

$$V_2\left(\frac{1}{R} + \frac{1}{2R}\right) = I_2$$

$$V_2\left(\frac{3}{2R}\right) = I_2$$

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0} = \frac{3\times 5}{2\times 6} = \frac{5}{4} = 1.25$$

Method 2

Step 1: Determine the Admittance (Y) Matrices of individual sub-networks The complex composite structure consists of two symmetric Π-networks connected directly in parallel. Let us compute the individual Y-parameter matrices for both systems.

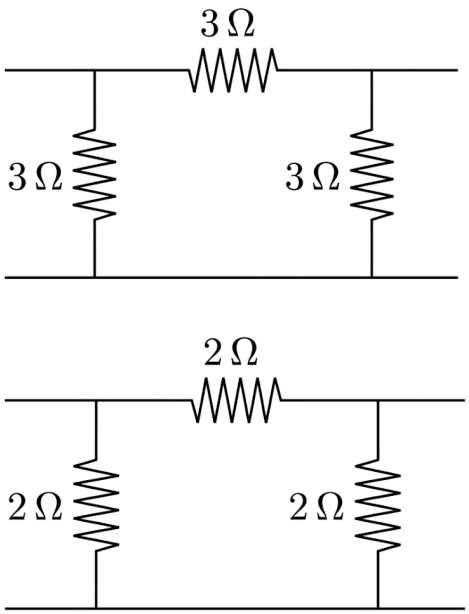


Fig. Solution Q6.14.1

For the upper sub-network containing 3 Ω resistors arranged in a Π-profile:

$$[Y_A] = \begin{bmatrix} \frac{1}{3}+\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3}+\frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix} \text{ S}$$

For the lower sub-network containing 2 Ω resistors arranged in a similar Π-profile:

$$[Y_B] = \begin{bmatrix} \frac{1}{2}+\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2}+\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} \text{ S}$$

Step 2: Combine Parallel Networks Since both networks share identical parallel node boundaries, the total short-circuit admittance matrix $[Y]$ is the direct algebraic sum of their individual matrices:

$$[Y] = [Y_A] + [Y_B]$$

$$[Y] = \begin{bmatrix} \frac{2}{3}+1 & -\frac{1}{3}-\frac{1}{2} \\ -\frac{1}{3}-\frac{1}{2} & \frac{2}{3}+1 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} & -\frac{5}{6} \\ -\frac{5}{6} & \frac{5}{3} \end{bmatrix} \text{ S}$$

From the complete combined admittance system matrix, we express the terminal current expressions as:

$$I_1 = \frac{5}{3}V_1 - \frac{5}{6}V_2 \quad \ldots(1)$$

$$I_2 = -\frac{5}{6}V_1 + \frac{5}{3}V_2 \quad \ldots(2)$$

Step 3: Calculate the Target parameter h_{22} To evaluate under the condition $I_1 = 0$, substitute this constraint directly into expression (1):

$$0 = \frac{5}{3}V_1 - \frac{5}{6}V_2 \implies \frac{5}{3}V_1 = \frac{5}{6}V_2$$

$$V_1 = \frac{3}{5} \times \frac{5}{6}V_2 = \frac{1}{2}V_2$$

Now, substitute the resulting node relationship $V_1 = \frac{1}{2}V_2$ directly into current expression (2):

$$I_2 = -\frac{5}{6}\left(\frac{1}{2}V_2\right) + \frac{5}{3}V_2$$

$$I_2 = \left(-\frac{5}{12} + \frac{5}{3}\right)V_2 = \left(-\frac{5}{12} + \frac{20}{12}\right)V_2$$

$$I_2 = \frac{15}{12}V_2 = \frac{5}{4}V_2$$

Thus, the short-circuit parameter output admittance value yields:

$$h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0} = \frac{5}{4} = 1.25 \text{ S}$$

Key Points

- **Parallel Interconnections:** Admittance (Y) matrices are linearly additive when two independent two-port networks are linked in a parallel-parallel alignment.

Mistakes to be avoided

- **Sign Accuracy during Matrix Addition:** Ensure precise adherence to standard Π-model parameter layouts; the transfer terms must retain their negative notations ($Y_{12} = Y_{21} = -Y_{series}$).

1.24 to 1.26

Q6.14.2 MCQ | 2014 | 2M | Ans: C | ● ○ ○

Method 1

The network is in the standard π form, for which we can compute the Y - parameters. The Y-parameters are calculated as:

$$Y_{11} = \frac{1}{10} + \frac{1}{30} = \frac{4}{30} = \frac{2}{15}\,\text{S}$$

$$Y_{21} = Y_{12} = -\frac{1}{30}\,\text{S}$$

$$Y_{22} = \frac{1}{60} + \frac{1}{30} = \frac{3}{60} = \frac{1}{20}\,\text{S}$$

Thus, the admittance matrix Y is:

$$Y = \begin{bmatrix} \frac{2}{15} & -\frac{1}{30} \\ -\frac{1}{30} & \frac{1}{20} \end{bmatrix}$$

To find the impedance parameters (Z-parameters), we take the inverse of the Y matrix:

$$Z = Y^{-1} = \frac{\text{adj}(Y)}{|Y|}$$

The determinant of the matrix Y can be found as follows :

$$|Y| = \left(\frac{2}{15} \times \frac{1}{20}\right) - \left(-\frac{1}{30} \times -\frac{1}{30}\right)$$

$$|Y| = \frac{2}{300} - \frac{1}{900} = \frac{6-1}{900} = \frac{5}{900} = \frac{1}{180}$$

Now, calculating the Z-matrix:

$$Z = 180 \begin{bmatrix} \frac{1}{20} & \frac{1}{30} \\ \frac{1}{30} & \frac{2}{15} \end{bmatrix} = \begin{bmatrix} 180 \times \frac{1}{20} & 180 \times \frac{1}{30} \\ 180 \times \frac{1}{30} & 180 \times \frac{2}{15} \end{bmatrix}$$

$$Z = \begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$$

The final Z-parameter matrix is:

$$\begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$$

Method 2

We can convert a π-network (delta) to a T-network (star) by star-delta conversion, and the individual star resistances are calculated as:

$$R_1 = \frac{30 \times 10}{30 + 10 + 60} = 3\,\Omega$$

$$R_2 = \frac{60 \times 30}{30 + 10 + 60} = 18\,\Omega$$

$$R_3 = \frac{60 \times 10}{30 + 10 + 60} = 6\,\Omega$$

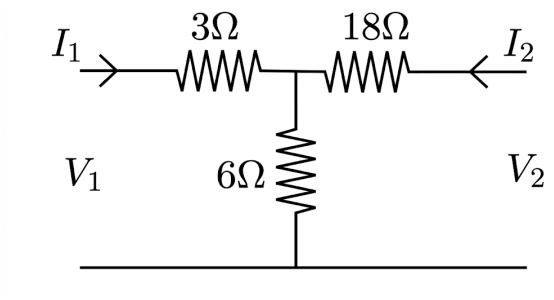


Fig. Solution Q6.14.2

Using Kirchhoff's Voltage Law (KVL) for the resulting T-network, the loop equations for the two ports are expressed as:

$$V_1 = 3I_1 + 6(I_1 + I_2) = 9I_1 + 6I_2$$

$$V_2 = 18I_2 + 6(I_1 + I_2) = 6I_1 + 24I_2$$

Comparing these equations with the standard impedance parameter (Z-parameter) matrix form:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Thus, the impedance matrix $[Z]$ is given by:

$$[Z] = \begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$$

Method 3

We can find the parameters by opening both ports one by one.

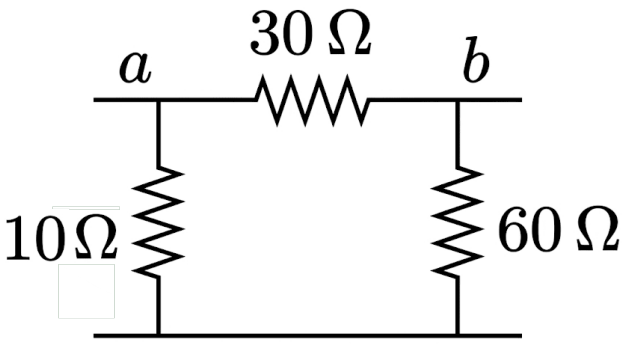


Fig. Solution Q6.14.2

i) Case 1: $I_1 = 0$ **(Port-1 open)** Applying KCL at node b:

$$\frac{V_2}{60} + \frac{V_2}{40} = I_2$$

$$\frac{2V_2 + 3V_2}{120} = I_2 \implies \frac{5V_2}{120} = I_2$$

$$\frac{V_2}{24} = I_2$$

$$z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} = 24\,\Omega$$

By voltage division rule, the voltage V_1 at node a is:

$$V_1 = V_2 \times \frac{10}{10+30} = V_2 \times \frac{10}{40} = \frac{V_2}{4}$$

Substituting $V_2 = 24I_2$:

$$V_1 = \frac{24I_2}{4} = 6I_2$$

$$z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = 6\,\Omega$$

ii) Case 2: $I_2 = 0$ **(Port-2 open)** Applying KCL at node a:

$$\frac{V_1}{10} + \frac{V_1}{30+60} = I_1$$

$$\frac{V_1}{10} + \frac{V_1}{90} = I_1$$

$$\frac{9V_1 + V_1}{90} = I_1 \implies \frac{10V_1}{90} = I_1$$

$$\frac{V_1}{9} = I_1$$

$$z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = 9\,\Omega$$

By voltage division rule, the voltage V_2 at node b is:

$$V_2 = V_1 \times \frac{60}{30+60} = V_1 \times \frac{60}{90} = \frac{2V_1}{3}$$

Substituting $V_1 = 9I_1$:

$$V_2 = \frac{2(9I_1)}{3} = 6I_1$$

$$z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = 6\,\Omega$$

Thus, the standard Z-parameter matrix is:

$$Z = \begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$$

$$\boxed{(c)\ \begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}}$$

Q6.12.1 MCQ | 2012 | 2M | Ans: C | ● ○ ○

Based on the experimental conditions given for the linear network, we can determine the Thevenin equivalent parameters (V_{th} and R_{th}) across the load terminals.

From (i): When the load resistance $R_L = 1\,\Omega$, the current through the load is $I = 3$ A for a source voltage $V_{DC} = 10$ V. Using Thevenin's formula:

$$V_{th} = I(R_{th} + R_L)$$

$$V_{th} = 3(R_{th} + 1)$$

$$V_{th} = 3R_{th} + 3 \quad \text{— (1)}$$

From (ii): When the load resistance is changed to $R_L = 2.5\,\Omega$, the current reduces to $I = 2\,\text{A}$ for the same source voltage $V_{DC} = 10\,\text{V}$.

$$V_{th} = 2(R_{th} + 2.5)$$

$$V_{th} = 2R_{th} + 5 \quad \text{— (2)}$$

Solving Equations (1) and (2): Equating the expressions for V_{th}:

$$3R_{th} + 3 = 2R_{th} + 5$$

$$R_{th} = 2\,\Omega$$

Substituting $R_{th} = 2\,\Omega$ back into equation (1):

$$V_{th} = 3(2) + 3 = 9\,\text{V}$$

When the load resistance is updated to $R_L = 7\,\Omega$:

$$I = \frac{V_{th}}{R_{th} + R_L}$$

$$I = \frac{9}{2+7} = \frac{9}{9} = 1\,\text{A}$$

$$\boxed{(c)\,1A}$$

Q6.12.2 MCQ 2012 2M Ans: C ●●○

Based on the provided imformation, the network contains dependent sources, as it is linear but non-reciprocal. Consequently, the Thevenin voltage V_{th} varies linearly with the independent applied voltage V, while the Thevenin resistance R_{th} remains constant at $2\,\Omega$ (as it does not depend on the independent voltage source magnitude since independent sources are short-circuited during its calculation).

As per problem, an independent voltage source $V_{DC} = 6\,\text{V}$ and load resistance $R_L = 1\,\Omega$, the current is given as $I = \frac{7}{3}\,\text{A}$. The Thevenin voltage for this specific condition is:

$$V_{th} = I(R_{th} + R_L)$$

$$V_{th} = \frac{7}{3}(2+1) = 7\,\text{V}$$

Since the network is linear and contains a dependent voltage source, V_{th} is a linear function of the applied voltage V:

$$V_{th} = aV + b$$

We have two data points to determine the constants a and b: 1. From the previous problem setup: When $V = 10\,\text{V}$, $V_{th} = 9\,\text{V}$.

$$9 = a(10) + b \quad \text{— (1)}$$

2. From the current condition: When $V = 6\,\text{V}$, $V_{th} = 7\,\text{V}$.

$$7 = a(6) + b \quad \text{— (2)}$$

Subtracting equation (2) from equation (1):

$$(10a + b) - (6a + b) = 9 - 7$$

$$4a = 2 \implies a = \frac{1}{2}$$

Substituting $a = \frac{1}{2}$ into equation (2):

$$7 = 6\left(\frac{1}{2}\right) + b \implies 7 = 3 + b \implies b = 4$$

Thus, the general equation for the Thevenin voltage is:

$$V_{\text{th}} = \frac{V}{2} + 4$$

Substituting $V = 8\,\text{V}$ into the linear relationship equation:

$$V_{\text{th}} = \frac{8}{2} + 4 = 4 + 4 = 8\,\text{V}$$

Key Points

- **Linear Dependency with Dependent Sources:** In networks containing both independent and dependent sources, the Thevenin open-circuit voltage V_{th} scales linearly with changes in the independent excitation source, taking the mathematical form $V_{\text{th}} = aV + b$.
- **Invariance of R_{th}:** The equivalent Thevenin resistance R_{th} is independent of the value of independent sources because all independent sources are deactivated (short-circuited for voltage sources, open-circuited for current sources) during its computation.

Mistakes to be avoided

- **Incorrect Assumption of Proportionality:** Do not assume V_{th} is directly proportional to V (i.e., $V_{\text{th}} = aV$) if dependent sources are present; internal independent elements can create a constant offset (b). Always solve for the full linear relationship $aV + b$.

$\boxed{(c) 8V}$

Q6.11.1 MCQ | 2011 | 2M | Ans: D | ● ○ ○

The standard admittance (Y) parameter equation for the output port current is:

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

(Note we do not need to consider the admittance (Y) parameter equation for the input port current). Given the parameter values from the problem statement:

$$I_2 = 0.01V_1 + 0.1V_2 \quad \text{— (1)}$$

From the output port termination with a load resistance $R_L = 100\,\Omega$, the voltage V_2 drop relative to the reference terminal satisfies Ohm's law with a negative sign due to the standard passive sign convention for two-port currents:

$$V_2 = -I_2R_L = -100I_2$$

$$I_2 = -\frac{V_2}{100}$$

Substituting the expression for I_2 back into equation (i):

$$-\frac{V_2}{100} = 0.01V_1 + 0.1V_2$$

Expressing $-\frac{1}{100}$ as a decimal (0.01):

$$-0.01V_2 = 0.01V_1 + 0.1V_2$$

$$-0.01V_2 - 0.1V_2 = 0.01V_1$$

$$-0.11V_2 = 0.01V_1$$

Rearranging the terms to determine the forward voltage gain ratio $\frac{V_2}{V_1}$:

$$\frac{V_2}{V_1} = -\frac{0.01}{0.11}$$

$$\frac{V_2}{V_1} = -\frac{1}{11}$$

$$\boxed{(d) -1/11}$$

Q6.08.1 MCQ 2008 2M Ans: C ● ● ○

When S_1 is open and S_2 is closed

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

Since S_1 is open, it implies I_1 is 0, as we can see from the ammeter reading. I_2 is given as 1A and V_1 as 4.5V.

$$4.5 = Z_{12} \times 1$$

$$\therefore Z_{12} = 4.5$$

And $V_2 = Z_{21}I_1 + Z_{22}I_2$ Similarly, since V_2 is 1.5V,

$$1.5 = Z_{22} \times 1 \implies Z_{22} = 1.5$$

When S_1 closed and S_2 open Now S_2 is open, it implies I_2 is 0, as we can see from the ammeter reading. I_1 is given as 4A and V_1 as 6V.

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$6 = Z_{11} \times 4 \implies Z_{11} = 1.5$$

And $V_2 = Z_{21}I_1 + Z_{22}I_2$ Similarly, since V_2 is 6V,

$$6 = Z_{21} \times 4 \implies Z_{21} = 1.5$$

Thus we can form the matrix as :

$$Z = \begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 1.5 \end{bmatrix}$$

$$\boxed{(c) \begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 1.5 \end{bmatrix}}$$

Q6.08.2 MCQ 2008 2M Ans: A ● ● ○

Method 1

When S_1 is open and S_2 is closed

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$4.5 = h_{12} \times 1.5 \text{ [Since } I_1 \text{ is 0A]}$$

$$\therefore h_{12} = 3$$

$I_2 = h_{21}I_1 + h_{22}V_2$ Similarly

$$1 = h_{22} \times 1.5$$

$$h_{22} = \frac{2}{3} = 0.67$$

When S_1 is closed and S_2 is open

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$6 = h_{11} \times 4 + 3 \times 6$$

$$h_{11} = -3$$

Similarly $I_2 = h_{21}I_1 + h_{22}V_2$

$$0 = h_{21} \times 4 + h_{22} \times 6$$

$$h_{21} = -1$$

Thus we can form the matrix as :

$$H = \begin{bmatrix} -3 & 3 \\ -1 & 0.67 \end{bmatrix}$$

Method 2

We have the z-parameter matrix as :

$$Z = \begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 1.5 \end{bmatrix}$$

From this we can write:-

$$V_1 = 1.5I_1 + 4.5I_2 \quad -(1)$$

$$V_2 = 1.5I_1 + 1.5I_2 \quad -(2)$$

From (2),

$$I_2 = \frac{V_2}{1.5} - I_1 \quad -(3)$$ Thus $h_{21} = -1$ and $h_{22} = 0.67$

Putting (3) in (1)

$$V_1 = 1.5I_1 + 4.5\left(\frac{V_2}{1.5} - I_1\right)$$

$$= -3I_1 + 3V_2$$

Thus $h_{11} = -3$ and $h_{12} = 3$.
Hence we write the matrix as:-

$$H = \begin{bmatrix} -3 & 3 \\ -1 & 0.67 \end{bmatrix}$$

Key Points

- Here the polarity of ammeter I_2 is such that it measures direction of current going inwards into the 2 port network to be positive, which is the conventional direction of I_2 in standard 2 port parameters. However, it must be noted that in case the ammeter polarity was marked to be opposite, a suitable adjustment had to be made (taking $-I_2$ in calculations).

$$\boxed{(a) \begin{bmatrix} -3 & 3 \\ -1 & 0.67 \end{bmatrix}}$$

Q6.06.1 MCQ 2006 2M Ans: B ● ○ ○

Given the circuit-derived mesh equations for the two-port network:

$$V_1 = r_e I_1$$

$$V_2 = (I_2 - \beta I_1) r_0$$

Expanding the expression for V_2 gives:

$$V_2 = -\beta r_0 I_1 + r_0 I_2$$

The standard open-circuit impedance (Z-parameter) equations for a two-port network are defined as:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

By comparing the coefficients of I_1 and I_2 from the circuit equations with the standard Z-parameter equations, we obtain:
For the V_1 equation:

$$Z_{11} = r_e$$

$$Z_{12} = 0$$

For the V_2 equation:

$$Z_{21} = -\beta r_0$$

$$Z_{22} = r_0$$

Key Points

- Other methods like opening one port at a time can also be used here.

$$\boxed{(b) 0 \text{ and } -\beta r_0}$$

Q6.06.2 MCQ 2006 2M Ans: D ● ● ○

To find the input impedance of a two-port transmission network characterized by its ABCD parameters, we begin with the standard transmission line equations:

$$V_1 = AV_2 - BI_2 \quad \text{— (1)}$$

$$I_1 = CV_2 - DI_2 \quad \text{— (2)}$$

Additionally, since Port 2 terminates in a load impedance R_L, the terminal condition dictates:

$$V_2 = -I_2 R_L \quad \text{— (3)}$$

Step 1: Express I_2 in terms of I_1 Substituting equation (3) into equation (2):

$$I_1 = C(-I_2 R_L) - DI_2$$

$$I_1 = -(CR_L + D) I_2$$

$$I_2 = \frac{-I_1}{CR_L + D} \quad \text{— (4)}$$

Step 2: Express V_1 in terms of I_2 Substituting equation (3) into equation (1):

$$V_1 = A(-I_2 R_L) - BI_2$$

$$V_1 = -(AR_L + B) I_2 \quad \text{— (5)}$$

Step 3: Derivation of Input Impedance (Z_{in}) Substituting equation (4) into equation (5):

$$V_1 = -(AR_L + B)\left(\frac{-I_1}{CR_L + D}\right)$$

$$V_1 = \frac{AR_L + B}{CR_L + D} I_1$$

The input impedance Z_{in} looking into Port 1 is the ratio of input voltage to input current:

$$Z_{in} = \frac{V_1}{I_1} = \boxed{\frac{AR_L + B}{CR_L + D}}$$

$$\boxed{(d)\frac{B + AR_L}{D + CR_L}}$$

Q6.05.1 MCQ | 2005 | 2M | Ans: D | ● ● ○

Method 1

To determine the hybrid (h) parameters of the given two-port network, we compare the circuit equations with the standard two-port h-parameter definitions.

The standard hybrid parameter network equations are defined as:

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Step 1: Formulate the First Network Equation From the given loop analysis, applying Kirchhoff's Voltage Law (KVL) yields:

$$V_1 - 10I_1 - V_2 = 0$$

$$V_1 = 10I_1 + V_2 \quad \text{— (1)}$$

Comparing equation (1) with the standard expression for V_1: $h_{11} = 10\,\Omega$ $h_{12} = 1$ Step 2: Formulate the Second Network Equation The node voltage relationship at the output port is given as:

$$V_2 = 20(I_1 + I_2)$$

Rearranging the terms to express output current I_2 as a function of I_1 and V_2:

$$\frac{V_2}{20} = I_1 + I_2$$

$$I_2 = -I_1 + \frac{1}{20} V_2 \quad \text{— (2)}$$

Comparing equation (2) with the standard expression for I_2: $h_{21} = -1$ $h_{22} = \frac{1}{20}$ ℧ (or S)

Step 3: Matrix Representation Combining the individual parameter values into the standard matrix format $\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}$$

Method 2

The standard defining equations for h-parameters are:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

From these equations, the individual parameters are defined under specific open-circuit or short-circuit conditions:

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0}, \quad h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}, \quad h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}, \quad h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

Thus, we need to investigate the network under two distinct operating conditions:

a) Shorting Port 2 ($V_2 = 0$)

b) Opening Port 1 ($I_1 = 0$)

Condition a): Shorting Port 2 ($V_2 = 0$)

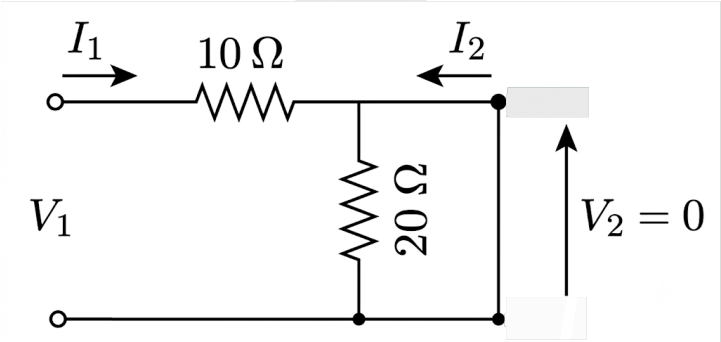


Fig. Solution Q6.05.1

When Port 2 is short-circuited, the parallel 20 Ω resistor is bypassed by the short, meaning no current flows through it. The entire current I_1 flows straight out through the short-circuited output terminal in opposition to the defined direction of I_2:

$$I_1 = -I_2 \implies h_{21} = \frac{I_2}{I_1} = -1$$

The input voltage V_1 drops entirely across the series input resistor:

$$V_1 = 10I_1 \implies h_{11} = \frac{V_1}{I_1} = 10\,\Omega$$

Condition b): Opening Port 1 ($I_1 = 0$)

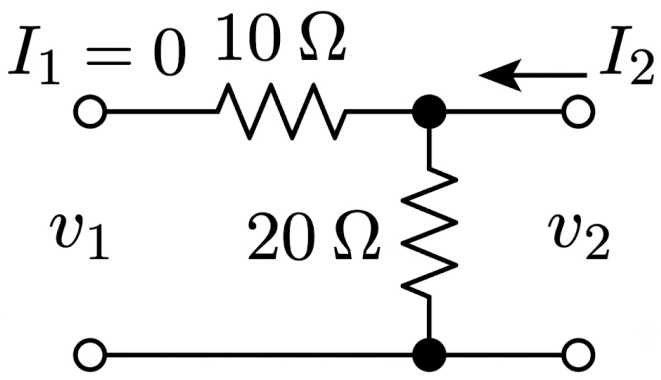


Fig. Solution Q6.05.1

When Port 1 is left open, the input current is zero ($I_1 = 0$). Since no current flows through the $10\,\Omega$ resistor, there is no voltage drop across it. Therefore, the open-circuit input voltage equals the voltage across the shunt resistor:

$$V_1 = V_2 \implies h_{12} = \frac{V_1}{V_2} = 1$$

The output current I_2 passes entirely through the $20\,\Omega$ shunt resistor, establishing the output voltage:

$$V_2 = 20I_2 \implies h_{22} = \frac{I_2}{V_2} = \frac{1}{20} = 0.05\,\text{S}$$

Assembling the computed values into the h parameter matrix:

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}$$

$$\boxed{(d)\ \begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}}$$

Q6.05.2 MCQ 2005 1M Ans: B ●●○

To find the transmission ($ABCD$) parameters of an ideal transformer, we analyze the relationship between the primary and secondary voltages and currents based on the turns ratio $n : 1$.

For an ideal transformer with a turns ratio of $n : 1$, the current relationship is:

$$\frac{I_2}{I_1} = \frac{n}{1} \implies I_1 = \frac{1}{n} I_2 \quad \text{— (1)}$$

Similarly, the voltage relationship is given by:

$$\frac{V_1}{V_2} = \frac{n}{1} \implies V_1 = nV_2 \quad \text{— (2)}$$

Step 1: Standard Two-Port Transmission Equations The standard $ABCD$ parameter equations for a two-port network are defined as:

$$V_1 = AV_2 - BI_2$$
$$I_1 = CV_2 - DI_2$$

In standard transmission line modeling, I_2 is defined as flowing *out* of the port. If we match equations (1) and (2) by considering the direction matching conventions for an ideal transmission element:

$$V_1 = nV_2 + 0(-I_2)$$
$$I_1 = 0V_2 + \frac{1}{n}(-I_2)$$

Step 2: Parameter Matrix Extraction Comparing the coefficients of equations (1) and (2) with the standard two-port transmission parameter equations yields: $A = n$, $B = 0$, $C = 0$, $D = \frac{1}{n}$

Expressing this relationship in matrix form:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & \frac{1}{n} \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

From the problem statement, evaluating the target variable x associated with the parameter element D:

$$x = \boxed{\frac{1}{n}}$$

$$\boxed{(b)\ \frac{1}{n}}$$

Q6.04.1 MCQ 2004 2M Ans: D ●●○

To find the open-circuit impedance (Z) parameters of the given symmetrical lattice network, we analyze the circuit under standard open-circuit terminal conditions.

The standard two-port network equations for Z-parameters are defined as:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Step 1: Open-Circuit Port 2 ($I_2 = 0$) When Port 2 is open-circuited ($I_2 = 0$), the input current I_1 enters Port 1 and divides equally between the two identical parallel lattice paths because each branch contains a series combination of the series arm impedance Z_b and diagonal arm impedance Z_a:

$$I_{\text{branch}} = \frac{I_1}{2}$$

Applying Kirchhoff's Voltage Law (KVL) to find the port voltages V_1 and V_2:

$$V_1 = \frac{I_1}{2}(Z_a + Z_b) \implies Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = \frac{Z_a + Z_b}{2}$$

$$V_2 = \frac{I_1}{2}(Z_a - Z_b) \implies Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0} = \frac{Z_a - Z_b}{2}$$

Step 2: Open-Circuit Port 1 ($I_1 = 0$) Due to the perfect structural symmetry of the lattice network, opening Port 1 ($I_1 = 0$) yields an identical current division for I_2:

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} = \frac{Z_a + Z_b}{2}$$

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = \frac{Z_a - Z_b}{2}$$

Step 3: Numerical Evaluation Given the branch impedance parameters from the problem notes:

$$Z_a = j2\,\Omega$$

$$Z_b = 2\,\Omega$$

Substituting these values into our derived parameter relationships: For the driving-point impedances (Z_{11} and Z_{22}):

$$\frac{Z_a + Z_b}{2} = \frac{j2 + 2}{2} = (1 + j)\,\Omega$$

For the transfer impedances (Z_{12} and Z_{21}):

$$\frac{Z_a - Z_b}{2} = \frac{j2 - 2}{2} = (-1 + j)\,\Omega$$

Assembling the parameters into the standard matrix format $[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$:

$$[Z] = \boxed{\begin{bmatrix} 1 + j & -1 + j \\ -1 + j & 1 + j \end{bmatrix}}$$

Mistakes to be avoided

- **Transfer Impedance Sign Convention:** Ensure correct branch loop tracking when deriving $V_2 = V_3 - V_4$: terminal 3 is reached from terminal 1 through the series arm Z_b and terminal 4 through the diagonal arm Z_a. Flipping the subtraction order to $(Z_b - Z_a)$ gives $(1 - j)$, which matches no option.

$$\boxed{(d)\ \begin{bmatrix} 1 + j & -1 + j \\ -1 + j & 1 + j \end{bmatrix}}$$

Q6.03.1 MCQ 2003 2M Ans: A ●●○

To find the open-circuit impedance parameters (Z_{11} and Z_{12}) of the given two-port network, we perform a delta-to-star (Δ-Y) transformation on the internal network elements to simplify the circuit configuration.

Step 1: Delta-to-Star (Δ-Y) Transformation The inner delta network consists of three resistors: $2\,\Omega$, $1\,\Omega$, and $1\,\Omega$. Converting this into an equivalent star network gives the star resistances R_1, R_2, and R_3:

$$R_1 = R_2 = \frac{2 \times 1}{1 + 1 + 2} = \frac{2}{4} = \frac{1}{2}\,\Omega$$

$$R_3 = \frac{1 \times 1}{1 + 1 + 2} = \frac{1}{4}\,\Omega$$

Step 2: Total Loop Impedance Simplification Combining the transformed star parameters with the existing external series elements ($2\,\Omega$ at Port 1 and $3\,\Omega$ at Port 2) provides the simplified T-network: Total input branch resistance:

$$2 + \frac{1}{2} = 2.5\,\Omega$$

Total output branch resistance:

$$3 + \frac{1}{2} = 3.5\,\Omega$$

Central shunt branch resistance:

$$R_3 = \frac{1}{4} = 0.25\,\Omega$$

Step 3: Loop Analysis using KVL Applying Kirchhoff's Voltage Law (KVL) to the input loop yields:

$$V_1 = 2.5I_1 + 0.25(I_1 + I_2)$$

$$V_1 = 2.75I_1 + 0.25I_2 \quad \text{— (1)}$$

Applying KVL to the output loop yields:

$$V_2 = 3.5I_2 + 0.25(I_1 + I_2)$$

$$V_2 = 0.25I_1 + 3.75I_2 \quad \text{— (2)}$$

Step 4: Parameter Extraction The standard parameter equations for an open-circuit impedance (Z) network are defined as:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Comparing equations (1) and (2) with the standard parameter expressions gives:

$$Z_{11} = \boxed{2.75\,\Omega}$$

$$Z_{12} = \boxed{0.25\,\Omega}$$

Key Points

- Other methods like opening one port at a time can also be used.

$$\boxed{(a) z_{11} = 2.75\Omega \text{ and } z_{12} = 0.25\Omega}$$

Q6.02.1 NAT | 2002 | 5M | Ans: * | ● ● ○

a) To find the short-circuit admittance (Y) parameters of the given two-port network containing a dependent current source, we perform nodal analysis at Node 1 (V_1) and Node 2 (V_2). The standard two-port network equations for Y-parameters are defined as:

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

From the circuit diagram, the controlling current I_x flowing through the 5 Ω resistor from Node 1 to Node 2 is given by:

$$I_x = \frac{V_1 - V_2}{5}$$

Step 1: Nodal Analysis at Node 1 Applying Kirchhoff's Current Law (KCL) at Node 1:

$$-I_1 + I_x + \frac{V_1 - V_2}{10} = 0$$

$$I_1 = I_x + \frac{V_1 - V_2}{10}$$

Substituting the expression for I_x:

$$I_1 = \frac{V_1 - V_2}{5} + \frac{V_1 - V_2}{10}$$

$$I_1 = (V_1 - V_2)\left(\frac{1}{5} + \frac{1}{10}\right)$$

$$I_1 = 0.3V_1 - 0.3V_2 \quad \text{— (1)}$$

Comparing equation (1) with the standard admittance equation for I_1: $Y_{11} = \boxed{0.3\,\text{S}}$ $Y_{12} = \boxed{-0.3\,\text{S}}$ Step 2: Nodal Analysis at Node 2 Applying Kirchhoff's Current Law (KCL) at Node 2:

$$-I_2 + 2I_x + \frac{V_2}{10} + \frac{V_2 - V_1}{10} - I_x = 0$$

$$I_2 = I_x + \frac{V_2}{10} + \frac{V_2 - V_1}{10}$$

Substituting the expression for I_x:

$$I_2 = \left(\frac{V_1 - V_2}{5}\right) + \frac{V_2}{10} + \frac{V_2 - V_1}{10}$$

Grouping the coefficients by V_1 and V_2:

$$I_2 = V_1\left(\frac{1}{5} - \frac{1}{10}\right) + V_2\left(\frac{1}{10} + \frac{1}{10} - \frac{1}{5}\right)$$

$$I_2 = 0.1V_1 + 0V_2 \quad \text{— (2)}$$

Comparing equation (2) with the standard admittance equation for I_2: $Y_{21} = \boxed{0.1\,\text{S}}$ $Y_{22} = \boxed{0\,\text{S}}$ The Y parameter matrix is : $\begin{bmatrix} 0.3 & -0.3 \\ 0.1 & 0 \end{bmatrix}$

b) This can be done using the following methods

Method 1

To find the open-circuit output driving-point impedance (Z_{22}) we put $I_1 = 0$ as

$$Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0} \quad \text{— (1)}$$

The standard system describing the network using short-circuit admittance (Y) parameters is:

$$I_1 = y_{11}V_1 + y_{12}V_2$$

$$I_2 = y_{21}V_1 + y_{22}V_2$$

Substituting the Port 1 open-circuit condition ($I_1 = 0$) into the first admittance equation yields:

$$0 = y_{11}V_1 + y_{12}V_2 \implies V_1 = -\frac{y_{12}}{y_{11}}V_2 \quad \text{— (2)}$$

Substituting equation (2) into the second admittance expression to isolate the terminal variable I_2:

$$I_2 = y_{21}\left[-\frac{y_{12}V_2}{y_{11}}\right] + y_{22}V_2$$

Factoring out the shared voltage term V_2 and establishing a common denominator:

$$I_2 = \left[\frac{y_{11}y_{22} - y_{12}y_{21}}{y_{11}}\right]V_2 \quad \text{— (3)}$$

The numerator of equation (3) represents the matrix determinant of the admittance parameters (ΔY):

$$\Delta Y = y_{11}y_{22} - y_{12}y_{21}$$

Rearranging equation (3) to calculate the ratio required by equation (1) provides the structural conversion formula:

$$Z_{22} = \frac{V_2}{I_2} = \frac{y_{11}}{y_{11}y_{22} - y_{12}y_{21}} = \frac{y_{11}}{\Delta Y}$$

Using the specific circuit admittance parameter values given in the reference solution notes ($y_{11} = 0.3\,\text{S}$ and $\Delta Y = 0.03\,\text{S}^2$):

$$Z_{22} = \frac{0.3}{0.03} = \boxed{10\,\Omega}$$

Method 2

We can find the Z-parameter Z_{22} from the inverse of the Y-parameter matrix.

$$[Z] = [Y]^{-1}$$

Now, the determinant of the admittance matrix $|Y|$ is calculated as:

$$|Y| = 0.3 \times 0 - (-0.3) \times (0.1)$$

$$|Y| = 0.03$$

The adjoint of the admittance matrix $\text{adj}(Y)$ is given by:

$$(\text{adj}\,Y) = \begin{bmatrix} 0 & 0.3 \\ -0.1 & 0.3 \end{bmatrix}$$

Evaluating the inverse matrix relation:

$$Z = Y^{-1} = \frac{(\text{adj}\,Y)}{|Y|} = \frac{1}{0.03}\begin{bmatrix} 0 & 0.3 \\ -0.1 & 0.3 \end{bmatrix}$$

From the resulting matrix, the driving-point parameter Z_{22} is extracted as:

$$Z_{22} = \frac{0.3}{0.03} = 10\,\Omega$$

Key Points

- We can use other methods like shorting each port one by one to calculate the admittance parameter matrix.

$$[Y] = \begin{bmatrix} 0.3 & -0.3 \\ 0.1 & 0 \end{bmatrix}, Z_{22} = 10\,\Omega$$

Q6.01.1 MCQ 2001 1M Ans: C

Method 1

By applying KCL at the node where E_1 is measured at the two-port network, we can determine y_{12}.

Step 1: Apply KCL at Node E_1 The total current entering node E_1 must equal the sum of the currents leaving through the parallel branches:

$$I_1 = \frac{E_1}{5} + \frac{E_1 - E_2}{20}$$

Grouping the coefficients for the node voltage variables E_1 and E_2:

$$I_1 = \left(\frac{1}{5} + \frac{1}{20}\right) E_1 - \frac{1}{20} E_2$$

$$I_1 = \frac{1}{4} E_1 - \frac{1}{20} E_2 \quad \text{— (1)}$$

Step 2: Compare with Standard Two-Port Y-Parameter Equations Comparing with

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

By mapping the calculated coefficients from equation (1) to the standard parameters, we extract Y_{12} directly:

$$Y_{12} = -\frac{1}{20}\,\text{S}$$

Method 2

y_{12} is given by

$$y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0} \quad \text{— (1)}$$

Thus we need to short Port 1 to make $V_1 = E_1 = 0$.

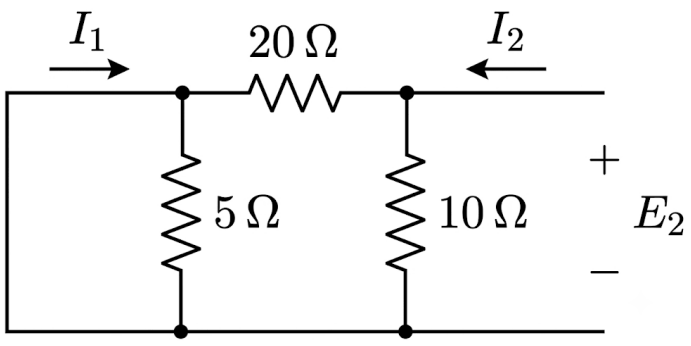


Fig. Solution Q6.01.1

From the figure,

$$I_1 = -\frac{E_2}{20} \implies y_{12} = -0.05S$$

$$\boxed{(c) - 0.05mho}$$

Q6.01.2 MCQ 2001 2M Ans: C ● ● ○

Applying Kirchhoff's Voltage Law (KVL) in the left mesh:

$$E_1 = 2I_1 + 4(I_1 + I_2) - 10E_1$$

$$11E_1 = 6I_1 + 4I_2$$

$$E_1 = \frac{6}{11}I_1 + \frac{4}{11}I_2 \quad \cdots\cdots (1)$$

Applying KVL in the right mesh:

$$E_2 = 4(I_1 + I_2) - 10E_1$$

Substituting the value of E_1 from equation (1) into the output loop equation:

$$E_2 = 4I_1 + 4I_2 - \frac{60}{11}I_1 - \frac{40}{11}I_2$$

$$E_2 = -\frac{16}{11}I_1 + \frac{4}{11}I_2 \quad \cdots\cdots (2)$$

Comparing equations (1) and (2) with the standard two-port network impedance (Z) parameter equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Where $V_1 = E_1$ and $V_2 = E_2$. By direct comparison, we get:

$$Z_{11} = \frac{6}{11}\Omega \quad \text{and} \quad Z_{21} = -\frac{16}{11}\Omega$$

$$\boxed{(c) z_{11} = \frac{6}{11}\Omega; z_{21} = -\frac{16}{11}\Omega}$$

Q6.01.3 NAT 2001 5M Ans: * ● ● ○

Given the 2-port admittance (Y) parameter equations:

$$I_1 = 2E_1 - 0.5E_2 \quad \cdots\cdots (1)$$

$$I_2 = 4.8E_1 + E_2 \quad \cdots\cdots (2)$$

For a load admittance $Y_L = 0.2\,\text{S}$ connected at port 2 the relationship between port 2 current and voltage is

$$I_2 = -E_2 \times 0.2$$

Substituting this value into equation (2):

$$-E_2 \times 0.2 = 4.8E_1 + E_2$$

$$-1.2E_2 = 4.8E_1$$

$$E_2 = -4E_1 \quad \cdots\cdots (3)$$

(a) Finding E_2 for $E_1 = 10\angle 0^\circ$ V: Substituting E_1 into equation (3):

$$E_2 = -4(10\angle 0^\circ) = -40\angle 0^\circ \text{ V}$$

(b) Finding E_2 for $I_1 = 10\angle 0^\circ$ A: From equation (3), we have $E_1 = \frac{-E_2}{4}$. Substituting this into equation (1):

$$I_1 = 2\left(\frac{-E_2}{4}\right) - 0.5E_2$$

$$I_1 = -E_2$$

Given $I_1 = 10\angle 0^\circ$ A:

$$E_2 = -10\angle 0^\circ \text{ V}$$

(c) Finding E_2 for the source with internal resistance:
Applying KVL to the input loop of the modified circuit considering the new specifications:

$$E_1 = 10\angle 0^\circ - 0.25I_1$$

$$I_1 = 40\angle 0^\circ - 4E_1$$

Equating this expression for I_1 with equation (1):

$$40\angle 0^\circ - 4E_1 = 2E_1 - 0.5E_2$$

$$40\angle 0^\circ = 6E_1 - 0.5E_2$$

Substituting $E_1 = \frac{-E_2}{4}$ from equation (3):

$$40\angle 0^\circ = 6\left[\frac{-E_2}{4}\right] - 0.5E_2$$

$$E_2\left[\frac{-3}{2} - \frac{1}{2}\right] = 40\angle 0^\circ$$

$$E_2 = -20\angle 0^\circ \text{V}$$

$$\boxed{(a)E_2 = -40\angle 0^\circ \text{ V}, (b)E_2 = -10\angle 0^\circ \text{ V}, (c)E_2 = -20\angle 0^\circ \text{V}}$$

Q6.99.1 MCQ | 1999 | 1M | Ans: A | ● ○ ○

Method 1

Given the open-circuit impedance (Z) parameter relationships of the two-port network:

$$V_1 = 2I_1R + I_2R \quad \cdots\cdots (1)$$

$$V_2 = I_1R + 2I_2R$$

To find the hybrid (h) parameter h_{21}, we need to express I_2 as a function of I_1 and V_2. Rearranging the second equation to solve for I_2:

$$2I_2R = -I_1R + V_2$$

Dividing both sides by $2R$:

$$I_2 = -\frac{I_1R}{2R} + \frac{V_2}{2R}$$

$$I_2 = -\frac{I_1}{2} + \frac{V_2}{2R} \quad \cdots\cdots (2)$$

The standard defining equation for the hybrid (h) parameters at port 2 is:

$$I_2 = h_{21}I_1 + h_{22}V_2$$

Comparing equation (2) with the standard hybrid parameter equation, the coefficient of I_1 gives the forward current gain $h_{21} = -\frac{1}{2}$

Method 2

By definition, the short-circuit forward current gain hybrid parameter h_{21} is given by:

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$$

Thus, we need to short-circuit port 2 to make $V_2 = 0$.

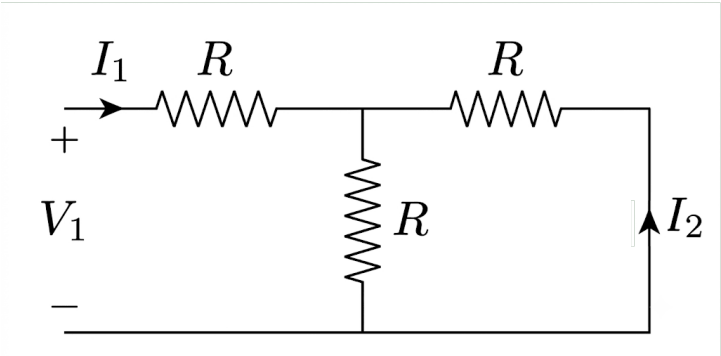


Fig. Solution Q6.99.1

Applying the current division rule at the central node where the total input current I_1 splits between the parallel branches: The short-circuit current flowing in the output branch is opposite in direction to the standard reference direction of I_2. Therefore:

$$I_2 = -I_1 \times \frac{R}{R+R}$$

$$I_2 = -\frac{I_1}{2}$$

Finding the ratio of $\frac{I_2}{I_1}$ under the condition $V_2 = 0$:

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}$$

$$\boxed{(a)\ -\frac{1}{2}}$$

Q6.98.1 MCQ 1998 1M Ans: B ●○○

Given the admittance (Y) matrix of the two-port network:

$$Y = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$$

From the matrix, the individual admittance parameters are:

$$Y_{11} = 0, \quad Y_{12} = -\frac{1}{2}$$

$$Y_{21} = \frac{1}{2}, \quad Y_{22} = 0$$

Since $Y_{12} \neq Y_{21}$ (as $-\frac{1}{2} \neq \frac{1}{2}$), the property of reciprocity is violated. Hence, the network is **non-reciprocal**.
The definitive terminal equations for the input current I_1 and output current I_2 are:

$$I_1 = Y_{11}V_1 + Y_{12}V_2 = -\frac{1}{2}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 = \frac{1}{2}V_1$$

Now, analyzing the network properties based on these parameters: Since Y_{12} is negative, the internal dependent source can supply power to external circuits under specific conditions, indicating that it is an **active network**.

(b) Non-reciprocal and Active

Q6.94.1 NAT 1994 1M Ans: *

For a network to be reciprocal the determinant of its transmission parameter matrix should be equal to 1.

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$|T| = AD - BC = 1$$

Condition for reciprocity: $AD - BC = 1$

Q6.93.1 NAT 1993 2M Ans: *

The composite circuit consists of two networks connected in parallel:

1. **First Network**: A series capacitor branch with admittance s (where $s = j\omega C$ and $C = 1\,\text{F}$).
2. **Second Network**: A resistive T-network containing $1\,\Omega$, $2\,\Omega$, and $3\,\Omega$ resistors.

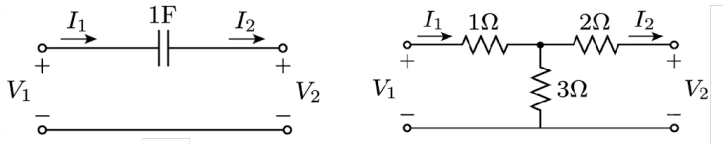


Fig. Solution Q6.93.1

Step 1: Analyze the Second Network (Resistive T-Network) To find its admittance (Y) parameters, we convert the T-network (Wye configuration) into an equivalent Π-network (Delta configuration). Given resistances in the wye configuration: $R_1 = 1\,\Omega$, $R_2 = 2\,\Omega$, and $R_3 = 3\,\Omega$.
The delta resistances are calculated as follows:

$$R_a = 1 + 3 + \frac{1 \times 3}{2} = \frac{11}{2}\,\Omega$$

$$R_b = 1 + 2 + \frac{1 \times 2}{3} = \frac{11}{3}\,\Omega$$

$$R_c = 3 + 2 + \frac{3 \times 2}{1} = 11\,\Omega$$

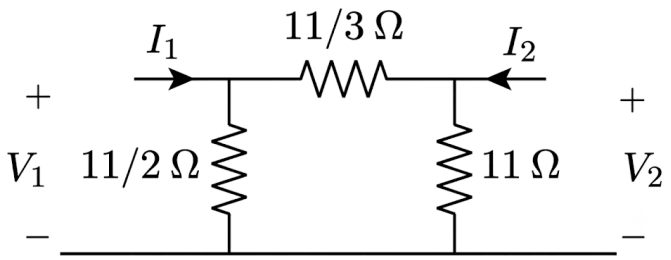


Fig. Solution Q6.93.1

Writing Kirchhoff's Current Law (KCL) at the input node (Node 1) and output node (Node 2) for this Π-network: At the input node:

$$-I_1 + \frac{V_1}{\frac{11}{2}} + \frac{V_1 - V_2}{\frac{11}{3}} = 0$$

$$I_1 = V_1\left(\frac{2}{11} + \frac{3}{11}\right) - \frac{3}{11}V_2$$

$$I_1 = \frac{5}{11}V_1 - \frac{3}{11}V_2 \quad \cdots\cdots (1)$$

At the output node:

$$-I_2 + \frac{V_2}{11} + \frac{V_2 - V_1}{\frac{11}{3}} = 0$$

$$I_2 = -\frac{3}{11}V_1 + V_2\left(\frac{1}{11} + \frac{3}{11}\right)$$

$$I_2 = -\frac{3}{11}V_1 + \frac{4}{11}V_2 \quad \cdots\cdots (2)$$

Thus, the admittance matrix $[Y_2]$ for the second network is:

$$[Y_2] = \begin{bmatrix} \frac{5}{11} & -\frac{3}{11} \\ -\frac{3}{11} & \frac{4}{11} \end{bmatrix} \text{S}$$

Step 2: Analyze the First Network (Capacitor Branch) For a single series element with admittance s:

$$I_1 = (V_1 - V_2)s = sV_1 - sV_2$$

$$I_2 = (V_2 - V_1)s = -sV_1 + sV_2$$

Thus, the admittance matrix $[Y_1]$ for the first network is:

$$[Y_1] = \begin{bmatrix} s & -s \\ -s & s \end{bmatrix}$$

Step 3: Combine Parallel Networks When two-port networks are connected in parallel, their individual admittance parameters are added to find the equivalent admittance matrix $[Y_{eq}]$:

$$[Y_{eq}] = [Y_1] + [Y_2]$$

$$[Y_{eq}] = \begin{bmatrix} \frac{5}{11} + s & -\frac{3}{11} - s \\ -\frac{3}{11} - s & \frac{4}{11} + s \end{bmatrix}$$

Key Points

- For parallel-connected two-port networks, the overall admittance matrix is the sum of the individual admittance matrices: $[Y_{eq}] = [Y_1] + [Y_2] + \cdots + [Y_n]$.
- A single series element with admittance Y connected between terminal pairs has a Y-parameter matrix defined as:

$$[Y] = \begin{bmatrix} Y & -Y \\ -Y & Y \end{bmatrix}$$

$$\boxed{[Y] = \begin{bmatrix} \frac{5}{11} + s & -\frac{3}{11} - s \\ -\frac{3}{11} - s & \frac{4}{11} + s \end{bmatrix}}$$

Q6.92.1 MSQ 1992 2M Ans: B,C ● ○ ○

The conditions for reciprocity include

- For impedance (Z) parameters : $z_{12} = z_{21}$
- For admittance (Y) parameters : $y_{12} = y_{21}$
- For hybrid (H) parameters : $h_{12} = -h_{21}$
- For inverse hybrid (G) parameters : $g_{12} = -g_{21}$
- For transmission (ABCD) parameters : $AD - BC = 1$

Out of the given options only (B) and (C) satisfy the conditions for reciprocity.

$$\boxed{(b) y_{21} = y_{12}, (c) h_{21} = -h_{12}}$$

Q6.91.1 MCQ 1991 1M Ans: D ● ○ ○

When two 2 port networks are connected in cascade, i.e. in a manner such that the output of the first network is the input of the next, the equivalent parameters are obtained by multiplication of their individual ABCD parameters.

For Network 1,

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

For Network 2,

$$\begin{bmatrix} V_1' \\ I_1' \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}$$

Now since $V_2 = V_1'$ and $-I_2 = I_1'$, therefore the equivalent transmission parameters include :

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}$$

Thus,

$$\begin{bmatrix} A_{eq} & B_{eq} \\ C_{eq} & D_{eq} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}$$

$$\boxed{(d) ABCD \text{ parameter matrix}}$$

Q6.90.1 MCQ | 1990 | 2M | Ans: A | ● ● ○

Writing Kirchhoff's Current Law (KCL) at the input node (Node 1):

$$I_1 = \frac{V_1}{1} + \frac{V_1 - V_2}{2}$$

$$I_1 = 1.5V_1 - 0.5V_2 \quad \cdots\cdots (1)$$

Writing KCL at the output node (Node 2), noting the presence of the dependent current source $3I_1$:

$$I_2 = 3I_1 + \frac{V_2 - V_1}{2}$$

Substituting the expression for I_1 from equation (1) into the output node equation:

$$I_2 = 3(1.5V_1 - 0.5V_2) + 0.5V_2 - 0.5V_1$$

$$I_2 = 4.5V_1 - 1.5V_2 + 0.5V_2 - 0.5V_1$$

$$I_2 = 4V_1 - V_2 \quad \cdots\cdots (2)$$

Expressing equations (1) and (2) in standard two-port admittance (Y) parameter matrix form:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1.5 & -0.5 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

To find the open-circuit impedance (Z) parameters, we compute the inverse of the admittance (Y) matrix:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1.5 & -0.5 \\ 4 & -1 \end{bmatrix}^{-1} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

First, find the determinant of the Y-matrix (ΔY):

$$|Y| = (1.5)(-1) - (-0.5)(4) = -1.5 + 2 = 0.5$$

Now, computing the inverse matrix elements:

$$[Z] = [Y]^{-1} = \frac{adj(Y)}{|Y|} = \frac{1}{0.5}\begin{bmatrix} -1 & 0.5 \\ -4 & 1.5 \end{bmatrix} = 2\begin{bmatrix} -1 & 0.5 \\ -4 & 1.5 \end{bmatrix}$$

$$[Z] = \boxed{\begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}}$$

Mistakes to be avoided

- **Dependent Source Substitution**: When working with dependent sources, eliminate internal variables like I_1 early by substituting them in terms of node voltages (V_1 and V_2) before arranging coefficients into the matrix.

$$\boxed{(a)\ \begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}}$$

Q6.89.1 MSQ 1989 1M Ans: C,D ● ○ ○

For a transfer function $H(s) = \frac{N(s)}{D(s)}$ of any stable, physical, linear time-invariant (LTI) two-port network, the location of poles and zeros in the complex frequency s-plane ($\sigma + j\omega$) must satisfy strict realizability and stability conditions.

Let us evaluate the validity of each given statement:

- **Statement A is incorrect**: For a standard network transfer function, zeros (roots of $N(s) = 0$) represent frequencies at which the network output is nullified. They can lie anywhere in the complex s-plane—including the right-half plane (RHP)—without violating causality or stability. Networks containing zeros in the right-half plane are termed *non-minimum phase networks*.
- **Statement B is incorrect**: The poles (roots of $D(s) = 0$) govern the natural response of the network. If any pole lies strictly in the right-half of the s-plane ($\sigma > 0$), the natural response contains an exponentially growing term ($e^{\sigma t}$), causing the physical system to be unstable. Thus, poles cannot lie "anywhere" in the s-plane.
- **Statement C is correct**: If a system has a repeated (multiple-order) pole on the imaginary axis ($j\omega$-axis), the time-domain response will include a term of the form $t \cdot \sin(\omega t)$ or $t \cdot \cos(\omega t)$. Because of the multiplying factor t, the amplitude grows infinitely over time, rendering the network unstable. Therefore, any poles lying on the imaginary axis must be simple (first-order) to yield a bounded, sustained sinusoidal response.
- **Statement D is correct**: A single, simple pole can reside at the origin ($s = 0$). This structurally represents a perfect integrator in system behavior (physically implemented by configurations like an ideal series capacitor or parallel inductor), which produces a stable bounded step response.

Therefore, the correct options representing properties of a physical two-port network are C and D.

(c) the poles lying on the imaginary axis must be simple, (d) a pole may lie at origin

Key Points

- **System Stability Summary**:

$$\begin{aligned} \text{Left-Half Plane (LHP) Poles} &\implies \text{Absolutely Stable (Exponential Decay)} \\ \text{Simple } j\omega\text{-axis / Origin Poles} &\implies \text{Marginally Stable (Sustained Oscillation / Constant)} \\ \text{Repeated } j\omega\text{-axis Poles} &\implies \text{Unstable (Linearly Ramp-Amplified Output)} \\ \text{Right-Half Plane (RHP) Poles} &\implies \text{Unstable (Exponential Growth)} \end{aligned}$$

- Zeros only alter the phase characteristics and minimum delay properties of the transfer network; their location does not jeopardize the bounded-input bounded-output (BIBO) stability framework.

Mistakes to be avoided

- **:** Never mix the constraints of poles and zeros. Remember that poles handle stability (restricted boundary), whereas zeros handle phase minimums (unrestricted boundary except in specific driving-point functions).
- **:** Do not assume all $j\omega$-axis poles mean instability. A single pole at the origin or a single pair of conjugate imaginary poles is perfectly permissible for a physical passive network. It only causes instability if those roots are repeated.

Q6.89.2 MCQ 1989 1M Ans: A ● ○ ○

When the value of the determinant of the transmission parameter matrix is 1, i.e., $|T| = 1$, or $|AD - BC| = 1$, it implies reciprocity of the network.

(a)reciprocal network

Q6.88.1 MCQ | 1988 | 1M | Ans: C | ● ○ ○

When two 2 port networks are connected in parallel, i.e. in a manner such that the net input and output currents are the sum of the input and output current of each port and input and output voltages of each port are equal, the equivalent parameters are obtained by addition of their individual Y parameters.

For Network 1,

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

For Network 2,

$$\begin{bmatrix} I_1' \\ I_2' \end{bmatrix} = \begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix} \begin{bmatrix} V_1' \\ V_2' \end{bmatrix}$$

Now since $V_{1eq} = V_1 = V_1'$,$V_{2eq} = V_2 = V_2'$, $I_{1eq} = I_1 + I_1'$ and $I_{2eq} = I_2 + I_2'$, therefore the equivalent admittance parameters include :

$$\begin{bmatrix} I_{1eq} \\ I_{2eq} \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} + \begin{bmatrix} I_1' \\ I_2' \end{bmatrix}$$

Thus,

$$\begin{bmatrix} I_{1eq} \\ I_{2eq} \end{bmatrix} = \left(\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} + \begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix} \right) \begin{bmatrix} V_{1eq} \\ V_{2eq} \end{bmatrix}$$

Hence, $[Y_{eq}] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} + \begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix}$

$\boxed{(c) y\text{-parameters}}$

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